

CYBER ELIAS ACADEMY

TEACHING MANUAL

Computer Repairs

Session Roadmap · Lesson Notes · Concept Diagrams · Answer Keys

READY TO TEACH

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OVERVIEW

Course Roadmap

What we cover in every session

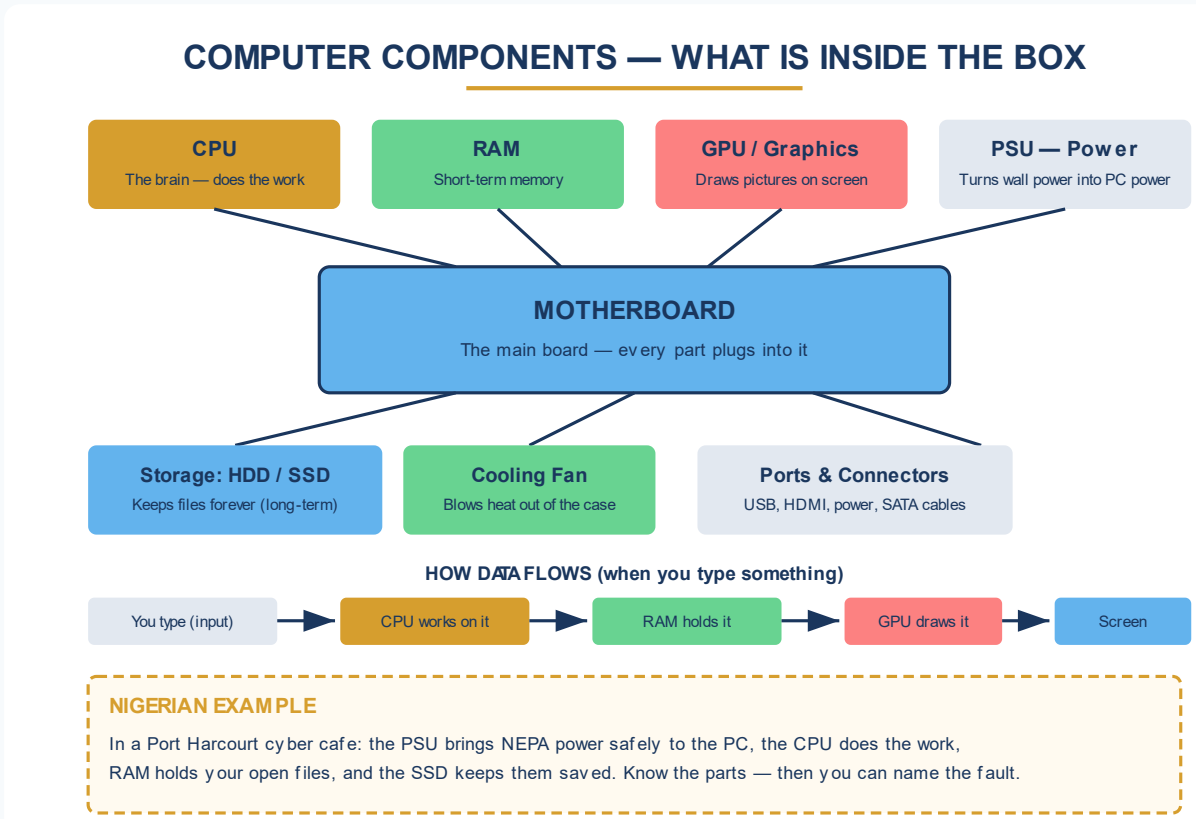
#	SESSION TOPIC	WEEK / MODULE	DURATION
01	Computer Components	Week 1 â€" Module: Components and Safety	105 minutes (1 hour 45 minutes)
02	Safety and Tools	Week 1 â€" Module: Components and Safety	105 minutes (1 hour 45 minutes)
03	Disassembly Reassembly	Week 1 â€" Module: Components and Safety	105 minutes (1 hour 45 minutes)
04	Diagnostic Workflow	Week 2 â€" Module: Diagnostics	105 minutes (1 hour 45 minutes)
05	Hardware Diagnostics	Week 2 â€" Module: Diagnostics	105 minutes
06	Software Diagnostics	Week 2 â€" Module: Diagnostics	105 minutes
07	Component Replacement	Week 3 â€" Module: Repair and Maintenance	105 minutes
08	Cleaning Maintenance	Week 3 â€" Module: Repair and Maintenance	105 minutes
09	OS Installation	Week 3 â€" Module 3: Repair and Maintenance	105 minutes
10	Customer Service	Week 4 â€" Module 4: Professional Practice	105 minutes
11	Service Reports	Week 4 â€" Module 4: Professional Practice	105 minutes
12	Final Project Capstone	Week 4 â€" Module 4: Professional Practice (Capstone)	105 minutes (plus a short certificate sign-off)

How to use this book: Each session below is a complete, ready-to-teach lesson note. Section A tells you what to prepare, B lists the outcomes, D–F are the teaching content, G–H are your demos and practicals, and the knowledge check with answer key is at the end. The concept diagram for each session is embedded as a visual aid.

SESSION 01 OF 12

Computer Components

CONCEPT DIAGRAM — SESSION 01



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 1 â€" COMPUTER COMPONENTS: WHAT IS INSIDE THE BOX

A. SESSION IDENTIFICATION

Item	Detail
Session number	1 of 12
Session title	Computer Components â€" What Is Inside the Box
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner (first contact with computer hardware)
Prerequisite knowledge	Basic computer skills: typing, saving files, opening programs. No hardware experience needed.

Item	Detail
Connection to previous session	This is the first session of the Computer Repairs course. It assumes only the Typing & Computer Basics course.
Connection to the course	This session gives every learner the vocabulary of the trade â€" the names and jobs of every part inside a computer. Every later session (safety, disassembly, diagnostics, repair, the capstone) depends on knowing these parts by name and function.
Preparation before class	Physical samples of motherboard, CPU (+ cooler), RAM, HDD, SSD, PSU, GPU, SATA cable, power cable; at least one powered-off desktop computer; projector or large screen; printed Component Identification Chart (one per learner); anti-static wrist straps; component name cards for a labelling race.

Safety note for this session: Learners will handle real components. Even though this is an "identification" session, every learner must wear an anti-static wrist strap or regularly touch the bare metal of the case before touching any board. The machine used for demonstration must be unplugged from the wall and the power button pressed once to discharge residual electricity. No component is opened, and no mains-powered work is done. All handling is done by the edges only. Where the academy has only one working demonstration machine, learners should be grouped and supervised.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Identify and name** all major internal components of a desktop computer â€" motherboard, CPU and its heatsink/fan, RAM, HDD, SSD, PSU, GPU, expansion cards, cooling fans, and the main ports and connectors.
2. **Explain the function** of each major component in their own words, using a familiar comparison.
3. **Describe how data flows** through a computer from a user's input to the screen, naming the components involved at each step.
4. **Distinguish clearly between RAM and storage**, and between HDD and SSD, including speed, durability and cost differences.
5. **Explain the relationship** between components â€" how they connect to the motherboard and why the system needs all of them to work.
6. **Compare a desktop and a laptop**, stating at least three differences in how components are arranged and serviced.
7. **Handle components safely** â€" holding boards by their edges, never touching gold contacts, and using anti-static protection.
8. **Identify components physically** on a real open machine or on labelled photographs and diagrams.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- Physical samples of each component: motherboard, CPU with heatsink/fan, RAM stick(s), one HDD, one SSD (SATA type if available), one M.2/NVMe stick if available, PSU, a dedicated GPU card, a spare expansion card (sound/network card), a case cooling fan, SATA data cable, SATA power connector, 24-pin ATX power connector, and a 8-pin CPU power connector.
- At least one desktop computer, powered off and unplugged, with its side panel removed or removable, for live identification.
- A laptop (any working or non-working unit) for the desktop-vs-laptop comparison.
- Projector or large screen connected to the instructor's computer for showing photographs and diagrams.

Software:

- A simple slideshow or photo gallery of computer internals (instructor's own photos or freely available images). No special software required.
- Optional: a free system-information viewer already built into Windows (Task Manager â†' Performance tab) to show live CPU, RAM and disk activity. This is only a demonstration aid and is not essential.

Safety gear:

- Anti-static wrist straps (one per learner, or one per pair).
- Anti-static mat for the demonstration table (one is enough).
- Clean, dry, well-lit table surface. No food or drink on the table.

Sample / fault components:

- A dead/faulty RAM stick and a dead hard drive (clearly labelled "FAULT â€" DO NOT FIT") to show what failed parts look like. These are for looking only.
- A burnt-looking PSU (clearly labelled) to introduce the idea that components can fail. Do not open it.

Worksheets (printed):

- Worksheet 1.1 â€" "Component Identification Chart" (one per learner): a table with columns for component name, what it does, one key fact, and one common problem.
- Worksheet 1.2 â€" "Data Flow Diagram" (one per learner): a blank flow diagram the learner fills in during the guided exercise.
- Component name cards (small cards with one component name each) for the opening labelling race.

Tools:

- A Phillips PH2 screwdriver (only for the instructor to remove a panel; learners do not unscrew anything this session).
- A small torch/flashlight to point out parts inside the case.

Alternatives for a learner or academy with limited equipment:

- If no real components are available, use printed or projected high-quality labelled photographs and diagrams. The identification exercise can be done entirely from labelled photos.
- If there is only one desktop machine, rotate small groups of three to four learners around it while the others work with photo sheets and the handling samples.
- If a learner has low vision or shaky hands, pair them with a careful partner and let them identify and describe while the partner holds the part.

D. OPENING ACTIVITY (6 minutes)

"What Is Inside the Box?"

Instructor setup: Bring a sealed cardboard box to the front. On a side table, hidden under a cloth, place the loose components (motherboard, RAM, PSU, etc.). Do not reveal them yet. On the projector, keep a blank screen.

What the instructor says:

> *"Good morning, everyone. Welcome to the Computer Repairs course. Before we open a single computer, I want to play a game. I have a box here. Inside this box is a mystery item. I want you to guess what it is. I will give you one clue: it is something you all use, some of you every single day, and most of you have never seen the inside of it."*

Take three or four guesses from around the room.

> *"You are all close. The answer is: it is a computer â€" or at least, all the parts of one. Now here is the shocking part. Look at Chinedu's phone, look at the cyber cafÃ© machines down the road, look at the school computer lab â€" inside, they are all made of the same handful of parts, just arranged differently.*
>
> *Here is my question for you: if you took a computer apart, could you name every single thing inside it? Raise your hand if you could."*

Most hands stay down.

> *"By the end of today, every hand in this room will be up. You will know the name, the job, and the feel of every major component. And here is why that matters in Port Harcourt: when a customer walks into your shop and says 'my system is*

slow,' the technician who knows the parts can ask the right questions and guess the right cause. The technician who only knows how to click does not.

Now â€" let us meet the parts."

Then reveal the hook: Pull the cloth off the side table and hold up the motherboard â€" the strangest-looking part. Ask: *"Is this a radio? A small road? A city map?"* Let them guess. Say:

> *"This is the **motherboard** â€" the main board every other part plugs into. Everything we learn today connects back to this single board. By the end of the session, you will be able to point to every part on this board and tell me what it does."*

Expected learner responses: Learners may guess "brain," "memory," "hard disk," or stay quiet. Accept all answers, because some will be partly right. Some will confuse the motherboard with a "chip" or the whole case. Note these misconceptions silently â€" you will address them in Section F.

Why this opener works: It creates curiosity about a familiar object, connects the lesson to real Port Harcourt life (cyber cafÃ©, school lab), and turns a dry list of parts into a mystery to be solved. It also sets the promise of the session clearly and measurably.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral round-robin. Put the questions on the projector. Ask different learners each time; do not let one person answer everything.

Question 1: "What is the difference between hardware and software?"

- Expected answer: Hardware is the physical parts you can touch (the keyboard, the screen, the parts inside). Software is the programs and instructions (Windows, a game, a browser) â€" you cannot touch it.
- Common gap: Some learners say software is "the internet." Gently correct: the internet is something software uses, not software itself.
- Bridge line: "Today is entirely about hardware â€" the parts you can touch."

Question 2: "When you type a letter on the keyboard and it appears on the screen, is anything physical happening?"

- Expected answer: Yes â€" electricity moves through the parts inside the computer to turn the keypress into a letter on the screen.
- Common gap: "It is magic" or "the computer just knows." Explain that the computer is following instructions, and the parts we learn today are what carry those instructions.

Question 3: "What happens to your typed document when the electricity goes off suddenly?"

- Expected answer: If you had not saved it, it is lost. Saved work stays. This introduces the idea of temporary vs permanent storage, which this session clarifies with RAM and HDD/SSD.
- Common gap: Some learners believe everything is always "saved." Use this as the hook for RAM vs storage.

Question 4: "Name one part inside a computer. Even if you are not sure, guess."

- Expected answer: Any of: memory, hard disk, fan, "the thing that makes it fast." Accept everything.
- Common gap: Many beginners name only the case or the screen. Reassure them that by the end of today they will name eight or more.

Bridge to today (say this):

> *"Think of a computer like a restaurant kitchen. The kitchen has a head chef who does the cooking â€" that is the **CPU**. It has a prep table where ingredients sit while being used â€" that is the **RAM**. It has a store room that keeps ingredients even overnight â€" that is the **storage**. It has a gas cylinder that supplies the fire â€" that is the **PSU**. It has a manager who coordinates everyone â€" that is the **motherboard**. Today, we meet the whole kitchen staff."*

If more than a third of the class cannot answer Question 1 or 2, spend an extra three minutes re-explaining hardware vs software before continuing; the rest of the session depends on that distinction.

F. FULL LECTURE NOTES

F.1 What a computer actually is

The idea. A computer is a machine that does three things over and over, extremely fast: it **takes in information** (input), it **works on that information** (processing), and it **produces a result** (output). It also **keeps information** for later (storage).

Plain-English definition. A computer is an electronic machine made of separate physical parts that work together. Each part does one main job, and no single part can do everything alone. When people say "the CPU is the brain," that is a comparison, not the whole truth â€” a computer needs its memory, storage, power and board just as much as a person needs lungs, stomach and heart, not only a brain.

Why the parts are separate. Have you ever noticed that you can buy a bigger hard disk, or add more memory, without buying a new computer? That is deliberate. By making the parts separate, a computer can be **repaired, upgraded and replaced part by part** instead of being thrown away whole. This modular design is the reason the computer repair trade exists â€” and the reason you have a job to learn.

Real-life comparison. A computer is like a **Nigerian market stall**. The **stall itself** (the case) holds everything. The **stall owner** (CPU) makes the decisions. The **table in front** (RAM) holds only what is being sold right now. The **store behind** (storage) keeps the goods overnight. The **generator** (PSU) supplies power. The **signboard** (GPU/monitor) shows customers what is on offer. If the generator dies, the whole stall stops â€” even if the owner and goods are fine. That is why one failed part can stop a whole computer.

Important limitation / misconception. Learners often think the "computer" is the screen. It is not. The screen (monitor) is an output device â€” a separate thing. The computer itself is the box (the system unit). Confusing the two causes wrong diagnoses later, for example blaming the monitor when the fault is inside the case.

F.2 The motherboard â€” the city where everything meets

Plain-English definition. The **motherboard** is the large flat green (or sometimes black, blue or red) circuit board fixed inside the case. It is the main board that every other internal component plugs into or connects to. A circuit board is a flat board that has tiny metal pathways printed on it, along which electricity travels like traffic on roads.

Why it exists. If every component tried to connect to every other component directly, there would be thousands of messy cables and the machine could never be built or repaired. The motherboard solves this by providing a single meeting point. Every part plugs into the board, and the board carries signals between them.

What it does. The motherboard:

- Holds the **CPU socket** â€” the special square socket where the processor sits.
- Provides **RAM slots** â€” long thin slots where memory sticks click in.
- Provides **expansion slots** (usually PCIe slots) where graphics cards and other add-on cards go.
- Provides **SATA ports** where hard drives and SATA SSDs connect by cable.
- Provides an **M.2 slot** (on modern boards) where NVMe SSDs slot in flat against the board.
- Carries the **main power connector** (the 24-pin ATX connector) where the PSU plugs in.
- Holds the **back I/O panel** â€” the row of ports at the back of the case for USB, network, audio and sometimes video cables.
- Often holds a small coin-shaped **CMOS battery** that keeps the clock and basic settings alive when the machine is off.

Real-life comparison. The motherboard is like **Rumuola junction** in Port Harcourt â€” the central roundabout where many roads meet. Cars (signals) come from all directions and pass through the junction. If the junction is blocked, traffic stops everywhere at once, even though every car is fine.

Realistic repair-shop scenario. A customer brings a desktop that powers on but never shows a picture. You open the case and notice the RAM stick is not clicked fully into its slot on the motherboard. You reseat it and the machine works. Nothing was "broken" â€” a connection on the main board simply was not complete. This is one of the most common quick fixes, and it is why knowing the board matters.

Common misconceptions.

- "The motherboard is the processor." No â€” the processor sits *on* the motherboard, but they are different parts.

- "All motherboards are the same." They are not. Different boards fit different CPU types and memory types, and they are not freely interchangeable. A beginner must never assume a random board fits a random CPU.

Limitations and safety. The motherboard is delicate. Its surface has tiny tracks and small components. Never place a board on a carpet or a metal surface; never bend it; hold it by the edges; and always discharge static before touching it. Some repairs on the board itself (soldering, replacing capacitors) are advanced and are **not** part of this course â€" a beginner should refer those to a specialist.

F.3 The CPU â€" the processor (the brain)

Plain-English definition. The **CPU (Central Processing Unit)** is the small square chip that carries out the instructions of every program. It does the arithmetic and the decision-making. It is often called the "brain" of the computer.

Why it exists. Someone has to actually *do* the work â€" add numbers, compare values, move data. The CPU is the part that executes those instructions, billions of them every second.

What it does. It reads instructions from programs, performs calculations, makes simple decisions (is this number bigger than that one?), and coordinates the rest of the parts by telling them what to do.

Key facts (plainly stated, no invented numbers):

- Two main makers of desktop CPUs are **Intel** and **AMD**.
- Speed is commonly described in **gigahertz (GHz)** â€" a measure of how many billions of cycles it can run per second. Generally a higher GHz means faster work, but it is not the only factor.
- A CPU runs **hot** when working. It therefore has a metal top (the **heat spreader**) and sits under a **heatsink and fan** (the cooler) that carries heat away.
- The cooler is critically important: without it, a CPU can overheat within seconds and shut the machine down, or be permanently damaged.

The heatsink and fan â€" the CPU's cooling team. A **heatsink** is a block of metal with many thin fins. It presses against the CPU to soak up heat. A **fan** blows air across the fins to carry that heat away. Between the CPU and the heatsink, a thin layer of **thermal paste** (a grey, slightly sticky compound) fills the tiny gaps so heat passes through properly. Too little, too old or dried-out paste is a common cause of overheating.

Real-life comparison. The CPU is like the **head chef** in a busy kitchen. The chef cooks very fast and gets very hot. The **heatsink and fan** are like the kitchen's extractor fan and cooling system â€" without them, the chef collapses. The **thermal paste** is like oil in a frying pan, helping heat pass evenly.

Realistic repair-shop scenario. A customer complains their laptop shuts itself off after about twenty minutes of use and the base gets very hot. You open it, find the fan clogged with dust and the thermal paste dried and crusty. You clean the fan and replace the paste, and the shutdown stops. The CPU was never faulty â€" it was simply being cooked. (Full cleaning is Session 8; today you only need to recognise the parts.)

Common misconceptions.

- "A faster CPU always makes the computer faster." Not always. If the computer has too little RAM or a slow hard drive, a faster CPU will not fix the slowness. The slowest part sets the pace, like a convoy that moves at the speed of its slowest vehicle.
- "The CPU is the whole computer." No â€" the CPU is one chip on the board.

Limitations and safety. Never touch the pins under a CPU or the contacts in the CPU socket. Even a tiny bend can stop the machine working. Never power on a computer with the CPU cooler removed. CPU installation (Session 7 covers replacement) requires great care and should be practised only on a machine you are authorised to repair.

F.4 RAM â€" the short-term worktable

Plain-English definition. RAM (Random Access Memory) is the computer's short-term, working memory. It holds the programs and data that the computer is using *right now*. While the power is on, RAM keeps this information; the moment the power goes off, everything in RAM is **erased**.

Why it exists. The CPU works incredibly fast, but permanent storage is much slower to read from and write to. If the CPU had to fetch every tiny piece of data straight from the hard disk, it would spend most of its time waiting. RAM solves this: it is a fast, temporary holding area close to the CPU.

What it does. When you open a program, the computer copies it from storage into RAM. As you work, the CPU reads from and writes to RAM. When you close the program or switch off, RAM is cleared.

Key facts:

- RAM comes in **sticks** (long, thin, flat modules) that click into the RAM slots on the motherboard.
- Capacity is measured in **gigabytes (GB)** â€" 8 GB, 16 GB and 32 GB are common in modern machines.
- More RAM lets more programs stay open at once without slowing down.
- RAM comes in **generations** called **DDR** (DDR3, DDR4, DDR5). Different generations are **not interchangeable** â€" a DDR4 stick will not fit a DDR5 slot, and forcing it will damage the board.

Real-life comparison. RAM is the **prep table in a kitchen**. Ingredients being used right now sit on the table. When the kitchen closes (power off), the table is cleared â€" nothing stays on it overnight. The **store room** (storage) is where ingredients are kept permanently.

Realistic repair-shop scenario. A cyber cafÃ© operator says their machines "start hanging when many browser tabs are open." You check and find only 4 GB of RAM. The fix is not a repair at all â€" it is an **upgrade** to more RAM. This is one of the most common and profitable jobs in a Port Harcourt repair shop, and it exists purely because people confuse "slow" with "broken."

Common misconceptions.

- "RAM and hard disk are the same thing." The single most common beginner mistake. RAM is temporary and small (GB) and empties on power-off; storage is permanent and large (hundreds of GB or TB).
- "More RAM makes the computer store more files." No â€" more RAM lets you *run more at once*. Storing files is the job of storage.
- "Adding RAM always speeds things up." Only if RAM was the bottleneck. If the slowness is from a failing hard drive or malware, more RAM will not help.

Limitations and safety. RAM sticks are delicate. Hold them by their edges and never touch the **gold contacts** â€" the shiny strip along the bottom. Oils from your fingers on those contacts cause poor connections. Always handle RAM with anti-static protection.

(Note: "GB" for RAM and "GB" for storage use the same letters but mean different jobs â€" one is the size of the worktable, the other is the size of the store room.)

F.5 Storage â€" the store room (HDD, SSD, NVMe)

Plain-English definition. Storage is where the computer keeps information **permanently** â€" the operating system, your programs, documents, photos and videos. Unlike RAM, storage keeps its contents when the power is off.

Why it exists. You need your files to survive when the machine is switched off. Storage is the long-term memory of the computer.

The three main types a beginner must know:

1. HDD â€" Hard Disk Drive. Inside an HDD are shiny magnetic platters that spin, with a read/write head that moves across them â€" like an old record player. Because parts move, an HDD is **slower**, can be **damaged by drops or knocks**, and makes soft whirring or clicking sounds while working. It is usually **cheaper per gigabyte** and comes in large sizes (for example 1 TB, 2 TB).

2. SSD â€" Solid State Drive. An SSD has **no moving parts** â€" it stores data in electronic memory chips. It is **much faster** than an HDD, **more durable** (nothing to break when dropped), uses less power, and is silent. It is usually **more expensive per gigabyte** than an HDD.

3. NVMe â€" Non-Volatile Memory Express. NVMe is the **fastest** type of SSD. It plugs flat into a special slot (the **M.2 slot**) directly on the motherboard, rather than connecting by cable like a SATA drive. It is commonly used for the operating system and the programs that must load fast.

SATA â€" mention this because learners will hear it constantly. SATA is a standard for connecting storage drives and is the name of both the cable and the port. A SATA SSD uses the SATA cable; an NVMe SSD slots straight into the board.

Real-life comparison. An HDD is like a **warehouse with a worker on a bicycle** fetching each item â€" it works, but takes time. An SSD is like a **warehouse where every item sits on a numbered shelf within arm's reach** â€" same goods, far faster to get. An NVMe is like a **shelf right beside your hand**.

Realistic repair-shop scenario. A customer says "my computer takes forever to start up â€" I have to wait five minutes." You find the machine runs on an old HDD. You clone the system onto an SSD, and startup drops to seconds. The customer thinks

you performed magic, but you simply moved the data from a spinning warehouse to an electronic shelf. Explaining this plainly is part of good customer service (Session 10).

Common misconceptions.

- "SSDs are just smaller hard drives." No â€" they are a completely different technology (electronic vs mechanical). Size is not the difference; technology is.
- "A 256 GB SSD holds less than a 256 GB HDD." Both hold about the same; they differ in speed and durability, not in how much they can hold for the same number.
- "I can drop an HDD because it is metal." Do not. The internal platters and head are extremely delicate; a fall can destroy it and your data.

Limitations and safety. HDDs must be handled gently and never shaken while running. SSDs are more robust but can still fail, and when they fail they often give little warning. This is why backups matter. "Storage fails" is one of the most common real repair cases, so treat data with respect from the very first session.

F.6 The PSU â€" the power supply unit (the gas cylinder)

Plain-English definition. The **PSU (Power Supply Unit)** is usually a metal box at the top or bottom-back of the case. It takes electricity from the wall socket (which in Nigeria is commonly around 220â€"240 volts alternating current) and converts it into the lower, steady **direct-current** voltages that the computer's parts need (commonly +3.3 V, +5 V and +12 V).

Why it exists. The wall socket supplies one kind of electricity; the delicate components need a different, carefully controlled kind. Without the PSU, the computer cannot run. A computer has no internal battery to fall back on (a desktop, unlike a laptop, has no built-in battery).

What it does. It converts and delivers power, through cables, to the motherboard (the big 24-pin connector), the CPU (the 4-pin or 8-pin connector), the drives and the graphics card.

Key facts:

- Capacity is measured in **watts (W)** â€" for example 400 W, 500 W, 650 W. The PSU must supply enough total power for all the components together.
- It provides several different voltages through different cables.
- A failing PSU can cause **random crashes, failure to start, sudden shutdowns, or damage to other components**.

Real-life comparison. The PSU is like the **generator or gas cylinder** that supplies the fire for the kitchen. If it is weak, everything in the kitchen suffers. If it is faulty, nothing can cook at all, no matter how good the chef (CPU) is.

Realistic repair-shop scenario. (This is the Session's case study, expanded here.) A customer's desktop will not power on at all â€" no lights, no fan spin. You check the wall socket and the cable first (always the simplest things first). Those are fine. You then test the PSU and find it has failed. You replace it. This is one of the most common reasons a computer shows "no sign of life."

Common misconceptions.

- "Any PSU will do; more watts is always better." A PSU must match the machine's needs and, importantly, must be of good quality. A cheap, badly built PSU can fail and damage everything else.
- "I can repair a PSU myself." **No.** Inside the PSU are components that can hold a dangerous charge even when unplugged. A beginner must **never open a PSU.** If it is faulty, replace it. This is a hard safety rule.

Limitations and safety.

- **Always unplug the computer from the wall before opening the case.**
- **Never open, drill, or tamper with the PSU itself** â€" the components inside can store a lethal charge.
- If you smell burning or see smoke, unplug immediately and do not touch it beyond that. Replace the PSU; never try to "fix" a burning one.
- This session does not test or replace the PSU. Those skills come later (Sessions 5 and 7), and even then, the PSU is replaced, not opened.

F.7 The GPU â€" the graphics card (the signboard painter)

Plain-English definition. The **GPU (Graphics Processing Unit)** handles images, video and everything shown on the screen. It turns the computer's data into the picture you see.

Two types:

1. **Integrated graphics** â€” built into the CPU or motherboard. Good enough for everyday work: browsing, typing, watching videos, light school work. It shares the main memory.
2. **Dedicated graphics card (GPU)** â€” a separate card that plugs into an expansion slot (PCIe) on the motherboard. It is far more powerful, has its own memory (called **VRAM**), usually has its own fan, and may draw extra power from the PSU. It is used for gaming, video editing, design and 3D work.

Why it exists. Modern screens show millions of tiny dots (pixels) that must be calculated and redrawn many times per second. The CPU can do this, but a dedicated GPU does it far faster and frees the CPU for other work.

Key facts:

- The two main makers of dedicated GPUs are **NVIDIA** and **AMD**.
- A dedicated card has its own memory and its own cooling.
- Some powerful cards need a dedicated power cable from the PSU.

Real-life comparison. If the CPU is the head chef, the GPU is the **artist who paints the signboard**. The chef cooks; the artist makes it look good to the customer. A busy restaurant needs both, and if the artist is weak, the signboard looks poor even if the food is perfect.

Realistic repair-shop scenario. A customer says their monitor shows "no signal" even though the computer seems to be running (fans spin, lights on). You check the display cable â€” it is plugged into the motherboard's port, not the graphics card's port. Because a dedicated card is fitted, the motherboard port is often disabled. You move the cable to the card and the picture appears. The customer was sure the computer was "dead," but it was a one-minute fix â€” because you understood the GPU.

Common misconceptions.

- "A graphics card is only for games." Not true â€” many design and video jobs need one, and even everyday machines use graphics (built into the CPU).
- "No display always means the GPU is broken." No â€” it may be a loose cable, an unseated RAM stick, or the cable in the wrong port. This is why Session 4 teaches a proper workflow.

Limitations and safety. A dedicated GPU can be large and heavy; support it when removing it so it does not bend in its slot. Some cards keep warm after use; let the machine cool before handling. Never touch the gold contact strip at the bottom of the card.

F.8 Expansion cards, cooling fans, ports and connectors

Expansion cards. An **expansion card** is any add-on card that plugs into an expansion slot to add a feature the board lacks â€” for example a **sound card**, a **network (Ethernet) card**, or a TV tuner card. Many modern motherboards have sound, network and graphics built in, so dedicated cards are less common than they once were, but you will still meet them, especially in older or specialist machines.

Cooling fans. A computer has several fans. The **CPU fan** sits on the CPU heatsink. The **case fans** move air in and out of the case. The **GPU fan** cools the graphics card. The **PSU fan** cools the power supply. **Purpose:** heat is the enemy of electronics; fans and heatsinks keep parts within safe temperatures. Dust is the enemy of fans â€” a dust-clogged fan cannot cool, and the machine overheats. (Cleaning is Session 8.)

Ports and connectors â€” where the outside world plugs in. At the back of the case on the motherboard's I/O panel (and often the front) you will find:

- **USB ports** â€” for keyboards, mice, flash drives, printers.
- **Network (Ethernet/RJ45) port** â€” for a network cable.
- **Audio jacks** â€” for speakers, headphones, microphone.
- **Display ports** â€” HDMI, DisplayPort or VGA, for monitors (on the motherboard if integrated graphics are used, or on the graphics card if one is fitted).
- **Power socket** â€” where the power cable plugs in (at the PSU).

Internal connectors to recognise:

- **24-pin ATX power connector** â€” the big one from the PSU to the motherboard.
- **4-pin/8-pin CPU power connector** â€” a smaller PSU cable near the CPU.
- **SATA data cable** â€” flat, with a small L-shaped connector, links drives to the board.

- **SATA power connector** â€" a wider, flat connector from the PSU to drives.
- **Fan connectors** â€" small 3-pin or 4-pin plugs for fans.
- **Front-panel connectors** â€" tiny plugs from the case's power button, reset button and front USB/audio to the board. These are fiddly and easy to misplace; we handle them carefully in Session 3.

Optical drive (legacy). An **optical drive** (CD/DVD drive) is a bay device that reads and writes discs. It is now largely **legacy** â€" modern computers, especially laptops, often have none. You will still see them in older machines. Recognise it as a storage device with a disc tray, connected by SATA like a hard drive.

Real-life comparison. Ports and connectors are the **doors and gates of a compound**. Each door is shaped for one purpose: the gate for cars (power), the pedestrian gate (USB), the network gate (Ethernet). You cannot drive a car through the pedestrian gate, and you cannot plug a USB cable into an Ethernet port â€" the shapes protect you from yourself.

Common misconceptions.

- "All USB ports are the same speed." They look alike but can differ in speed and purpose. Do not invent exact numbers; simply teach that they are not identical.
- "The VGA/HDMI port is on the wall." Some monitors and older machines use VGA; modern ones use HDMI/DisplayPort. Learn to look for the port on the machine and match it to the cable.

Limitations and safety. Never force a connector â€" every plug has one correct orientation. USB plugs only fit one way (though USB-C fits either way). Forcing a connector bends pins and damages ports. Power off and unplug before plugging or unplugging internal cables.

F.9 How it all fits together â€" the relationship and data flow

The relationship in one paragraph. The **motherboard** connects everything. The **CPU** does the thinking and sits under a **heatsink and fan** to stay cool. The **RAM** is the fast workspace the CPU uses right now. The **storage** keeps everything permanently. The **PSU** feeds electricity to all of them. The **GPU** paints what you see. **Fans** keep the heat down. **Ports and connectors** bring the outside world in and out. Remove any one and the whole system fails or runs badly â€" they are a team, not a list.

How data flows â€" follow one keypress from finger to screen. This is the single most useful mental model a repair technician can carry.

1. **Input:** You press the letter **A** on the keyboard.
2. The signal travels through the **USB port** and along the motherboard to the CPU.
3. The **CPU** processes the instruction: "show an A."
4. The CPU uses **RAM** as its workspace while it decides what to do.
5. The CPU sends drawing instructions to the **GPU**.
6. The **GPU** turns that into the picture and sends it out through a **display port** (HDMI/DisplayPort/VGA).
7. The **monitor** shows the letter **A**.
8. Meanwhile, if the program is saving the document, the CPU writes the data to **storage** so it survives when the power goes off.
9. Throughout all of this, the **PSU** supplies the electricity and the **fans** keep everything cool.

Why this matters in the shop. When a customer says "my typing is delayed," you now have a map: is the keyboard/port at fault (step 2)? The CPU overloaded (step 3)? Too little RAM (step 4)? The GPU or cable (step 6)? The storage busy (step 8)? A good technician walks the flow, one step at a time. That is exactly the skill Session 4 builds into a formal workflow.

Real-life comparison. Think of sending a **letter through the Nigerian postal system**. You write it (input), it goes to the local post office (motherboard), the sorting clerk reads the address (CPU), uses a desk to sort it (RAM), the delivery rider takes it out (GPU), and it arrives at the house (monitor). If the rider is missing, the letter never arrives, even though you wrote it perfectly.

Common misconceptions.

- "The CPU does everything." No â€" the CPU directs, but RAM, storage, GPU and PSU each do their own jobs.
- "Data flows only one way." It flows constantly in both directions: input comes in, output goes out, and the CPU reads and writes memory and storage continuously.

Limitations and safety. Do not carry this table too far into advanced territory (buses, chipset, cache) â€" that is beyond a beginner course. The goal is a working mental map, not an engineering degree.

F.10 Desktop versus laptop â€” same parts, different arrangement

A **laptop** contains the same core parts as a desktop (CPU, RAM, storage, motherboard, cooling, power) but everything is **smaller, tightly packed and often harder to reach and replace**.

Aspect	Desktop	Laptop
Size and spacing	Large case, parts spaced out, easy to reach	Compact, tightly packed, few screws
Upgrades	Easy â€” RAM, storage, GPU, PSU often replaceable	Limited â€” RAM and storage sometimes replaceable; CPU and GPU usually fixed to the board
Power	PSU inside the case, fed from wall	Internal battery plus a charger; runs on battery when unplugged
Cooling	Larger fans and heatsinks	Small fans and heat pipes; clogs with dust more easily
Screen and input	Separate monitor, keyboard, mouse	Built in
Repair difficulty	Beginner-friendly	More delicate; needs care and sometimes special tools
Common faults	Failed PSU, dust, loose RAM, dead HDD	Overheating/dust, worn battery, broken screen/hinge, keyboard faults

Real-life comparison. A desktop is like a **bungalow** â€” roomy, easy to walk around, easy to change furniture. A laptop is like a **well-packed suitcase** â€ everything fits, but to find one item you may have to disturb the others.

Safety note specific to laptops: Laptops are more delicate and their batteries can be a hazard if punctured or swollen. This course's Week 1 disassembly practical (Session 3) uses **desktops**. Laptop disassembly should be attempted only with supervision and only once you have practised on desktops. Never open a laptop battery or pierce a swollen battery â€ report it to the instructor.

Why this matters in Port Harcourt. Many customers bring laptops because that is what they own. Recognising the differences helps you set honest expectations: "Madam, this part is soldered to the board, so it cannot simply be swapped â€ let me check what the real fault is before I promise anything." Honest assessment is part of the professional standard (Session 11).

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The unplugged desktop with side panel removed, on the anti-static mat, on a clean, well-lit table.
- Loose component samples in a row, each with a name card placed face-down or hidden so learners must recall.
- The wrist strap, drawn on and clipped to bare metal of the case.
- The projector showing a labelled motherboard diagram.
- The dead/fault parts (RAM, HDD, PSU) clearly labelled "FAULT â€ LOOK ONLY."
- A torch to point inside the case.

G.2 Demonstration 1 â€ live identification (8 minutes)

Say: "I am going to open up a real computer and name every part inside it. Follow my finger with your eyes, then write the name on your chart."

Do this in order, pointing with the torch (never with a screwdriver, to avoid slips):

1. Point to the big board: "This is the **motherboard** â€ the main board everything plugs into."
2. Point to the square chip under the fan: "This is the **CPU**, under its **heatsink and fan**. The CPU is the brain; the fan keeps it cool."
3. Point to the long thin sticks: "These are **RAM** sticks. Short-term memory. Notice the shiny **gold contacts** along the bottom â€ we never touch those."
4. Point to the boxy drive: "This is a **hard disk drive** â€ permanent storage, with moving parts inside."
5. Point to an SSD or M.2 stick: "This is an **SSD** or **NVMe** â€ permanent storage with no moving parts, much faster."

6. Point to the metal box with many cables: "This is the **power supply** â€" it feeds electricity to everything. **We never open this box.**"
7. Point to the large card with its own fan: "This is the **graphics card** â€" it paints what you see on the screen."
8. Point to the spinning fan: "This is a **cooling fan** â€" it moves heat out of the case."
9. Point to the row of coloured plugs on the board: "These are **connectors** â€" power, SATA and front-panel cables plug in here."

Expected result: Learners can repeat each name and a one-line job for each part.

Common errors to watch: The instructor must not use a metal tool to point at a powered board (it is unplugged, but the habit matters). Do not let learners touch the CPU pins or the RAM gold contacts.

Questions to ask during the demo:

- "Which part stores your files permanently?" (Storage.)
- "Which part forgets everything when the power goes off?" (RAM.)
- "Which part would you change to make games run better?" (The GPU â€" or add RAM; accept both with reasoning.)
- "Why does the CPU need a fan?" (It gets hot when working.)

G.3 Demonstration 2 â€" correct, safe component handling (7 minutes)

Say: "Now I will show you how to hold these parts so you do not damage them."

Do this on the projector, slowly:

1. Put on the anti-static wrist strap and clip it to bare metal. "This keeps static electricity from jumping into the parts."
2. Pick up the RAM stick. "I hold it **by its edges**. I do not touch the gold contacts at the bottom."
3. Pick up the GPU. "I hold it by the metal bracket and the edges, and I support its weight so it does not bend."
4. Pick up the HDD. "I hold it firmly by the sides and I do **not** drop it â€" the inside is fragile."
5. Pick up the CPU (if a spare is available). "I hold it by the edges. I never touch the pins underneath or the contacts in the socket."
6. Point to the "FAULT" parts. "Remember, static damage can be invisible. A part can be dead and look perfect. That is why we protect every part, every time."

Expected result: Learners can demonstrate the correct grip for RAM, GPU, HDD and CPU.

Reasoning to state aloud: "I wear the strap because a static spark you cannot even feel can kill a chip. I hold boards by the edges because oils on my fingers corrode the gold contacts. I never open the PSU because it can hold a deadly charge even when unplugged."

Common errors to watch: Learners grabbing the RAM contacts, touching the CPU pins, or resting boards on carpet or metal. Correct immediately and kindly.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "Component Identification Stations"

Objective: With the instructor guiding step by step, learners identify each major component and write its name, function and one key fact on Worksheet 1.1.

Materials per learner/group: Worksheet 1.1 (Component Identification Chart), Worksheet 1.2 (Data Flow Diagram), pencil, and access to the stations below.

Setup (instructor arranges four stations before class):

- Station A: the open desktop computer (unplugged, wrist strap provided).
- Station B: loose samples â€" CPU + cooler, RAM, expansion card.
- Station C: loose samples â€" HDD, SSD, NVMe.
- Station D: loose samples â€" PSU (sealed/closed), fans, SATA/power cables, ports photo sheet.

Procedure (instructor calls each step; learners move in small groups):

1. Put on your anti-static wrist strap and clip it to bare metal at the station. **If you have no strap, touch bare metal of the case before handling any part, and again before each new part.**
2. At **Station A**, point to and name: motherboard, CPU and cooler, RAM, storage, PSU, GPU, fans. Write each on your chart.
3. At **Station B**, pick up the RAM stick **by its edges** and describe what it does. Pick up the CPU and describe the role of the heatsink/fan and thermal paste.
4. At **Station C**, hold the HDD and the SSD. Write **two differences** between them (speed, moving parts, durability). Do not shake or drop the HDD.
5. At **Station D**, identify the PSU, the fan, the SATA data cable, the SATA power connector and the 24-pin power connector. **Never open the PSU.**
6. On Worksheet 1.2, fill in the **data-flow diagram**: keyboard â†' port â†' motherboard â†' CPU â†' RAM/GPU â†' monitor, with storage and PSU noted.
7. When finished, raise your hand for the instructor to check your chart.

Pause-and-predict points (instructor calls these aloud):

- After the group has seen RAM and storage: "*Predict: the power goes off while a document is open and unsaved. What is lost â€" the file in RAM or the file on storage?*" Expected: the unsaved work in RAM is lost; anything already saved to storage survives.
- After the group has seen HDD and SSD: "*Predict: which one would break if dropped while working?*" Expected: the HDD, because it has moving parts.
- After the data-flow diagram: "*Predict: if the GPU is removed, what happens to the picture?*" Expected: no picture on a dedicated-card setup, or lower performance if integrated graphics take over â€" accept reasoning.

Expected result: Every learner completes a chart naming at least eight components with a correct function and one key fact each, and a filled data-flow diagram.

Troubleshooting while walking the room:

- Learner grabs RAM contacts â†' gently redirect to the edges; explain the oil problem.
- Learner says "RAM stores my files" â†' correct with the store-room/prep-table comparison.
- Learner calls the CPU "the whole computer" â†' point out it is one chip under the cooler.
- Learner cannot tell HDD from SSD â†' have them look for moving parts and the label; explain electronic vs mechanical.

Completion checklist (learner ticks each):

- ☐ Motherboard identified
- ☐ CPU + heatsink/fan identified
- ☐ RAM identified, held by the edges
- ☐ HDD identified and handled gently
- ☐ SSD (and NVMe if present) identified
- ☐ PSU identified (not opened)
- ☐ GPU identified
- ☐ At least one cooling fan and the main connectors identified
- ☐ Function and one key fact written for each
- ☐ Data-flow diagram completed
- ☐ Wrist strap worn or bare-metal touch done before handling

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I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: "The Repair Shop Intake"

Scenario: You are the newest technician at **Elias Computer Clinic, Diobu, Port Harcourt**. A young man walks in carrying a desktop computer and complaining. He does not know the names of any parts. Your job is to translate his plain complaints into the correct technical component, and explain what each part does so he trusts your diagnosis.

Instructions: For each complaint below, write (a) **which component** is most likely involved, (b) **what that component does**, and (c) **one question** you would ask to find out more. Work alone.

Complaints given to you:

1. "The machine makes a soft ticking sound and sometimes the files take long to open."
2. "It starts but the screen stays black, and I can hear the fan running."
3. "When I have Facebook, WhatsApp and music open at the same time, everything freezes."
4. "Nothing happens at all when I press the power button. No light, no sound."
5. "It gets very hot and switches itself off after some time."
6. "It works, but it is slow to start up in the morning and programs take long to open â€" it has never been upgraded."

Requirements:

- Answer all six complaints.
- For each, give a one-sentence plain-English explanation a non-technical customer would understand.
- No copying the instructor's demonstration wording â€" use your own words.

Constraints:

- Assume this is a normal desktop computer from around a cyber cafÃ©.
- Keep each explanation to two or three sentences.
- Use correct component names.

Time allowance: 15 minutes.

Success criteria:

- All six complaints matched to a component with correct reasoning.
- Explanations use correct names and are understandable to a non-technical person.
- At least one question written for each complaint.

Hints if stuck:

- Hint 1: "No power at all, no light" points to the part that supplies electricity to everything.
- Hint 2: "Freezes when many programs are open" points to the fast temporary memory.
- Hint 3: "Slow to start up, never upgraded" points to permanent storage with moving parts.
- Hint 4: "Gets hot and switches off" points to the cooling parts and the CPU.

Instructor answer key (do not show learners):

1. **HDD (hard disk drive).** It stores files permanently using spinning platters and a moving head; a soft ticking and slow file access suggests a failing drive. Question: "When did the ticking start, and have you backed up your important files?" (Possible backup needed before any work â€" data safety first.)
2. **No display with fans running â†'** likely RAM not seated, GPU/cable issue, or display cable in the wrong port. Component focus: RAM / GPU / display connection. Question: "Did anything change before it happened â€" a move, a power cut, new parts?" (Accept a well-reasoned answer that names RAM, GPU or the cable and explains why.)
3. **RAM.** It is the fast temporary workspace; when too little, the machine slows badly. Question: "How much memory does it have, and does it happen only when many programs are open?"
4. **PSU (power supply).** It converts wall electricity to the voltages the parts need; if it fails, there is no sign of life. Question: "Have you checked the wall socket and the power cable, and does any light come on at all?" (Rule out the simplest things first.)
5. **Cooling system â€" heatsink/fan and thermal paste, plus the CPU.** Dust or dried paste causes overheating and shutdown. Question: "When did it last have its dust cleaned, and does the base or back feel very hot?"
6. **Storage â€" an old HDD.** Slow startup and program opening on a never-upgraded machine suggests an HDD; an SSD upgrade would help. Question: "How old is the machine, and how much free space is left on the drive?"

Marking guidance (out of 12 â€" 2 per complaint):

- Correct component named (1)
- Correct, plain explanation (1)
- Bonus credit within the same mark if a sensible question is also given.

J. REAL-WORLD CASE STUDY

Situation. *Chinedu* runs a small business centre near the University of Port Harcourt. One Monday morning his main desktop will not come on at all. He presses the power button repeatedly â€" nothing: no fan, no light, not even a flicker. He has a queue of customers waiting to print, type and browse. He calls a technician friend, who says over the phone, "It is a motherboard problem, you need a new computer" â€" without opening the machine. Chinedu is about to spend money he cannot afford. Before he does, he carries the machine to your shop.

Relevant facts.

- The computer worked perfectly on Saturday evening.
- There was a heavy thunderstorm and a power "surge" (voltage spike) on Sunday night in the area.
- The machine is a normal desktop with an internal PSU; no battery.
- The customer has important files on the hard drive that he needs for his business.

Decision points.

1. Should you trust the phone diagnosis of "motherboard" and sell him a new machine, or investigate?
2. Where do you start â€" the most expensive part or the simplest checks?
3. What component is most consistent with "worked Saturday, power surge Sunday, completely dead Monday"?
4. What must you protect before doing anything else?

Possible approaches.

1. Accept the phone diagnosis and replace the whole computer (or motherboard). Fast, but expensive and possibly wrong.
2. Follow a basic order: check the wall socket and power cable, check the PSU, then look deeper. Cheap and methodical.
3. Try to repair the PSU yourself. **Unsafe and forbidden** â€" the PSU is replaced, never opened.

Recommended solution (for this stage). Approach 2. Check the simplest things first (socket, cable, that the machine is actually receiving power), then test the PSU. Given the surge, a failed PSU is the most likely cause of a completely dead machine. **Replace** the PSU (never open it). Before returning the machine, confirm his hard drive is healthy so his business files are safe. Only if the machine is still dead after a known-good PSU is fitted would you suspect the motherboard â€" and even then, you confirm by testing, not guessing.

Trade-offs.

- Approach 1 is fast but risks replacing a healthy expensive board and charging the customer for the wrong repair â€" bad for reputation and for the customer.
- Approach 2 takes a little longer but is safe, cheap and usually correct, and it protects the customer's data and your reputation.
- Trying to repair the PSU is dangerous: the charge inside can injure or kill, and it is outside the scope of this course.

Professional connection. This case is exactly why component knowledge matters. The technician who knows that the PSU converts and supplies *all* the power immediately suspects it after a surge, and checks it early and safely. The technician who only knows "it won't on, so buy a new one" costs the customer money and loses trust. In Port Harcourt, where power surges are common, this single insight saves customers thousands of naira â€" and wins you loyal customers who tell their friends.

Transfer exercise. "A customer's computer runs but the display is blank and fans are spinning. Before blaming the GPU, what three simple things would you check?" (Display cable connected; cable in the correct port â€" motherboard vs graphics card; RAM seated fully.)

Safety reminder. Everything above is done with the machine **unplugged**. The PSU is replaced, never opened. Working on mains-powered equipment or opening a PSU is never permitted in this course.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the live troubleshooting guide during the practical. When a learner struggles, ask them to check the relevant row rather than giving the answer immediately.

K.1 Touching the gold contacts on RAM or expansion cards

Symptoms: A part that was working now fails to detect; intermittent faults; an upgrade that "does not show up." **Likely causes:** Finger oils or dirt on the gold contacts create a poor connection. The cause is *confirmed* only after you inspect the contacts and reseat the part. **How to investigate:** Look at the contacts for smudges or dullness; check the part seats fully with clips closed. **Safe corrective steps:** (Instructor-supervised) remove the part with the machine unplugged, wipe the contacts gently with **isopropyl alcohol (90%+)** on a lint-free cloth if available, let it dry fully, and reseat holding only the edges. **How to confirm resolved:** The part is detected again and the machine behaves normally. **How to prevent recurrence:** Always hold boards by the edges; never touch the contacts.

K.2 Confusing RAM with storage

Symptoms: The learner says "add a bigger RAM to store more videos" or "the RAM is full of files." **Likely cause:** The two concepts were never separated in the learner's mind. **How to investigate:** Ask the learner what happens to a document on the "RAM" when the power goes off. If they say "it stays," they are confusing RAM with storage. **Corrective steps:** Re-teach with the prep-table and store-room comparison. RAM empties on power-off; storage keeps data. **How to confirm resolved:** The learner can state the difference in one sentence using their own example. **How to prevent recurrence:** Use the comparison every time the topic returns.

K.3 Handling an HDD roughly or while it is running

Symptoms: Clicking or grinding sounds after a knock; the drive stops being detected; files disappear. **Likely cause:** The internal platters or head were damaged by a shock; a mechanical failure. (Confirmed by the noise and by a drive test in a later session.) **How to investigate:** Listen for clicking, check whether the drive is detected, and (later, with tools) read its health. **Safe corrective steps:** Stop using the drive immediately; if there is important data, **do not keep power-cycling it** â€" refer for data recovery. Replace the drive (Session 7). **How to confirm resolved:** A replacement drive works normally. **How to prevent recurrence:** Never shake, drop or move a running HDD.

K.4 "No display always means the GPU is dead"

Symptoms: Learner immediately blames the graphics card when the screen is blank. **Likely cause:** Skipping the simple checks (cable, port, RAM seating). **How to investigate:** Check the display cable is connected and in the correct port; check whether RAM is fully seated; test with a different monitor/cable if authorised. **Corrective steps:** Fix the simple cause first; only then consider the GPU. **How to confirm resolved:** The display returns. **How to prevent recurrence:** Always follow a simplest-first order (this becomes the Session 4 workflow).

K.5 Forgetting anti-static protection

Symptoms: No immediate symptom â€" the danger is invisible. Later, a part mysteriously fails. **Likely cause:** Static electricity discharged into a component, causing latent (delayed) damage. **How to investigate:** You usually cannot see it; that is the point. Prevent it rather than detect it. **Corrective steps:** Always use the wrist strap or touch bare metal before handling parts. **How to confirm resolved:** You cannot always confirm â€" prevention is the only reliable answer. **How to prevent recurrence:** Make the strap part of the routine, like wearing gloves before handling a hot pot.

K.6 Forgetting to unplug before opening the case

Symptoms: Risk of electric shock, short circuit, or damage; a fan may start unexpectedly. **Likely cause:** Rushing. **How to investigate:** N/A â€" this is a discipline issue. **Corrective steps:** Always unplug from the wall and press the power button once to discharge residual electricity before opening the case. **How to confirm resolved:** The machine is verified unplugged and discharged. **How to prevent recurrence:** Build the habit into every session's checklist.

Distinguishing likely vs confirmed: In every row above, the "likely cause" becomes a "confirmed cause" only after you *check* it â€" see the contacts, hear the click, test the drive, reseat the part and see the fault disappear. Teach learners to confirm before they "fix."

L. SESSION ASSIGNMENT

Assignment title: "My Component Reference Card"

Purpose: To reinforce component knowledge by having the learner build a personal reference they will keep using all course. This becomes the foundation of the capstone service report.

Realistic scenario: A customer walks into your new shop and asks, "Please, what exactly is inside this computer, and which part is likely causing my problem?" You must be able to answer confidently and simply. Your reference card is the tool you will use to explain it.

Detailed instructions:

1. Create a document in Microsoft Word or Notepad. Title it **"My Component Reference"**.
2. For **each** of these components, write:
   - The **name** of the component.
   - A short **description** of what it looks like (a sentence or a rough drawing; you may sketch on paper and photograph it).
   - Its **function** in your own words.
   - **One key fact** (for example, "RAM empties when the power is off").
   - **One common problem** linked to it (for example, "a dust-clogged fan causes overheating"). Components to include: motherboard, CPU (with heatsink/fan), RAM, HDD, SSD, PSU, GPU, cooling fan, ports/connectors.
3. Add a short paragraph (3â€"5 sentences) titled **"How data flows"** describing the path from a keypress to the screen.
4. Save the file as **"My Component Reference"** on the Desktop.
5. Show the saved file to the instructor before leaving, and be ready to answer one question about any component in it.

Required deliverables:

- One saved document with all nine components and the data-flow paragraph.

Submission format: Saved on the Desktop, file named exactly `My Component Reference`. If the learner has no computer at home, they may complete it in the academy lab and show it before leaving.

Deadline: Before Session 2 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
All components included	All nine present	Seven or eight present	Six or fewer
Function accuracy	All functions accurate	Mostly accurate	Several errors
Descriptions	Clear and detailed	Mostly clear	Vague or missing
Common problems	Relevant and accurate	Mostly correct	Incorrect or missing
Data-flow paragraph	Correct, ordered, clear	Mostly correct	Missing or wrong order
File saved correctly	Saved on Desktop, correct name	Saved, minor naming issue	Not saved

(Reduce the "All components" column to a 5/3/1 scale and keep total 25 by treating each row as equal weight.)

Common mistakes to avoid: confusing RAM with storage; touching/claiming to touch contacts; calling the CPU "the whole computer"; leaving out the cooling system; saving the file in the wrong place.

Instructor reference solution (summary, do not show learners beforehand):

- Motherboard: main board; connects all parts; common problem = loose cable/expansion card.
- CPU (+heatsink/fan): carries out instructions; gets hot; common problem = overheating from dust or dried thermal paste.
- RAM: fast temporary memory; empties on power-off; common problem = not fully seated, or too little for heavy use.
- HDD: permanent storage with moving parts; common problem = mechanical failure (clicking).
- SSD: permanent electronic storage, faster; common problem = can fail with little warning â€" hence backups.
- PSU: converts wall power to needed voltages; common problem = total failure (no sign of life) â€" and it is never opened.
- GPU: renders images; common problem = no display (often a cable/port issue, not the card).
- Cooling fan: moves heat out; common problem = dust clogging.
- Ports/connectors: connect outside devices; common problem = bent pins from forcing plugs.
- Data-flow paragraph: keyboard â†' USB port â†' motherboard â†' CPU â†' RAM works with CPU â†' GPU â†' monitor; storage and PSU noted.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below â€” do not read the key until you have attempted all questions.)

Q1. Short answer. In one sentence each, say what the motherboard, the CPU and RAM do.

Q2. Multiple choice. Which component stores your files **permanently**, even when the power is off? a) RAM b) CPU c) Storage (HDD/SSD) d) PSU

Q3. Explain in your own words. Why is RAM described as "short-term" and storage as "permanent"? Give your own everyday example.

Q4. Practical identification. A customer's computer makes a soft ticking sound from a spinning part inside and files open slowly. Which component is most likely involved, and why?

Q5. Error detection / truth-check. A learner says: "I will add more RAM so I can store more videos." Find the mistake and correct it.

Q6. Scenario. A desktop powers on â€” the fans spin and a light shows â€” but the screen stays black. List three things you would check, in order from simplest to most complex.

Q7. Short answer. Why must the PSU never be opened, even when it is faulty?

INSTRUCTOR ANSWER KEY (Q1â€“Q7)

A1. The **motherboard** is the main circuit board that connects all the other components; the **CPU** carries out the instructions of programs (the "brain"); **RAM** is the fast temporary workspace that holds data currently in use and empties when the power goes off.

A2. (c) Storage (HDD/SSD).

A3. RAM holds only what is being used right now and loses everything when the power goes off, like items on a kitchen prep table cleared at closing. Storage keeps files even with no power, like goods in a store room. Everyday example: an open, unsaved document lives in RAM; once you save it, it goes to storage and survives a power cut.

A4. The **HDD (hard disk drive)**. It has moving parts â€” spinning platters and a read/write head â€ so a ticking sound and slow file access suggest a mechanical problem. (Advise protecting the customer's data before further use.)

A5. The mistake is confusing storage with RAM. RAM does not store files; it holds programs and data being used right now and empties on power-off. To store more videos you need a bigger **storage** drive (or an external drive), not more RAM. (More RAM helps when many programs are open at once.)

A6. Accept any sensible, simplest-first order such as: (1) check the display cable is connected and in the correct port (motherboard vs graphics card); (2) check the monitor has power and is switched on; (3) reseat the RAM (a common cause of no display). GPU comes only after these.

A7. The PSU contains components that can hold a dangerous electrical charge even when unplugged from the wall. Opening it risks a serious or fatal shock, and repairing it is outside the scope of a beginner course. If it is faulty, it is **replaced**, never opened.

Marking: Award up to 2 marks per question (Q3 and Q6 up to 3 each). Total â‰ˆ 16â€“18; scale proportionally. Full credit requires the correct component name and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned what is inside a computer and what each part does. You met the **motherboard** (the board that connects everything), the **CPU** (the part that does the work, kept cool by a heatsink and fan), **RAM** (fast temporary memory that empties when the power goes off), **storage** (permanent memory â€ HDD and SSD), the **PSU** (the part that supplies electricity and must never be opened), the **GPU** (the part that paints the screen), the **cooling fans**, and the **ports and connectors** that bring the outside world in and out. You also learned how data flows from a keypress to the screen, how a desktop differs from a laptop, and how to handle parts safely by their edges with anti-static protection.

Five important takeaways:

1. A computer is a **team of parts**, each with one main job â€” none works alone.

2. **RAM is temporary; storage is permanent.** Never mix them up.
3. **HDD is mechanical and slower; SSD is electronic and faster; NVMe is the fastest SSD.**
4. The **PSU supplies all the power and is never opened** â€" replace it, never repair it.
5. Always **handle parts by the edges**, protect against static, and unplug before opening any case.

Questions to help you recognise what you can now do:

- Can I name at least eight components inside a computer without looking?
- Can I explain RAM vs storage to a customer in one sentence?
- Can I trace how a keypress becomes a letter on the screen?
- Can I state why the PSU must never be opened?

Connection to the next session: Now that you know the parts, next session you will learn how to work around them **safely** â€" how to protect yourself from electricity and from ESD, how to choose and use the right tools, and how to set up a clean, safe workspace. Knowing the parts without knowing safety is dangerous; Session 2 makes you safe.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner names all major components without prompting and gives each a correct one-line function.
- The learner spontaneously holds RAM/boards by the edges and does not touch gold contacts.
- The learner correctly separates RAM from storage and HDD from SSD in their own words.
- The learner can trace the data flow from input to output and name the parts at each step.
- The learner treats the PSU with respect and states it must never be opened.

Common misconceptions to watch for and correct:

- "The monitor is the computer." â†' The system unit is the computer; the monitor is output only.
- "RAM and hard disk are the same." â†' Temporary vs permanent; GB of workspace vs GB of store room.
- "The CPU is the whole computer / the case." â†' The CPU is one chip under the cooler.
- "A graphics card is only for games." â†' Used for design and video too; and integrated graphics exist.
- "The PSU can be fixed." â†' It is replaced; opening it is dangerous and forbidden.

Suggested remedial exercises for learners who struggle:

1. **Component flash cards:** write each component on one card and its function on another; match them, then shuffle and repeat.
2. **Draw-and-label:** sketch a simple desktop with three parts and label them; repeat from memory.
3. **Data-flow relay:** learners physically stand in a line holding component name cards (keyboard, port, CPU, RAM, GPU, monitor) and pass a paper "signal" along the line in the correct order.
4. **Analogy practice:** ask the learner to invent their own kitchen/market analogy for three components.

Differentiation:

- **Level 1 (supported):** provide a partially completed Component Chart (names filled, functions to finish) and work alongside the learner at the stations; pair with a careful partner.
- **Level 2 (standard):** complete the exercise as written, independently with the group.
- **Level 3 (challenge):** research and explain one additional component (for example a sound card, network card, optical drive, or the CMOS battery), and describe the data flow for saving a file, not just typing a letter.

Timing guidance: The lecture (Section F) is long by design; if the room is slow, compress F.8 and F.10 to quick summaries and protect the hands-on time in Section H â€" the practical identification is the priority and is directly assessed. Keep Section J as a 3â€"4 minute story if time is short.

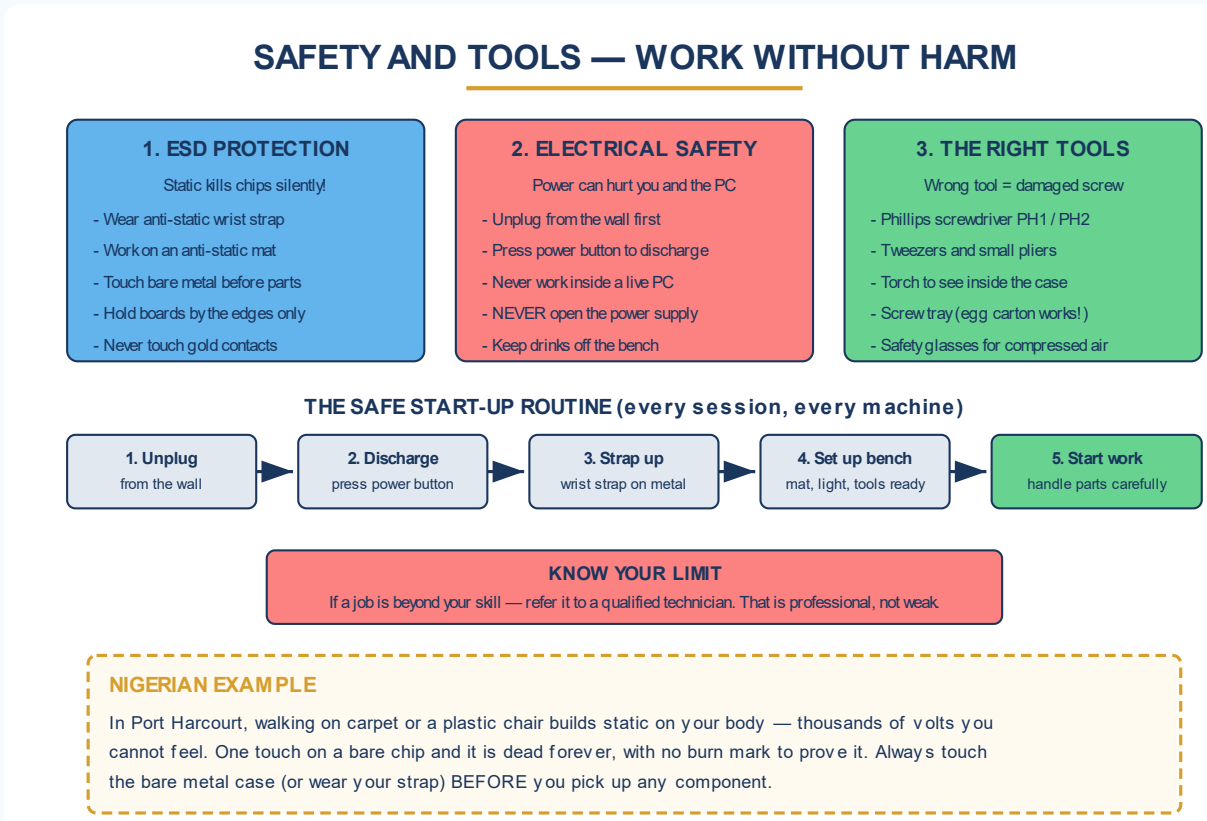
END OF SESSION 1

Next: Session 2 â€" Safety and Tools: Working on Computers Without Harming Yourself or the Machine.

SESSION 02 OF 12

Safety and Tools

CONCEPT DIAGRAM — SESSION 02



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 2 â€" SAFETY AND TOOLS: WORKING WITHOUT HARMING YOURSELF OR THE MACHINE

A. SESSION IDENTIFICATION

Item	Detail
Session number	2 of 12
Session title	Safety and Tools
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner

Item	Detail
Prerequisite knowledge	Session 1 â€" Computer Components. The learner must already be able to name the major components.
Connection to previous session	Session 1 named the parts. This session teaches how to be safe around those parts: electrical safety, ESD (electrostatic discharge) protection, the correct tools, and a safe workspace.
Connection to the course	Every later session â€" disassembly (3), diagnostics (4â€"6), repair (7â€"8), the capstone (12) â€" requires these safety habits and tool skills. This is the session that makes the rest of the course possible without injury or damage.
Preparation before class	Tool kits per learner/group (Phillips PH1 and PH2, flathead, tweezers, torch, needle-nose pliers); anti-static wrist straps (one per learner); anti-static mat; safety glasses; a workspace table; a powered-off desktop to demonstrate on; printed Safety Procedures Checklist and Tool Identification Chart; a balloon for the opening ESD demonstration; optional multimeter (demonstration only).

Safety note for this session: This session is *about* safety, so model it perfectly. Nothing is opened on mains power. The demonstration machine must be unplugged and discharged before any handling. The PSU is never opened. Learners must not test anything on a live socket. The multimeter is used only by the instructor on a low-voltage example (for example a battery) and never on mains wiring. If a learner has any medical condition that makes electrical work risky, speak to them privately and provide alternative observation roles.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain ESD (electrostatic discharge)** in plain words, describe how it damages components, and explain why damage can be invisible.
2. **Demonstrate the correct ESD-safe routine** â€" wearing and connecting an anti-static wrist strap, working on an anti-static mat, and touching bare metal before handling parts.
3. **Apply electrical safety rules** â€" unplugging from the wall, discharging residual electricity, never working inside a powered machine, and never opening the PSU.
4. **Identify and correctly name the common computer repair tools**, and state what each is used for.
5. **Use a screwdriver of the correct type and size** and explain why the wrong size damages screws.
6. **Set up a safe, organised workspace** with good lighting, screw organisation and safe handling.
7. **Practise safe lifting and handling** of computers and components.
8. **State clearly when a repair is beyond their skill level** and must be referred to a qualified technician or supervisor.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- Tool kits, one per learner or pair: Phillips screwdrivers in sizes **PH1 and PH2**, a small flathead screwdriver, a pair of tweezers, a small torch/flashlight, and needle-nose pliers (optional pliers, one per table).
- Anti-static wrist straps â€" **one per learner**.
- An anti-static mat for each workspace table.
- A powered-off desktop computer to demonstrate safe setup on.
- A small container or screw tray for organising screws (an egg carton or ice-cube tray works well and is cheap).
- Safety glasses (one pair per learner where available; required when using compressed air).
- A balloon (for the ESD demonstration) and a woollen or synthetic jumper/cloth (optional).

Software:

- None required for this session. Safety and tools are physical skills.

Safety gear:

- Anti-static wrist straps and mat (as above).
- Safety glasses.
- A first-aid kit available in the room.
- A fire-safety reminder: know where the electrical point is and how to switch off the room's power in an emergency (point it out to learners).

Sample / fault components:

- A dead motherboard or dead expansion card labelled "FAULT â€" DEMO ONLY" to show a component that looks fine but is dead â€" useful for discussing invisible ESD damage.
- A stripped screw and a stripped screw hole to show what happens when the wrong driver is forced.
- A **closed, unopened** faulty PSU labelled "DO NOT OPEN" to reinforce the rule.

Worksheets (printed):

- Worksheet 2.1 â€" "Tool Identification Chart" (one per learner): a table with the tool name, what it is used for, and a "safe/unsafe for beginners" column.
- Worksheet 2.2 â€" "Safety Procedures Checklist" (one per learner): a step-by-step list the learner ticks during practice.
- Worksheet 2.3 â€" "Safe or Unsafe?" scenario sheet (10 short scenarios).

Tools (instructor demonstration kit):

- A complete tool kit, laid out neatly on the anti-static mat.
- A multimeter (instructor-only demonstration; used only on a low-voltage battery, never on mains).
- A can of compressed air (used outdoors or with good ventilation and safety glasses).

Alternatives for a learner or academy with limited equipment:

- If there are few tools, run the session as rotating stations: tool-identification station, ESD-routine station, workspace-setup station.
- If no anti-static straps are available, the fallback is the "touch bare metal before each part" method â€" but explain that a strap is far better and should be bought as soon as possible.
- If no real tools are available, use printed photographs of tools and have learners name and describe them; then practise with whatever basic screwdrivers exist.

D. OPENING ACTIVITY (6 minutes)

"The Invisible Danger"

Instructor setup: Stand near a light switch or the front of the room. Hold a balloon. Rub it briskly on your hair or on a woollen cloth for several seconds. In a dark corner, you may even see tiny sparks. Then hold the balloon near a small, light piece of paper or your hair to show attraction.

What the instructor says:

> *"Watch this. [Rub the balloon.] I have just made electricity â€" but not from a socket. I made it by rubbing. You cannot see it on the balloon, but it is there. Now watch what it does to paper. [Bring the balloon near the paper; it jumps.] That invisible thing is called **static electricity**.*
>
> *Now here is the serious part. Today, when we rub things together â€" our clothes, a plastic bag, our shoes on a carpet â€" we build up this same invisible charge on our bodies. It can be thousands of volts. When you touch something metal, that charge jumps across in a spark. Usually you feel a tiny shock, or nothing at all.*
>
> *But inside a computer, that same invisible spark â€" one you cannot even feel â€" can kill a chip. Dead. And you will not find out until weeks later, when the part fails for 'no reason.'*
>
> *We call this danger **ESD** â€" electrostatic discharge. It is the number-one silent killer of computer parts in Nigeria and everywhere. And there is more than one danger today: plain electricity from the wall, tools used wrongly, and lifting*

machines carelessly.

Today you learn how to touch a computer without harming it â€" or yourself. This is the most important session in the whole course, because if you are not safe, your skills do not matter."

Then show the invisible damage: Hold up the dead board labelled "FAULT â€" DEMO ONLY." Say:

> *"This board looks perfectly fine, does it not? Shiny. Clean. No burn marks. And yet it is dead. Static electricity leaves no mark. That is why we do not rely on our eyes â€" we rely on our **habits**."*

Expected learner responses: Learners are often surprised that ESD damage is invisible and that they may have killed parts before without knowing. Some may say "but my phone works when I touch it" â€" accept this and explain that phones and finished products are more protected than exposed circuit boards during repair. Some may mention carpets at home. Praise that, because carpet is a real risk.

Why this opener works: It makes an invisible, abstract danger physical and memorable using a simple balloon, and it immediately connects to the expensive consequences (dead parts, lost money). It also previews the whole session's theme: safety first.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral review. Ask different learners each time.

Question 1: "Name at least six components inside a desktop computer and say what one of them does."

- Expected answer: Motherboard, CPU, RAM, HDD/SSD, PSU, GPU (and fans, connectors). Any correct function.
- Common gap: learners forget the cooling fan or connectors. Remind them that the fan is a component too.
- Bridge: "Today you will not just name these parts â€" you will learn to handle them without killing them."

Question 2: "Which component must we never open, and why?"

- Expected answer: The PSU (power supply). It can hold a dangerous charge even when unplugged.
- Common gap: learners think "unplugged means safe" for everything. Clarify: unplugging makes the machine safe to work in *except* inside the PSU, where a stored charge can remain; that is why we never open it.
- Bridge: "That rule starts today and stays with you for life."

Question 3: "How should you hold a RAM stick, and why?"

- Expected answer: By the edges; do not touch the gold contacts, because finger oils damage the connection.
- Common gap: learners may not know *why*. Explain the oil problem.
- Bridge: "Correct handling is part of being safe â€" safe for the component. Today we also learn safe for you."

Question 4: "What is the difference between RAM and storage?"

- Expected answer: RAM is temporary, empties on power-off; storage is permanent.
- Common gap: still mixing the two. Correct briefly; it comes up again in the practical.
- Bridge: "Keep this clear, because today's case study involves a mistake with physical parts, not memory."

Question 5: "Have you ever received a small electric shock touching a car door or a metal tap? What caused it?"

- Expected answer: Static electricity discharging.
- Common gap: some say "faulty wiring." Clarify the difference between harmless static and dangerous mains electricity â€" both are real, but they behave differently, and today we handle both.

Bridge to today (say this):

> *"Think about a **hospital operating theatre**. The surgeon has the best skills, but none of that matters if the equipment is not sterile and the theatre is not safe. A computer repair bench is our operating theatre. Today we set the rules and the tools so that every repair we do later is safe â€" for us and for the machine. Safety is not a lecture before the real work; safety **is** the real work."*

If more than a third cannot answer Question 2 or 3, spend an extra three minutes re-teaching the PSU rule and the handling rule before continuing, since both are safety-critical.

F. FULL LECTURE NOTES

F.1 ESD â€" the invisible killer

Plain-English definition. ESD (electrostatic discharge) is the sudden jump of a small electric charge from one object to another when they touch. It is the same little "snap" you feel when you touch a car door or a metal gate after walking on a rug.

Why it happens. When two different materials rub together (your shoes and carpet, your shirt and a chair, a plastic bag and your hand), tiny charges build up on your body. Your body becomes like a tiny battery. When you then touch something that conducts electricity â€" a metal object, or a circuit board â€" the charge jumps across in an instant.

How it damages components. Computer chips and boards are designed for very small voltages. A static discharge that you cannot even feel can carry a voltage far higher than a chip can survive. Two things can happen:

1. **Immediate damage:** the chip dies instantly. The machine fails right away.
2. **Latent (delayed) damage:** the chip is weakened but still works. Weeks or months later, it fails "for no reason." This is the most dangerous kind, because you may blame the wrong part or the wrong technician.

Why you cannot rely on your eyes or your senses. A static shock that you can *feel* is only a small portion of the danger. You can damage a component with a discharge **you never feel and never see**. As the demonstration showed, a dead board can look perfect.

Important limitation â€" do not invent numbers. It is tempting to say "10 volts kills a chip" or "your body holds 3,000 volts." Different sources quote different figures, and the exact numbers vary with the situation. The safe, honest teaching is this: **static electricity from ordinary daily activity can cause voltages far beyond what a chip can survive, even when you feel nothing.** Teach the *principle* and the *habit*, not a made-up exact figure.

Real-life comparison. ESD is like **sand in an engine**. One grain is nothing; you cannot always see it going in; but over time, or at the wrong moment, it destroys the engine. You never allow sand near an engine, even though it looks harmless. In the same way, you never allow static near a board, even though you cannot see it.

Realistic repair-shop scenario. A technician in a Port Harcourt shop reseats a customer's RAM on a hot, dry afternoon while standing on a tiled floor and wearing a synthetic shirt. Three weeks later the customer returns: the same machine now crashes randomly. The RAM had a latent ESD failure from that day. The technician blamed the customer. The truth is that a 30-second habit â€" wearing the strap â€" would have prevented it. This is why ESD is taught before any tool is picked up.

Common misconceptions.

- "ESD is not real; I have never destroyed anything." You may have â€" you just never connected the failure to your handling. Invisible damage is the whole problem.
- "Living in a humid climate, I am safe." Humidity reduces static buildup but does **not** remove the risk. Port Harcourt's humidity helps, but strap up anyway.
- "Only cheap parts are at risk." All exposed boards are at risk; expensive ones hurt the most when they die.

Safety implications. ESD protection is a rule, not an option. No learner handles a board without the routine in F.2. A component that a learner is unsure about should be handled only under supervision.

F.2 How to prevent ESD â€" the routine

Principles. The aim is to keep your body at the *same* electrical level as the machine, so that no charge jumps when you touch it. This is called **grounding** or **bonding** yourself to the equipment.

The ESD-safe routine (teach learners as a fixed sequence):

1. **Wear an anti-static wrist strap.** It is a band that goes on your wrist with a coiled wire that clips to bare metal of the computer case (or the anti-static mat). It slowly and safely drains any static from your body.
2. **Work on an anti-static mat** where possible. The mat is designed to carry charge away safely rather than let it build up.
3. **Touch bare metal of the case before handling any component** â€" every time, and again each time you come back to the machine after moving around. Unpainted metal, not the plastic or painted panels.

4. **Handle all boards by their edges.** Never touch gold contacts, chips or pins.
5. **Avoid building up static:** do not shuffle on carpets, do not rub balloons or plastic, do not work in socks on a synthetic rug, and avoid wearing loose woollen or synthetic clothing if possible.
6. **Keep components in anti-static bags** when they are not installed. The shiny grey or pink bags you receive parts in are antistatic; do not throw them away.
7. **Never place a board on a carpet, a foam mattress, a plastic tablecloth or a metal surface.** Use the mat, or the antistatic bag, or the original box.

Why the wrist strap must be clipped to the case, not to a wall socket. The strap grounds you to the *equipment you are working on*, so that you and the equipment are at the same level. Never, ever clip an anti-static strap or any wire to a mains socket or a live conductor â€" that is extremely dangerous. The strap is a safety tool, not a power wire.

Real-life comparison. Wearing the strap is like **sharing a plate of food with a friend** â€" you are both at the same table, so nothing "jumps" between you. Without it, you are like a person holding a live electric fence and then touching your friend; the spark finds the path.

Common misconceptions.

- "I only need the strap when I am touching RAM." No â€" any board, and every time, including when you first sit down.
- "A cloth or aluminium foil works as well." No. Use a proper anti-static strap and mat.
- "If I keep the machine plugged in, it is grounded." Partly true for some systems, but the safe rule for beginners is: **unplug the machine first, then use the strap.** Never assume that being plugged in makes you safe â€" it can make you less safe.

Limitations and safety. The strap protects equipment, not from mains electricity. It must be worn **in addition** to unplugging, never as a substitute. A damaged strap (torn wire, broken clip) must be replaced.

F.3 Electrical safety â€" the rules that keep you alive

Plain-English idea. Inside the computer, the PSU and some parts use mains-derived electricity (from the wall, around 220â€"240 V in Nigeria). Touching certain parts while the machine is live can give a serious or fatal shock. The rules below are absolute.

The rules:

1. **Always unplug the power cable from the wall before opening the case.** Pull the plug out, do not just switch off at the socket â€" switch off first to be safer, then pull the plug.
2. **Press the power button once after unplugging.** This is called **discharging residual electricity**. Some parts (especially capacitors) can hold a small stored charge for a short time. Pressing the power button with the plug removed helps drain it so there is no surprise.
3. **Never work inside a powered-on computer.** Not even to "look." Not even to "just plug the fan." If the machine is on, stop and switch off.
4. **Never open the PSU.** It holds a dangerous charge even when unplugged, and mending it is beyond this course. If it is faulty, replace it â€" with supervision as you learn.
5. **Never use metal tools near live circuits.** No jewellery, no loose metal, no tools resting inside a live machine. Remove rings, watches and metal bangles before working.
6. **Keep liquids away.** Water and electricity do not mix. No drinks on the bench; if there is a spill, switch off at the wall before doing anything.
7. **If you smell burning, see smoke, or hear a loud pop â€" stop.** Switch off / unplug from a safe distance if possible, do not touch the machine, and call the instructor. Never move or open a burning or smoking machine that is still plugged in.
8. **Know the emergency switch.** Point out the room's main switch or fuse board to every learner. In an emergency, cut the power there â€" do not grab the machine.
9. **Watch the cables.** A trailing power cable or a laptop charger is a trip hazard. Keep the floor clear.
10. **Know your limits.** If a job involves mains wiring, the inside of the PSU, wall sockets or anything you are not confident about â€" stop and refer it to a qualified electrician or technician. No repair is worth your life.

Real-life comparison. Electricity in a computer is like **a caged lion**. While the cage is closed (unplugged), the lion is still real and still dangerous if you poke into the wrong cage (the PSU), but the area around it is safe. Once the cage opens (live power), everything changes. Our rule is simple: we work only when the lion is asleep and the area is checked.

Realistic repair-shop scenario. A technician in a hurry opens a case to swap a hard drive while the machine is still plugged in, intending to "just be quick." His metal wristwatch touches the exposed PSU connector. The result is a burn and a ruined

machine. Ten seconds of unplugging would have prevented it. This is a true-to-life pattern â€” hurry causes more injuries than difficulty.

Common misconceptions.

- "The power switch on the PSU or the case makes it safe." No â€” always pull the plug from the wall and discharge.
- "Low voltage = no danger." The motherboard and internal parts run on low voltage, but the **PSU input side and mains cable are high voltage**, and stored charges can still bite. Respect the whole machine.
- "I will just hold the tool carefully." Caution does not replace unplugging. The plug always comes out first.

Limitations and safety. This course never asks you to work on mains wiring or inside a PSU. Those are outside scope and dangerous. When in doubt, stop and ask the instructor.

F.4 The correct tools â€” and what each one is for

Why tools matter. The right tool makes a job safe and clean; the wrong tool strips screws, damages boards and causes injury. A professional technician's skills begin with good habits with basic tools.

1. Screwdrivers â€” the most important tool.

- **Phillips screwdriver (cross-shaped tip).** Used for most computer screws. Sizes **PH1** and **PH2** cover most jobs; **PH2** is the everyday size for case screws. Using the correct size matters: too small, and you slip and chew up the screw head; too large, and it will not seat. A magnetised tip helps hold screws in tight corners.
- **Flathead screwdriver (flat blade).** Used for prying open plastic clips and gentle levering, and for some slotted screws. **Use with great care** near boards â€” the flat blade can slip and scratch or cut circuit tracks. Never use a flathead as a chisel or to force a screw.
- **Precision (small) screwdrivers.** Needed for some laptops and small screws. If you do not have the correct size, do not force it â€” get the right driver.

Why screw size and type matter (teach this clearly). Computer screws come in different lengths and thread types. A long screw forced into a short hole can damage the board. Always keep screws organised (see workspace) and always match screw to hole.

2. Tweezers. Fine-point tweezers are used to retrieve dropped screws, hold small jumpers and connectors, and place tiny screws in awkward spots. Always use them with the machine unplugged, and never force a screw out with tweezers â€” you may bend the board.

3. Torch / flashlight. A small torch is essential for seeing inside a dark case, checking for dust, and reading labels. It also helps spot a bent pin or a loose cable.

4. Needle-nose pliers. Used for gripping, straightening and bending in tight spaces, e.g., carefully retrieving a part, or holding a nut while you turn a screw. **Never use pliers to force a screw head** â€” that is how screws get destroyed. Insulated handles are safer.

5. Anti-static wrist strap and mat. Already covered â€” the ESD tools. Treat them as tools you use every single time.

6. Compressed air (canned air). Used to blow dust out of heatsinks, fans and vents. **Important safety:** hold the can upright, never tilt it (liquid propellant can spray out and chill or damage parts), keep a safe distance, do not spin a fan with the air jet (which can over-speed and damage the bearing), and **wear safety glasses and use it in a well-ventilated area (ideally outdoors)** because dust can be harmful to breathe. Do not use a vacuum cleaner on electronics â€” it can create static.

7. Isopropyl alcohol (90% or higher). Used to clean gold contacts, remove old thermal paste, and clean stubborn grime. It evaporates quickly and leaves no residue. **Do not use water**, and do not use petrol or kerosene. Use on a lint-free cloth or a cotton bud.

8. Microfibre / lint-free cloth. Used for cleaning screens, contacts and surfaces without leaving fibres.

9. Thermal paste. A grey compound applied between the CPU and its heatsink. Only used when a cooler is removed and refitted (Session 8). Not a beginner's first tool, but you should recognise it and know its purpose.

10. Cable ties / velcro straps. Used to tidy cables so air can flow and the machine looks professional. Good cable management is part of good work.

11. Screw tray / container (or an egg carton). Used to keep screws organised and separated by location, so you can put each one back in the right place.

12. Multimeter (instructor demonstration only). A multimeter measures voltage, current and resistance. Professionals use it to test a PSU's output and check continuity. **In this course, the multimeter is demonstrated by the instructor only, on a safe low-voltage example (such as a battery), never on mains wiring.** Beginners do not test live circuits. Recognising the tool and knowing its purpose is the goal for now.

Real-life comparison. Tools are like a **carpenter's plane and chisels**. The right tool for the right job produces clean work; using a chisel as a hammer ruins the wood and the tool. In computer repair, a wrong-size screwdriver is the most common self-inflicted damage.

Common misconceptions.

- "Any screwdriver will do." No â€" the wrong size strips screws and damages boards.
- "A bigger screwdriver gives more power." No â€" it gives more damage. Match the size.
- "A vacuum cleaner is fine for dust." No â€" it can generate static and zap parts. Use compressed air, carefully.
- "Water cleans contacts." No â€" isopropyl alcohol only.

Limitations and safety. Do not use damaged tools (rounded tips, cracked handles, loose heads). Do not use a tool you do not understand. Ask before using anything unfamiliar. Keep tools dry and away from liquids.

F.5 Workspace setup and organisation

Why it matters. A safe workspace prevents accidents and mistakes. On a cluttered, badly lit bench, screws vanish, cables get mixed, and slips happen.

A good computer-repair workspace has:

1. A **clean, flat, sturdy table** at a comfortable height (so you are not stooping).
2. An **anti-static mat** under the work.
3. **Good lighting** â€" a desk lamp plus overhead light, so you can see into the case.
4. A **screw tray or egg carton**, ideally with the compartments labelled by location (case, motherboard, drive, bracket).
5. **Tools laid out within reach**, not scattered where you might knock them into the case.
6. **Enough free space** to lay out removed components safely, ideally on the antistatic mat or on antistatic bags.
7. A **hard floor** or a non-static mat under your feet; avoid working directly on carpet if possible.
8. A **container for loose screws and small parts**, so nothing rolls onto the floor and into a fan.
9. **Good ventilation** if you will be using compressed air or cleaning fluids.
10. **No food or drink** on the bench.

Real-life comparison. Setting up the bench is like a **surgeon preparing the operating theatre** before the patient arrives. Everything must be in its place, clean and ready, before the delicate work begins.

Common misconceptions.

- "Organisation slows me down." The opposite â€" it speeds you up, because you are not hunting for screws or re-opening a case.
- "Screws are all the same, one pile is fine." Different screws go in different places. One mixed pile means wrong screws in wrong holes and possible damage.

F.6 Safe lifting and handling

Why it matters. Computers and monitors can be heavy and awkward, and many benches are at the wrong height. Back injuries are a real workplace hazard.

Rules for safe lifting and handling:

1. **Assess the weight** before lifting. A desktop tower or a CRT monitor can be heavier than expected.
2. **Bend your knees, keep your back straight**, and lift with your legs, not your back.
3. **Hold the load close to your body** â€" do not stretch out your arms.
4. **Get help for heavy or awkward items** â€" two people for big monitors or full towers.
5. **Plan your path** â€" make sure the way is clear of cables and obstacles before you lift.
6. **Place the machine down gently** â€" never drop it on the bench.

7. **For components:** handle by the edges, support large cards and boards with two hands, and never carry a hard drive loosely where it can be knocked.
8. **Keep the bench tidy as you go**, so you do not trip over tools or cables.

Real-life comparison. Lifting a computer is like lifting a **bag of cement**. If you bend at the waist and jerk it up, your back pays. If you bend your knees and lift smoothly, you stay safe.

Common misconceptions.

- "It is small, so it is light." Some small items (a heavy GPU, a full tower PSU) are deceptively heavy. Assess first.
- "I can carry three things at once to save a trip." That is how things get dropped. Take one at a time on the bench.

F.7 Working only on machines you are authorised to repair

Plain-English rule. A technician must only work on equipment they have been **given permission** to open and repair. This protects the customer's property, the machine's warranty, and you.

Why it matters:

- A machine may still be under a **warranty**; opening it yourself can void the warranty and cause the customer to lose money.
- A machine may belong to a **school, church, company or government office**, where only approved staff may open it. Opening it without permission can cost you and the academy serious trouble.
- A customer's **data** is private. You must not browse their files, copy them, or share them. You only work on what is needed to fix the fault.
- If you damage an unauthorised machine, you may be personally liable.

The rule in practice:

1. Get the **owner's permission** before opening any machine.
2. Record what you received and its condition (with the intake process taught in Session 10).
3. Never open a machine labelled as belonging to someone else without the instructor's instruction.
4. Never plug a customer's drive into your own computer to "look" at their files without a good, authorised reason.
5. If you are unsure whether you are allowed, **ask before you touch.**

Real-life comparison. It is like a **church keyboard or the school's projector**. You may use it, but you do not open it and take it apart unless the person in charge says you may. Respect for others' property is both professional and legal.

Common misconceptions.

- "It is a broken machine, nobody will mind." Someone owns it. Always get permission.
- "The customer said 'fix it,' so I can do anything." Fixing it does not mean rifling through private files or changing things they did not ask for.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- A clean bench with the anti-static mat, the powered-off desktop, the full tool kit laid out, the screw tray, and safety glasses.
- The wrist strap ready to demonstrate.
- The balloon (from the opener) and the dead board.
- A bottle of isopropyl alcohol and a lint-free cloth (demonstration only).
- A can of compressed air (demonstrate outdoors or at a window with ventilation; learners wear safety glasses).

G.2 Demonstration 1 â€” the ESD-safe routine (6 minutes)

Say: "I am going to handle a component the safe way. Copy my habits exactly, because these are the habits that will protect every computer you ever touch."

Do this slowly:

1. "First, I **unplug** the computer from the wall. I do not just switch it off â€” I pull the plug out."

2. "Then I **press the power button once**. This discharges residual electricity, like letting the last water drain from a pipe."
3. "Now I put on my **anti-static wrist strap** and clip it to **bare metal** on the case."
4. "I **touch the metal** of the case once more before I pick up any part."
5. "I pick up the RAM **by its edges**. My fingers never touch the gold contacts."
6. "If I walk away and come back, I **touch the metal again** before resuming."
7. "When a part is not installed, it lives in an **anti-static bag**, never on the carpet or a plastic sheet."

Show the wrong way, then correct it (hands-on contrast):

- Ruffle your hair, then pretend to grab a board by its contacts. Freeze. "Wrong." Now do it the right way. "Right."
- Ask: *"Why do we clip the strap to the case and not to anything else?"* Expected: to keep us and the machine at the same electrical level â€” and never to a socket, which would be deadly.

Expected result: Learners can recite and copy the seven-step routine.

Common errors to watch: clipping the strap to a wall socket (stop at once), touching the case's painted surface instead of bare metal, forgetting to re-touch metal after walking away.

G.3 Demonstration 2 â€” tools, one by one (6 minutes)

Say: "Now, the tools of the trade. For each one, I will hold it up, name it, say what it is for, and say the main danger."

Hold up and explain, in this order:

1. **Phillips PH2** â€” "Most computer screws. The danger is using the wrong size and stripping the screw."
2. **Phillips PH1** â€” "Smaller screws. Match the size to the head."
3. **Flathead** â€” "For prying clips. The danger is slipping and scratching a board â€” use gently."
4. **Tweezers** â€” "For picking up dropped screws. Never force with them."
5. **Torch** â€” "For seeing inside the case. Always use light."
6. **Needle-nose pliers** â€” "For gripping and bending. Insulated handles are safer."
7. **Wrist strap** â€” "ESD protection â€” every single time."
8. **Compressed air** â€” "For dust. Upright, never tilt; wear glasses; ventilate; do not spin the fan with the jet."
9. **Isopropyl alcohol** â€” "For contacts and thermal paste. Never water."
10. **Multimeter** â€” "For measuring voltage. I demonstrate it **only** on a low-voltage battery. We never test mains wiring."

Then demonstrate the multimeter safely: Show the dials and probes. Measure a **small battery** only. Say: "This is a demonstration. In this course, you do not test live circuits. Later, with supervision, you will learn to test a PSU safely â€” never inside it."

Expected result: Learners can name each tool and its use, and remember at least one danger per tool.

Questions to ask during the demo:

- "Which screwdriver is the danger for scratching a board?" (The flathead.)
- "What is the golden safety rule with compressed air?" (Hold it upright, never tilt; ventilate; wear glasses.)
- "What do we never do with the PSU?" (Never open it.)

G.4 Demonstration 3 â€” workspace setup (3 minutes)

Say: "Watch how I set the bench. Notice everything has a place."

1. Wipe the table clean; place the anti-static mat.
2. Position the lamp so light falls into the case.
3. Place the tool kit to one side, screw tray in front-centre.
4. Unplug and discharge the machine; set it on the mat.
5. Lay out antistatic bags ready for removed parts.
6. Keep drinks off the bench and the floor clear.

Expected result: Learners replicate the layout at their own bench.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "My Safe Bench and My Tool Kit"

Objective: With the instructor guiding each step, learners set up a fully safe workspace, don the ESD protection, identify every tool, and complete the safety checklist and tool chart.

Materials per learner/group: A bench, anti-static mat, anti-static wrist strap, a tool kit, a screw tray, a powered-off practice desktop, Worksheet 2.1 (Tool Identification Chart), Worksheet 2.2 (Safety Procedures Checklist), safety glasses.

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **Clean and prepare** your bench: clear everything, wipe the surface, and place the anti-static mat.
2. **Position the light** so it shines into the case area. Place your tools to one side and the screw tray at the front.
3. **Unplug the practice computer** from the wall. Do not just switch it off â€" remove the plug.
4. **Press the power button once** to discharge residual electricity.
5. **Put on the anti-static wrist strap** and clip it to bare metal of the case.
6. **Touch bare metal once more**, then place the machine on the mat.
7. **Lay out every tool.** On Worksheet 2.1, write each tool's name and what it is used for. Tick the "safe for beginners" column only where appropriate (note: the multimeter and PSU work are **not** beginner tasks).
8. **Practise the correct screwdriver choice:** find a loose screw (or a bolt on the practice stand) and pick the **correct size** Phillips driver. Test that it seats fully without wobbling.
9. **Practise safe handling:** pick up a loose RAM stick (if available) by its edges; return it to its anti-static bag.
10. **Complete Worksheet 2.2** (Safety Procedures Checklist), ticking each item as you do it: unplugged, discharged, strap on, mat in place, tools identified, bench tidy, drinks away.
11. **Walk-away test:** step back two metres, return, and show the instructor that your **first action** is to touch bare metal before handling any part.
12. When finished, raise your hand for the instructor to check your bench and both worksheets.

Pause-and-predict points:

- After step 4: *"Why do we press the power button after unplugging? Predict what could happen if we did not."* Expected: to drain stored charge so we do not get a surprise or damage a part.
- After step 8: *"Predict what happens if I use a PH1 driver on a PH2 screw and force it."* Expected: the screw head gets stripped and may become impossible to remove.
- After step 11: *"Predict: what is the risk if I skip touching the metal after walking on carpet?"* Expected: a static discharge into the part when I touch it.

Expected result: Every learner has a safe bench, a completed tool chart, a completed safety checklist, and can demonstrate the ESD routine from memory.

Troubleshooting while walking the room:

- Learner still has the machine plugged in â†' stop and correct immediately; this is a critical safety error.
- Learner touches RAM contacts â†' redirect to the edges; explain the oil problem.
- Learner cannot name a tool â†' have them look at the chart, then cover the chart and try again.
- Learner uses the wrong driver size â†' show them the wobble, then the correct fit.

Completion checklist (learner ticks each):

- ☐ Bench cleaned and anti-static mat in place
- ☐ Light positioned; tools and screw tray laid out
- ☐ Computer unplugged (plug removed) and power button pressed
- ☐ Anti-static wrist strap worn and clipped to bare metal
- ☐ Bare metal touched before handling
- ☐ All tools named and their uses written on Worksheet 2.1
- ☐ Correct screwdriver size selected and tested
- ☐ RAM handled by the edges and returned to its bag
- ☐ Worksheet 2.2 completed

- ☐ Walk-away test passed (touched metal first on return)

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: "The Safety Audit"

Scenario: You are now a junior technician at **Elias Computer Clinic, Diobu, Port Harcourt**. Your supervisor asks you to inspect the shop bench and a colleague's habits before you begin a repair, and to write a short audit that will be posted on the wall as a reminder for everyone.

Instructions: Read the ten situations below. For each, decide whether it is **SAFE** or **UNSAFE**. If unsafe, write the correction. Then write the **single most important rule** you would put at the top of the wall poster.

Situations:

1. A technician unplugs the power cable but does not press the power button before opening the case.
2. A technician works barefoot on a thick woollen carpet while handling a motherboard.
3. A technician clips the anti-static strap to a screw on the case, but leaves the shop's power cable plugged into the wall "just for the light."
4. A technician uses a PH1 screwdriver on a large PH2 case screw and pushes hard when it slips.
5. A technician smells a burning smell from a customer's computer and immediately opens the case, which is still plugged in, to look.
6. A technician picks up a RAM stick by its gold contacts to show it to a customer.
7. A technician holds the case fan still with a pencil while blowing it with compressed air, and wears safety glasses.
8. A technician using compressed air tilts the can sideways to reach under a heatsink.
9. A technician is unsure how to replace a power supply, so they ask their supervisor to supervise the job.
10. A technician opens a school's computer without asking anyone, saying "it was switched off anyway."

Requirements:

- Mark all ten SAFE/UNSAFE.
- Give the correction for every unsafe item.
- Write your single top rule.
- Write one sentence explaining *why* your top rule matters most.

Constraints:

- Use correct safety language: ESD, residual electricity, unplug, ground, anti-static.
- Keep each correction to one or two sentences.

Time allowance: 15 minutes.

Success criteria:

- All ten situations correctly judged.
- Corrections are accurate and specific.
- The top rule captures the spirit of the session.

Hints if stuck:

- Hint 1: Anything done while the machine is still plugged in is automatically unsafe.
- Hint 2: Static comes from carpets, woollen clothes and rubbing â€” anything that builds charge.
- Hint 3: The wrist strap grounds you to the *machine*, never to a socket.

Instructor answer key (do not show learners):

1. **UNSAFE.** After unplugging, press the power button once to discharge residual electricity.
2. **UNSAFE.** Do not work on carpets or barefoot on wool; work on a hard floor or anti-static mat and wear the strap.
3. **UNSAFE.** The machine should be unplugged while working. The strap is clipped to the case for grounding â€” but the mains plug must still be removed, and never rely on the machine being "plugged for the light." (If the "light" needs power, use a separate lamp.)
4. **UNSAFE.** Use the correct size driver (PH2 for that screw) and never force; the wrong size strips the screw head.

5. **UNSAFE.** With burning smell/smoke, unplug first (from a safe distance if needed) and do not open a live machine. Stop and get the instructor.
6. **UNSAFE.** Hold RAM by the edges; never touch the gold contacts (finger oils damage the connection).
7. **SAFE.** Holding the fan still while blowing and wearing safety glasses is correct â€" it prevents over-spinning the bearing and protects the eyes. (The machine should, of course, be unplugged.)
8. **UNSAFE.** Hold the compressed-air can upright; tilting can spray liquid propellant and damage parts.
9. **SAFE.** Knowing your limits and asking for supervision is exactly right.
10. **UNSAFE.** Never open equipment you are not authorised to repair, even if it is switched off. Get permission.

Top rule (accept any strong, defensible answer): "Unplug the power and protect against static before you touch anything inside a computer." **Why it matters most:** because these two habits prevent the two biggest causes of injury and damage â€" electric shock and ESD.

Marking guidance (out of 12): one mark per correct SAFE/UNSAFE (10), one mark for a strong top rule, one mark for a correct explanation.

J. REAL-WORLD CASE STUDY

Situation. *Tunde* is a talented young technician in a busy Port Harcourt shop. He has good hands and learns fast, but he has developed two dangerous shortcuts: he opens machines while they are still plugged in "to save time," and he rarely wears his anti-static strap because "Port Harcourt is humid and it never causes problems." One Tuesday, a customer brings in a laptop to have its RAM upgraded and its dust cleaned. Tunde opens the case with the charger still connected, drops a metal screwdriver near the PSU area, and then, without a strap, grabs the RAM by its contacts. The machine boots after the upgrade. The customer pays and leaves. Three weeks later, the customer returns furious: the laptop now shuts down randomly. Tunde insists the RAM "must be faulty from the shop."

Relevant facts.

- The charger was connected while Tunde worked (mains power present).
- No anti-static strap was worn; the RAM was touched by its gold contacts.
- A metal tool was used near a live PSU area.
- The failure appeared weeks later â€" the classic timing of a **latent** fault (ESD or a poor contact), not a part that was "faulty from the shop."

Decision points.

1. What are the *probable* causes of the weeks-later shutdown â€" and how would you confirm rather than guess?
2. Was the machine safe to open the way Tunde did it?
3. How should Tunde have handled the RAM?
4. Who is responsible for the failure, and what should the shop do for the customer?

Possible approaches.

1. Blame the RAM supplier and demand a replacement â€" but the evidence points elsewhere.
2. Re-open the machine safely, check the RAM seating and contacts, test the RAM properly, and inspect for damage. Honest and methodical.
3. Continue as before, since "it has always worked" â€" this is how a reputation dies.

Recommended solution. Approach 2. A good technician would admit the machine was handled unsafely, re-open it **unplugged and with the strap**, reseat the RAM cleanly (checking for oily contacts), test the memory properly (a skill from Session 5), and confirm the fault. A latent ESD failure of the RAM stick is likely; the shop should replace the damaged part at the shop's cost and apologise honestly. Crucially, the shop should then **change its habits**: always unplug, always strap up, never bring metal tools near a live board.

Trade-offs.

- Approach 1 is fast and shifts blame, but it is dishonest and will not fix the machine if the cause is a poor contact or latent damage from handling.
- Approach 2 is slower but honest, and it protects the customer and the shop's reputation. It also teaches the lesson.
- Approach 3 is what actually causes shops to lose customers â€" word travels fast in Port Harcourt.

Professional connection. In the repair trade, your **reputation is your business**. A technician who is careful earns trust and repeat customers; a technician who cuts corners saves seconds and loses customers. The two rules in this case â€" unplug, and strap up â€" cost nothing and prevent nearly all of the serious damage a beginner can cause. This is why CEA teaches safety before tools and tools before repair.

Transfer exercise. "Name two habits from this case that, if Tunde had followed them, would have prevented the fault, and explain how each one helps." (Unplug â€" prevents shock and accidental short damage; strap and edge-handling â€" prevents ESD damage and contact contamination.)

Safety reminder. This case is only to be investigated with the machine **unplugged and discharged**, with a strap, and under supervision. Never open a machine that is on, never open the PSU, and never test live mains.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this as the live troubleshooting guide. Ask learners to find the relevant row rather than telling them the answer.

K.1 Working on a plugged-in machine

Symptoms: Possible electric shock, a spark, a sudden fan spin, or a part that dies. In the worst case, serious injury. **Likely cause:** Rushing, or assuming "switched off" is the same as "unplugged." Caused by habit, not difficulty. **How to investigate:** There is no safe investigation â€" the rule is prevention. Confirm the plug is out before work begins. **Safe corrective steps:** Stop immediately. Unplug from the wall. Press the power button once to discharge. Only then continue. **How to confirm resolved:** The machine is verified unplugged and discharged. **How to prevent recurrence:** Make "unplug and discharge" the first item on every checklist.

K.2 Skipping the anti-static strap

Symptoms: No immediate sign. Weeks later, random failures, a part that "died for no reason." **Likely cause:** ESD damage, often latent. Confirmed only indirectly â€" there is usually no visible mark. **How to investigate:** Look for the pattern (recent handling, dry conditions) and test the suspect part properly (later sessions). You cannot see ESD damage. **Safe corrective steps:** Adopt the routine: strap, mat, touch metal, hold by edges, antistatic bags. **How to confirm resolved:** Prevention, not repair â€" you cannot undo ESD damage. **How to prevent recurrence:** Treat the strap as non-negotiable, every time.

K.3 Using the wrong screwdriver size

Symptoms: A screw that will not turn, a chewed-up screw head, or a screw stuck forever. **Likely cause:** Wrong driver size or type (PH1 on a PH2 screw, flathead on a Phillips). **How to investigate:** Try the correct size; if it does not seat snugly, the head may already be damaged. **Safe corrective steps:** Use the correct driver; if the head is rounded, use a proper screw extractor (instructor-level) â€" do not keep forcing. **How to confirm resolved:** The screw turns cleanly with the correct driver. **How to prevent recurrence:** Identify the screw type and match the driver before turning.

K.4 Touching gold contacts / mishandling boards

Symptoms: A part not detected, an upgrade that "does not work," intermittent faults. **Likely cause:** Finger oils on gold contacts, or physical damage from bending. **How to investigate:** Inspect contacts; reseat holding only the edges. **Safe corrective steps:** Clean contacts gently with isopropyl alcohol (unplugged, supervised) and reseat. **How to confirm resolved:** The part is detected and works. **How to prevent recurrence:** Always hold by edges; never touch contacts.

K.5 Compressed air used wrongly

Symptoms: Liquid spray on the board, a part that suddenly chills, a fan bearing that fails later, or dust in the eyes/lungs. **Likely cause:** Tilted can, too close, spinning the fan with the jet, or no eye protection/ventilation. **How to investigate:** Look for residue; check the fan spins freely by hand (with the machine off). **Safe corrective steps:** Hold the can upright; keep distance; hold the fan still; wear glasses; ventilate (ideally outdoors). **How to confirm resolved:** No residue; fan spins freely. **How to prevent recurrence:** Follow the compressed-air safety rules every time.

K.6 Opening a machine you are not authorised to repair

Symptoms: A warranty voided, an angry owner, a data-privacy complaint, or damage you must pay for. **Likely cause:** No permission obtained before opening. **How to investigate:** Check with the owner or instructor before touching. **Safe corrective**

steps: Stop, apologise, and get permission. Record the machine's condition. **How to confirm resolved:** You have documented permission and the machine is handled properly. **How to prevent recurrence:** Always ask first â€" owner, warranty, permission.

Distinguishing likely vs confirmed: Most safety problems are *prevented*, not diagnosed. Where a fault has already happened (a dead board, a stripped screw), the "likely cause" becomes "confirmed" only when you check it and the evidence matches. Teach learners that in safety, the goal is to never need the diagnosis.

L. SESSION ASSIGNMENT

Assignment title: "My Safety and Tools Reference"

Purpose: To build a personal, practical safety and tools guide that the learner will keep on the bench for the rest of the course and their career. This supports the capstone's emphasis on safe, professional work.

Realistic scenario: You have just been hired as a junior technician. On your first day, the shop owner hands you a bench and says: "Before you touch any customer's computer, I want your personal safety rules and your tool list written down." This is that document.

Detailed instructions:

1. Create a document in Microsoft Word or Notepad titled **"My Safety and Tools Reference"**.
2. Write a section **"ESD Prevention"** with at least five numbered points (for example: wear a strap, work on a mat, touch bare metal, hold by edges, use antistatic bags).
3. Write a section **"Electrical Safety Rules"** with at least six numbered points (unplug, discharge, never work live, never open the PSU, insulated tools/no jewellery, know your limits).
4. Write a section **"Tool Chart"** as a table: for each of at least eight tools, give the **name**, what it is **used for**, and one **safety warning**.
5. Write a section **"Workspace Setup Checklist"** with at least six items (clean table, anti-static mat, lighting, screw tray, ventilation, no drinks).
6. Write a section **"When to Call a Professional"** with at least four clear situations (PSU internals, mains wiring, anything you are unsure of, a job requiring special tools).
7. Save the file as **"My Safety and Tools Reference"** on the Desktop.
8. Be ready to explain, in one sentence, why each of your top three rules matters.

Required deliverables: One saved document containing all five sections.

Submission format: Saved on the Desktop with the exact file name. If a learner has no home computer, complete it in the academy lab and show it before leaving.

Deadline: Before Session 3 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
ESD prevention section	5+ accurate points with reasons	3â€"4 points	Fewer than 3 or inaccurate
Electrical safety section	6+ clear rules	4â€"5 rules	Fewer than 4
Tool chart	8+ tools, uses and warnings all correct	6â€"7 tools, mostly correct	Fewer than 6 or many errors
Workspace checklist	6+ complete items	4â€"5 items	Fewer than 4
Professional referral section	4+ clear situations	2â€"3 situations	Fewer than 2

Common mistakes to avoid: confusing static electricity with mains electricity; saying "switch off" when you mean "unplug"; leaving out the PSU danger; forgetting the compressed-air warnings; copying the session notes word for word instead of using own words.

Instructor reference solution (summary, do not show learners beforehand):

- ESD: strap worn and clipped to bare metal; anti-static mat; touch metal before handling; hold by edges, never contacts; components kept in antistatic bags; avoid carpets/wool.
- Electrical: unplug from wall; press power button to discharge; never work on a live machine; never open the PSU; no metal jewellery/tools near live circuits; no liquids on the bench; know the emergency switch; know your limits.
- Tool chart (sample): Phillips PH2 â€" most computer screws â€" wrong size strips screws; flathead â€" prying clips â€" can slip and scratch; tweezers â€" retrieve screws â€" never force; torch â€" see inside â€" keep dry; wrist strap â€" prevent ESD â€" check for damage; compressed air â€" remove dust â€" upright, ventilated, glasses; isopropyl alcohol â€" clean contacts â€" never water; multimeter â€" measure voltage â€" instructor demonstration only, never mains.
- Workspace: clean flat table; anti-static mat; good lighting; screw tray; ventilation; tools within reach; no food/drink; clear floor.
- Professional referral: PSU internals; mains/wall wiring; unknown faults beyond experience; jobs needing special tools or training.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below.)

Q1. Short answer. What does ESD stand for, and why is it dangerous to computer parts?

Q2. Multiple choice. What is the correct first thing to do before opening a computer case? a) Put on safety glasses b) Unplug the power cable from the wall c) Remove the side panel d) Touch the RAM

Q3. Explain in your own words. Why can ESD damage be worse than damage you can see?

Q4. Practical identification. Name the tool and its purpose: (a) a cross-shaped screwdriver used for most computer screws; (b) a fine pair of pincers for picking up dropped screws; (c) a small lamp used to see inside a dark case.

Q5. Error detection. A learner says: "After I unplug the computer, I do not need to press the power button, because no electricity is flowing." Find the mistake and correct it.

Q6. Scenario. A friend asks you to open their church's computer to clean the dust, but tells you it is still under warranty. What should you do, and why?

Q7. Short answer. Give two safety rules for using compressed air.

INSTRUCTOR ANSWER KEY (Q1â€"Q7)

A1. ESD = **electrostatic discharge** â€" a sudden jump of static electricity from your body (or another object) to a component. It is dangerous because it can carry a voltage far higher than a chip can survive, and the damage can be **invisible and delayed**, so the part may fail weeks later.

A2. (b) Unplug the power cable from the wall. (Then press the power button to discharge.)

A3. Visible damage (a burn, a crack) can be seen and dealt with immediately. ESD damage often leaves **no visible mark** and may be latent â€" the part works at first and fails later, so the technician blames the wrong cause and the customer loses trust. Invisible, delayed damage is the hard part.

A4. (a) Phillips screwdriver (PH1/PH2) â€" used for most computer screws; (b) tweezers â€" for retrieving small screws and holding small parts; (c) torch/flashlight â€" for seeing inside the case.

A5. The mistake is believing that unplugging removes all stored charge instantly. Some parts (capacitors) can hold a **residual charge** for a short time. The correction: after unplugging, **press the power button once** to help discharge residual electricity before handling anything.

A6. Refuse to open it yourself and explain that opening a machine still under warranty may **void the warranty**, costing the owner money. Advise them to take it to the warranty provider (or get the owner's documented permission and check the terms). Getting permission and protecting the owner's property are the professional rules.

A7. Any two of: hold the can **upright** (never tilt); use in a **well-ventilated** area (ideally outdoors); wear **safety glasses**; keep a safe distance; do **not spin the fan** with the jet; never use a vacuum cleaner on electronics.

Marking: up to 2 marks each (Q3, Q6 up to 3). Total â‰ˆ 16; scale proportionally. Full credit requires correct names and reasoning.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned how to work on computers without harming yourself or the machine. You learned that **ESD (electrostatic discharge)** is an invisible danger that can damage parts without any sign, and how to prevent it with a **wrist strap, an anti-static mat, touching bare metal, handling boards by the edges, and using antistatic bags**. You learned the **electrical safety rules**: always unplug and discharge, never work on a live machine, and **never open the PSU**. You learned the **tools of the trade** and what each is for, why the **correct screwdriver size** matters, and the safe use of compressed air and isopropyl alcohol. You set up a **safe, organised workspace**, learned **safe lifting and handling**, and learned to work only on machines you are **authorised** to repair.

Five important takeaways:

1. **ESD is invisible and can kill parts silently** â€" strap up every time.
2. **Unplug from the wall and press the power button** before opening any case.
3. **Never open the PSU** â€" replace it, never repair it.
4. **Use the correct tool and the correct screwdriver size** â€" the wrong one causes damage.
5. **Know your limits and get permission** â€" refer tricky or unauthorised jobs.

Questions to help you recognise what you can now do:

- Can I demonstrate the ESD-safe routine from memory?
- Can I explain in one sentence why ESD is dangerous?
- Can I name at least eight tools and one danger per tool?
- Can I state the rule about the PSU and the wall plug without hesitation?

Connection to the next session: You now know the parts and how to be safe around them. Next session, you put both skills together: you will **disassemble and reassemble a real desktop computer** correctly â€" removing the panel, disconnecting cables safely, extracting and refitting components, recording screws and cables, and verifying the machine powers on. Everything you learned about safety today is used in every step of Session 3.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner unplugs and discharges **without being reminded**.
- The learner puts on the wrist strap and touches bare metal as a habit before each session of handling.
- The learner selects the correct screwdriver size by inspecting the screw first.
- The learner handles boards by the edges and never touches contacts.
- The learner can name all common tools and their purposes, with at least one danger each.
- The learner states the PSU rule and the authorisation rule confidently.

Common misconceptions to watch for and correct:

- "Static electricity is not dangerous / Port Harcourt is too humid." â†' Humidity reduces but does not remove risk; strap up anyway.
- "Switched off means safe." â†' Only **unplugged and discharged** is safe.
- "Any screwdriver will do." â†' The wrong size strips screws and damages boards.
- "I can repair the PSU." â†' Never open it; replace it.
- "If it is broken and off, I can open anyone's machine." â†' Get permission; respect warranty and data privacy.

Suggested remedial exercises for learners who struggle:

1. **Routine drill:** have the learner perform the unplug-discharge-strap-touch routine five times until it is automatic.
2. **Tool matching cards:** match tool names to pictures and to uses, then to dangers.
3. **Scenario sorting:** give ten cards (safe/unsafe) and have the learner sort them into two piles, explaining each.
4. **Screw-size drill:** present several screws and drivers and have the learner match them correctly without forcing.

Differentiation:

- **Level 1 (supported):** provide the worksheets partially filled and walk the learner through the bench setup step by step; pair with a steady-handed partner.
- **Level 2 (standard):** complete the exercise and the Safety Audit independently.
- **Level 3 (challenge):** research additional ESD/safety equipment (ESD gloves, ionisers, heel straps) and explain when each is used; also explain the difference between ESD safety and mains safety in detail.

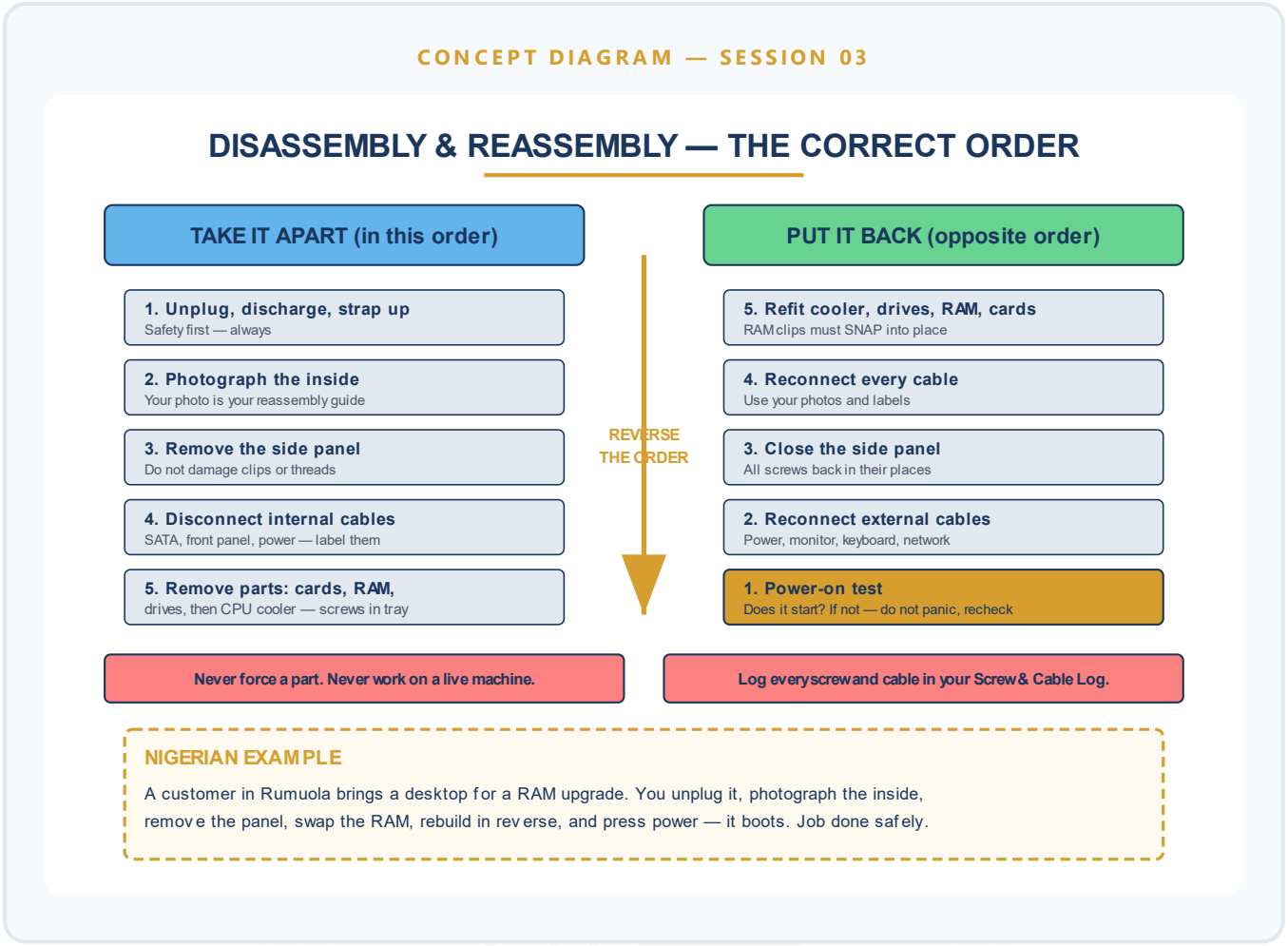
Timing guidance: Safety cannot be rushed. If the room is slow, compress the lecture sections F.5â€"F.7 into quick summaries, but **never** shorten the ESD and electrical rules (F.1â€"F.3) or the practical routine (H). The practical demonstration of the correct routine is the assessed priority.

END OF SESSION 2

Next: Session 3 â€" Disassembly and Reassembly: Opening a Computer Safely and Putting It Back Correctly.

SESSION 03 OF 12

Disassembly Reassembly



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 3 â€" DISASSEMBLY AND REASSEMBLY: OPEN IT AND PUT IT BACK CORRECTLY

A. SESSION IDENTIFICATION

Item	Detail
Session number	3 of 12
Session title	Disassembly and Reassembly
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner, but hands-on and delicate; needs close supervision

Item	Detail
Prerequisite knowledge	Sessions 1â€"2: component names and functions; ESD, electrical safety, tools.
Connection to previous session	The learner can name the parts and work safely. This session applies both skills by opening a real desktop, removing components in the correct order, and rebuilding it so it works.
Connection to the course	Disassembly and reassembly are the physical foundation of all repair. Session 7 (component replacement) and the Session 12 capstone both require exactly these skills. This session also completes the Week 1 mini-project, "My First Teardown."
Preparation before class	At least two practice desktop computers (ideally non-working or expendable units) openable by simple screws; tool kits; anti-static wrist straps per learner; anti-static mats; screw trays; a smartphone/camera for photographing connections; printed Disassembly and Reassembly checklists; tape and a marker for labelling cables; a working reference machine to test against.

Safety note for this session: This is the first session where learners physically open a computer. Every rule from Session 2 applies and is enforced. The machine must be **unplugged from the wall and discharged** (power button pressed) before the case is opened. **No mains-powered work.** Never open the PSU. Never force a component. All work stops immediately if a learner is unsure. Learners work in pairs or small groups with an instructor circulating. Laptops are **not** dismantled in this session; desktop practice only. If a practice machine is a working one, test power-on only briefly and with the instructor present.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Prepare a machine safely** â€" unplug, discharge, set up the anti-static mat, wear the strap, and photograph the internals before starting.
2. **Safely disconnect all external cables** from the back of a desktop computer.
3. **Remove the side panel** without damaging clips or threads.
4. **Disconnect internal cables** â€" SATA data and power, front-panel, and power connectors â€" safely and without straining them.
5. **Remove components in the correct order** (expansion card, RAM, storage drives, then cooler) and set them aside safely.
6. **Reinstall components in the correct order**, applying the correct technique (for example, RAM clips snapping into place).
7. **Reconnect all cables correctly**, using photographs and labels to guide them.
8. **Verify the machine powers on** after reassembly, and know what to do if it does not.
9. **Record screw locations and cable positions** so reassembly is accurate and repeatable.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- At least two practice desktop computers per class (non-working or expendable preferred; working units acceptable under close supervision).
- Tool kits: Phillips PH1 and PH2 screwdrivers, tweezers, torch, needle-nose pliers.
- Anti-static wrist straps (one per learner) and anti-static mats.
- Screw trays or labelled containers (egg cartons/trays) for keeping screws organised by location.
- A torch/desk lamp for each bench.
- A smartphone or camera (one per group) for photographing connections. If no camera is available, provide paper and pencil for sketching and labelling.

Software:

- None required. (Optional: a system-info tool is not needed for this session.)

Safety gear:

- Anti-static wrist straps and mats.
- Safety glasses if compressed air is used (optional this session).
- A first-aid kit in the room.

Sample / fault components:

- A spare dead hard drive and a spare RAM stick for practice handling, in case a group finishes early.
- A machine deliberately set up with a "loose RAM" fault for the troubleshooting discussion (optional, instructor-prepared).

Worksheets (printed):

- Worksheet 3.1 â€" "Disassembly Checklist" (one per learner).
- Worksheet 3.2 â€" "Reassembly Checklist" (one per learner).
- Worksheet 3.3 â€" "Screw and Cable Log" (a table: step, screw type/location, cable label, notes).
- Cable labels (small stickers or masking tape + marker).

Alternatives for a learner or academy with limited equipment:

- If only one machine is available, rotate small groups (3â€"4 learners) around it, while others watch the system-recorded demonstration or work with photo sheets and the ordering exercise.
- If no machine can be opened, run the session as a "dry" simulation: learners sequence cards showing each step, plus a photo-based identification of cable and screw types, and a walk-through of the demonstration machine.
- If a learner has shaky hands or low confidence, pair them with a careful partner: one removes, one records (writes the screw/cable log and takes photos), then swap.

D. OPENING ACTIVITY (6 minutes)

"The Teardown Challenge"

Instructor setup: Place a fully assembled desktop computer on the front bench, connected to a monitor. Power it on and let the class see it working. Then power it off properly and unplug it. Keep the tool kit nearby.

What the instructor says:

> *"Look at this computer. It is working. It has taken engineers years to design and factories hours to build. Now watch this."*

Unplug the machine and hold up a single screwdriver.

> *"With one small screwdriver and the right knowledge, I am going to take this whole thing apart â€" and then put it back together so that it still works. Anybody can break a computer by opening it. It takes a technician to open it and give it back working.*
>
> *Here is your challenge for today: by the end of this session, you will do exactly what I am about to do. You will open a real desktop, remove its parts, put them back, and press the power button â€" and it must come on. This is the skill that separates a technician from somebody who just 'carries' computers to the repair shop.*
>
> *But before we touch anything, I want you to remember Session 2. What is the first thing we do before opening any case?"*

Pause for answers.

> *"Correct â€" we unplug it. Then we press the power button to discharge. Then we strap up. Safety first, always. Today, safety and skill walk together."*

Then create the question: Point to the inside (once the panel is off) and say:

> *"Tell me â€" if you had to remove all these parts, which would you take out first? The RAM? The big card? The hard drive?*

Why does the order even matter? By the end of today, you will know the correct order and the reason for it."

Expected learner responses: Learners may guess "biggest first" or "whatever is easiest." Accept the answers. Some will remember the safety steps from Session 2 (unplug, discharge, strap) and should be praised. Note their guesses â€" you will correct the reasoning in the lecture.

Why this opener works: It frames a mechanical task as a challenge with a clear, observable win (the machine works at the end), connects to the previous session's safety rules, and creates a genuine question (what order, and why) that the lecture answers.

E. PREREQUISITE REVIEW (7 minutes)

Run as a quick oral review; ask different learners.

Question 1: "What are the first two things you must do before opening a computer case?"

- Expected answer: Unplug the power cable from the wall; press the power button to discharge residual electricity.
- Common gap: learners say "switch it off." Correct: **unplug** from the wall, then discharge. Off is not enough.
- Bridge: "Today, that is step one of everything."

Question 2: "What is ESD, and name two ways you prevent it."

- Expected answer: Electrostatic discharge; prevent with an anti-static wrist strap clipped to bare metal, an anti-static mat, touching bare metal, holding by edges, antistatic bags.
- Common gap: learners remember the strap but forget to *touch bare metal* before handling. Emphasise.
- Bridge: "You will handle real boards today, so this matters every minute."

Question 3: "Name three essential tools for opening a desktop."

- Expected answer: Phillips screwdriver (correct size), tweezers, torch (and anti-static wrist strap, pliers, screw tray).
- Common gap: not knowing the screwdriver size matters. Remind them: wrong size strips screws.
- Bridge: "Choosing the correct screwdriver is the first skill of disassembly."

Question 4: "Where is the power supply, and what is the golden rule about it?"

- Expected answer: The metal box that supplies electricity to all parts; **never open it** â€" replace it if faulty.
- Common gap: thinking the PSU can be repaired. Correct firmly.
- Bridge: "You will meet the PSU today but you will never open it."

Question 5: "Why do we keep screws organised and take photos?"

- Expected answer: So we know which screw goes where and how the cables were connected, making reassembly correct.
- Common gap: learners think it wastes time. Explain it saves time and prevents damage.
- Bridge: "Today, taking photos and logging screws is not optional â€" it is how professionals work."

Bridge to today (say this):

> *"Think of disassembling a computer like **dismantling a rented room you must hand back exactly as you found it**. Every item has a place; every wire has a route. If you throw everything in a pile and forget where the wires went, you will struggle to rebuild it. So the professional's secret is not strength or speed â€" it is **recording and order**. Today you learn the order, and the discipline of recording."*

If more than a third cannot answer Question 1, spend an extra three minutes re-teaching the unplug-and-discharge routine, because it is safety-critical.

F. FULL LECTURE NOTES

F.1 Why order matters â€" the idea of reversibility

The idea. Disassembly is the process of taking a computer apart; reassembly is putting it back together. The single most important principle is this: **reassembly is the reverse of disassembly**. If you removed A, then B, then C, you reinstall C, then B, then A.

Why it exists. Components are often stacked or overlapping. A long graphics card may sit over the RAM clips; a cooler may sit over the CPU; a drive cage may block a cable. If you try to remove things in the wrong order, you will find you cannot â€" or you will force something and break it. The order is not arbitrary; it follows the physical layout.

Real-life comparison. Think of **dressing a child for school**. You put on the vest first, then the shirt, then the jumper. To undress, you reverse it: jumper, shirt, vest. If you tried to remove the vest first, you would fail. A computer is no different.

Realistic repair-shop scenario. A new technician at a Port Harcourt shop removes the CPU cooler first, before the RAM. He knocks the cooler into a RAM stick and bends a pin in a slot. Now, instead of a simple cleaning job, the board is damaged. Order is not a formality; it prevents damage.

Common misconceptions.

- "Order does not matter; I will just take out whatever I can reach." It matters â€" the layout dictates the order.
- "Faster is better." In repair, **correct** is faster, because mistakes cost far more time than care.

Limitations and safety. Different machines have different layouts. The order below is the standard, but always **look first**, take photos, and adapt to your specific machine. If you are unsure, stop and ask the instructor.

F.2 Safety preparation before any disassembly

The non-negotiable pre-flight checklist (do this every time):

1. **Get permission.** Only open machines you are authorised to repair (Session 2, F.7).
2. **Record the machine.** Note the model, the condition, and any existing damage. Take a photo of the outside.
3. **Switch off** the operating system properly (do not just yank the power while Windows is running, unless it is already dead).
4. **Unplug** the power cable from the wall.
5. **Press the power button once** to discharge residual electricity.
6. **Set up the bench:** anti-static mat, good light, screw tray, tools within reach, no drinks.
7. **Wear the anti-static wrist strap** and clip it to bare metal of the case.
8. **Touch bare metal** before handling any internal part.
9. **Photograph the inside** before disconnecting anything. Take several photos from different angles, especially of the cable connections.
10. **Label cables** with tape and a marker if the machine is unfamiliar or complex.

Why each step matters:

- Permission protects the owner, the warranty and you.
- Recording protects you if something is already damaged.
- Proper shutdown protects the data and the operating system.
- Unplugging and discharging protect you from shock and the parts from surges.
- The mat, strap and bare-metal touch protect the parts from ESD.
- Photos and labels protect you from forgetting where things go.

Real-life comparison. This is like the **pre-flight checklist a pilot runs before take-off**, no matter how many times they have flown. The checklist is not a sign of inexperience; it is what professionals rely on so that nothing is forgotten.

Common misconceptions.

- "I have done this before; I do not need the checklist." Experience is exactly why professionals *use* checklists â€" to stay consistent when they are confident and could become careless.
- "I can skip photos for a simple job." Even simple jobs go wrong; photos cost nothing and save everything.

Safety implications. Never proceed without steps 4, 5, 7 and 8. These are the life-and-machine rules. If you are interrupted, re-touch bare metal before resuming, and re-check the plug is out.

F.3 Disassembly â€" the correct order, step by step

Below is the standard desktop disassembly order. Each step lists **what to do** and **why**.

Step 1 â€" External cables. Disconnect everything from the back: monitor cable, keyboard, mouse, USB devices, network (Ethernet) cable, speakers, and the power cable (already unplugged). Label any that are unclear. *Why:* You must free the machine so you can move it to the bench safely. Cable labels help you reconnect exactly.

Step 2 â€" Move to the bench and open the panel. Place the machine on its side on the anti-static mat (or stood up, per its design). Remove the screws holding the side panel (usually two at the back). Some cases use thumb screws (turn by hand) or slide-and-lift latches. *Why:* The side panel gives access. Keep the panel screws in the screw tray â€" do not put them on the floor.

Step 3 â€" Photograph and note the layout. Before touching cables, photograph the whole inside, then close-ups of the connections: the 24-pin power connector, the CPU power connector, the SATA data cables, the SATA power cables, and the front-panel wiring. *Why:* This is your map for reassembly. The front-panel wires (power switch, reset, front USB/audio) are the most easily forgotten.

Step 4 â€" Disconnect internal cables. Work carefully and grip the connectors, not the wires (pulling wires can break them):

- Unplug the **SATA data cables** from the drives (press the small latch if present).
- Unplug the **SATA power connectors** from the drives.
- Unplug the **power connectors** from the motherboard only if you will remove the board â€" otherwise leave them.
- If removing the board, note the **front-panel connectors** carefully. *Why:* Cables must be out of the way before components can be removed. Grip the connector to avoid damaging the wire and the socket.

Step 5 â€" Remove expansion cards (the graphics card first). If a graphics card is fitted, it is usually the first component to remove because it often lies over other parts. Remove the screw holding its bracket to the case, release the clip at the back of the PCIe slot, and pull the card straight up by its edges, supporting its weight. *Why:* The card overlaps and blocks other parts; removing it early frees access. Supporting it prevents bending it or cracking the slot.

Step 6 â€" Remove RAM. Push down the retaining clips at both ends of each RAM slot; the stick will lift up. Pull the stick straight out by its edges and place it in an anti-static bag. *Why:* RAM is easy to remove and blocks nothing else, so it comes after the card and before the drives. Holding by the edges protects the contacts.

Step 7 â€" Remove storage drives. Remove the screws holding each drive (or the drive cage screws), disconnect any remaining cables, and slide the drive out. Handle HDDs gently. *Why:* Drives sit in bays that may need the cables cleared first. Gentle handling protects the fragile HDD.

Step 8 â€" Remove the CPU cooler (only if the task requires). Unplug the fan's power connector from the board. Then unscrew or release the cooler's clips. Gently twist the cooler slightly to break the seal of old thermal paste, then lift it off. Place it aside, fan-side up. *Why:* The cooler sits on top of the CPU; it comes off late because it is the most delicate and requires handling paste. Do this only when needed (for example cleaning or CPU work). *Warning:* With the cooler off, **never** power on the machine â€" the CPU can overheat in seconds.

Step 9 â€" Remove the motherboard (only if required). Remove all screws holding the board to the case (note their positions), disconnect any remaining cables, and lift the board out carefully, holding by its edges. *Why:* This is the last step because the board is the foundation to which everything else attaches. Beginners usually leave the board in place unless the task demands otherwise.

A note on laptops. Laptops follow the same *principle* (reverse order) but the parts are far smaller and more delicate, and many are held by hidden clips and tiny screws. **Do not dismantle laptops in this session.** Only attempt laptops under supervision and after you are confident on desktops.

Real-life comparison. Disassembly is like **unpacking a well-packed suitcase**: you take out the top layers first, and you keep note of how things were folded. Reassembly is packing it the same way so it closes properly.

Common misconceptions.

- "The CPU cooler is always the first thing to remove." No â€" it is near the last, because it sits on top and handling is delicate.
- "I can pull cables by the wire." No â€" always grip the connector.
- "If it does not move, pull harder." Never. Check for a screw, clip or cable.

Limitations and safety. Some machines use proprietary screws and clips (common in branded desktops). If you meet resistance you cannot explain, stop and ask. Never force.

F.4 Reassembly â€" the reverse order, with extra care

Reassembly is the reverse of disassembly, but with two extra emphases: **care with paste** and **verification before closing**.

Step 1 â€" Motherboard (if removed) first. Place the board back, align it with the case's standoffs, and fit the I/O shield correctly at the back. Replace all screws evenly, **without over-tightening**.

Step 2 â€" CPU cooler. If the cooler was removed, the old thermal paste must be cleaned off both the CPU and the cooler with isopropyl alcohol, and a **small pea-sized amount** of fresh paste applied to the centre of the CPU. Refit the cooler, tighten or clip it evenly (tighten screws in a cross pattern, a little at a time), and reconnect the fan cable. *Why:* Old, dried paste does not transfer heat; too much paste is also bad; even pressure seats the cooler properly.

Step 3 â€" Storage drives. Slide each drive into its bay and secure the screws. Reconnect SATA data and SATA power cables, checking your photos.

Step 4 â€" RAM. Open the clips, align the notch on the stick with the key in the slot (they only fit one way), and press down firmly and evenly at both ends until the clips **snap** into place. Listen and feel for the click. *Why:* A RAM stick not fully seated is one of the most common causes of a computer that powers on but shows nothing. The snap tells you it is seated.

Step 5 â€" Expansion card (graphics card). Remove the case's expansion slot cover if needed, align the card with the PCIe slot, press down firmly until the slot clip engages, and replace the bracket screw. Reconnect any power cable the card needs.

Step 6 â€" Internal cables. Reconnect the SATA data and power cables, the motherboard power connectors if removed, and â€" most carefully â€" the **front-panel cables** (power switch, reset, USB, audio), using your photos and labels. The power switch wire is the one that lets the power button work; if it is misconnected, the machine will not start from the button.

Step 7 â€" Check before you close. Before replacing the panel, **do a careful visual check**: Are all cables firmly seated? Is the RAM clipped? Are there any loose screws inside? Is the CPU cooler connected? Are the drive cables correct? A screw left loose inside can short-circuit the board. *Why:* This silent check catches most reassembly errors before power-on.

Step 8 â€" Close and reconnect. Replace the side panel and its screws. Reconnect the external cables (monitor, keyboard, mouse, power). Plug in the power cable.

Step 9 â€" Power-on verification. Press the power button. Check: Do the fans spin? Does a light show? Does the monitor display? Does it reach the operating system (or the BIOS/POST screen)? *Why:* This is the moment of truth. If it does not start, you diagnose (Section K) â€" you do not panic.

Real-life comparison. Reassembly is like **reassembling a bicycle after fixing a puncture**. You fit the inner tube, the tyre, the wheel, then check the brakes before you ride. You always check before you trust it â€" you do not just close the case and hope.

Common misconceptions.

- "Any amount of thermal paste is fine." No â€" a pea-sized amount, not a thick layer.
- "The RAM will seat with a light push." Not always â€" it needs a firm, even push until it clicks. But never force it in the wrong orientation.
- "I can close the case and test later." Test as soon as possible, before fully working the machine, so faults are easier to trace.

Limitations and safety. Do not over-tighten screws (they strip and can crack the board). Do not power on without the CPU cooler. If the machine does not start, **unplug again** before re-opening.

F.5 Recording screws, cables and the mirror-image rule

Recording screws. Different screws have different lengths and threads. A long screw forced into a short hole can damage the board. Use a screw tray with compartments labelled by location (case, motherboard, drive, bracket) and place each screw where it belongs as you remove it. Keep a written **Screw and Cable Log** (Worksheet 3.3).

Recording cables. Photograph before disconnecting, label awkward cables, and note the front-panel layout. The front-panel connectors are the most common source of "the power button does nothing" after reassembly.

The mirror-image rule. Disassembly order and reassembly order are exact mirrors. Write them down as numbered lists so you can simply read one backwards to do the other.

Real-life comparison. Recording is like a **tailor taking measurements before cutting cloth**. The few minutes spent measuring save the whole garment. In repair, a few photos and a screw log save the whole repair.

Common misconceptions.

- "I will remember where everything goes." You will not â€" especially on an unfamiliar machine, and especially after a break.
- "Photos are for beginners." Professionals photograph and label constantly, especially on complex jobs.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- One practice desktop on the bench with the anti-static mat, wrist strap, tool kit and screw tray.
- A phone/camera to demonstrate photographing.
- Tape and marker to demonstrate labelling.
- The projector showing the machine (or a second camera) so all learners can see.
- The Disassembly and Reassembly checklists on the wall or projected.

G.2 Demonstration â€" full teardown and rebuild, with narration (15 minutes)

Say: "Watch every step, and listen to the reason behind it. I will disassemble, then reassemble, and the machine will work at the end."

Do and say (slowly, narrating reasons):

1. "I confirm this machine is mine to work on (permission). I note its condition."
2. "I shut it down properly, **unplug it from the wall**, and **press the power button** to discharge."
3. "I set up the mat, put on my **strap** and clip it to bare metal. I **touch bare metal** again."
4. "I remove the **side panel** screws and place them in the tray. Compare â€" the correct driver seats perfectly."
5. "I **photograph** the inside and the connections. This is my map."
6. "I disconnect the **SATA data** and **power** cables, gripping the connectors, not the wires."
7. "I remove the **graphics card** first â€" screw, slot clip, straight up by the edges, supporting it."
8. "I remove the **RAM** â€" clips down, straight out by the edges, into an anti-static bag."
9. "I remove the **storage drive** â€" screws out, slide out, handle gently."
10. "I set the components safely on the mat/antistatic bags."
11. "Now I reassemble in **reverse order**: drive first, then RAM (listen for the click), then the card."
12. "I reconnect the cables using my photo â€" especially the **front-panel** wires."
13. "Before closing, I **check**: all cables seated, RAM clipped, no loose screws inside, cooler connected."
14. "I close the panel, reconnect external cables, plug in, and **press power**. Fans spin, light on, display shows. It works."

Expected result: Learners see the complete cycle and understand each reason. The machine powers on.

Reasoning to state aloud at key points:

- "I remove the graphics card first because it lies over everything else."
- "I grip connectors, not wires, because wires tear out of their plugs."
- "I press RAM until it clicks because a half-seated stick gives no display."
- "I check before closing because a loose screw can short the board."

Errors to watch and deliberately show (then correct):

- Not fully seating the RAM (simulate a loose stick) â†' "See how I check the clips. If they are not snapped, no picture."
- Forgetting the front-panel wire â†' "If I leave this off, the power button will do nothing."
- Cross-threading a screw â†' "I turn backwards gently first to find the thread, then tighten. Never force."

Questions to ask during the demo:

- "Why do we remove the card before the RAM?" (It blocks access.)
- "What should you do if a component will not come out easily?" (Check for a screw or clip; never force.)
- "How do you know the RAM is fully seated?" (The clips snap.)
- "What is the last check before closing the case?" (All cables seated, RAM clipped, no loose screws, cooler connected.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "My First Teardown"

Objective: With the instructor guiding each step, learners disassemble and reassemble a practice desktop without damage and verify it powers on. This is the Week 1 mini-project.

Materials per group: Practice desktop, anti-static mat and wrist strap, tool kit, screw tray, torch, camera/phone, Worksheets 3.1, 3.2 and 3.3, cable labels, and a safe place for removed components.

Procedure (instructor calls each step; learners follow; instructor circulates continuously):

1. **Prepare:** Confirm permission. Note the machine's condition. Shut it down.
2. **Safety:** Unplug from the wall. Press the power button to discharge. Put on the wrist strap, clip it to bare metal, and touch bare metal.
3. **Bench:** Place the machine on the anti-static mat. Lay out tools and the labelled screw tray.
4. **Record:** Photograph the outside, then the inside, then each cable connection. Write the Screw and Cable Log (Worksheet 3.3).
5. **Open:** Remove the side panel screws (into the tray) and remove the panel. Set it aside safely.
6. **Disconnect:** Grip the connectors and disconnect SATA data, SATA power, and any power connectors you need to. Note the front-panel wiring.
7. **Remove the expansion card first** (if present), then the **RAM** (clips down, by the edges, into an antistatic bag), then the **storage drive** (gently).
8. **Set components safely** on the mat or antistatic bags. Do not stack boards.
9. **Reassemble in reverse order:** drive â†' RAM (push until the clips **snap**) â†' expansion card â†' reconnect all cables using your photos.
10. **Pre-close check:** All cables seated? RAM clipped? No loose screws inside? Cooler/fan connected?
11. **Close and reconnect:** Panel on, screws in, external cables connected, power cable plugged in.
12. **Power-on verification:** Press the power button. Confirm fans spin, lights show, and the display appears (or the BIOS/POST screen).
13. **Show the instructor** the working machine and your completed checklists.
14. **Reverse the teardown politely:** Power off properly, unplug, and leave the bench as you found it. If the machine is to remain assembled, leave it working.

Pause-and-predict points:

- After removing the graphics card: *"Predict: what happens if I reinstall the card but forget the bracket screw?"* Expected: it may not seat fully and could work loose or damage the slot.
- Before seating the RAM: *"Predict: what will happen if the clips do not snap?"* Expected: no display / the machine will not POST.
- Before closing the case: *"Predict: what is the danger of a loose screw inside?"* Expected: it can short-circuit the board or jam a fan.
- Before power-on: *"Predict: if the front-panel power wire is off, what happens when you press the button?"* Expected: nothing â€" the button will not start the machine.

Expected result: The machine powers on after reassembly, and both checklists plus the screw/cable log are complete.

Troubleshooting while walking the room:

- Component will not move â†' check for a screw or clip; never force.
- RAM will not click â†' check the orientation (the notch/key); do not force the wrong way.
- Machine will not start â†' unplug again, re-open, check that RAM is clipped, the front-panel power wire is connected, and the power connectors are seated.
- A screw is left over â†' re-open and find its home before finishing; do not ignore loose screws.
- Learner rushes â†' slow them down; use the checklist aloud.

Completion checklist (learner ticks each):

- ☐ Permission confirmed and machine condition recorded
- ☐ Machine shut down properly
- ☐ Unplugged from the wall and discharged (power button pressed)
- ☐ Wrist strap worn and clipped; bare metal touched
- ☐ Machine on the anti-static mat; tools and screw tray ready
- ☐ Photos taken before disconnecting

- [ ] Screw and Cable Log completed
- [ ] Side panel removed without damage
- [ ] Internal cables disconnected by the connectors
- [ ] Expansion card removed first
- [ ] RAM removed by the edges, into an antistatic bag
- [ ] Storage drive removed gently
- [ ] Components reinstalled in reverse order
- [ ] RAM clips snapped into place
- [ ] All cables reconnected correctly (front panel included)
- [ ] Pre-close check done (no loose screws, cables seated)
- [ ] Side panel replaced and external cables reconnected
- [ ] Computer powers on and displays
- [ ] Bench left tidy

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: "Write the Shop Procedure"

Scenario: You have been promoted to senior junior-technician at **Elias Computer Clinic, Diobu, Port Harcourt**. New apprentices are joining, and the owner wants a clear, numbered, step-by-step procedure pinned on the wall for opening and rebuilding a desktop computer. You must write it yourself, in your own words, including safety at every stage.

Instructions: Write two numbered procedures:

1. **Disassembly procedure** (at least nine steps, from permission and safety to removing the components).
2. **Reassembly procedure** (at least seven steps, ending with power-on verification).

Then answer these questions in the same document:

- **Q-A:** Why is reassembly the reverse of disassembly?
- **Q-B:** Why do we remove the graphics card before the RAM and drives?
- **Q-C:** What are the three most important checks before closing the case?
- **Q-D:** What should an apprentice do if a component will not come out easily?

Requirements:

- Safety reminders at every stage where they apply (unplug, discharge, strap, bare metal, no forcing).
- Correct order throughout.
- Written in your own words (do not copy the instructor's demonstration sentences).
- A short "common mistakes" list at the end (at least four).

Constraints:

- Assume the machine is a standard desktop with a graphics card, two RAM sticks and one hard drive.
- Keep each step to one or two clear sentences.
- Include the photo/labelling step.

Time allowance: 15 minutes.

Success criteria:

- Both procedures are complete and in the correct order.
- Safety is built into the steps, not listed separately only.
- The four questions are answered correctly.
- The "common mistakes" list is realistic.

Hints if stuck:

- Hint 1: Start disassembly from the outside (cables, panel) and finish at the deepest part (cooler/motherboard).
- Hint 2: Reassembly simply reads your disassembly list backwards.

- Hint 3: The pre-close checks are: cables seated, RAM clipped, no loose screws (and cooler/fan connected).

Instructor answer key (do not show learners):

Disassembly procedure (model answer):

1. Confirm permission; record the machine's model and condition; shut it down properly.
2. Unplug from the wall; press the power button to discharge; wear the wrist strap and touch bare metal.
3. Set up the anti-static mat, tools, and a labelled screw tray.
4. Photograph the inside and each cable connection; label awkward cables.
5. Remove the side panel screws (into the tray) and remove the panel.
6. Disconnect internal cables by gripping the connectors (SATA data, SATA power, power connectors; note front-panel wiring).
7. Remove the graphics card first (screw, slot clip, straight up by edges, supporting it).
8. Remove the RAM (clips down, by the edges, into an antistatic bag).
9. Remove the storage drive (screws, slide out gently).
10. (If required) remove the CPU cooler â€" unplug the fan, release the clips, twist gently, lift off.
11. (If required) remove the motherboard â€" all screws noted, cables disconnected, lift by the edges.
12. Place all components safely on the mat/antistatic bags.

Reassembly procedure (model answer):

1. (If the board was removed) refit the motherboard, aligning standoffs and I/O shield; screws evenly, not over-tight.
2. Refit the CPU cooler with fresh thermal paste (pea-sized) if it was removed; reconnect the fan.
3. Refit the storage drive and reconnect SATA data and power using your photos.
4. Refit the RAM, aligning the notch, pressing until the clips **snap**.
5. Refit the graphics card into the PCIe slot until the clip engages; replace the bracket screw.
6. Reconnect all internal cables, especially the front-panel wires, following your photos.
7. Pre-close check: cables seated, RAM clipped, no loose screws, cooler/fan connected.
8. Replace the side panel and screws; reconnect external cables and the power cable.
9. Press the power button; verify fans spin, lights show, and the display appears.

Q-A: Because the layout is stacked â€" the part removed last is the foundation and must be placed first; reversing guarantees nothing is trapped or blocked. **Q-B:** Because a graphics card often lies over the RAM clips and drives, so it blocks access; removing it first frees the area and prevents bending parts or slots. **Q-C:** All cables firmly seated (including front panel); RAM clips fully snapped; no loose screws inside (and the cooler/fan connected). **Q-D:** Stop and check for a screw, clip, or cable you have missed. Never force it; ask the instructor if unsure.

Common mistakes (accept any four): forcing components; not taking photos; mixing up screws; touching gold contacts; over-tightening screws; forgetting the I/O shield; not seating RAM fully; forgetting the front-panel wire; leaving a loose screw inside; powering on without the CPU cooler.

Marking guidance (out of 15): Disassembly order and completeness (4); reassembly order and completeness (3); safety integrated (3); four questions (4); common-mistakes list (1).

J. REAL-WORLD CASE STUDY

Situation. *Adaeze* is a careful learner who opens her home desktop to clean out the thick dust of Port Harcourt's dry season. She unplugs the machine, discharges it, straps up, and takes photos â€" good habits. She removes the side panel, the graphics card, the RAM and the drive, and cleans everything thoroughly with compressed air. Happy with her work, she rebuilds the machine: drive, RAM, graphics card, cables, panel. She plugs in and presses the power button. The fans spin and the lights come on, but the **screen stays black**. Nothing on the monitor. Her heart sinks â€" did she break it?

Relevant facts.

- The machine powers on (fans and lights) but there is no display.
- Adaeze removed and refitted the RAM and the graphics card.
- She used compressed air and cleaned thoroughly.
- The monitor cable was connected to the graphics card's port when she started.

Decision points.

1. Is the machine "dead," or is this a common, fixable reassembly issue?
2. What is the most likely cause given she removed and refitted RAM and the card?
3. What should she check, and in what order?
4. What could have damaged the machine â€” if anything?

Possible approaches.

1. Assume the worst and rush it to a shop for an expensive repair.
2. Panic and start removing everything again randomly.
3. Stay calm, unplug, and check the most likely, simplest causes first: RAM seating, card seating, display cable and port.

Recommended solution. Approach 3. "Powers on but no display" after a teardown is very often a **RAM stick not fully seated** or the **graphics card not fully in its slot**, or the **display cable** in the wrong port or loose. The correct steps: unplug and discharge, re-open the case, press the RAM sticks down until the clips snap, press the graphics card firmly into its slot until the clip engages, check the bracket screw, and confirm the monitor cable is firmly in the **graphics card's** port (not the motherboard's). Close, reconnect, and test. In Adaeze's case, one RAM stick had not clicked home. With an even push, it snapped in, and the picture returned.

Trade-offs.

- Approach 1 wastes money and time on a fault she could fix in two minutes.
- Approach 2 risks causing real damage by random handling.
- Approach 3 is systematic, free, and teaches the exact lesson the course is built on: **check the simple, likely physical causes first.**

Professional connection. "Powers on but no display" is one of the most common situations in any real repair shop. The technician who methodically reseats RAM, checks the card, and checks the cable solves it quickly and cheaply; the one who guesses replaces parts the customer did not need. This case is a direct bridge to **Session 4 (Diagnostic Workflow)** â€” the same logic, made formal.

Transfer exercise. "List, in order, the first three things a technician should check when a freshly reassembled desktop powers on but shows no display." (RAM fully seated; graphics card fully seated; display cable connected to the correct port and monitor powered on.)

Safety reminder. All of this checking is done with the machine **unplugged and discharged**, on the anti-static mat, with the strap worn. Never re-open or poke inside a machine that is still plugged in.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use as the live guide. Ask learners to find the relevant row.

K.1 Component will not come out

Symptoms: A part feels stuck; pushing harder seems to work but you hesitate. **Likely cause:** A screw, clip, latch or cable you have not released. Cause is confirmed once you find and release it. **How to investigate:** Stop. Look all around the part (front, back, under) with a torch. Check for hidden screws and slot clips. **Safe corrective steps:** Release the missed fastener, then lift the part straight out. Never use force. **How to confirm resolved:** The part lifts out smoothly with no resistance. **How to prevent recurrence:** Study the machine's layout and take photos before removing; work slowly.

K.2 Powers on but no display after reassembly

Symptoms: Fans spin, lights on, but the monitor shows nothing. **Likely cause:** RAM not fully seated, graphics card not fully seated, display cable loose or in the wrong port, or monitor off. **How to investigate:** Unplug and discharge; re-open; check RAM clips, the card's seating and clip, and the display cable/port. **Safe corrective steps:** Reseat RAM until it clicks; reseat the card; move the display cable to the graphics card's port; confirm the monitor is powered. **How to confirm resolved:** The display appears. **How to prevent recurrence:** Always check seating and cables before closing (Step 7 in reassembly).

K.3 Power button does nothing after reassembly

Symptoms: Machine completely silent; no fans, no lights. **Likely cause:** The front-panel power switch connector is loose or misconnected, or the main power connectors are not seated, or the wall plug/socket is off. **How to investigate:** Check the wall socket and plug; check the 24-pin and CPU power connectors; check the front-panel power wire against your photo. **Safe corrective steps:** Reseat the front-panel power connector and the power connectors (machine unplugged). **How to confirm resolved:** The power button starts the machine. **How to prevent recurrence:** Photograph front-panel wiring before disconnecting; check it in the pre-close step.

K.4 Loose screw left inside the case

Symptoms: A rattle when moving the machine; or a sudden short/failure on power-on. **Likely cause:** A screw fell inside during work. Confirmed by finding it. **How to investigate:** With the machine **unplugged and discharged**, tip and inspect carefully; use a torch. **Safe corrective steps:** Retrieve it with tweezers or by gently tilting; never shake a machine containing drives violently. **How to confirm resolved:** No rattle; the screw is found and put in its place. **How to prevent recurrence:** Always count screws; keep them in the tray; do the pre-close check.

K.5 Stripped or cross-threaded screws

Symptoms: The screw turns but does not tighten, or bites the wrong way and jams. **Likely cause:** Wrong driver size, or forcing a screw into the wrong hole/angle. **How to investigate:** Try the correct size; inspect the head and the hole. **Safe corrective steps:** Turn the screw slowly backwards first to find the thread, then tighten gently. If it is badly stripped, get instructor help (a screw extractor may be needed). **How to confirm resolved:** The screw tightens cleanly without force. **How to prevent recurrence:** Match driver to screw; never over-tighten; start screws by hand.

K.6 Forgetting to reconnect a cable

Symptoms: A specific function fails (e.g., no power from the button, no front USB, a drive not detected), while the rest works. **Likely cause:** A cable left disconnected or mis-plugged. Confirmed by comparing with your photo. **How to investigate:** Compare the current connections with your pre-disassembly photos. **Safe corrective steps:** Unplug and discharge, re-open, reconnect the correct cable. **How to confirm resolved:** The function works again. **How to prevent recurrence:** Use photos and labels; do the pre-close check.

Distinguishing likely vs confirmed: In every case, the "likely cause" becomes "confirmed" only when you actually check it (find the missed screw, see the RAM clip snap, trace the cable against the photo). Teach learners to check before "fixing."

L. SESSION ASSIGNMENT

Assignment title: "My Disassembly and Reassembly Guide"

Purpose: To produce a complete, safety-aware, step-by-step guide that the learner can follow on any standard desktop, and that reinforces the Week 1 skills needed for the capstone.

Realistic scenario: A friend is starting a small repair business and has asked you for a written procedure he can pin above his bench. He is a beginner, so your guide must be clear and safe enough for him to follow without you present.

Detailed instructions:

1. Create a document titled **"My Disassembly and Reassembly Guide"**.
2. Write a **numbered disassembly procedure** (at least nine steps), including permission, recording the machine, and safety at every stage.
3. Write a **numbered reassembly procedure** (at least seven steps), ending with power-on verification and the pre-close check.
4. Include a **Safety Reminders** section (at least five points) that applies to the whole process.
5. Include a **Common Mistakes** section with at least five mistakes and how to avoid each.
6. Include a **Troubleshooting** section: what to do if (a) the machine will not power on, (b) it powers on but shows no display, (c) a component will not come out.
7. Save the file as **"My Disassembly and Reassembly Guide"** on the Desktop.
8. Be ready to explain one step in detail if asked.

Required deliverables: One saved document with all sections.

Submission format: Saved on the Desktop with the exact file name. If the learner has no home computer, complete it in the academy lab and show it before leaving.

Deadline: Before Session 4 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Disassembly steps	Complete, correct order, 9+	Mostly correct, 6â€"8	Incomplete or wrong order
Reassembly steps	Complete, correct order, 7+	Mostly correct, 5â€"6	Incomplete or wrong order
Safety reminders	At every critical step	At most steps	Few or none
Common mistakes	5+ realistic, with solutions	3â€"4 mistakes	Fewer than 3
Troubleshooting	3 realistic faults solved correctly	2 faults	1 or none

Common mistakes to avoid: forgetting to unplug and discharge; jumping straight to reassembly without photos; forcing components; over-tightening screws; forgetting the front-panel wire; not seating RAM fully.

Instructor reference solution (summary, do not show learners beforehand):

- Disassembly: permission/record â†' shut down â†' unplug/discharge/strap â†' bench setup â†' photograph/label â†' remove panel â†' disconnect cables (by connector) â†' remove card first â†' remove RAM â†' remove drive â†' (cooler/board only if needed) â†' set parts aside safely.
- Reassembly: board first (if removed) â†' cooler with fresh paste â†' drive â†' RAM until clips snap â†' card until clip engages â†' reconnect cables (front panel per photo) â†' pre-close check â†' panel/external cables â†' power-on verification.
- Safety: unplug and discharge always; strap and bare-metal touch; hold by edges; no forcing; grip connectors not wires; never open the PSU; do not power on without the cooler; no liquids.
- Common mistakes: no photos; mixed screws; touching contacts; over-tightening; forget I/O shield; RAM not seated; missing front-panel wire; loose screw inside; forcing.
- Troubleshooting: (a) no power â†' check wall/plug, main power connectors, front-panel power wire; (b) no display â†' reseat RAM, reseat card, check display cable/port; (c) stuck part â†' find the missed screw/clip; never force.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below.)

Q1. Short answer. What is the correct order for removing the main components during disassembly, and why?

Q2. Multiple choice. Why do we take photographs before disconnecting internal cables? a) To show the customer b) To know where to reconnect them during reassembly c) To make the job look professional d) It is not necessary

Q3. Explain in your own words. Why is reassembly described as "the reverse of disassembly"?

Q4. Practical identification. How do you know a RAM stick is correctly installed?

Q5. Error detection. A learner says: "If a component will not come out, I should pull harder." Find the mistake and correct it.

Q6. Scenario. After reassembling a desktop, the fans spin and the lights come on, but the monitor shows nothing. List, in order, the first three things you would check.

Q7. Short answer. Why should you never power on a computer with the CPU cooler removed?

INSTRUCTOR ANSWER KEY (Q1â€"Q7)

A1. Remove the **graphics card first**, then the **RAM**, then the **storage drives**, and only later (if required) the **CPU cooler** and motherboard. The card often lies over other parts, so it must go first to free access; the RAM and drives come next; the cooler and board are last because they are the foundation and the most delicate.

A2. (b) To know where to reconnect them during reassembly. (Photos are your map, especially for the front-panel wires.)

A3. Because components are stacked/overlapping: whatever was removed last is the foundation and must be placed first, and whatever was removed first (the top layer) is placed last. Reversing the order guarantees nothing is trapped or blocked, and that everything fits as before.

A4. The retaining **clips at both ends snap into place**, and the stick sits level and firm in the slot. (You hear/feel the click.)

A5. The mistake is using force. The correction: **stop and check** for a screw, clip, latch or cable you have missed; a part that will not move easily is being held by something. Never force it â€" forcing causes damage.

A6. (1) Confirm the machine is unplugged/discharged, then check the **RAM is fully seated** (clips snapped); (2) check the **graphics card is fully seated** and its clip engaged; (3) check the **display cable is firmly connected to the correct port** (the graphics card, not the motherboard) and the monitor is powered on.

A7. Because the CPU produces heat very quickly and relies on the cooler (heatsink/fan + thermal paste) to remove it. With the cooler off, the CPU can overheat within seconds and shut down or be permanently damaged.

Marking: up to 2 marks each (Q1, Q6 up to 3). Total â‰ˆ 16; scale proportionally. Full credit requires correct order, correct component names, and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary. Today you opened a real computer and put it back together. You learned the **safety preparation** (permission, recording, unplugging, discharging, strap, bare-metal touch), the **correct disassembly order** (external cables â†' panel â†' photograph â†' disconnect cables â†' card â†' RAM â†' storage â†' cooler only if needed), the **mirror-image rule** (reassembly is the reverse), the **correct reassembly technique** (RAM seated until it clicks, fresh thermal paste when the cooler is refitted, cables reconnected using photos), the **pre-close check**, and **power-on verification**. You also learned to record screws and cables so the job can be repeated correctly.

Five important takeaways:

1. **Safety first, every time:** unplug, discharge, strap, bare metal.
2. **Disassembly order matters**, and reassembly is its **mirror image**.
3. **Take photos and log screws** â€" they are your map and your memory.
4. **Never force a component**; check for the missed screw, clip or cable.
5. **RAM must click into place** and you must **check before you close** the case.

Questions to help you recognise what you can now do:

- Can I prepare a machine for work safely without reminders?
- Can I state the correct removal order and the reason for it?
- Can I reassemble a machine and get it to power on and display?
- Can I list the three checks to make before closing the case?

Connection to the next session: You can now open, rebuild and safely test a machine. Next session you begin **Week 2: Diagnostics**. You will learn the **systematic diagnostic workflow** â€" how to gather symptoms, form a hypothesis, test one thing at a time, isolate the fault, verify the fix and document it. The "no display after reassembly" logic you used today is the seed of that whole method.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner unplugs, discharges, straps and touches bare metal **without prompting**.
- The learner photographs and labels before disconnecting, and keeps a screw log.
- The learner follows the correct removal and refit order.
- The learner grips connectors (not wires) and never forces a part.
- The learner seats RAM until the clips snap and performs the pre-close check unprompted.
- The learner verifies power-on and knows the first checks if there is no display.

Common misconceptions to watch for and correct:

- "Order does not matter." â†’ It does; the layout dictates it, and wrong order damages parts.
- "Force it if it is stuck." â†’ Never; find the missed fastener.
- "Testing after reassembly is optional." â†’ Always verify before declaring the job done.
- "Photos are for beginners." â†’ Professionals photograph and label, especially on unknown machines.
- "Any amount of thermal paste works." â†’ A pea-sized amount, applied correctly.

Suggested remedial exercises for learners who struggle:

1. **Order cards:** give the steps on shuffled cards and have the learner sequence them, then check against the checklist.
2. **One-component drill:** have the learner remove and refit just the RAM, then just the drive, until each is confident and clean.
3. **Photo-and-rebuild:** have the learner photograph a machine, then rebuild a second identical machine using only the photos.
4. **Fault-chasing drill:** the instructor creates a loose-RAM or disconnected-front-panel fault for the learner to find methodically.

Differentiation:

- **Level 1 (supported):** provide a partially disassembled machine for the learner to finish (for example, remove and refit only the RAM and drive), with a partner handling the trickier parts; use the checklists step by step.
- **Level 2 (standard):** complete the full disassembly and reassembly as written.
- **Level 3 (challenge):** disassemble and reassemble a **different model** of desktop (for example one with a tool-less case or a different drive bay), and/or remove and refit the CPU cooler with fresh thermal paste under supervision.

Timing guidance: The practical (Section H) is the heart of this session and is directly assessed, so protect it. If the class is slow, compress the lecture sections F.4â€"F.5 into summaries and keep the demonstration and practical. If a group is fast, give them the healthy-machine-plus-simulated-fault challenge. Never shorten the safety steps.

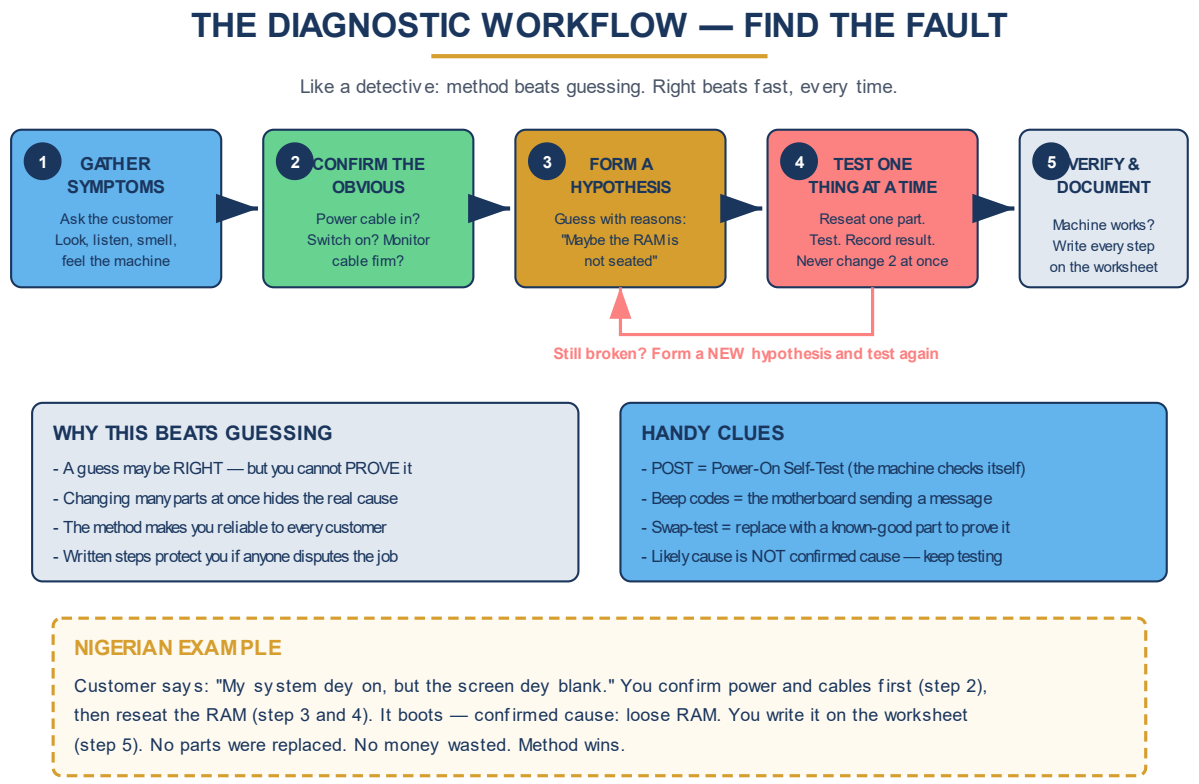
END OF SESSION 3 â€" END OF WEEK 1 (COMPONENTS AND SAFETY)

Next: Session 4 â€" The Diagnostic Workflow: Finding the Fault Systematically.

SESSION 04 OF 12

Diagnostic Workflow

CONCEPT DIAGRAM — SESSION 04



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 4 â€" THE DIAGNOSTIC WORKFLOW: FINDING THE FAULT SYSTEMATICALLY

A. SESSION IDENTIFICATION

Item	Detail
Session number	4 of 12
Session title	The Diagnostic Workflow
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner (first contact with formal fault-finding)

Item	Detail
Prerequisite knowledge	Sessions 1â€"3: component names and functions; ESD and electrical safety; tools; disassembly and reassembly.
Connection to previous session	In Session 3 the learner met the classic "powers on but no display" problem after reassembly. This session turns that informal troubleshooting into a **structured, repeatable method**.
Connection to the course	This workflow is the backbone of the whole course. Sessions 5 (hardware diagnostics), 6 (software diagnostics), 7 (replacement), and the Session 12 capstone all run through this method. The capstone is assessed *partly on whether the learner follows it*.
Preparation before class	Scenario cards (at least six faults); at least one practice computer with a **simulated fault** (for example RAM not fully seated) that can power on but show no display; a working reference machine; printed Symptom-to-Cause Reference and Diagnostic Questions Checklist; whiteboard for drawing the workflow; optional: a machine that beeps on power-on (to demonstrate POST beeps) or a diagram of BIOS beep codes.

Safety note for this session: All physical diagnosis is done on **unplugged and discharged** machines, with the anti-static strap worn and boards handled by the edges. Never reseat or swap a component while the machine is powered. When testing power-on, the instructor supervises and only brief tests are made. Never open the PSU. Because this session involves deliberate faults, students must be told clearly that these are **practice machines** and that the same actions must never be attempted on a customer's machine without permission.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **State and follow the five-step diagnostic workflow** â€" gather symptoms, confirm the obvious, form a hypothesis, test one thing at a time, and verify and document.
2. **Gather symptoms systematically** by asking effective diagnostic questions and observing look/listen/smell/feel.
3. **Confirm the obvious first** â€" power, cables and switches â€" before assuming a deep fault.
4. **Explain POST (power-on self-test)** and what it means when a machine fails it.
5. **Interpret basic BIOS beep codes and no-display symptoms** as diagnostic clues (recognising that codes vary by maker and must not be invented).
6. **Use swap-testing** to isolate a faulty component safely.
7. **Apply a safe boot troubleshooting order** for common "no display" and "no power" situations.
8. **Document each step** of a diagnosis clearly, including what was tested, the result, and the conclusion.
9. **Explain why a structured method beats guessing**, using the two ideas of "likely cause" and "confirmed cause."

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- At least one practice computer with a **simulated fault** that powers on but shows no display (for example, RAM deliberately left unseated, or the display cable in the wrong port).
- A working reference machine to compare against (very useful for swap-testing).
- One or two spare known-good components for swap-testing, e.g., a tested RAM stick, a tested display cable.
- A desktop for the instructor demonstration.
- Anti-static wrist straps and mats; tool kit; screw tray; torch.

Software:

- None essential. Optional: a free system-information viewer (Windows Task Manager â†' Performance) to show live activity, but the session can run entirely without software. Note: no statistical or hardware-specific claims should be made.

Safety gear:

- Anti-static wrist straps and mats.
- First-aid kit available.

Sample / fault components:

- Known-good RAM stick for swap-testing.
- A known-good display cable and/or a second monitor if available.
- Clearly-labelled "FAULT â€" DEMO" parts where relevant.

Worksheets (printed):

- Worksheet 4.1 â€" "Symptom-to-Cause Reference" (one per learner): a table of symptoms and likely causes.
- Worksheet 4.2 â€" "Diagnostic Questions Checklist" (one per learner): the seven questions.
- Worksheet 4.3 â€" "Diagnostic Worksheet / Service Report Draft" (one per learner): a structured page with fields for symptoms, observations, questions, hypothesis, tests and results, resolution, and verification.
- Scenario cards (at least six) for the guided and independent exercises.

Alternatives for a learner or academy with limited equipment:

- The workflow can be taught almost entirely through **paper scenarios** if no faulty machine is available. Use the scenario cards and the diagnostic worksheet; role-play the customer and technician.
- If only one faulty machine exists, run it as a whole-class demonstration and let learners fill in the worksheet together, then give each learner different scenario cards to work through independently.
- If a learner is uncomfortable with hands-on fault work, let them be the "diagnostic recorder" and the "customer", which still practises the workflow and the questions.

D. OPENING ACTIVITY (6 minutes)

"The Detective Analogy"

Instructor setup: On the projector, show a simple picture or drawing of a detective with a magnifying glass and a notebook, beside a computer with a question mark over it. Keep the faulty practice machine hidden or off to the side.

What the instructor says:

> *"Imagine a detective arrives at a crime scene. Does the detective walk in and immediately say, 'It was the gardener'? No. That is a guess. What does a good detective do? They look at the scene. They ask witnesses questions. They form a theory. They test the theory â€" fingerprints, alibi, evidence. And only when the evidence is solid do they say who did it.*
>
> *A computer technician is a detective. When a customer brings a faulty computer, the customer has given you a clue â€" a **symptom** â€" like 'it will not come on' or 'the screen is black.' Your job is to find the **cause**. And here is the trap: the worst technicians are the ones who **guess**. They open the case, see a dirty fan, and say, 'Ah, it is the fan!' Maybe. Or maybe the real problem is somewhere else entirely.*
>
> *Today you learn the detective's method. You will learn a fixed set of steps â€" the same steps every single time â€" that stops you from guessing and leads you to the true cause.*
>
> *Let me ask you: which is better, a technician who guesses fast, or a technician who is right?"*

Pause for answers.

> *"Correct. Right beats fast, every time. A wrong guess costs the customer money and costs you your reputation. A method makes you reliable."*

Then create the hook: Reveal the faulty practice machine. Say:

> *"This computer is the crime scene. It powers on â€" you can see the fans spin â€" but there is no picture. Somewhere in here is the cause. By the end of today, using the method, you will find it â€" without guessing, and without replacing a single*

part."

Expected learner responses: Learners often suggest "the screen is broken" or "we need to check the graphics card." Accept all answers but point out that these are **hypotheses**, not yet confirmed. Some learners may guess correctly â€" praise them but insist on the method: "You may be right, but how will you *prove* it? That is the skill."

Why this opener works: It uses a familiar role (detective) to make an abstract method concrete, draws a sharp contrast between guessing and method, and gives the class a real, solvable problem to look forward to.

E. PREREQUISITE REVIEW (7 minutes)

Run as a quick oral review; ask different learners.

Question 1: "Name the major components and one thing each does."

- Expected answer: Motherboard (connects everything), CPU (does the work), RAM (temporary memory), storage (permanent), PSU (power), GPU (images), fans (cooling).
- Common gap: learners forget the PSU in the list, yet it is central to "no power" diagnosis. Remind them.
- Bridge: "All of these are suspects for today's detective."

Question 2: "What is ESD, and why does it matter when we swap parts?"

- Expected answer: Electrostatic discharge; invisible damage; prevented by strap, mat, bare-metal touch, edge handling.
- Common gap: forgetting to re-touch bare metal between parts. Emphasise when swapping.
- Bridge: "When we swap components to test, we handle many parts â€" so the ESD habit matters even more."

Question 3: "What are the first two things to check before opening a case?"

- Expected answer: Unplug from the wall; press the power button to discharge.
- Common gap: confusion between the power button and the wall plug. Clarify order.
- Bridge: "And note â€" the very *first* diagnostic clue is often outside the case entirely."

Question 4: "From Session 3: if a freshly reassembled computer powers on but shows no display, what were the first things to check?"

- Expected answer: RAM seated (clips snapped); graphics card seated; display cable in the correct port; monitor powered.
- Common gap: jumping to "the GPU is broken."
- Bridge: "Today we turn that instinct into a proper order of operations."

Question 5: "What is the difference between a 'likely' cause and a 'confirmed' cause?"

- Expected answer: A likely cause is a guess or probability; a confirmed cause is proven by a test or observation.
- Common gap: learners treat their first idea as confirmed. Clarify the difference â€" it is the heart of today.
- Bridge: "This distinction is the whole point of the method: confirm before you fix."

Bridge to today (say this):

> *"Think of diagnosis like **tracing a fault in the NEPA/PHED line** after an outage. Would you climb the pole and start replacing cables immediately? No â€" you would first check whether the whole area is out, then whether your neighbour has light, then whether it is only your compound, then your meter, then your wiring. You follow an order from the biggest, simplest possibility to the smallest. Today's method is exactly that discipline, applied to a computer."*

If more than a third cannot answer Question 5, spend an extra three minutes on likely-vs-confirmed, as it underpins the whole session.

F. FULL LECTURE NOTES

F.1 Why a method beats guessing

The idea. Diagnosis is the process of finding the cause of a fault. A **workflow** is a fixed sequence of steps you follow every time, so that you never skip something important and never jump to a conclusion.

Plain-English definition. The diagnostic workflow is a step-by-step method: gather the symptoms, confirm the obvious, form a hypothesis, test one thing at a time, then fix and verify. It replaces "I think" with "I have proven."

Why it exists. Computers have many parts that can cause similar symptoms. "No display," for example, can come from RAM, the graphics card, the cable, the monitor, the motherboard, or even the power supply. Guessing wastes time and money and often replaces healthy parts. A method narrows the possibilities in an efficient, safe order.

What it does. It makes your work:

- **Repeatable** â€" the same fault is diagnosed the same way every time, by anyone.
- **Reliable** â€" you confirm the cause instead of assuming it.
- **Safe** â€" you test simple, safe things before complex or risky ones.
- **Accountable** â€" because you document each step, you can explain your reasoning to a customer or a colleague.

Real-life comparison. The workflow is like a **doctor's consultation**. The doctor asks about symptoms, checks the basics (temperature, blood pressure), forms a suspicion, orders a test, confirms the diagnosis, treats it, and then checks you again. A doctor who prescribed medicine without checking would be dangerous. A technician who replaces parts without checking is the same.

Realistic repair-shop scenario. Two technicians receive the same "no display" computer. The first guesses "graphics card" and fits a new one â€" the fault remains, and the customer is charged for nothing. The second follows the method: checks the cable and port, reseats the RAM (which was loose), and the display returns in two minutes for free. Same fault, opposite results â€" the difference is the method.

Common misconceptions.

- "Experienced technicians do not need a method; they just know." Experienced technicians *have internalised* the method; they still follow it. When a fault is unusual, even they return to the steps.
- "The most obvious cause is always the true one." Sometimes the obvious cause is a **red herring**. You only know by testing.

Limitations and safety. The workflow guides you, but some faults (intermittent, or requiring special tools) need more time or specialist help. Recognising limits is part of the method â€" Step 5 explicitly includes referring onward when needed.

F.2 The five-step diagnostic workflow

Step 1 â€" Gather the symptoms (observe and ask). Collect every clue before you touch anything.

Observe (your own senses):

- **Look:** Is there visible damage? Swelling (swollen capacitors or batteries)? Burn marks? Dust? A loose cable? A fan not spinning? Any light at all?
- **Listen:** Do you hear beeps (POST codes)? Clicking or grinding (often a failing HDD)? Rattling (a loose fan or screw)? Silence (no power)?
- **Smell:** Is there a burning or "hot" smell (often an overheated part or a failing PSU)? **If you smell burning, stop and unplug safely â€" do not continue.**
- **Feel (carefully, only after unplugging):** Is a part unusually hot? Is a connector loose?

Ask the customer (the seven questions are detailed in F.4): What happened, when, what were they doing, what changed, does it happen always or sometimes, have they tried anything, is there any unusual sound/smell/damage.

Step 2 â€" Confirm the obvious (the simple things first). Before assuming a deep fault, rule out the simple, common causes:

- Is the machine actually **plugged in** and is the **wall socket working** (test with a known-good device, or a lamp)?
- Is the **power switch** (on the PSU, on a laptop, on a power strip) in the correct position?
- Is the **monitor switched on** and set to the correct input (HDMI/VGA/DisplayPort)?
- Are the **external cables** connected (display, keyboard, power)?
- Is there **power in the building** (a blackout affects everything)? Many "faults" are none of these being right. Always check them first.

Step 3 â€" Form a hypothesis (an educated guess). Based on your observations and the customer's answers, write down the **two or three most likely causes, in order of probability and ease of testing**. A hypothesis is not a conclusion; it is a smart

guess to be tested. Example: "Given the machine powers on but shows no display, the most likely causes are (1) RAM not seated, (2) display cable/port, (3) graphics card not seated."

Step 4 â€" Test one thing at a time (and change one thing at a time). Test your hypotheses, starting with the **simplest and most likely**. Crucially, **change only one thing per test**, then observe the result. If you change three things at once, you will not know which one fixed (or caused) the problem. Common safe tests:

- **Reseat** the suspected component (RAM, card, cable).
- **Swap-test:** replace a suspected component with a **known-good** one (for example, a tested RAM stick). See F.6.
- **Compare** with a working machine. After each test, note the result. If the fault is gone, you have likely found the cause; confirm it (Step 5).

Step 5 â€" Resolve, verify and document. Fix the confirmed cause (or, at this stage, identify and record it). **Verify** the fix: does the machine now work fully? Test more than just the original symptom â€" for example, restart it, and check it behaves normally under use. Then **document** everything: the symptoms, the questions, your hypothesis, the tests and results, the cause, and the fix. Documentation protects the customer and you, and it is the basis of the professional service report (Sessions 11 and 12).

The golden rule of testing: *One change at a time.* This single rule transforms confusing diagnosis into clear, logical progress.

Real-life comparison. The five steps are like **finding which bulb has blown in a house**. You notice the lights are off (symptom). You check whether the whole street is off and whether the switch is on (confirm the obvious). You suspect a fuse or a bulb (hypothesis). You try one bulb at a time and one fuse at a time (test one thing). You confirm the light comes on (verify) and note what you changed (document). You would never replace every bulb in the house at once.

Common misconceptions.

- "Test many things at once to save time." This does the opposite â€" it creates confusion. One change at a time.
- "Fix it and move on." Always verify and document; a fix that is not verified may not be a fix at all.
- "Hypothesis means I already know." No â€" a hypothesis is only a guess until a test confirms it.

Limitations and safety. Never test by working on a powered machine. All component reseating and swapping happens with the machine **unplugged and discharged**, wearing the strap. If a test requires mains-level work (for example, testing a PSU's output), that is a supervised, later-session activity and is not attempted here on live circuits.

F.3 POST â€" the computer's start-up self-check

Plain-English definition. POST (Power-On Self-Test) is a basic self-check the computer runs automatically the moment it is switched on, before it tries to load the operating system. It is built into the motherboard's firmware (called the **BIOS** or, on newer machines, the **UEFI**). Firmware is small built-in software that runs before Windows does.

Why it exists. Before handing control to the operating system, the computer checks that its essential parts are present and working: it detects the CPU, the RAM, the graphics, and the storage, and confirms basic settings. If something essential is missing or faulty, the POST fails, and the machine signals the problem instead of pretending everything is fine.

What the POST does.

- It checks that basic components are detected and respond.
- If the check **passes**, the machine usually continues to the BIOS/UEFI screen briefly and then boots the operating system (or shows a "no bootable device" message if there is no bootable system, which is a different issue from a dead machine).
- If the check **fails**, the machine can signal with a **beep code** (a pattern of beeps) if a small internal speaker is fitted, and/or a display message, and/or codes on the motherboard. Many modern machines have no internal speaker, so they cannot beep â€" instead they may show LED indicators or simply show no display.

How POST helps diagnosis. When a computer **powers on but does not reach its operating system**, you can often tell *where* it stopped:

- **No display, no beep, fans spinning:** often a failure very early â€" RAM, graphics, or the POST itself.
- **Beep pattern:** a code that (on machines that do beep) points to a particular subsystem.
- **Reaches the BIOS screen but cannot find the OS:** the hardware is basically fine, but the boot drive or the operating system has a problem (a software or storage issue, not a dead board).

A crucial honesty note â€" beep codes vary. BIOS beep codes are **not universal**. Different makers (AMI, Award, Phoenix, and different manufacturers) use different beep patterns, and modern machines often have no speaker at all. Therefore, **do not memorise invented "universal" beep codes.** The professional approach is:

1. Note the **make and model** of the machine or motherboard.
2. Look up that maker's **documented** beep codes (from the official manual or the manufacturer's website).
3. Interpret the beep against that specific documentation. Teach learners this discipline: *the code is only meaningful with the maker's own chart.* Do not invent or quote specific patterns as if they were universal.

Real-life comparison. POST is like the **security check at the airport gate**. Before anyone boards, staff verify the plane's essential systems and passengers. If a check fails, the flight does not proceed until it is fixed. The beep code is like the specific alarm message: "gate 3 sensor fault" â€" but you must read *that airport's* manual to know what gate 3 means, because every airport numbers things differently.

Common misconceptions.

- "Beeps mean the same thing on every computer." No â€" they differ by maker. Always consult the maker's chart.
- "No beep means it is definitely broken." Not always â€" many machines have no speaker, so silence is normal on POST failure.
- "If it beeps, the POST passed." Not necessarily â€" some beeps indicate a fault.

Limitations and safety. POST only checks the basics, not everything; a machine can pass POST and still have a failing drive or software fault. And you must never rely on invented codes. If unsure, use the method: confirm simple things, then test components one at a time.

F.4 The seven diagnostic questions

The idea. What the customer tells you is often the single biggest clue. Asking the right questions saves hours.

The seven questions:

1. **"When did the problem start?"** â€" Ties the fault to a time or event (a power cut, a move, a new update).
2. **"What were you doing when it happened?"** â€" Context reveals the trigger (gaming â†' overheating or GPU; a certain program â†' software or driver).
3. **"Has anything changed recently?"** â€" New hardware, a new program, an update, a new cable, a different room.
4. **"Does it happen all the time or only sometimes?"** â€" Intermittent faults are harder; consistent faults are usually easier to reproduce and test.
5. **"Have you tried anything to fix it?"** â€" You need to know what was already done or changed, so you do not undo it or chase your tail.
6. **"Is there any unusual sound, smell, or visible damage?"** â€" Physical clues (burning, clicking, swelling, a spill) are often decisive.
7. **"Does it happen with specific programs, or all the time?"** â€" Helps separate a software problem from a hardware one.

How to ask well. Ask in plain, respectful language a customer will understand. Do not make the customer feel stupid. Repeat back what they say to confirm. Write down the answers â€" memory is unreliable, and notes become your report.

Real-life comparison. These questions are like a **doctor asking about your symptoms and history**. The more accurate the information, the faster and more accurate the diagnosis.

Common misconceptions.

- "I will just open it and see." You will miss vital clues. Ask first.
- "The customer does not know anything." The customer may not know the *cause*, but they always know *what changed* and *when* â€" that is gold.

F.5 Safe boot troubleshooting order for "no display" and "no power"

When a machine shows **no picture** (or **no power at all**), follow a fixed, safe order from the outside inward and from simplest to most complex. This is an application of the workflow.

A. No display, but the machine powers on (fans spin, lights on). Work in this order:

1. **Monitor and cable first:** Is the monitor powered on? Is the cable firmly connected at both ends? Is the monitor set to the correct input? (These are free and easy.)
2. **Cable port:** If a dedicated graphics card is fitted, is the display cable in the **graphics card's** port, not the motherboard's? (A very common mistake.)

3. **Signal check:** Does the monitor say "no signal" or is it completely dark? "No signal" points to a source/cable issue; complete darkness may point to the monitor itself.
4. **RAM reseating:** Unplug and discharge; reseat the RAM until the clips snap. (One of the most common no-display causes.)
5. **Graphics card reseating:** Reseat the card firmly into its slot until it clicks; check the bracket screw.
6. **Swap-test the cable and monitor** with known-good ones if available.
7. **Only then** consider deeper causes (motherboard, RAM fault, etc.), ideally with supervision.

B. No power at all (completely dead: no fans, no lights). Work in this order:

1. **Wall power:** Is the socket working? Test with a known-good lamp or device. Is there power in the building?
2. **Power cable:** Is it firmly connected at both ends? Try a known-good cable.
3. **Power switch:** Is the PSU switch (if present) on? Is a power strip switched on?
4. **Front-panel power button:** Is the front-panel power connector seated on the board? (A common result of reassembly.)
5. **PSU test:** With supervision, and using a safe method, check whether the PSU is delivering power. **Never open the PSU.** At this stage, swap-testing with a known-good PSU is done **only under supervision**.
6. **Motherboard/board-level:** If a known-good PSU still gives nothing, suspect the board â€" but confirm carefully, do not guess.

The principle behind the order. Always start with the **cheapest, simplest, safest, most likely** checks and move toward the **expensive, complex, riskier** ones. This saves money, avoids damage, and usually finds the fault sooner.

Real-life comparison. This order is like **checking your phone will not charge**: first you check the socket (is there power?), then the cable (is it damaged?), then the charger, and only last the phone's charging port. You would not open the phone first.

Common misconceptions.

- "No display means the GPU is dead." The cable, port or RAM is more often the cause. Check those first.
- "No power means the PSU is dead." It may be the socket, the cable, the switch, or the front-panel button. Confirm first.

Limitations and safety. Never test power-related faults while working inside a live machine. PSU testing is a supervised activity and the PSU is never opened. If you smell burning, stop and unplug safely.

F.6 Swap-testing â€" isolating the faulty component

Plain-English definition. Swap-testing (also called substitution testing) means replacing a **suspected** component with a **known-good** one to see whether the fault moves or disappears. If the fault disappears after the swap, the suspected part was faulty. If the fault remains, the problem is elsewhere.

Why it works. It turns a doubt into a fact. Instead of guessing whether the RAM is faulty, you prove it by fitting a stick you know works.

How to swap-test safely.

1. Identify the component you suspect, based on the workflow.
2. **Power off, unplug and discharge** the machine.
3. Wear the strap and touch bare metal. Handle parts by the edges.
4. Remove the suspected part and fit a **known-good** part (from a working machine or a tested spare).
5. Power on and observe. **One component per test.**
6. If the fault is gone, you have strong evidence the original part was faulty. If it remains, put the original part back and move to the next hypothesis.

Critical cautions.

- Only swap-test components you are **authorised** to handle, and only on **practice machines** in this course.
- Never swap a part from a customer's working machine into another machine without permission â€" you could introduce a fault or lose data.
- When swapping storage drives, be extremely careful: a drive may contain important data. Do not swap drives around casually.
- After swap-testing, restore the machine to its correct configuration â€" do not leave a "spare" part installed and forgotten.

Real-life comparison. Swap-testing is like **trying a different phone charger** to see if your phone charges. If the new charger works, your old charger was faulty. If it still does not charge, the problem is the phone, not the charger.

Common misconceptions.

- "If the new part fixes it, the old part is definitely dead." Usually true, but confirm â€” sometimes the act of reseating (not the part itself) fixes a poor connection.
- "Swap many parts at once to be quick." Never â€” you will not know which change did what. One at a time.

Limitations and safety. Swap-testing needs known-good spares. Without them, rely on the other steps. Never swap a PSU or open it without supervision. Keep track of which part went where.

F.7 Documenting the diagnosis (the beginning of the service report)

Why document. A diagnosis you cannot explain is of little professional value. Documentation:

- Records what you did, so you or a colleague can pick up the job later.
- Protects you if the customer claims you did something you did not.
- Becomes the **service report** the customer receives (Sessions 11â€“12).
- Helps you learn â€” patterns you record today become experience tomorrow.

What to record (a simple structure):

- **Customer and machine details:** name, contact, model, any visible condition.
- **Reported symptoms:** in the customer's words.
- **Your observations:** look/listen/smell/feel.
- **Questions and answers:** the seven questions.
- **Hypotheses:** the likely causes you listed.
- **Tests and results:** what you tested, one at a time, and what happened.
- **Confirmed cause:** the root cause once proven.
- **Fix/plan:** what was done or recommended.
- **Verification:** how you confirmed the machine is now working.

Real-life comparison. Documentation is like a **doctor's case notes**. If you see a different doctor next time, the notes tell them your whole history. Without notes, every visit starts from zero.

Common misconceptions.

- "Documentation is paperwork that wastes time." It saves time, prevents repeated mistakes, and is what separates a professional from a tinkerer.
- "I will remember." You will not â€” especially after several jobs in a day.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The practice desktop with a **simulated fault** (recommended: RAM left unseated so the machine powers on but shows no display; alternative: display cable plugged into the motherboard port while a dedicated card is fitted).
- A working reference machine nearby, plus a known-good RAM stick and a spare display cable for swap-testing.
- The whiteboard with the five steps drawn.
- The Diagnostic Worksheet (4.3) on the projector to show how to fill it in.
- The wrist strap, mat and tools.

G.2 Demonstration â€” diagnosing the simulated "no display" fault (15 minutes)

Say: "Watch me follow the five steps exactly. Notice that I do not guess, and I test one thing at a time."

Do and say, step by step:

Step 1 â€” Gather symptoms.

- "I power on the machine. The fans spin; a light comes on; the monitor says 'no signal.'"
- "I listen â€” no beeps. I look â€” no burn marks, no loose drive. I smell â€” nothing burning."

- "I ask the 'customer' (I will play both roles): 'When did it start? What were you doing? Has anything changed?' The customer says: 'It worked yesterday; today I opened it to clean dust and now there is no picture.'"

Step 2 â€" Confirm the obvious.

- "Is the monitor on? Yes. Is the cable connected? I check both ends. Is it on the correct input? Yes. Is the cable in the right port? I check â€" the machine has a graphics card, and I see the cable is in the graphics card port, which is correct."
- "So the simple things are fine. Now I move inward."

Step 3 â€" Form a hypothesis.

- "Given it worked until the case was opened for cleaning, and now there is no display, my likely causes are: (1) RAM not fully seated, (2) graphics card disturbed, (3) a cable knocked loose. I write these down in order."

Step 4 â€" Test one thing at a time.

- "I **power off, unplug and discharge**. I wear the strap and touch bare metal."
- "Test 1: I reseat the **graphics card** first? No â€" I start with the most likely and simplest: the **RAM**. I press each stick until the clips **snap**." (If the simulated fault is loose RAM, one stick was not clipped.)
- "I check the graphics card is seated â€" it is fine."
- "I make sure I changed only the RAM seating. Nothing else."

Step 5 â€" Resolve, verify, document.

- "I close, reconnect, and power on. The display appears. The fault is gone."
- "I verify beyond the symptom: I restart it; the display returns and the machine behaves normally."
- "I document: symptoms, the customer's answer, my hypothesis, the test (reseat RAM), the result (display returned), the confirmed cause (RAM not fully seated), and the verification."

Expected result: The machine shows a display; the class sees the method find the fault without any part being replaced.

Reasoning to state aloud at key points:

- "I start with the cable and monitor because they are outside the machine, free, and easy."
- "I reseat RAM before touching the graphics card because in a case that was just cleaned, RAM is very commonly nudged loose, and reseating it is simple and safe."
- "I change one thing at a time so I know exactly what fixed it."
- "I verify by restarting, not just by seeing one picture."

Errors to watch and mention:

- Do not test by opening the machine while it is plugged in â€" narrate the unplug/discharge clearly.
- Do not change two things at once â€" demonstrate the discipline.
- Do not declare success after one picture â€" verify.

Questions to ask during the demo:

- "Why did I check the monitor and cable first?" (Simplest, outside the machine, free.)
- "After reseating the RAM, how do I know the fix is real?" (The display returns and it restarts normally â€" verified.)
- "What would I do if reseating the RAM had not worked?" (Put it back, move to the next hypothesis â€" reseat/test the card, then swap-test.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "Diagnostic Scenarios"

Objective: With the instructor guiding, learners apply the five-step workflow to written scenarios and (if a faulty machine is available) to a real simulated fault, filling in a Diagnostic Worksheet for each.

Materials per learner/group: Worksheets 4.1 (Symptom-to-Cause), 4.2 (Questions Checklist) and 4.3 (Diagnostic Worksheet); scenario cards; access to the faulty practice machine (rotating); a working reference machine for comparison.

Procedure (instructor guides each part; learners work in pairs; instructor circulates):

1. **Review the worksheet structure.** The instructor shows Worksheet 4.3 on the projector: fields for symptoms, observations, questions, hypothesis, tests/results, cause, and verification.
2. **Real machine (whole class, guided):** The class diagnoses the instructor's simulated fault together, using the worksheet. The instructor calls the steps; learners fill in each field and predict what the instructor will do next.
3. **Written scenarios (pairs):** Each pair takes **two scenario cards** and completes a full Diagnostic Worksheet for each, using all five steps.
4. **Swap and check:** Pairs swap worksheets with another pair and check each other's logic (did they confirm the obvious? did they test one thing at a time? did they verify?).
5. **Report back:** Two or three pairs read out their best diagnosis. The class gives feedback.
6. **Instructor checks** each pair's worksheets before they finish.

Scenario cards (the six core scenarios):

- **Card 1:** Computer powers on (fans spin) but shows no display.
- **Card 2:** Computer shows a blue error screen (BSOD) occasionally, then restarts.
- **Card 3:** Computer is very slow to start and programs take a long time to open.
- **Card 4:** Computer makes a clicking noise from the drive area.
- **Card 5:** Computer shuts down randomly after about thirty minutes of use.
- **Card 6:** Nothing happens at all when the power button is pressed (no fans, no lights).

Pause-and-predict points:

- After reading Card 1: "*Before touching anything inside, what three obvious things will you confirm?*" (Monitor on; cable connected and in the correct port; monitor input correct.)
- After forming a hypothesis on Card 5: "*If a machine shuts down after about thirty minutes, what is your leading hypothesis and why?*" (Overheating/cooling â€" because it happens after time under load; but confirm, do not assume.)
- After a test on Card 6: "*If the socket and cable are fine and there is still no power, what is your next safety-aware step, and what must you never do?*" (Check the front-panel power wire and, under supervision, consider PSU testing; never open the PSU.)

Expected result: Each pair completes two full, sensible Diagnostic Worksheets, and the class completes the real-machine worksheet, each showing all five steps.

Troubleshooting while walking the room:

- Learner jumps to a cause without observing â†' point them back to Step 1 and ask for the clues.
- Learner tests several things at once â†' remind them of "one change at a time."
- Learner forgets safety when handling the machine â†' stop and reinforce unplug/discharge/strap.
- Learner declares success without verifying â†' ask "how do you know the fault is truly gone?"
- Pair disagrees â†' encourage them to state the evidence for each idea.

Completion checklist (learner ticks each):

- ☐ Worksheet 4.3 fields understood (symptoms, observations, questions, hypothesis, tests/results, cause, verification)
- ☐ Real-machine diagnosis followed all five steps
- ☐ Two scenario cards completed with full worksheets
- ☐ For each scenario: symptoms gathered and obvious things confirmed
- ☐ For each scenario: at least two hypotheses listed in order
- ☐ For each scenario: tests described one at a time
- ☐ For each scenario: cause stated as likely or confirmed, with evidence
- ☐ For each scenario: verification described
- ☐ Peer-check completed and feedback given
- ☐ Safety followed while handling the real machine

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: "The Cyber CafÃ© Fault"

Scenario: You are the technician at **Elias Computer Clinic, Diobu, Port Harcourt**. A cyber cafÃ© owner, *Mama Ngozi*, brings in one of her ten computers. She says: "*This one just stops working after some time. The customer will be browsing, and then the screen freezes and it restarts by itself. It happens more when many people are using the cafÃ© at once, and in the afternoon when the place is hot. In the morning it works fine for a long time.*" She wants to know the cause before she pays for any parts.

Your task: Apply the **five-step diagnostic workflow** to Mama Ngozi's computer and produce a written diagnosis.

Requirements:

1. **Symptoms:** List the reported symptoms and the observations you would make yourself (look/listen/smell/feel).
2. **Obvious checks:** State which simple things you would confirm first.
3. **Questions:** Write at least **five** diagnostic questions you would ask Mama Ngozi, and the likely answer you are hoping to hear.
4. **Hypotheses:** List at least **three** likely causes ranked from most to least likely, with a reason for each.
5. **Test plan:** Describe, one test at a time, how you would confirm or rule out your top hypothesis. Include one swap-test.
6. **Safety:** State every safety step you would take before handling components.
7. **Documentation:** Complete a one-page diagnostic note (you may use Worksheet 4.3).
8. **Conclusion:** State your **leading** cause, and whether it is *likely* or *confirmed*, and how you would confirm it.

Constraints:

- This is a computer that must keep serving customers, so your tests must be safe and quick.
- Do not replace any part without confirming the cause first.
- Never open a live machine or the PSU.
- Use the workflow steps as headings in your answer.

Time allowance: 15 minutes.

Success criteria:

- All five workflow steps are present and correctly applied.
- Symptoms are distinguished from confirmed observations.
- At least three ranked hypotheses with reasons.
- A test plan with one change at a time and a swap-test.
- A correct safety section.
- A clear leading cause marked likely or confirmed, with a confirmation method.

Hints if stuck:

- Hint 1: "Happens in the afternoon when it is hot" and "worse under heavy use" both point toward **heat**.
- Hint 2: A very common cause of a computer shutting down under load is **overheating from dust, a failing fan, or dried thermal paste.**
- Hint 3: A safe, quick test for overheating is to observe whether it fails sooner with the case side off in a cool spot with a fan directed at it â€" but always under supervision, and the machine must be handled safely.
- Hint 4: A swap-test here could be testing with a known-good case fan or checking the CPU fan spins; do not swap the CPU casually.

Instructor answer key (do not show learners):

1. **Symptoms:** Freezes and restarts by itself; worse in the hot afternoon and under heavy use; fine in the morning and light use. Self-observations: check for dust-clogged heatsink/fan; listen for a weak or noisy fan; feel (after shutdown) for unusual heat at the vents; look for error messages on restart.
2. **Obvious checks:** Confirm the power supply from the wall is steady (a cyber cafÃ© may have many machines and a weak extension); confirm the machine is in a well-ventilated spot; confirm nothing is blocking the vents.
3. **Questions (any five):** When did it start? What were you doing? Has anything changed (new machines, new programs, an update)? Does it happen all the time or only when busy/hot? Does the machine feel very hot at the back/top? Did you see any message when it restarted? Have you cleaned it recently? (Accept well-chosen questions.)
4. **Hypotheses (ranked):** (1) **Overheating** â€" dust in the heatsink/fan or dried thermal paste, worsened by heat and load; (2) a **failing power supply** struggling under load (common where many machines share a weak supply); (3) a **failing RAM stick** or (3) **drivers/malware** causing restarts under certain tasks. (Any sensible ranked list with reasons is acceptable, but overheating should be prominent given the pattern.)

5. **Test plan:** First, with the machine off, inspect and clean the dust from the heatsink and fan (Session 8 skill; in this session, describe it). Confirm the CPU fan spins when powered. Monitor the behaviour in a cooler environment. Swap-test: if a known-good PSU is available, test the machine with it under supervision. Test one thing at a time and observe.
6. **Safety:** Unplug from the wall and discharge before opening; wear the anti-static wrist strap and touch bare metal; handle boards by the edges; never open the PSU; never work inside a live machine; ensure ventilation when using compressed air.
7. **Documentation:** A filled Worksheet 4.3 with the above.
8. **Conclusion:** Leading cause: **overheating** (likely). Confirm by cleaning and observing that the shutdown stops, and/or by showing temperature-related behaviour; if it persists after cleaning, test the PSU and RAM by swap-testing. Because a single observation can mislead, the cause remains *likely* until the fix removes the fault reproducibly.

Marking guidance (out of 15): five steps applied (5); symptoms vs observations (2); ranked hypotheses with reasons (3); test plan with one-at-a-time and a swap-test (3); safety completeness (2).

J. REAL-WORLD CASE STUDY

Situation. A customer brings a desktop to the shop with a complaint that sounds familiar: "It works fine for a few minutes, then the picture freezes or the machine restarts." A young technician at the shop tests the RAM and finds nothing wrong. He tests the hard drive and finds nothing wrong. He tests the power supply and finds nothing wrong. Frustrated, he tells the customer: "Your machine is fine â€" I cannot find any problem." The customer leaves annoyed, and the fault continues. A day later, the same customer returns, even more upset.

Relevant facts.

- The fault is **intermittent** â€" it does not appear during casual testing.
- The machine works for a while, then fails.
- Standard component tests (RAM, drive, PSU) show "nothing wrong."
- The young technician tested for the fault without understanding **when** and **why** it happens.

Decision points.

1. Why did "testing everything" fail to find the fault?
2. What did the technician fail to do in Step 1 (gather symptoms)?
3. What question should he ask that would change everything?
4. How should he confirm the cause once he has a hypothesis?

Possible approaches.

1. Keep testing components randomly, replacing parts until something works.
2. Give up and tell the customer "it is fine."
3. Return to Step 1: gather better symptoms, especially *what task* triggers the fault, then form a targeted hypothesis.

Recommended solution. Approach 3. Intermittent faults are diagnosed by **reproducing** them and by asking sharper questions â€" not by testing parts at random. The technician should ask: *"Does it happen when you are doing a particular thing, like watching videos or playing games, or when the room is hot?"* Suppose the customer says: *"Yes â€" it happens when I watch videos for a while, especially in the afternoon."* That points strongly to **overheating under load**. The technician then monitors the machine's behaviour under load in a controlled way and finds the CPU temperature climbing and the fan struggling/blocked. **Confirmed cause:** overheating from a dust-clogged fan and dried thermal paste. **Fix:** clean the cooling system and replace the thermal paste (Session 8 skill), then verify by running the machine under load for a longer period and confirming the fault no longer occurs.

Trade-offs.

- Approach 1 (random replacement) wastes the customer's money and may introduce new faults. It is guesswork disguised as testing.
- Approach 2 (giving up) destroys trust; the fault is real, even if it is intermittent.
- Approach 3 (better symptoms â†' targeted hypothesis â†' reproduce â†' confirm) is slower at the start but finds the true cause and builds a lasting reputation.

Professional connection. Intermittent faults are common and are where a real technician earns respect. The skill is patience and structure: ask **when** and **under what conditions**, reproduce the fault, test one variable at a time, and confirm before charging

the customer for anything. This is also where the "likely vs confirmed" distinction matters most â€” you must *reproduce* the fault and see it disappear after the fix.

Transfer exercise. "Why are intermittent problems harder to diagnose than problems that happen every time, and what two techniques help?" (They do not appear during casual testing; techniques: ask sharper questions to find the trigger and reproduce the fault deliberately; and test one variable at a time, often over a longer period or under load.)

Safety reminder. Any handling of the machine during this diagnosis is done **unplugged and discharged**, with the strap worn, and without opening the PSU. If testing under load, it is supervised, and the machine is never left unattended while running a stress test on a practice unit.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use as the live guide. Ask learners to find the relevant row.

K.1 Jumping to conclusions (guessing)

Symptoms: Replacing a part that was probably fine; the fault returns; the customer is charged for the wrong work. **Likely cause:** Skipping Steps 1â€“3 and going straight to a favourite guess. Confirmed when the fault persists after the "fix." **How to investigate:** Ask: what evidence did I actually have? Did I confirm the simple things first? **Safe corrective steps:** Return to Step 1. Gather symptoms, confirm the obvious, list hypotheses, and test one at a time. **How to confirm resolved:** The fault disappears **and** you can explain the evidence. **How to prevent recurrence:** Always write the hypothesis list before touching a part.

K.2 Not asking enough questions

Symptoms: A diagnosis that fits the symptom but misses the context (like "only when hot" or "only when busy"). **Likely cause:** Rushing to open the machine. Confirmed when the customer reveals a key detail you never asked about. **How to investigate:** Use the seven questions; write the answers. **Safe corrective steps:** Re-interview the customer and revise the hypothesis. **How to confirm resolved:** The new information explains the pattern. **How to prevent recurrence:** Make the seven questions a fixed script.

K.3 Testing the most complex cause first

Symptoms: Wasted time and money on deep checks while a simple cause is ignored. **Likely cause:** Excitement, or a bias toward a favourite part. Confirmed when a simple check (like a loose cable) fixes it. **How to investigate:** Always start from the outside in, simplest first. **Safe corrective steps:** Follow the safe boot order (F.5). **How to confirm resolved:** The simple cause is found and fixed. **How to prevent recurrence:** Memorise the "cheapest, simplest, safest, most likely first" rule.

K.4 Changing more than one thing at a time

Symptoms: The fault disappears but you do not know why, or it returns unpredictably. **Likely cause:** Multiple changes in one test. Confirmed by the confusion when it fails again. **How to investigate:** Recall exactly what you changed; re-test one variable at a time. **Safe corrective steps:** Redo the tests, one change each time. **How to confirm resolved:** You can point to the single change that fixed the fault. **How to prevent recurrence:** Adopt the golden rule: one change per test.

K.5 Not verifying the fix

Symptoms: The machine seems fine, then the fault returns; the customer loses trust. **Likely cause:** Declaring success after one observation or a single picture. Confirmed by the return of the fault. **How to investigate:** Test the machine properly â€” restart it, use it, run it under the conditions that caused the fault. **Safe corrective steps:** Verify under real conditions before returning the machine. **How to confirm resolved:** The fault does not return after proper verification. **How to prevent recurrence:** Make verification the final, mandatory step.

K.6 Not documenting

Symptoms: You cannot explain your work later; a colleague must start from scratch; the customer disputes what was done. **Likely cause:** Believing documentation is a waste of time. Confirmed when memory fails. **How to investigate:** N/A â€” it is a habit. **Safe corrective steps:** Fill in the Diagnostic Worksheet during the job, not after. **How to confirm resolved:** Your notes completely describe the job. **How to prevent recurrence:** Make documentation part of the workflow, not an extra.

K.7 Misreading beep codes / inventing universal codes

Symptoms: A wrong diagnosis based on a beep pattern from the wrong maker. **Likely cause:** Assuming all BIOS beeps are universal. Confirmed when the maker's actual chart contradicts the assumption. **How to investigate:** Identify the make/model and consult that maker's documented beep codes. **Safe corrective steps:** Interpret the beep using the maker's own chart; if unavailable, fall back to the workflow. **How to confirm resolved:** The interpretation matches the maker's documentation and the tested fault. **How to prevent recurrence:** Never memorise invented "universal" beep codes.

Distinguishing likely vs confirmed: Every "likely cause" above becomes "confirmed" only when checked and verified. In diagnosis, an unverified conclusion is just another hypothesis.

L. SESSION ASSIGNMENT

Assignment title: "My Diagnostic Reference and Worked Case"

Purpose: To build a reusable diagnostic reference and to practise applying the five-step workflow to a complete written case, laying groundwork for the Week 2 mini-project, "My Diagnostic Report."

Realistic scenario: You have just started work at a repair shop. Your supervisor hands you a reference book to maintain: a symptom-to-cause table, the workflow, the questions, and one full worked example that new apprentices can study.

Detailed instructions:

1. Create a document titled **"My Diagnostic Reference and Worked Case"**.
2. **Symptom-to-Cause Table:** list at least **eight** symptoms, each with its likely causes (for example: no power; no display; blue screen; slow performance; overheating/shutdown; strange drive noise; random restarts; no internet).
3. **The Five-Step Workflow:** write the five steps in your own words, with one sentence each on why the step matters.
4. **The Seven Diagnostic Questions:** list all seven, and for each, write what kind of clue it gives you.
5. **Safe Boot Order:** write the safe order for (a) "no display but powers on" and (b) "no power at all."
6. **Worked Case:** choose **one** scenario (you may invent a realistic one), and write a full diagnosis using the five steps as headings, including symptoms, observations, questions, ranked hypotheses, a test plan (one change at a time, with a swap-test), the likely-or-confirmed cause, and the verification method.
7. Save the file as **"My Diagnostic Reference and Worked Case"** on the Desktop.
8. Be ready to explain your worked case aloud in the next session.

Required deliverables: One saved document containing all six parts.

Submission format: Saved on the Desktop with the exact file name. (If no home computer, complete in the lab and show before leaving.)

Deadline: Before Session 5 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Symptom-to-cause table	8+ symptoms, accurate	5â€“7 symptoms	Fewer than 5 or inaccurate
Five-step workflow	Complete and clear, with reasons	Mostly complete	Incomplete or vague
Seven questions	All seven with useful clues	5â€“6 with clues	Fewer than 5
Safe boot order	Both cases correct and in order	One correct	Both unclear or wrong
Worked case	All five steps applied, verified, coherent	Mostly applied	Missing steps or illogical

Common mistakes to avoid: confusing symptoms with causes; listing hypotheses as if they were confirmed; testing several things at once; forgetting the verification step; omitting safety; inventing universal beep codes.

Instructor reference solution (summary, do not show learners beforehand):

- Symptom-to-cause examples: no power â†’ wall/cable, front-panel switch, PSU, board; no display â†’ cable/port, RAM seating, card seating, monitor; blue screen â†’ RAM, driver, OS, overheating; slow â†’ malware, full disk, insufficient RAM,

failing drive, overheating; shutdown under load â†' overheating/cooling, PSU; drive noise â†' failing HDD; random restart â†' RAM, PSU, overheating, driver; no internet â†' cable/router/driver.

- Workflow: gather symptoms; confirm the obvious; form a hypothesis; test one thing at a time; resolve, verify and document.
- Seven questions with their clues (as in F.4).
- Safe boot order (as in F.5).
- Worked case: any coherent five-step diagnosis with a clear likely/confirmed cause and a verification method.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below.)

Q1. Short answer. List the five steps of the diagnostic workflow in order.

Q2. Multiple choice. Why should you confirm the obvious things (power, cables, switches) first? a) Because they are the most interesting b) Because they are the simplest, cheapest, safest and often the cause c) Because the customer likes it d) Because it is a legal requirement

Q3. Explain in your own words. What is POST, and what does it mean when a machine "fails POST"?

Q4. Practical identification. A computer powers on (fans spin, lights on) but shows no display. Give the first three things you would check, in order.

Q5. Error detection. A learner says: "I will replace the RAM, the graphics card and the cable all at once to save time." Find the mistake and correct it.

Q6. Scenario. A customer says their computer restarts by itself, only when the room is hot. Which part of the diagnostic workflow did most technicians forget, and what would you do next?

Q7. Short answer. Why must you not treat BIOS beep codes as universal?

INSTRUCTOR ANSWER KEY (Q1â€"Q7)

A1. (1) Gather the symptoms (observe and ask); (2) confirm the obvious (power, cables, switches); (3) form a hypothesis; (4) test one thing at a time; (5) resolve, verify and document.

A2. (b) Because they are the simplest, cheapest, safest and often the actual cause â€" and checking them first prevents wasting time and money on deeper faults.

A3. POST (Power-On Self-Test) is the basic self-check the computer runs when switched on, before loading the operating system, to confirm its essential parts are present and working. If a machine "fails POST," it means this early check did not pass â€" often shown by a beep code (on machines with a speaker), LED indicators, or simply no display. It tells you the problem is very early, before the system loads.

A4. (1) Monitor and cable (is the monitor on; is the cable connected at both ends and in the correct port â€" graphics card vs motherboard; is the input correct); (2) reseat the RAM until the clips snap; (3) reseat the graphics card. (Order of the first two may vary; the point is simplest/outside-first.)

A5. The mistake is changing more than one thing at a time. The correction: **change one thing per test**, observe the result, then move on. Otherwise, if the fault changes, you will not know which change caused it. The golden rule is one change at a time.

A6. The technician forgot **Step 1 â€" gather symptoms** properly (especially the context: "only when hot"). Next: ask more targeted questions and **reproduce** the fault under the trigger conditions, form the hypothesis of **overheating**, and confirm it by checking the cooling system and observing whether the fix removes the fault. (This is a context-rich, intermittent fault.)

A7. Because BIOS/UEFI beep codes differ between makers (AMI, Award, Phoenix, and manufacturers), and many modern machines have no speaker at all. A beep pattern is only meaningful when read against the **specific maker's documented chart**. Do not memorise invented "universal" codes.

Marking: up to 2 marks each (Q1, Q4, Q6 up to 3). Total â‰ˆ 16â€"18; scale proportionally. Full credit requires correct steps, correct order, and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned to diagnose faults like a professional â€” not by guessing, but by a method. The **five-step workflow** is: **gather the symptoms** (observe and ask), **confirm the obvious** (power, cables, switches), **form a hypothesis**, **test one thing at a time**, and **resolve, verify and document**. You learned about **POST (Power-On Self-Test)** â€” the computer's automatic start-up self-check â€” and how to interpret basic clues (remembering that **beep codes differ by maker** and must not be treated as universal). You learned a **safe boot order** for "no display" and "no power," the technique of **swap-testing** to isolate a faulty component, and the discipline of **documentation**. Above all, you learned the difference between a **likely** cause and a **confirmed** cause.

Five important takeaways:

1. **Follow the five-step workflow every time** â€” method beats guessing.
2. **Confirm the obvious first** â€” power, cables and switches are often the cause.
3. **Test one thing at a time** (and swap-test with known-good parts).
4. **Beep codes are not universal** â€” always check the maker's own chart.
5. **Verify and document** â€” an unverified fix is not a fix, and undocumented work is not professional.

Questions to help you recognise what you can now do:

- Can I list and explain the five steps of the workflow?
- Can I gather symptoms using the seven questions?
- Can I give the safe order of checks for "no display" and "no power"?
- Can I explain the difference between a likely and a confirmed cause?

Connection to the next session: You now have the method. Next session you put it to work with **hardware diagnostics** â€” using tools and software to test RAM (for example MemTest86), check a hard drive's health (for example CrystalDiskInfo), monitor temperatures (for example HWMonitor), and check a power supply. The workflow you learned today decides *when* and *why* each of those tools is used.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner states and follows the five steps without prompting.
- The learner gathers symptoms before opening anything, and asks good questions.
- The learner confirms the obvious (power/cables/switches) first.
- The learner tests one thing at a time and uses swap-testing correctly.
- The learner distinguishes "likely" from "confirmed" causes.
- The learner documents each step clearly.

Common misconceptions to watch for and correct:

- "The obvious cause is always right / guessing is fine." â†’ Form a hypothesis and test it; the obvious cause can be a red herring.
- "Experienced technicians skip the method." â†’ They follow it faster; it is what makes them reliable.
- "Test many things at once." â†’ One change at a time.
- "Once it works, stop." â†’ Always verify; and always document.
- "Beep codes are universal." â†’ They are maker-specific; consult the maker's chart.

Suggested remedial exercises for learners who struggle:

1. **Step-by-step prompting:** give a single scenario and walk the learner through the five steps aloud, one at a time, until the sequence is automatic.
2. **Question drill:** role-play the customer while the learner practices the seven questions and takes notes.
3. **Likely vs confirmed sorting:** give ten statements and have the learner mark each as "likely" or "confirmed" and justify it.
4. **Safe-order drill:** shuffle the checks for "no display" and "no power" and have the learner re-sequence them.

Differentiation:

- **Level 1 (supported):** provide a partially completed Diagnostic Worksheet with the symptoms pre-filled and the steps labelled, and work through one scenario together before letting the learner try independently.
- **Level 2 (standard):** complete the guided and independent exercises as written.
- **Level 3 (challenge):** diagnose a **real intermittent** fault on a practice machine (for example a partially blocked fan), timing how long it takes to reproduce the fault, and produce a full written report; or design a new scenario card and model answer for the class.

Timing guidance: The value of this session is the method, so protect the guided practical (Section H) and the independent exercise (Section I). If the room is slow, compress the lecture sections on swap-testing (F.6) and documentation (F.7) to summaries, but do **not** shorten the five steps (F.2), safe boot order (F.5), or the POST honesty note (F.3). Keep the case study as a short story if time is tight.

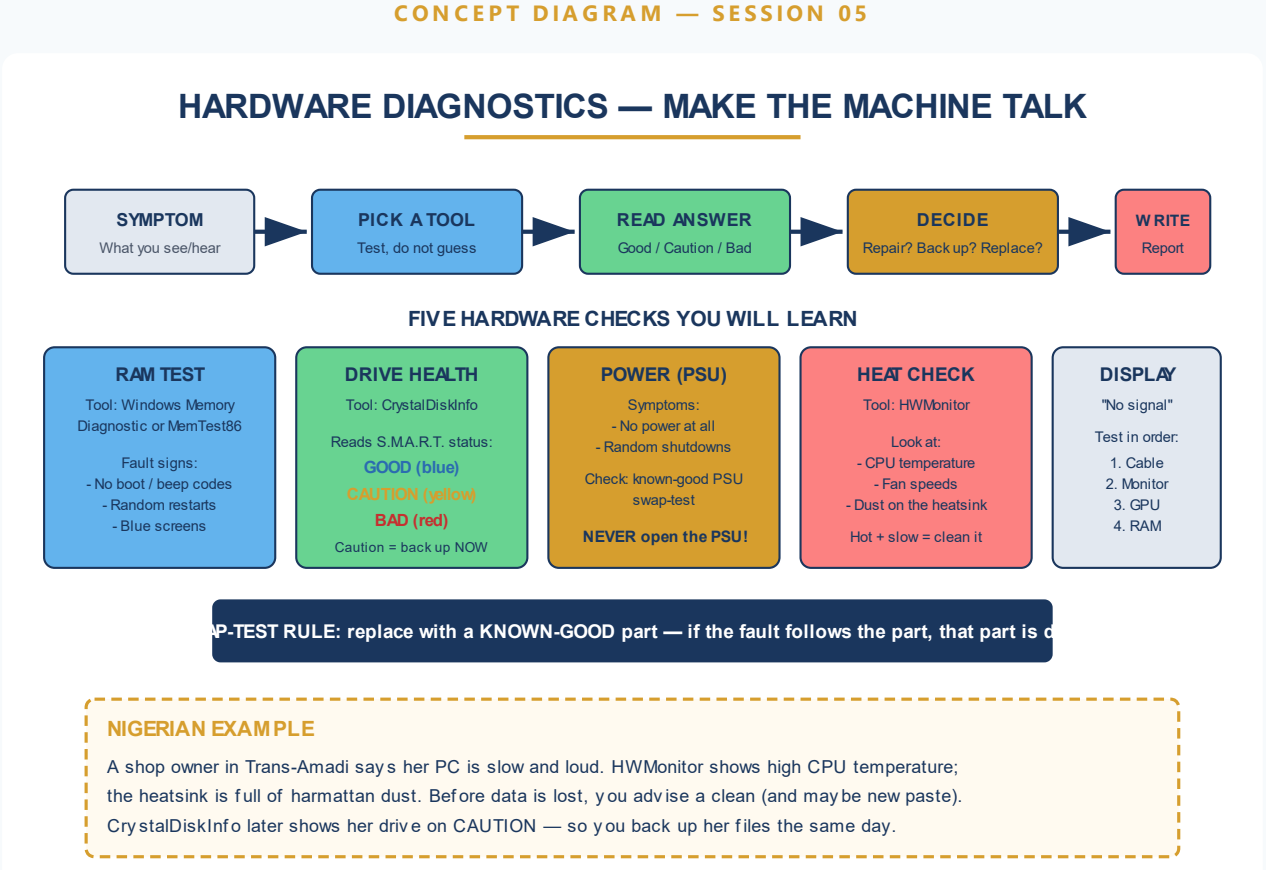
END OF SESSION 4

Next: Session 5 â€" Hardware Diagnostics: Testing RAM, Storage, Temperatures and the Power Supply.

SESSION 05 OF 12

Hardware Diagnostics

CONCEPT DIAGRAM — SESSION 05



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 5 â€" HARDWARE DIAGNOSTICS

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	5 of 12
Session title	Hardware Diagnostics
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes
Difficulty level	Beginner (first structured hardware fault-finding)

Item	Detail
Prerequisite knowledge	Sessions 1â€"4: the six major components, safe handling and tools, disassembly/reassembly, and the five-step diagnostic workflow (Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve). If a learner cannot name the six major components or list the workflow, run the 10-minute refresher in Appendix 5A before this session.
Connection to previous work	Session 4 gave you the *workflow*. This session gives you the *tools and tests* you apply at the "Test" step when you suspect the fault is hardware.
Preparation before class	Practice desktops on the benches; diagnostic software installed (CrystalDiskInfo, HWMonitor); a bootable MemTest86 USB prepared by the instructor; one known-good and one known-faulty RAM stick; one known-good and one failing hard drive; one known-good PSU; a working monitor and one spare VGA/HDMI cable; anti-static wrist straps; a PSU tester if the academy has one (optional); printed Diagnostic Report Forms (one per learner); Component Identification Charts on the wall.

A note on wording used in this course. In CEA materials the word "diagnostics" means *the act of finding out what is wrong*, not only software. A **diagnostic tool** is any device or program that helps you find a fault. This session is about hardware: physical parts that can fail. Session 6 is about software.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain**, in plain language, what a hardware fault is and how it differs from a software fault.
2. **Describe** the common symptoms of a faulty RAM stick â€" failure to boot, beep codes, random restarts, blue screens â€" and run a memory test using Windows Memory Diagnostic or MemTest86.
3. **Read** a storage drive's S.M.A.R.T. health in CrystalDiskInfo and decide whether the drive is Good, Caution or Bad; state when to back up and when to replace.
4. **List** the basic checks for a power supply fault (no power, random shutdowns) and explain why the PSU must never be opened.
5. **Identify** an overheating problem by checking temperatures with HWMonitor and inspecting dust, fans and thermal paste.
6. **Diagnose** a "no signal" display fault by testing the cable, the monitor, the GPU and the RAM in a sensible order.
7. **Apply** the swap-testing and isolation method to prove which component is faulty.
8. **Record** findings on a Diagnostic Report Form using clear, plain language a customer can understand.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least two practice desktop computers set up for diagnostics (they may be opened during the practical).
- One **known-good** RAM stick and one **known-faulty** RAM stick (instructor keeps these labelled; learners are not told which is which).
- One **known-good** hard drive and one drive that shows wear (the instructor can supply a drive with a rising reallocated-sector count, or simulate a "Caution" reading using a training drive).
- One known-good power supply and, if available, one PSU that has failed.
- A working monitor plus one spare display cable (VGA, HDMI or DisplayPort, matching the practice machines).
- Anti-static wrist straps (one per learner) and a clean, hard work surface.
- Screwdrivers, tweezers, flashlight, and small parts trays.
- Bootable MemTest86 USB drive (prepared by the instructor). One USB can be shared around the class.
- Optional: a PSU tester (a small device that plugs onto the PSU's main connector and lights up the voltage lines). This is a luxury item; the course works without it.

Software (provided by CEA)

- CrystalDiskInfo (free) â€" reads the health of a storage drive.
- HWMonitor (free) â€" shows temperatures, fan speeds and voltages.
- Windows Memory Diagnostic â€" already built into Windows (no install needed).
- MemTest86 (free, on the USB drive) â€" a deeper memory test run from outside Windows.

Printed materials

- Diagnostic Report Form (one per learner).
- Symptom-to-Cause quick reference card (one per learner).

Safety equipment

- Anti-static wrist straps, safety glasses, first aid kit.

Internet

- Not required. Optionally, the instructor may show a search for a beep-code meaning at the very end, but do not depend on it.

Alternative for learners who lack a facility

- Any working desktop can be used for the temperature and storage checks. Learners without a personal machine share a bench machine in pairs: one person runs the tool, the other records the findings, then they swap.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: On the projector, show a photo or drawing of a car dashboard with the **check-engine light** on. Do not explain yet.

What the instructor says:

> *"Good morning. When your car's check-engine light comes on, you do not open the bonnet and start changing parts at random. You take it to a mechanic. The mechanic plugs in a small computer and it gives codes â€" it tells him exactly which sensor is complaining. He does not guess; the tool does the guessing for him.*
>
> *A computer is the same. When a computer misbehaves, we do not start replacing parts blindly. We use tools to ask the machine a question and read the answer. Sometimes the machine answers with a **beep** instead of words. Sometimes it answers with a **temperature reading**. Sometimes it answers with a **health status: Good, Caution or Bad**.*
>
> *Today you become the mechanic. You will learn the tools that make a computer tell you what is wrong with its hardware."*

Expected learner response: Some may mention "blue screen", "it beeps", "loud fans" or "the screen just shows nothing". Accept all answers and write them on the board. Say: "*These are all symptoms. Today we turn symptoms into a diagnosis.*"

Then demonstrate the hook: On the projector, open CrystalDiskInfo. Point at the **blue "Good"** status and say:

> *"See this word â€" Good. That is the drive telling us it is healthy. If I open a drive that is failing, this turns yellow, or even red. One colour is the difference between a fine computer and five years of somebody's photos lost forever. By the end of today, you will be able to read this yourself."*

Why this opener works: It connects a familiar everyday tool (the mechanic's code reader) to computer work, it sets the expectation that we *test rather than guess*, and it ends with a concrete, exciting promise (reading drive health) that the learners will actually do in the practical.

E. PREREQUISITE REVIEW (7 minutes)

Run this as an oral round-robin. Put the questions on the projector. Keep it fast â€" this is revision, not new teaching.

Question 1: "Name the five steps of the diagnostic workflow."

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
- Common gap: learners remember "test" but forget "ask the customer" or "document". Remind them that a diagnosis you cannot explain is not finished.

Question 2: "What are the six major components inside a computer?"

- Expected answer: **Motherboard, CPU, RAM, storage (hard drive/SSD), PSU, GPU.**
- Common gap: omitting the PSU. Probe: *"What turns the wall electricity into the voltages the parts need?"*

Question 3: "Why do we always start by checking the simplest possible cause first?"

- Expected answer: Because the simplest cause is quick to check and is very often the real one. We save time and money by ruling out easy things (a loose cable, a dead socket) before opening the case.
- Common gap: learners jump straight to "replace the hard drive". Correct this â€" a loose monitor cable looks exactly like a dead GPU, and costs â‚¦0 to fix.

Question 4: "A computer fails to start and you hear repeated beeps. What is the machine trying to tell you?"

- Expected answer: The motherboard lights up and runs a self-test before loading Windows; if something basic fails, it beeps a pattern instead of showing a picture. Beeps usually point to RAM, the GPU, or another key part.
- Common gap: learners think the beeps are "just a noise". Introduce the term **POST** now (Power-On Self-Test) â€" we define it properly in Section F.

Question 5: "Why must we document every finding, even when we are not yet sure?"

- Expected answer: So the work can be verified later, so a colleague can continue where you stopped, and so the customer gets an honest, clear report. Also, comparing "before" and "after" readings proves the repair worked.

Bridge to today (say this):

> *"In Session 4 you learned to think like a technician. Today you learn to test like one. Think of a doctor. The workflow is the doctor asking questions and forming a theory. The diagnostic tools today are the stethoscope and the blood test â€" the things that turn a theory into a fact."*

Gap handling: If more than a third of the class cannot answer Question 1 or 2, pause and run the Appendix 5A refresher now (10 minutes), then continue. Do not push on with learners who cannot name the parts â€" today depends on it.

F. FULL LECTURE NOTES

F.1 Hardware fault versus software fault â€" what is the difference?

The idea. Everything that goes wrong with a computer is either **hardware** or **software**. Getting this right early saves hours.

Plain-English definition.

- A **hardware fault** is a problem with a physical part: a stick of RAM that is damaged, a hard drive that is wearing out, a fan that has stopped spinning.
- A **software fault** is a problem with the instructions or data: a virus, a corrupted Windows file, a wrong driver.

Why the distinction exists. The word "fault" describes *what is wrong*, but hardware and software are fixed in completely different ways. Hardware faults are usually fixed by **replacing** or **repairing a part**. Software faults are usually fixed by **cleaning, correcting or reinstalling** the software. If you fix the wrong type, you waste time and money.

Analogy. A car is hardware; the driver's bad habit of always pressing the brake is "software". If the brake pedal itself is broken, that is a hardware fault â€" you replace the pedal. If the pedal is fine but the driver keeps stalling, that is a "software" fault â€" you have to change the behaviour. Replacing a perfectly good pedal because of a driving habit would be a waste.

Realistic Port Harcourt scenario. A customer in Rumuokoro says his laptop "is very slow and sometimes beeps and restarts". The "slow" part sounds like software (too many startup programs, or a virus). But the "beeps and restarts" part sounds like **hardware** (RAM or overheating). This session's job is the second half â€" the hardware. Session 6 handles the first half. A good technician separating the two is what makes a diagnosis fast.

Important misconception. "If Windows is slow, it must be a virus." Not always. A failing hard drive and too little RAM both make a computer slow, and both are hardware. Always *test* before you decide.

Key limitation. Some faults look like one type but are the other. Example: a failing hard drive can corrupt Windows files, so you see software symptoms caused by a hardware problem. This is why we test the hardware *first* â€" if the foundation (RAM, drive, power, cooling) is bad, no amount of software fixing will hold.

F.2 POST and beep codes â€" the very first test

The idea. When you press the power button, before Windows ever appears, the computer runs its own quick self-check.

Plain-English definition. POST stands for **Power-On Self-Test**. It is a tiny set of instructions stored on the motherboard that checks that the basic parts â€" CPU, RAM, graphics, keyboard â€" are present and responding before handing over to the operating system.

Why it exists. Imagine a bus company that will not move the bus until it has checked that the engine, brakes and doors work. POST is the computer doing that check on itself. If a critical part fails the check, the computer cannot show a picture, so it uses **beeps** (or a light on the motherboard, depending on the model) to tell you which basic part failed.

What it does. It runs in the first one or two seconds after power-on. If everything passes, you hear a single short beep (on some models) and the screen shows the maker's logo. If something basic fails, you hear a pattern of beeps, or nothing at all.

Analogy. POST is like the fuel-pump priming sound you hear for a moment when you switch on many cars before the engine starts â€" the car checking itself quietly before it lets you drive.

Realistic Port Harcourt scenario. A cyber cafÃ© PC will not show anything on screen and beeps three times. The cafÃ© owner assumes the monitor is dead and is about to buy a new monitor. A technician recognises the beep pattern as a **memory (RAM)** warning, reseats the RAM stick, and the machine works â€" for free.

Important note on beep meanings. The beep pattern depends on the **make of the motherboard's BIOS/UEFI** (for example, the number and length of beeps). Because this differs between makers, the correct professional method is: **count the beeps, then look up that maker's beep code in the manual or on the maker's support page.** Do not memorise one pattern and assume it fits every computer. In this course we teach you to *count and look up*, not to guess.

Important misconception. "No beep means the computer is completely dead." Not always. Many modern motherboards have no built-in speaker, so they cannot beep at all. On those, a small **debug light** or **LED code** on the board does the same job. Also, some working computers never beep. Absence of a beep is a clue, not a verdict.

Limitation. POST only checks very basic things. It does not test your storage deeply, it does not test the display cable, and it cannot tell you a drive is slowly failing. For those, we use the tools below.

F.3 Diagnosing RAM faults

The idea. RAM (Random Access Memory) is the fast scratch-pad where the computer holds whatever it is working on right now. When RAM goes bad, the computer becomes unstable in a very particular way.

Plain-English definition. RAM faults are failures in the memory sticks â€" the long, flat strips plugged into the motherboard. A stick may be entirely dead, or it may work most of the time and fail only occasionally (called an **intermittent fault**, one of the hardest to catch).

Why RAM matters. Windows and every program keep their current work in RAM. If the memory returns a wrong value even once, calculations go wrong, programs crash, or Windows shows a blue screen.

Common symptoms of faulty RAM:

- Computer will not start, or starts then restarts again and again ("boot loop").
- Repeated **beep patterns** during POST.
- **Blue screen** errors (in Windows, the "Blue Screen of Death" or **BSOD**) â€" especially stop codes that mention memory.
- Programs closing suddenly with no warning.
- Files becoming corrupted for no clear reason.
- Random freezes under load.

Analogy. RAM is like the work-table of a tailor in the market. If one corner of the table is warped, most sewing is fine, but every time you lay cloth on that corner the measurement comes out wrong. The whole table is not useless â€" only that spot â€" but it spoils everything that touches it.

How we test RAM â€” Tool 1: Windows Memory Diagnostic.

1. Save all work and close all programs.
2. Press the **Windows key**, type **Windows Memory Diagnostic**, and open it.
3. Choose **Restart now and check for problems**.
4. The computer restarts and runs the test outside Windows. This can take 10â€“30 minutes.
5. When Windows restarts, the results appear (also visible later in Event Viewer).

What it does: it writes known patterns into RAM and reads them back, looking for any mismatch.

How we test RAM â€” Tool 2: MemTest86. MemTest86 is the deeper, more trusted test. It runs from a USB drive before Windows starts.

1. Insert the MemTest86 USB prepared by the instructor.
2. Restart and boot from the USB (the exact key to enter the boot menu â€” often F12, F10, F2 or Del â€” depends on the make; the instructor will show it on the practice machines).
3. The test starts automatically and shows a screen of moving numbers.
4. **Let it complete at least two full passes** (this can take 30+ minutes, so in class we usually start it and observe the first pass only).
5. **Any red line or "errors" count above zero means the RAM is faulty.**

Isolating the bad stick (swap testing). If the machine has more than one RAM stick:

1. Remove one stick, leaving only stick A, and run the test.
2. If errors appear, stick A is bad. If not, remove stick A and test stick B alone.
3. Test one at a time until you find the faulty one.
4. You can also **swap** a known-good stick in: if the faults disappear, the original stick was the problem.

Important misconception. "The computer counts 8 GB of RAM, so the RAM is fine." Capacity is not health. A stick can be fully detected and still return wrong values occasionally. Only a test proves health.

Limitation. Memory testers cannot find every intermittent fault, because a bad spot may only fail when hot or under a certain load. If a memory test passes but the symptoms strongly suggest RAM, an experienced technician may still swap the stick as a test. Also, a blue screen can be caused by a bad **driver** rather than bad RAM â€” Session 6 covers that.

F.4 Diagnosing storage faults (hard drives and SSDs)

The idea. Storage is where your files live permanently. When a spinning hard disk is wearing out, it usually warns you before it dies â€” if you know how to read the warning.

Plain-English definition.

- A **hard disk drive (HDD)** has spinning metal platters and a moving read/write head. Being mechanical, it can wear out and make noise.
- A **solid-state drive (SSD)** has no moving parts; it stores data in memory chips. It is faster and quieter, but it can still fail, often with less warning.
- **S.M.A.R.T.** stands for **Self-Monitoring, Analysis and Reporting Technology** â€” a built-in system where the drive continuously measures its own health and reports it.

Why it exists. Drive makers realised that most drives die slowly. S.M.A.R.T. was created so that a technician's tool could read these self-measurements and warn the owner *before* total failure â€” giving time to back up.

What it reports (the important parts).

- An overall **Health Status**: **Good (blue)**, **Caution (yellow)** or **Bad (red)**.
- **Temperature** â€” a drive running too hot may need better airflow or cleaning (see F.6).
- **Reallocated sector count** â€” sectors the drive has quietly set aside because they became unreadable. **Any number above zero is a warning sign**; a rising number means the drive is dying.
- **Power-on hours** â€” how long the drive has been running in its life, useful context.

Analogy. S.M.A.R.T. is like the oil-warning light in a car. It does not fix the engine, but it tells you "the engine is not happy" early enough for you to act. Ignore it, and you get a broken-down car and a big bill.

Tool: CrystalDiskInfo.

1. Open CrystalDiskInfo.
2. Read the **Health Status** word and colour at the top.
3. Read the **Temperature**.
4. Scroll the attribute list and look at **Reallocated Sectors Count** and any item flagged in yellow or red.
5. If it says **Good** â€" continue normal checks. If **Caution** â€" **back up the customer's data immediately** and recommend replacement. If **Bad** â€" stop writing to the drive, back up if it still reads, and replace it.

Symptoms a storage drive is failing:

- The computer became very slow, especially at start-up or when opening files.
- Clicking, ticking or grinding noises from the drive (HDD only).
- The drive is **not detected** by the BIOS or by Windows.
- "No boot device" or "operating system not found" errors.
- Files and folders disappearing or becoming corrupted.

Clicking noise â€" what to do. A repetitive **clicking** sound from an HDD almost always means the head is failing to read the platter. **Do not keep powering it on and off trying to make it work** â€" each attempt can scrape more data away. If the data is important, power it down and take it to a **data recovery** specialist. Explain to the customer, honestly, that data recovery is not something a general repair technician should attempt casually, and that it can be expensive.

Important misconception. "SSDs do not fail." They do fail. They are more shock-resistant than HDDs, but they can still stop working, often suddenly. Backups matter for both.

Limitation of CrystalDiskInfo. A drive can show "Good" and still fail suddenly (especially some SSDs). S.M.A.R.T. is a warning tool, not a guarantee. The only true protection is a **backup**. Say this to every customer: "*Health tools reduce risk; only a backup removes it.*"

F.5 Diagnosing power supply (PSU) faults

The idea. The **PSU (Power Supply Unit)** converts the wall electricity (240 V in Nigeria) into the low voltages (12 V, 5 V, 3.3 V) the components use. If it is weak or dead, the whole computer is affected.

Plain-English definition. The **PSU** is the metal box, usually at the top or bottom-back of the case, that the power cable plugs into. A **PSU fault** means it is not delivering correct, stable power.

Why it exists. Components cannot use 240 V directly â€" they would be destroyed instantly. The PSU safely steps the voltage down and steadies it.

Common symptoms of a PSU fault:

- The computer does not power on at all (no lights, no fans).
- The computer powers on for a second then dies.
- **Random shutdowns** or restarts, especially when the computer is working hard.
- Burning smell, or scorch marks near the PSU.
- The fans twitch but the machine will not start.

Analogy â€" and this analogy is a warning. The PSU is like the transformer that feeds a whole street. If your mains voltage is unstable, every appliance in the house behaves strangely â€" flickering lights, a fridge that hums then dies. The PSU is the same for the computer. But unlike a street transformer, you must **never** open the PSU to check inside.

> ***SAFETY â€" READ ALOUD AND POST ON THE WALL.*** *Never open a PSU case. Even when the computer is unplugged and has been off for hours, the large **capacitors** inside a PSU can store a **lethal electrical charge**. Touching the wrong part can kill you. A PSU is not a repairable item for a general technician â€" it is a **replace-only** item.*

Basic PSU checks a beginner may perform (all with the computer unplugged unless stated):

1. **Visual check (unplugged):** look for bulging capacitor tops, burn marks, melted plastic or a strong burnt smell. A bulging or burnt PSU means replace it, full stop.
2. **Smell test:** a sharp burning smell around the PSU = failed, replace it.
3. **Connection check (unplugged):** make sure the big 24-pin motherboard cable and the CPU power cable are fully seated. A loose connector causes no-power symptoms.

4. **Socket check:** confirm the wall socket actually has power â€” test it with another appliance or a known-good lamp. This sounds obvious, but it is the most common "no power" cause.
5. **Swap test (the safest and most reliable):** replace the suspect PSU with a **known-good PSU** of suitable wattage. If the computer then works, the original PSU was the fault.

The paperclip test â€” advanced and optional. There is a way to force a PSU on by bridging two pins on its main connector with a bent paperclip, to see if the fan spins. This test:

- Should only ever be done by the **instructor**, on a bench, as a demonstration; it is **not** a learner task.
- Requires care because you are working with a live PSU once it starts.
- Proves only that the PSU powers up; it does **not** prove the voltages are within spec.

Because of the danger and its limited usefulness, **this course uses the swap test as the standard method.** If the academy has a **PSU tester** (a small device that plugs onto the main connector and shows the voltage readings), the instructor may demonstrate it â€” it is safer and more informative than the paperclip method.

Important misconception. "A bigger wattage PSU always fixes random shutdowns." Wattage is a *capacity*, not a *health measure*. A failing 700 W PSU is still failing. Also, choosing the correct wattage matters (Session 7 covers this).

Limitation. A PSU can fail in subtle ways that basic checks miss â€” delivering slightly wrong voltages that cause random faults. When symptoms strongly suggest PSU and the swap test confirms it, replace it; do not try to "tune" it.

F.6 Diagnosing overheating

The idea. Every chip produces heat when it works. If that heat cannot escape, performance drops and, eventually, parts are damaged.

Plain-English definition. Overheating means a component is running hotter than it should. The usual cause is dust, a failed fan, dried-out thermal paste, or blocked airflow.

Why it matters. Modern CPUs and GPUs can slow themselves down on purpose when hot (called **thermal throttling**) â€” that is why a dusty computer *feels slow*. If it gets hotter still, the computer shuts itself off to protect the part (a **thermal shutdown**).

Analogy. Imagine wearing a thick coat (northern-style) in the Port Harcourt heat while doing hard labour. You cannot work well, and if you keep going you collapse. The computer "collapses" by shutting down.

Tool: HWMonitor. This free tool displays **temperatures**, **fan speeds** and **voltages** for the sensors it can find on the machine.

- **Normal ranges (rough guide):**
  - **CPU:** around 30â€“50 Â°C when idle; up to about 80 Â°C under heavy load.
  - **GPU:** around 30â€–60 Â°C idle; up to about 85 Â°C under load.
  - **Hard drive:** roughly 25â€–45 Â°C.
  - **Motherboard:** roughly 30â€–50 Â°C.
- These are guides, not exact rules; different models differ. What matters is: a reading well above the guide, or a big gap between "before" and "after" cleaning, tells you overheating is happening. If a value keeps climbing toward 90 Â°C or above, treat it as a fault.

Common symptoms of overheating:

- Random shutdowns after some minutes of use.
- Fans running loudly all the time.
- The case or the base of a laptop feeling very hot.
- Slowness that gets worse the longer the computer runs.

Basic checks (with the computer off and unplugged):

1. **Dust:** open the case and look at heatsinks and fan blades. Thick dust "felt" between the fins blocks airflow (the "dust blanket").
2. **Fans:** check that every fan spins freely when the machine is running and is not grinding. A stopped CPU fan is serious â€” the computer may run for a minute then shut down.
3. **Thermal paste:** the grey paste between the CPU and its heatsink can dry out over years and lose its ability to transfer heat. Re-applying it (Session 8) restores cooling.
4. **Airflow:** confirm intake and exhaust vents are not blocked by a wall, a curtain, or a laptop sitting on a soft bed.

Important misconception. "If the fan is loud, the fan is broken." Often the fan is loud because it is working hard to fight overheating â€" the *cause* is dust, not the fan. Clean first, then judge the fan.

Limitation. Not every machine exposes every temperature sensor, and some older machines report only a few values. Absence of a reading is not proof that temperature is fine. Where sensors are missing, rely on symptoms (shutdowns, hot case) and physical inspection.

F.7 Diagnosing display (no signal) faults

The idea. "No signal" or a black screen does not mean the computer is dead. It means the picture is not reaching the screen. There are several possible breaks between the computer and your eyes.

Plain-English definition. A **display fault** is anything that stops a correct picture appearing: the cable, the port, the GPU, the monitor, or the computer not finishing POST.

Analogy. Think of a DSTV or GOtv setup. If the TV shows "no signal", you do not immediately buy a new TV. You check the cable, the decoder, the dish. The same order of thinking applies to a computer screen.

Order of checks (cheapest and simplest first):

1. **Is the monitor on, and on the right input?** Many monitors have several inputs (HDMI1, HDMI2, VGA). The input may have been switched by mistake. Check the monitor's menu.
2. **Cable:** unplug and firmly reseat the display cable at **both** ends. Try a **known-good spare cable**. A damaged or loose cable is a very common "no signal" cause.
3. **Try a different port:** if the computer has both a motherboard video output and a GPU output, try the other. On a desktop with a dedicated GPU fitted, the monitor should be connected to the **GPU**, not the motherboard â€" connecting to the wrong one can give no signal.
4. **Does the computer finish POST?** If the fans spin but there is no picture and there is a beep pattern, the fault may be RAM or GPU, not the screen. Test RAM as in F.3.
5. **Swap the monitor:** try a **known-good monitor**. If the picture appears, the original monitor was faulty.
6. **Swap the GPU:** on a desktop, fit a **known-good GPU** (or use the motherboard output) to test. If the picture appears, the original GPU was faulty.

Common symptoms and what they suggest:

- "No signal" but the PC fans spin and lights come on â†' cable, monitor, GPU or POST failure.
- Picture appears only after wiggling the cable â†' damaged cable or port.
- Screen shows strange colours or lines â†' often the monitor or cable (sometimes the GPU).
- Laptop screen dark but an external monitor works â†' the laptop's screen or its ribbon cable, not the GPU.

Important misconception. "Black screen = dead computer." Very often the computer is running fine and only the display path is broken. Also, on a laptop, a very dim screen may just be backlight or brightness â€" shine a torch at an angle to see if a faint image exists; that points to backlight rather than a dead machine.

Limitation. Diagnosing laptop screens in detail (screen replacement, ribbon cables) is beyond this session; here we focus on isolating *whether* it is the screen and passing that finding on.

F.8 Motherboard and laptop battery issues

The idea. The motherboard is the main circuit board everything plugs into. The laptop battery stores power so the laptop can run without the charger.

Motherboard â€" plain-English definition and symptoms. The **motherboard** is the large board that carries the CPU socket, RAM slots, expansion slots and connectors. Motherboard faults are usually the hardest to diagnose because the board controls so much.

- Symptoms: no POST at all, no beeps, no display, no power lights; sometimes power comes on for one second and dies; sometimes some ports dead while others work.
- Common causes: a failed capacitor, physical damage, a power surge, or liquid spillage.
- **Beginner rule:** because the motherboard interacts with everything else, you must first rule out the *other* parts (RAM, PSU, GPU, display) using swap tests. Only when every other part is known-good do you conclude "the motherboard is faulty".

Motherboard-level repair (soldering, chip replacement) is **advanced work** and outside the scope of this certificate â€" the honest action is to replace the board or advise the customer.

Laptop battery â€" plain-English definition and symptoms. The **battery** is the rechargeable pack that powers a laptop when it is not plugged in. A worn battery is not a "fault" in the same way, but it causes problems:

- Symptoms: laptop only works when plugged in; battery percentage jumps; the laptop shuts down suddenly; the battery swells (bulges) and may push the trackpad out.
- **Safety warning:** a **swollen battery** is a fire risk. Stop using the laptop immediately, do not press on it, do not puncture it, and advise the customer to have it replaced by a technician. Do not attempt to force a swollen battery out of a laptop without proper care and supervision.
- Diagnostic note: a "not charging" laptop might be the battery, the charger cable, or the charging port. Try a **known-good charger** first (cheapest test), then consider the battery, then the port.

Important misconception. "A swollen battery just looks a bit fat, it is still fine." It is not fine. Swelling means the cells are failing and can catch fire. Treat it as a priority safety issue.

Limitation. Battery health detail (cycle count, capacity percentage) can be read with some built-in laptop tools, but tool names and menus differ by make. In this course we rely on the clear physical signs (only works on charger, swollen pack) rather than model-specific tools.

F.9 Swap testing and isolation â€" the professional method

The idea. The fastest, safest way to prove which part is faulty is to **change one thing at a time**.

Plain-English definition. Swap testing means replacing a suspect part with a **known-good** part to see if the fault disappears. **Isolation** means removing or disconnecting parts to narrow down where the fault is.

Why it exists. Guessing wastes money. If a customer is told "buy a new GPU" and the real fault was a â‚¦500 cable, you lose the customer's trust. Swap testing turns a guess into a **confirmed diagnosis**.

Rules of swap testing:

1. **Change only one thing at a time.** If you change two parts and it works, you do not know which one fixed it.
2. **Use a known-good part** â€" one you are sure works. If your "test" part is also faulty, you will reach a wrong conclusion.
3. **Power off and unplug** before swapping internal parts, and wear your anti-static wrist strap.
4. **Record each step** on the Diagnostic Report Form: what you changed, and what happened.
5. **Return the machine to its original state** if the swap does not fix it, before trying the next test.

Analogy. A doctor testing for malaria does not treat for five diseases at once. She tests one thing, reads the result, then decides the next test. Swap testing is the same discipline.

Important misconception. "I do not need to use a known-good part; any spare will do." No. Your conclusion is only as good as your test part. Always use a part you trust.

Limitation. Some parts are hard to swap quickly (a soldered laptop battery, an integrated screen). For those, isolation (disconnecting, checking connectors, using external equivalents) replaces swapping.

F.10 Summary of the session

- Every fault is **hardware** or **software**; today we prove hardware faults.
- **POST** runs first; **beep codes** (count, then look up) point to basic failures.
- **Faulty RAM** â†' boot loops, beeps, blue screens, crashes. Test with **Windows Memory Diagnostic** or **MemTest86**; isolate one stick at a time.
- **Failing storage** â†' slowness, clicks, not detected. Read **S.M.A.R.T.** in **CrystalDiskInfo** (Good / Caution / Bad). Back up on "Caution".
- **PSU faults** â†' no power, random shutdowns, burning smell. **Never open a PSU.** Use the **swap test**.
- **Overheating** â†' shutdowns, loud fans, slowness. Check dust, fans, paste, airflow and temperatures in **HWMonitor**.
- **Display faults** â†' check monitor, cable, port, GPU, POST, in that order.
- **Motherboard / laptop battery** â€" rule out everything else first; swollen batteries are a fire risk.
- **Swap testing + isolation** turns guesses into confirmed diagnoses.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- One practice desktop, opened on the bench, with an anti-static wrist strap.
- One known-good RAM stick and one stick designated "suspect" (the class is not told which).
- CrystalDiskInfo and HWMonitor installed and openable.
- One attempt at Windows Memory Diagnostic queued (or MemTest86 USB ready).
- A known-good monitor and a spare display cable.
- The Diagnostic Report Form on the projector.

G.2 Demonstration 1 â€” Reading drive health with CrystalDiskInfo (4 minutes)

Say: "Watch me ask this drive a question and read its answer."

Steps:

1. Open **CrystalDiskInfo**. Point to the top line: the **Health Status** word and colour.
2. Read the **Temperature**. Say: "38 Â°C â€” that is cool and normal."
3. Scroll to **Reallocated Sectors Count**. Say: "It is zero. That is what we want â€” zero means the drive has not yet set aside any bad spots."
4. Now open CrystalDiskInfo on the **training drive** that shows wear. Point at **Caution (yellow)** and the non-zero reallocated count.
5. **Say:** "Caution means this drive is telling us it is getting sick. Our action is simple and urgent: **back up the data today**, and tell the customer to plan for a replacement."

Expected results: "Good" (blue) on the healthy drive, "Caution" (yellow) with a non-zero reallocated sector count on the training drive.

Reasoning to say aloud: "I am not guessing. The drive measured itself and I am reading the measurement. That is the whole philosophy of diagnostics."

Common errors to show and correct:

- Reading only the word and ignoring the attribute list. Show that Caution can appear even if the summary word is slow to update.
- Confusing a warm drive with a failing drive â€” temperature and health are different columns.

Questions to ask during the demo:

- "If the status is Caution, what do we do *first* â€” back up or replace?" (Back up first, so we do not lose data.)
- "Is a drive with zero reallocated sectors guaranteed to live forever?" (No â€” that is why we back up.)

G.3 Demonstration 2 â€” Temperature check with HWMonitor (3 minutes)

Say: "Now the health of the cooling system."

Steps:

1. Open **HWMonitor**. Point to the **CPU** temperature and the **fan speed** rows.
2. Say: "CPU at 45 Â°C idle â€” normal for these machines."
3. Point to fan speeds. Say: "These are the CPU and case fan speeds. If a fan shows zero while the machine is on, that fan is not spinning â€” investigate."
4. Optional: gently run a heavy task (or just say what *would* happen) to show the CPU number rising under load.

Expected results: a CPU reading within the normal idle guide, and fan speeds above zero.

Questions to ask: "If the CPU sits at 85 Â°C while doing nothing, what are the first two things to check?" (Dust and fan rotation.)

G.4 Demonstration 3 â€” RAM test and swap isolation (5 minutes)

Say: "The hardest fault to catch is the one that only happens sometimes. Memory is famous for this."

Steps:

1. Show the RAM slots. Point out the **clips** at each end.
2. Put in one **suspect** stick. Say: "Let us say the customer complains of random restarts."
3. Start **Windows Memory Diagnostic** (or insert the MemTest86 USB and begin booting it). Explain that it runs outside Windows so it can test all of memory.
4. While it runs (or while describing the result), reveal the known-faulty stick result.
5. Perform the **isolation**: power off, unplug, strap on, remove the suspect stick, fit the known-good stick, power on.
6. **Say:** "The fault is gone. We have not guessed â€" we have **proven** the original stick was bad by swapping in a known-good one."

Expected results: the memory test reports errors on the faulty stick; the machine behaves normally with the known-good stick.

Reasoning to say aloud: "One change at a time. Known-good part. Record every step. That is how we avoid blaming the wrong component."

Common errors to show and correct:

- Forcing the stick in the wrong way (show the **notch** alignment).
- Forgetting to unplug and to wear the wrist strap.
- Concluding "RAM is bad" from a single blue screen without any test.

Questions to ask:

- "Why do we run the test outside Windows?" (So Windows is not using the memory we are testing.)
- "If there were two sticks and only one were bad, how would we find it?" (Test/isolation one at a time.)

G.5 Demonstration 4 â€" No-signal display isolation (3 minutes)

Say: "Black screen â€" do not panic, and do not buy a monitor yet."

Steps:

1. With the machine running and the monitor showing "No Signal", first check the monitor's **input menu**.
2. **Reseat** the display cable at both ends. Still no signal.
3. Swap in the **known-good spare cable**. Picture appears.
4. **Say:** "We found it: a bad cable. Cost to the customer: the price of a cable, not a monitor or a GPU. This is why we test from cheapest to most expensive."

Expected results: picture returns after the cable swap.

Questions to ask: "Why do we try the cheap cable before swapping the expensive GPU?" (Cost, and the cheapest cause is very often the real one.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding, each learner (or pair) runs the hardware diagnostic tools on a practice machine and records findings on a Diagnostic Report Form.

Safety first â€" read aloud before starting:

- **Switch off and unplug** the computer before opening the case or touching any internal part.
- Wear your **anti-static wrist strap** whenever you handle internal components.
- **Never open the PSU.**
- Keep drinks away from the bench.
- Handle RAM and drives **by the edges**; do not touch the gold contacts.
- If anything smells burnt or you see a swollen battery, **stop** and call the instructor.

Procedure (instructor calls out each step; learners follow; instructor walks the room):

1. On the **running** machine, open **CrystalDiskInfo**. Record: Health Status word and colour; drive temperature; and the **Reallocated Sectors Count** value.

2. Open **HWMonitor**. Record the **CPU temperature**, the **GPU temperature** (if listed) and the **fan speeds**. Note any fan showing zero.
3. **Pause and predict:** "Before we start a memory test, what would you expect to see if this stick is healthy?" (Zero errors, machine passes.) Write your prediction before running.
4. Start **Windows Memory Diagnostic** (or insert the MemTest86 USB and begin the boot). If time is short, observe the first pass and note the running errors count. Do **not** force the machine to keep testing if it is needed for the next group â€” coordinate with the instructor.
5. **Display isolation drill:** with the machine running, deliberately loosen the display cable at one end (or start from a "no signal" setup) and follow the order: check input â†’ reseat cable â†’ known-good cable â†’ known-good monitor. Record which step fixed it.
6. **Swap-test drill (instructor supplies parts):** power off, unplug, strap on. Remove the suspect RAM stick and fit the known-good stick. Note the difference in behaviour (or in the memory-test result).
7. **Overheating inspection:** with the machine off and unplugged, open the case and inspect the CPU heatsink, fans and vents for dust. Record what you see (light dust / heavy dust / clear).
8. Complete the **Diagnostic Report Form**, writing each finding in plain language a customer could understand.

Expected results:

- CrystalDiskInfo shows the drive health word and a temperature.
- HWMonitor shows at least one CPU temperature and fan speeds.
- The memory test reports a pass (or, on the faulty stick, an error count).
- The display isolation drill identifies the specific fault (input/cable/monitor/GPU).
- The swap test changes the machine's behaviour in the expected direction.
- A completed report form.

Points to pause on and predict:

- After step 1: "If the status is Caution but the count is zero, what could still be wrong?" (Temperature, or a different flagged attribute.)
- After step 5: "We changed two things â€” which one fixed it?" (Correct answer: we must change one at a time to *know*.)
- After step 6: "If the machine still fails with the known-good stick, what does that tell us?" (The RAM was probably not the fault â€” look elsewhere.)

Troubleshooting while walking the room:

- CrystalDiskInfo shows no drives â†’ the tool may need to be run as administrator; also confirm the drive cable is connected.
- HWMonitor shows very few values â†’ this machine has few sensors; record what is shown and say so.
- Windows Memory Diagnostic will not restart â†’ save work and make sure no other user is logged in.
- MemTest86 USB does not boot â†’ confirm the boot key and that "USB boot" is enabled; try another USB port.
- Learner forces RAM â†’ stop immediately, show the notch, re-seat gently.

Completion checklist (learner ticks each):

- ☐ CrystalDiskInfo opened; health status, temperature and reallocated count recorded
- ☐ HWMonitor opened; temperatures and fan speeds recorded
- ☐ A memory test started and its result noted
- ☐ Display isolation drill completed and the cause identified
- ☐ Swap test completed (one change at a time, known-good part)
- ☐ Overheating inspection done (dust, fans, vents noted)
- ☐ Diagnostic Report Form completed in plain language
- ☐ All safety steps followed (power off, unplugged, strap worn, PSU never opened)

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: Diagnose three **simulated hardware faults** without copying the instructor. For each one, use the correct tool, decide whether the component is healthy or faulty, and write your conclusion in plain language.

Scenario cards (each learner or pair receives these; the instructor can set up each as a bench station).

Card A â€" "The Beeping Cyber CafÃ© PC". A desktop will not display a picture. When switched on, it beeps three times, then pauses, then beeps three more times. All fans spin. The drive and PSU are known-good. There are two RAM sticks fitted.

Card B â€" "The Slow Office Computer". A desktop started fine but now boots to Windows very slowly and sometimes says "operating system not found". CrystalDiskInfo is available. The machine's drive makes a soft ticking sound occasionally.

Card C â€" "The Machine That Dies". A desktop runs for about 20 minutes, then suddenly switches off with no warning. It restarts when you press the power button, then dies again after some minutes. Fans spin during use. CrystalDiskInfo and HWMonitor are available.

Requirements:

1. For each card, state the **most likely faulty component**.
2. State **which tool or test** you used (or would use) to confirm it.
3. State the **next safe action** for a technician (e.g., back up, clean, replace).
4. Write each answer as a short paragraph in plain English.

Time allowance: 15 minutes.

Constraints:

- Do not open the PSU.
- Use a **known-good part** for any swap test.
- Record the finding on the Diagnostic Report Form.

Success criteria:

- Each card is matched to the correct component.
- The confirmation test chosen makes sense.
- The recommended action is safe and honest.

Hints available if stuck:

- Hint 1 (Card A): repeated beeps before any picture usually point to one basic component that the machine needs before it can show anything.
- Hint 2 (Card B): ticking plus "operating system not found" points to a mechanical component that stores data.
- Hint 3 (Card C): a machine that dies after a set time, then dies again, is the classic sign of a heat or power problem.

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

Card A â€" Most likely faulty component: RAM. Three repeating beeps during POST usually indicate a memory fault; all fans spinning means the PSU is probably fine and the machine is trying to start.

- Confirmation test: **MemTest86** (or Windows Memory Diagnostic), and **isolation** by removing one stick at a time, testing each alone.
- Next safe action: identify the bad stick, replace it with a matching known-good stick, re-test. Do not buy a new motherboard first.

Card B â€" Most likely faulty component: the storage drive (HDD) is failing. Slow boot, "operating system not found" and ticking all point to a dying mechanical drive.

- Confirmation test: **CrystalDiskInfo** â€" look for **Caution** and a **non-zero reallocated sector count**; note the ticking (mechanical failure).
- Next safe action: **back up the customer's data immediately** if the drive still reads; then replace the drive (an SSD upgrade is ideal â€" see Session 7) and reinstall/the-clone the system. Warn the customer that a ticking drive may fail fully at any moment.

Card C â€" Most likely faulty components: overheating (dust/dried paste) or a weakening PSU. Dying after a fixed time is the signature of heat buildup; random shutdowns can also be a PSU problem.

- Confirmation test: **HWMonitor** for temperatures under load; inspect **dust, fan rotation and thermal paste**; then a **PSU swap test** with a known-good PSU.
- Next safe action: clean the dust and check fans/paste first (free and safe); if it still dies, perform the PSU swap test and replace the PSU if confirmed. Never open the PSU.

Marking guidance (out of 12):

- Correct component identified for each card (3 Ã— 2 = 6)
- Sensible confirmation test chosen for each (3 Ã— 1 = 3)
- Safe, honest next action for each (3 Ã— 1 = 3)
- Plain-language writing and completed report form (bonus up to +1, cap at 12)

Accept any answer that shows correct testing logic, even if the wording differs. Penalise unsafe actions (such as opening the PSU or "replace everything") heavily.

J. REAL-WORLD CASE STUDY

Situation. *Emeka* owns a small business centre along Ikwerre Road, Port Harcourt. His main office computer holds his customer invoices, his accounting records and the software he uses for printing. For two weeks it has been slow and sometimes restarts on its own. He ignores it â€" "it is still working." One Friday evening, the machine refuses to boot: "Boot device not found." He calls a technician.

Relevant facts.

- The computer is seven years old and still has its original mechanical hard drive.
- It had begun to make a soft ticking noise some weeks earlier.
- Nobody had ever run a health check on the drive.
- No backup exists.

The decision point. The technician opens CrystalDiskInfo. The drive shows **Bad (red)**, with a high reallocated sector count and repeated read errors. The shop's invoice and accounting data were only on this one drive.

Possible approaches.

1. Keep trying to boot the drive repeatedly to copy files.
2. Power the failing drive down, back up what can still be read with care, then replace the drive.
3. Send the drive for data recovery immediately.

Recommended solution (for this stage). Approach 2, with honest limits. The technician explains that a drive in this state may copy some files but is at high risk of dying completely at any moment, and each attempt shortens its life. He backs up the most critical data first (invoices and accounts), then attempts the rest. He advises Emeka that if any important data cannot be copied, only a professional **data recovery** service should open the drive â€" not the shop technician.

Trade-offs.

- Approach 1 risks destroying the remaining data faster; repeatedly powering a dying drive is dangerous.
- Approach 2 preserves the best chance of saving data and is affordable; but some files may already be lost.
- Approach 3 is expensive and unnecessary if the important files are still readable â€" it should be the last resort.

Professional connection. Real repair shops earn trust by doing exactly this: reading S.M.A.R.T. health, being honest that a failing drive must be replaced, and telling the customer to **back up**. The course certificate says a graduate can "diagnose common hardware faults" â€" this is precisely that skill, plus the professional honesty that goes with it.

Transfer exercise (ask the class): "If a customer's drive shows Caution, and the customer says '*I will bring it back next month,*' what should the technician do or say?"

Suggested answer: explain clearly, in writing, that Caution means the drive is already showing wear, that it can fail before next month, and that the data should be backed up **now**. Record the advice on the service sheet so the customer cannot later say they were not warned.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide. When a learner hits a problem, ask them to check the relevant row first.

K.1 Black screen / "No Signal" but the machine seems to run

- **Symptoms:** fans spin, keyboard lights come on, monitor shows "No Signal".
- **Likely causes:** monitor on the wrong input; loose or damaged display cable; monitor connected to the motherboard instead of the GPU; a POST fault (RAM/GPU); or a dead monitor.
- **How to investigate:** check the monitor input menu; reseat the cable at both ends; try a known-good cable; try the other video output; check for beeps; try a known-good monitor.
- **Safe corrective steps:** work **from cheapest to most expensive** â€" input, cable, port, then monitor/GPU. Power off before changing cables.
- **How to confirm resolved:** a picture appears and stays stable.
- **Prevention:** keep a spare known-good cable and monitor on the bench.

K.2 The computer does not power on at all

- **Symptoms:** no lights, no fans, nothing.
- **Likely causes:** wall socket dead; power cable loose or faulty; PSU switch (on the back) off; PSU failed.
- **How to investigate:** test the socket with another appliance; check the power cable and the PSU switch; then try the swap test with a known-good PSU.
- **Safe corrective steps:** replace the power cable, use a working socket, swap the PSU. **Never open the PSU.**
- **How to confirm resolved:** the machine powers on and completes POST.
- **Prevention:** use a surge protector (Session 8); label good cables.

K.3 Random shutdowns and restarts

- **Symptoms:** works, then dies; sometimes restarts by itself.
- **Likely causes:** overheating (dust, dead fan, dried paste); weak PSU; loose power connectors; faulty RAM.
- **How to investigate:** check temperatures in HWMonitor; inspect dust and fans; reseat the 24-pin and CPU power connectors; run a memory test; do a PSU swap test.
- **Safe corrective steps:** clean cooling first (free); re-seat connectors; swap PSU. Power off and unplug for all internal checks.
- **How to confirm resolved:** the machine runs a heavy task (or a long memory test) without dying.
- **Prevention:** regular cleaning (Session 8), surge protection, correct PSU wattage.

K.4 Memory test shows errors

- **Symptoms:** MemTest86 or Windows Memory Diagnostic reports errors above zero.
- **Likely causes:** a faulty RAM stick; a dirty or faulty RAM slot; RAM not fully seated; incompatible/mismatched RAM.
- **How to investigate:** reseat the stick; clean the slot with compressed air; test one stick at a time; test each stick in each slot.
- **Safe corrective steps:** remove the faulty stick; if the slot itself is bad, use a different slot; replace the faulty stick.
- **How to confirm resolved:** re-run the test and get zero errors.
- **Prevention:** hold by the edges, never force a stick, power off first.

K.5 CrystalDiskInfo shows Caution or Bad

- **Symptoms:** yellow or red health status; non-zero reallocated sectors; read errors.
- **Likely causes:** the drive is wearing out or already failing.
- **How to investigate:** read the full attribute list; listen for clicks; note the temperature.
- **Safe corrective steps: back up the customer's data immediately**; stop writing new data; replace the drive. For clicks or unreadable important data, hand over to a data recovery specialist.
- **How to confirm resolved:** the new drive shows Good and the system runs normally.
- **Prevention:** regular health checks (Session 8) and a backup habit.

K.6 Distinguishing likely from confirmed

In every row above, a *likely* cause becomes a *confirmed* cause only after a test proves it: a swap test that changes the behaviour, a S.M.A.R.T. reading, a temperature reading, or a memory-test error count. **Teach learners never to write "confirmed" on a report based on a guess.** Saying "likely cause" honestly is professional; overclaiming is not.

L. SESSION ASSIGNMENT

Assignment title: "Hardware Diagnostic Report on a Real Machine"

Purpose: To practise the full hardware diagnostic process â€" inspecting, testing with tools, isolating with swap tests, and writing findings in plain language â€" on a real computer.

Realistic scenario: A customer brings you a desktop computer and says: *"It is slow, sometimes it restarts on its own, and last week the screen went black once."* You must produce a professional diagnostic report before any repair is done.

Detailed instructions:

1. Choose a real computer you have access to (a practice machine at CEA, a family computer, or a machine at your work). **Get permission before opening any machine that is not yours.**
2. **Observe and record:** note the make/model if visible, the symptoms, and anything the "customer" told you.
3. **Run the software checks** (safe, no case opening needed): CrystalDiskInfo (health, temperature, reallocated sectors) and HWMonitor (CPU/GPU temperatures, fan speeds).
4. **Run a memory test** (Windows Memory Diagnostic or MemTest86) and record the result.
5. **Physical inspection** (only if you may open the machine): with it **off and unplugged**, note dust levels, fan condition, and whether the case has good airflow. **Do not open the PSU.**
6. **Write your conclusion:** state the most likely faulty component (or that all tested parts look healthy), and your recommended next action.
7. Complete the **Diagnostic Report Form** and hand it in.

Required deliverables:

- One completed Diagnostic Report Form (the CEA template), including all readings.
- A one-paragraph plain-language summary for the "customer".

Submission format: Written report on the CEA template, submitted to the instructor (paper or the class folder).

Deadline: Before Session 6.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Observation & symptom record	3	Clear, specific symptoms and customer description
CrystalDiskInfo findings	3	Health status, temperature and reallocated count all recorded
HWMonitor findings	3	CPU (and GPU if available) temperatures and fan speeds recorded
Memory test result	3	Test named, run, and result clearly stated
Physical inspection (if permitted)	2	Dust, fans and airflow noted honestly
Conclusion & recommendation	4	Logical, safe, honest conclusion with a sensible next action
Plain-language summary	2	A customer could understand it; no unexplained jargon

Common mistakes to avoid: writing "confirmed" without a test; confusing temperature with health; recommending "replace everything"; opening the PSU; forgetting to record the actual numbers.

Instructor reference solution (abbreviated example):

> ***Machine:*** *CEA practice desktop (7th-gen CPU, 4 GB RAM, 500 GB HDD).* ***Symptoms:*** *slow boot; one random restart last week; one black screen.* ***CrystalDiskInfo:*** *Health* ***Caution (yellow)****; temperature 41 Â°C; Reallocated Sectors Count* ***18*** *(and rising).* ***HWMonitor:*** *CPU 52 Â°C idle; CPU fan 1,250 RPM; one case fan reads 0 RPM.* ***Memory test:*** *Windows Memory Diagnostic â€"* ***0 errors*** *(passes).* ***Physical inspection:*** *heavy dust between CPU heatsink fins; one case fan not turning.* ***Conclusion â€" likely cause:*** *the* ***hard drive is wearing out*** *(Caution, non-zero reallocated sectors) and the* ***cooling is compromised*** *(dust + one dead case fan).* ***Recommendation:*** *back up the customer's data* ***today****; replace the hard drive (an SSD will also make it much faster); clean the dust and replace/repair the stopped case fan; re-test*

temperatures after cleaning. ***Plain-language summary:*** *"Your computer's storage is showing early signs of failure, and dust is making it run hot. Please let us back up your files now, then we will replace the storage and clean the cooling."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer. What does **POST** stand for, and why does it use beeps instead of the screen? **Q2. Multiple choice.** A drive in CrystalDiskInfo shows **Caution (yellow)** with a non-zero reallocated sector count. What is the correct *first* action? a) Replace the drive immediately without copying anything. b) Back up the customer's data immediately. c) Ignore it â€" Caution is normal. d) Open the drive to look inside. **Q3. True or false, with a reason.** "A PSU can be safely opened to check the capacitors as long as it is unplugged." **Q4. Practical identification.** A desktop has two RAM sticks, shows random restarts and produces repeated beeps at start-up. Describe, step by step, how you would find the faulty stick. **Q5. Scenario.** A laptop shuts down after about 30 minutes every time. Give **two** likely causes and the test you would use for each. **Q6. Explain in your own words.** Why do we change **only one thing at a time** during swap testing?

INSTRUCTOR ANSWER KEY (Q1â€"Q6)

A1. POST = **Power-On Self-Test**. It uses beeps because it runs *before* the graphics/display is set up, so it cannot show a message on screen. If a basic part fails, a beep pattern (or a debug light) tells the technician where to look. (The beep pattern depends on the motherboard maker, so we count and look it up.)

A2. (b) Back up the customer's data immediately. Caution means the drive is already showing wear and could fail soon. Data first, then replace the drive.

A3. False. Even unplugged, the large capacitors inside a PSU can hold a **lethal charge** for a long time. A PSU is replace-only for a general technician.

A4. Step-by-step: power off and unplug; wear the anti-static strap. Remove one stick, leaving only stick A; run MemTest86 (or Windows Memory Diagnostic); note errors. If errors appear, stick A is bad. If not, power off and test stick B alone. Alternatively, swap in a **known-good** stick; if the faults disappear, the original stick was bad. Test one stick (one change) at a time and record each result. (Accept any answer that follows isolation + known-good swapping.)

A5. Likely causes: **overheating** (dust, failed fan, dried thermal paste) and a **weakening PSU**. Test for overheating with **HWMonitor** while the machine runs (watch the temperature climb before shutdown) and by inspecting dust/fans/paste. Test for the PSU with a **swap test** using a known-good PSU of suitable wattage. (Also accept "faulty RAM" if supported by a memory test.) Never open the PSU.

A6. Because if you change two things at once and the fault disappears, you cannot tell **which** change fixed it. One change at a time gives a **confirmed** answer, saves money, and produces an honest report.

Marking: award for correct concept even if wording differs. Deduct for unsafe answers (opening the PSU, "replace everything").

N. REFLECTION AND RECAP

Plain-English summary. Today you learned to prove hardware faults instead of guessing them. You saw how a computer checks itself at start-up (**POST**) and tells you via **beeps** when a basic part fails. You learned the symptoms and tests for faulty **RAM** (boot loops, blue screens â€" tested with Windows Memory Diagnostic or MemTest86), failing **storage** (slowness, clicks, "not found" â€" read with **CrystalDiskInfo** using **S.M.A.R.T.**), **PSU** faults (no power, random shutdowns â€" **swap test only**, never open it), **overheating** (shutdowns, loud fans â€" **HWMonitor**, dust, fans, paste), **display** faults (no signal â€" check input, cable, port, monitor), and **motherboard / laptop battery** issues (rule out everything else; swollen batteries are a fire risk). Above all, you learned **swap testing and isolation**: change one thing at a time, use a known-good part, and record every step.

Five important takeaways:

1. Test, do not guess â€" a diagnosis you cannot prove is not finished.
2. **Never open the PSU**; capacitors can hold a lethal charge.
3. **Caution** in CrystalDiskInfo means back up **now**, then replace.

4. Change **one thing at a time** with a **known-good** part.
5. Always **back up** before replacing storage, and always **record** your findings.

Questions to help you recognise what you can now do:

- Can I read a drive's health and decide the next action?
- Can I find a faulty RAM stick using isolation?
- Can I explain why we never open a PSU?
- Can I check temperatures and spot overheating?
- Can I find a "no signal" fault from cheapest to most expensive cause?

Safety self-check (tick honestly):

- ☐ I powered off and unplugged before opening any machine.
- ☐ I wore my anti-static wrist strap when handling parts.
- ☐ I never opened a PSU.
- ☐ I backed up before touching storage data.
- ☐ I stopped and called the instructor when unsure.

Connection to the next session. Today you proved faults in the **hardware**. But many "broken" computers have perfectly healthy parts â€" the problem is the **software**. Next session, Session 6 â€" Software Diagnostics, you will diagnose slow performance, viruses, boot problems and driver faults using Windows' own tools, and learn how to tell a software fault from a hardware one.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner checks **cheapest/simplest causes first** (socket, cable, monitor input) before opening the case.
- The learner reads the **actual number** in CrystalDiskInfo (reallocated count, temperature) rather than only the colour.
- The learner uses a **known-good** part for a swap test and changes **one thing at a time**.
- The learner states findings in plain language and distinguishes "likely" from "confirmed".
- The learner automatically powers off, unplugs, and wears the wrist strap.
- The learner refuses unsafe actions (opening the PSU, poking a swollen battery).

Common misconceptions to watch for and correct:

- "Black screen means the computer is dead." â†' Often just the display path or POST.
- "Caution is fine." â†' Caution means back up now.
- "A loud fan is a broken fan." â†' Usually it is fighting overheating; fix the dust/heat first.
- "A bigger PSU fixes shutdowns." â†' Wattage is capacity, not health.
- "I can open the PSU if it is unplugged." â†' Never. Lethal stored charge.

Suggested remedial exercises for learners who struggle:

1. **Tool-drill:** with a working machine, repeatedly open CrystalDiskInfo and HWMonitor and read aloud the health, temperature and fan values until fluent.
2. **Swap-test tracing:** give a written scenario where two parts were changed at once and ask "what can we conclude?" (Answer: nothing certain â€" repeat one at a time.)
3. **Symptom sorting:** hand out ten symptom cards and ask learners to sort them into "RAM / storage / PSU / heat / display" buckets.

Differentiation:

- **Level 1 (supported):** provide a step-by-step tool card and a partially completed report form; pair the learner with a stronger partner for the swap test.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** introduce a **mixed** fault (e.g., a failing drive *and* overheating at once) and ask the learner to diagnose both and prioritise which to fix first, justifying the order.

Timing guidance: This is a practical-heavy session. If the room is slow, protect the guided practical (Section H) and trim the case study to a two-minute mention. Memory tests can be time-consuming â€" start the test early, let it run while other checks happen, and do not let one test consume the whole session. If a real memory test cannot finish in class, record the running result and note honestly that a full pass is required.

APPENDIX 5A â€" 10-MINUTE REFRESHER (only if needed)

If learners cannot name the six major components or the workflow, run this before Section D.

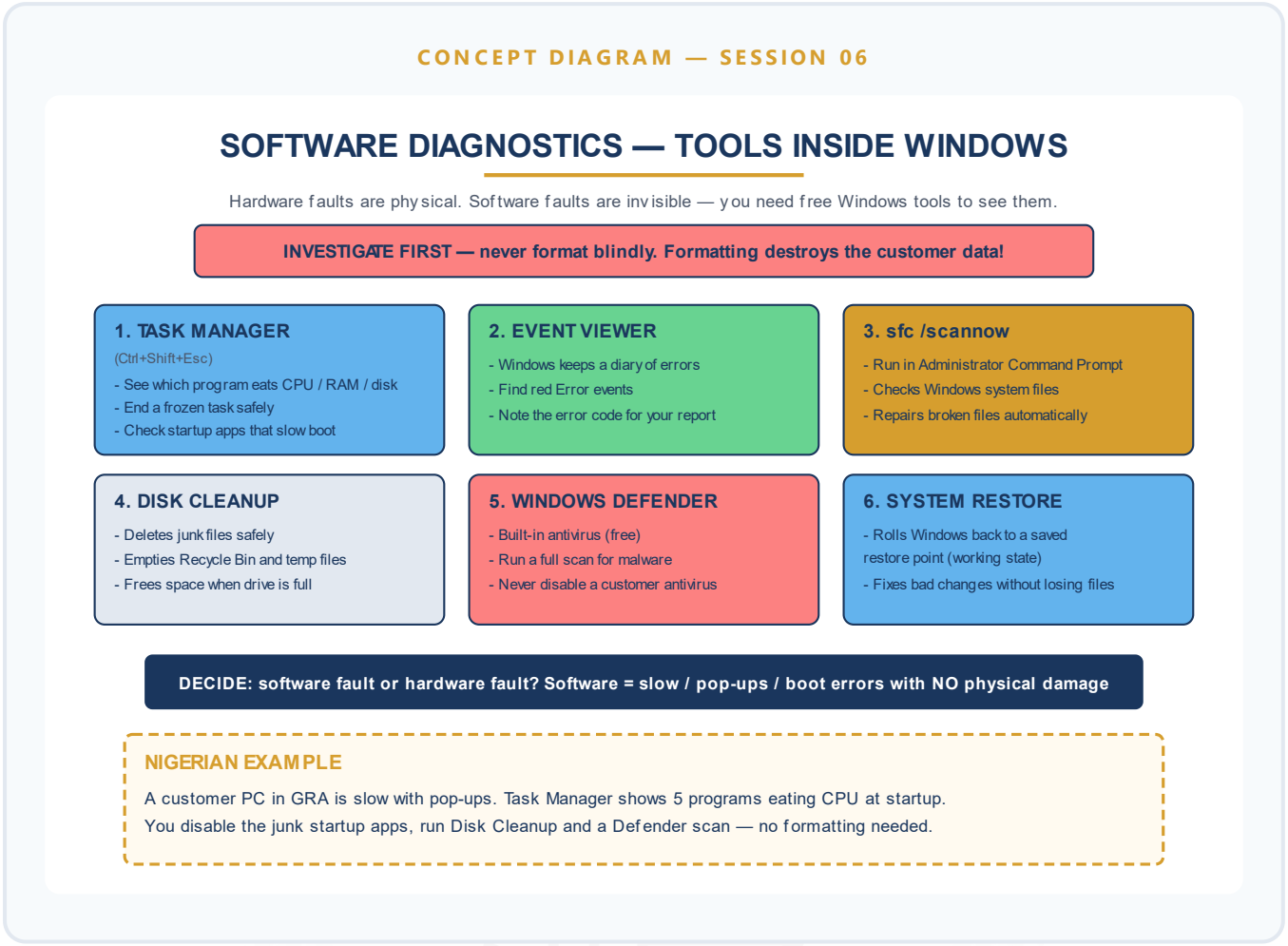
1. **The six major components** â€" point to each on a real machine or chart and have the class repeat: **Motherboard, CPU, RAM, Storage, PSU, GPU.**
2. **The five-step workflow** â€" write on the board and make the class chant: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
3. **Safety** â€" rehearsal: "Before I touch anything inside, I do two things: I _____ (power off and unplug) and I _____ (wear my anti-static wrist strap). I never open the _____ (PSU)." Confirm each learner can answer before continuing.

END OF SESSION 5

Next: Session 6 â€" Software Diagnostics.

SESSION 06 OF 12

Software Diagnostics



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 6 â€" SOFTWARE DIAGNOSTICS

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	6 of 12
Session title	Software Diagnostics
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes
Difficulty level	Beginner

Item	Detail
Prerequisite knowledge	Sessions 1â€"5: components, safety and tools, disassembly/reassembly, the five-step diagnostic workflow, and hardware diagnostics (RAM, storage, PSU, heat, display). If a learner cannot explain the difference between a hardware and a software fault, do the 5-minute recap in Appendix 6A before this session.
Connection to previous work	Session 5 proved faults in the **hardware** using physical tools. This session diagnoses the **software** â€" slow computers, viruses, boot problems and driver faults â€" using Windows' own built-in tools. Together the two sessions let you decide whether a fault is hardware or software.
Preparation before class	Practice computers with Windows 10/11 ready; at least one machine pre-loaded with a **safe simulated software fault** (e.g., several unnecessary startup programs enabled, a full Recycle Bin, a large unnecessary file, or a deliberately misconfigured/disabled device driver â€" do **not** use real malware); Windows Defender active and updated; Command Prompt access tested; Event Viewer accessible; a restore point created before the simulation so it can be undone. Keep a **second, clean** machine untouched for comparison.

Important safety and ethics note. In this session we *simulate* software faults safely for teaching. We do **not** intentionally download viruses, and we never disable a customer's antivirus. Where the course talks about malware, it teaches **recognition and safe removal**, not creation. All work is on academy machines with permission.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** the difference between a software fault and a hardware fault, and give an example of each.
2. **List** the common software faults â€" slow performance, malware infection, boot failure, driver issues, full disk â€" and match each to its usual symptoms.
3. **Use Task Manager** to identify programs using too much CPU, memory or disk, and safely end a misbehaving task.
4. **Use Event Viewer** to find error events and note their codes.
5. **Run** `sfc /scannow` in an Administrator Command Prompt to check and repair system files.
6. **Run Disk Cleanup** to free storage space safely.
7. **Describe** the safe approach to malware removal, including when to use a full scan and when a second-opinion scanner is appropriate.
8. **Use System Restore / recovery options** to return a machine to a previous working state when appropriate.
9. **Decide** whether a fault is more likely software or hardware, and explain the reasoning.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one practice desktop or laptop running Windows 10 or Windows 11 for each pair of learners.
- One machine pre-loaded with the **simulated** software faults described in Section A.
- One clean machine for comparison.
- A powered network is **not** required, but if available it helps update Defender's definitions.

Software (all built into Windows unless noted)

- **Task Manager** (Ctrl+Shift+Esc).
- **Event Viewer**.
- **Command Prompt** (run as Administrator) for `sfc /scannow` and `chkdsk`.
- **Disk Cleanup**.
- **System Restore / Recovery** options.
- **Device Manager** (for driver issues).

- **Windows Defender** (built-in antivirus) or the academy's approved antivirus.
- Optional second-opinion scanner (e.g., Malwarebytes â€" instructor only, pre-installed and updated, used for demonstration.
- `msconfig` / **Startup Apps** settings.

Printed materials

- Software Diagnostic Report Form (one per learner).
- Windows Tools quick-reference card (one per learner).

Safety equipment

- None special for software, but keep drinks away from machines and never disable antivirus.

Internet

- Optional. Needed only if the instructor chooses to show a driver download from a manufacturer's site at the end. Do not depend on it.

Alternative for learners who lack a facility

- Any Windows computer works for Task Manager, Disk Cleanup and Event Viewer; `sfc` needs Administrator rights. Learners without a personal machine share a bench machine and swap roles.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: On the projector, open the pre-prepared **simulated faulty** machine. Have several browser-like pop-up windows already sitting on the desktop (harmless images or text files the instructor arranged), and open Task Manager to show a high CPU figure. Keep it controlled and safe.

What the instructor says:

> *"Good morning. Look at this computer. The screen is crowded with pop-ups. It is crawling â€" everything takes forever. Programs close on their own. The owner is panicking.*
>
> *Last week we learned to test hardware with tools. But tell me something: if I open this case right now, will I find a broken part? No. The RAM is fine, the drive is fine, the power supply is fine. And yet the computer is very sick.*
>
> *This computer is sick in its **software**. Software problems are just as common as hardware problems â€" and do you know what makes them harder? You cannot see them. There is no clicking noise, no beep, no hot part you can feel. Today you learn to see software problems using Windows' own tools â€" the tools that are already inside every computer, free."*

Expected learner response: Learners may shout "virus!", "it needs formatting", or "delete everything". Accept all answers. Write them on the board. Say:

> *"Maybe. But a good technician does not format first and ask questions later â€" that destroys the customer's data. We investigate first. That is what today is about."*

Why this opener works: It creates a clear contrast with Session 5 (hardware is physical and testable; software is invisible), it captures a very common real-world situation for Nigerian computer users, and it immediately challenges the destructive "just format it" habit by promising a *safer, investigative* approach.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1: "Name the five steps of the diagnostic workflow."

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
- Common gap: forgetting to **document**. Remind learners: not documented = not done.

Question 2: "What hardware fault causes a computer to run slowly and make a soft clicking sound?"

- Expected answer: a **failing hard drive**. Note that this is a **hardware** cause of a symptom people usually blame on software.

Question 3: "What does it mean if CrystalDiskInfo shows 'Caution'?"

- Expected answer: the drive is showing wear; **back up now** and plan to replace it.

Question 4: "Why do we never open the PSU?"

- Expected answer: **lethal stored charge** in the capacitors, even when unplugged.

Question 5: "A computer beeps repeatedly and shows no picture. Is that more likely hardware or software â€" and why?"

- Expected answer: **hardware** â€" software has not even loaded yet when POST runs. This bridges neatly into today.

Bridge to today (say this):

> *"Session 5 was about a sick **body**. Today is about a sick **mind**. A computer can have perfect parts and still behave badly â€" because the instructions it is following are wrong, infected or broken. Your job today is to use the tools that let you look inside the mind."*

Gap handling: If learners cannot clearly separate a hardware from a software example (Question 5), spend up to three extra minutes with the Appendix 6A recap, because this distinction is the backbone of today's session.

F. FULL LECTURE NOTES

F.1 What a software fault is â€" and why it is different

The idea. A computer is hardware **plus** the instructions that run on it. Those instructions and data are the **software**. When the instructions or data go wrong, the machine misbehaves even though every physical part is fine.

Plain-English definition. A **software fault** is anything wrong with the programs, the operating system, the drivers, or the files â€" as opposed to the physical parts. Examples: a virus, a corrupted Windows file, an out-of-date or wrong driver, a full disk, or too many programs starting automatically.

Why the distinction exists. Hardware faults are fixed by **repairing or replacing a part**. Software faults are fixed by **cleaning, correcting, updating, restoring or reinstalling the software**. The two need completely different tools and skills, so step one of any diagnosis is to decide which type you are facing.

Analogy. Think of a football team. The **players** are the hardware. The **coach's instructions** are the software. If a player breaks a leg (hardware), no new instructions will help â€" you need a new player or rest. If all the players are strong but the coach gave the wrong formation (software), replacing players will not help â€" you must fix the instructions. Same team, same "hardware"; the problem is elsewhere.

Realistic Port Harcourt scenario. A student in Elelenwo buys a "free" cracked version of a design program from a friend's USB flash drive. Two weeks later her laptop is slow, opens strange pages in her browser, and shows adverts on the desktop. The parts are all fine; the machine is infected. A technician who understands software faults knows to scan, remove, and teach the student about trusted sources â€" not to replace her hard drive.

Important misconception. "Software problems are the easy ones â€" just restart." Sometimes a restart clears a temporary glitch, but many software faults (viruses, corruption, failing drivers) persist and get worse. Do not rely on restarts as a fix.

Key limitation. Software faults often *look like* hardware faults. A corrupted system file can cause a blue screen; a failing drive can corrupt system files. So a good technician still runs the **hardware** checks (Session 5) when symptoms are severe, to make sure a hidden hardware problem is not the root cause.

F.2 Slow performance â€" the most common complaint

The idea. "My computer is slow" is the number one thing customers say. It has several possible causes, and the skill is to find *which* one.

Plain-English definition. Slow performance means the computer takes much longer than expected to start, open programs, or respond.

The main causes of a slow computer (each with its own symptom pattern):

Cause	Type	Typical sign
Too many **startup programs**	Software	Slow only at start-up; settles after a few minutes
Full or nearly full disk	Software/storage	Slow saves; "low disk space" warnings; slow everywhere
Malware	Software	Slow *plus* pop-ups, redirects, strange behaviour
Failing drive	Hardware	Slow *plus* clicking/errors; check S.M.A.R.T. (Session 5)
Too little RAM	Hardware	Slow with many programs open; disk light constantly busy
Overheating	Hardware	Slow *plus* loud fans, hot case, random shutdowns
Fragmented HDD / old hardware	Both	Generally slow but healthy; an SSD upgrade helps

How we investigate:

- **Task Manager** (Ctrl+Shift+Esc) â†' **Processes** tab shows what is using CPU, memory and disk *right now*.
- **Startup Apps** tab (or `msconfig`) shows what runs automatically at start-up.
- **Disk space** check (This PC) shows how full the drive is.
- **CrystalDiskInfo** (Session 5) rules out a failing drive.
- **HWMonitor** (Session 5) rules out overheating.

Analogy. A slow computer is like a bus in Port Harcourt traffic. Sometimes the bus is fine but too many *passengers* (programs) are crammed in. Sometimes the *engine* (drive) is old. Sometimes the *road* (disk space) is blocked. To fix the delay you must know which it is.

Important misconception. "Slow means I need a new computer." Often a cleanup, more RAM, or moving Windows onto an SSD makes an old machine feel new again â€" far cheaper than replacing it (Session 7).

Limitation. Task Manager shows the *present moment*. A program that spikes the CPU only sometimes may look innocent when you glance. Watch it for a while, and use Event Viewer for the history.

F.3 Malware â€" viruses, spyware and ransomware

The idea. Malware is any software written to harm you or to take your money or data. A "virus" is only one type.

Plain-English definitions (keep these simple for beginners):

- **Virus:** malicious code that spreads by attaching itself to files and programs.
- **Spyware:** software that secretly watches what you do (passwords, banking, browsing).
- **Ransomware:** software that locks your files and demands money (a "ransom") to unlock them.
- **Adware / unwanted pop-ups:** programs that flood you with adverts, often bundled with "free" software.
- **Trojan:** a harmful program disguised as something useful (e.g., a "free game" that is really spyware).

Why it exists and how computers get infected:

- Downloaded from untrusted websites, especially "cracked" or "free" versions of paid software.
- From an **infected USB flash drive** shared between computers.
- From email attachments or links that look real (called **phishing**).
- From clicking fake pop-ups that pretend to be "your antivirus found 5 viruses â€" click here".

Common symptoms of infection:

- Pop-ups and adverts appearing even when the browser is closed.
- The browser sending you to strange websites (redirects).
- The computer suddenly slow, with high CPU/disk even at idle.
- Strange programs opening, or unfamiliar icons/programs appearing.
- Files, folders or settings changing on their own.
- Antivirus switching itself off and refusing to turn back on.
- For ransomware: files suddenly renamed or encrypted and a "pay us" note.

The safe removal approach (step by step):

1. **Do not panic and do not format yet.** First preserve the customer's data.
2. **Disconnect from the internet** (unplug the network cable / turn off Wi-Fi) to stop the malware communicating or spreading.
3. **Run a full scan** with the updated antivirus (Windows Defender or the academy's approved tool). A **Full Scan** checks the whole system; a Quick Scan checks only common locations.
4. **Review the results**, then **quarantine** or **remove** detected threats. Quarantine moves the threat to a safe place rather than deleting it immediately.
5. **Restart** if prompted, and **scan again** to confirm nothing remains.
6. **Second-opinion scan (optional):** if symptoms continue but the first scan found nothing, run a trusted second scanner (e.g., a reputable free malware-removal tool). Different tools catch different things.
7. **Change important passwords** afterwards if spyware was suspected (banking, email), because they may have been captured.
8. **Educate the customer:** explain how the infection happened (usually an untrusted download) and how to avoid it next time.

Analogy. Malware is like rats in a store-room. Poisoning (scanning) is not enough if you leave the door (the unsafe download habit) open. You must remove the rats *and* close the door.

Important misconception. "One antivirus is enough, forever." Antivirus **definitions** (the list of known threats) must be kept **updated**. An out-of-date antivirus is like a watchman with an old list of faces â€" new intruders walk straight past.

Limitation. No antivirus catches every threat, especially brand-new ones. Also, some "vaccines" people download are themselves malware ("fake antivirus"). Only install antivirus from the built-in Windows Defender or a well-known, trusted brand. Never pay a random pop-up that claims to clean your PC.

F.4 Boot problems â€" when Windows will not start

The idea. Sometimes the computer powers on (so hardware and POST are fine) but Windows does not finish loading. That is a **boot problem**.

Plain-English definition. A **boot problem** is a fault in starting Windows. The machine passes POST, but Windows does not reach the desktop â€" either it hangs, shows an error, restarts, or offers a repair screen.

Common symptoms:

- A black screen with a spinning dots animation that never finishes.
- An error such as "Boot device not found" (often **hardware** â€" check S.M.A.R.T. first).
- "Your PC ran into a problem and needs to restart" (a **blue screen**).
- An automatic "Automatic Repair" / "Recovery" screen.
- A **boot loop**: Windows starts to load, then restarts, over and over.

Why it happens: a corrupted system file, a bad update, a wrong or corrupted driver, a power cut during an update, or an underlying failing drive.

Recovery options (what each one does):

- **Start-up Repair:** Windows tries to repair common start-up problems automatically.
- **Safe Mode:** Windows starts with only the basic drivers and services. It is how you get in to fix a problem when normal mode fails. *(Safe Mode is defined properly in F.5.)*
- **System Restore:** returns system files and settings to an **earlier point** without touching the customer's personal files (see F.7).
- **Reset / reinstall:** the last resort. **Always back up first.**

Analogy. A boot problem is like a car that starts (engine turns) but the gear will not engage. The engine is fine; the path from engine to wheels is broken. You do not replace the engine â€" you find and fix the broken link.

Important misconception. "If Windows will not start, the computer is finished â€" format it." Not true, and formatting too quickly loses data. "Boot device not found" is often a hardware problem (or a loose cable), and a blue screen is often a driver or file problem. Investigate before you wipe.

Limitation. Some boot faults are actually hardware â€" a failing drive or bad RAM. That is why every boot problem starts with the Session 5 checks.

F.5 Safe Mode â€” the rescue room

The idea. When normal Windows will not start, **Safe Mode** gives you a stripped-down Windows so you can still get in and fix the problem.

Plain-English definition. Safe Mode starts Windows with only the **essential** drivers and services. Extra programs and fancy drivers are left out, so the system is more likely to start even when something is wrong.

Why it exists. If a bad driver or a program that starts automatically is crashing Windows, Safe Mode does not load it, so the system comes up and you can remove or disable the culprit.

Analogy. Safe Mode is like a hospital's **emergency room**. Everything non-essential is switched off so the doctors can focus on the problem. Normal mode is the whole hospital running at once â€” noisier, busier, harder to work in when something is broken.

When to use it:

- To remove stubborn malware that will not delete in normal mode.
- To uninstall a program or roll back a driver that broke Windows.
- To create a System Restore point or run recovery tools.

How to reach it (Windows 10/11): If Windows will not start, it often offers a **Recovery environment** after failing to boot two or three times. From the recovery screen choose **Troubleshoot â†' Advanced options â†' Startup Settings â†' Restart**, then select **Enable Safe Mode**. (Exact wording is stable but the paths depend on the Windows version; the instructor demonstrates on the practice machine.)

Important misconception. "I can work normally in Safe Mode forever." No â€" Safe Mode is temporary and limited. It is a tool for getting in and fixing the cause, not a place to stay.

Limitation. Some problems (a failing drive, bad RAM) are not fixed by Safe Mode. Safe Mode proves the fault is *software* â€" a useful clue in itself.

F.6 Driver issues â€" the translator problem

The idea. A **driver** is a small piece of software that tells Windows how to talk to a specific piece of hardware.

Plain-English definition. A **driver** is the translator between Windows and a hardware device (printer, graphics card, Wi-Fi adapter, sound chip). Each device needs the right driver to work properly.

Why drivers exist. Windows cannot contain instructions for every device ever made. Instead, each device's maker writes a driver, and Windows loads it. New hardware needs a driver; that is why after a fresh Windows install "some things may not work until drivers are installed."

Common symptoms of a driver problem:

- A device does not work at all (no sound, no Wi-Fi, printer not found).
- A yellow triangle with an exclamation mark next to a device in **Device Manager**.
- Poor performance (especially graphics: the GPU works but slowly, or with wrong resolution).
- Crashes or blue screens after installing or updating a driver.

Tool: Device Manager. Open it by pressing the Windows key and typing **Device Manager**. It lists every device. A problem device shows a **yellow warning triangle**.

- **To update a driver:** right-click the device â†' **Update driver** â†' search automatically, *or* download the correct driver from the **manufacturer's official website** and install it. (Manufacturer's website is preferred for graphics, sound and Wi-Fi.)
- **To roll back a driver:** if a problem started after an update, right-click the device â†' **Properties** â†' **Driver** tab â†' **Roll Back Driver**. This returns to the previous version.
- **To identify an unknown device:** right-click â†' Properties â†' Details â†' note the hardware ID and look it up (or search it online) to find the correct driver.

Analogy. A driver is like a person who speaks both languages. Windows speaks "Windows"; the printer speaks "printer". Without a translator, neither understands the other, and nothing prints. Updating the driver is getting a better translator.

Important misconception. "Always get drivers from any site that offers a free 'driver updater'." Avoid random driver-updater websites and tools; they often bundle unwanted software. Get drivers from **Windows Update** or the **device manufacturer's**

official site.

Limitation. Not every old device has a driver that works on the latest Windows; sometimes the honest answer is "this device is too old for this version of Windows." Also, the *newest* driver is not always the *best* â€" if the machine was stable before an update and broke after, rolling back is the right move.

F.7 The Windows diagnostic tools

These tools are already inside Windows. Learn what each does before you use it.

1. Task Manager (Ctrl+Shift+Esc).

- Shows **Processes**: every running program and how much **CPU**, **memory**, **disk** and **network** it is using.
- The **Startup Apps** tab shows what launches automatically at start-up (see F.8).
- Use it to spot the culprit: a program using 80â€"100% CPU or disk while you do nothing is suspicious.
- To stop a frozen or suspect program: select it â†' **End task**. *Safety:* end tasks only for programs you recognise as misbehaving; ending a Windows system process can unbalance the machine.

2. Event Viewer.

- A **log** of everything Windows has recorded: information, **warnings** and **errors**.
- Look under **Windows Logs â†' System** and **Windows Logs â†' Application**.
- **Errors** show a red circle with an "X". Note the **error code** and the **source** and search the exact code online for its meaning.
- Use it to find *what happened* when a crash occurred, even after the machine restarted.

3. System File Checker (`sfc /scannow`).

- A built-in tool that scans Windows' own system files and repairs any that are corrupted, using a cached good copy.
- Run it in a **Command Prompt opened as Administrator**: press Windows key â†' type **cmd** â†' right-click **Command Prompt** â†' **Run as administrator** â†' type `sfc /scannow` â†' Enter.
- It takes roughly 10â€"20 minutes. When finished it reports one of: no problems found; problems found and repaired; or problems found that could not be repaired (then you may need to check `DISM` , an advanced repair â€" instructor demonstration only).
- *Safety:* never run it on a failing drive without backing up first â€" repairs write to the disk.

4. Chkdsk (`chkdsk`).

- Checks a **disk** for file-system errors and bad sectors. `chkdsk C: /f` fixes errors on drive C.
- Advanced, writes to the disk; back up first and use under supervision.

5. Disk Cleanup.

- Press Windows key â†' type **Disk Cleanup** â†' choose the drive â†' tick categories (Temporary files, Recycle Bin, Windows Update leftovers, etc.) â†' **OK** to delete.
- Frees storage space safely. Larger cleanup ("Clean up system files") can also remove old Windows update files.

6. System Restore.

- Press Windows key â†' type **Create a restore point** â†' in the window, click **System Restore**.
- Choose a **restore point** from *before* the problem began â†' follow the prompts.
- Restores system files, drivers and settings to that earlier time. It **does not delete the customer's personal documents** â€" but warn the customer that programs installed after the restore point may need reinstalling.

7. Startup Apps / `msconfig` .

- **Startup Apps** (in Task Manager) lists programs that start automatically. Disabling unnecessary ones speeds up boot time.
- `msconfig` (System Configuration) is an older, more powerful tool; use it under supervision. Disabling critical services can break Windows.

Important misconception. "`sfc /scannow` fixes everything." It repairs *system files* only. It does not remove viruses, it does not fix a failing drive, and it does not replace missing drivers. It is one tool among many.

Limitation. These tools work only if Windows can start at least in Safe Mode. If Windows cannot start at all, you use the recovery environment (F.4) first.

F.8 Startup programs and disk space

Startup programs. Many programs (chat apps, updaters, "helper" tools) add themselves to the start-up list. Each one slows boot and uses memory. Review the **Startup Apps** tab in Task Manager and **disable** anything the user does not need at start-up. To disable: select the app â†' **Disable**. *Safety:* do not disable items you do not recognise if they appear related to security, audio, or trackpad â€" ask first.

Disk space. Windows needs **free space** to work. When a drive is nearly full, everything slows. Check free space in **This PC**. Free it with **Disk Cleanup**, emptying the **Recycle Bin**, uninstalling unused programs, and moving personal files to an external drive or cloud. A common real-world cause of a "slow" office PC is a drive that is 98% full.

Analogy. Startup programs are like guests who arrive at your shop every morning and occupy chairs before you open. A few are helpful; dozens block the shop. Disk space is the shop floor: if it is packed to the ceiling with boxes, you cannot work.

Important misconception. "Deleting files always frees space permanently." Files in the Recycle Bin still occupy space until the bin is emptied.

F.9 Software fault or hardware fault? â€" deciding

The idea. The most valuable skill in Week 2 is choosing the correct category before you act.

Quick decision guide:

Symptom	Lean towards	First check
Beeps, no picture, no power	Hardware	POST, PSU, RAM (Session 5)
Slow + clicking + "not found"	Hardware	CrystalDiskInfo S.M.A.R.T.
Slow + hot + loud fans	Hardware	HWMonitor; dust/fans
Pop-ups, redirects, strange behaviour	Software	Antivirus full scan
Blue screen after installing a program/driver	Software	Safe Mode; roll back driver
Device not working (sound/Wi-Fi/printer)	Software	Device Manager
"Low disk space", slow saves	Software	Disk Cleanup
Random restarts even in Safe Mode	Hardware	RAM, PSU, heat

Why this matters. If you format a machine whose real problem is a failing drive, the fault returns and you have destroyed data. If you replace a drive whose real problem is a virus, you have wasted the customer's money. Deciding correctly saves both.

Analogy. A doctor first decides whether the problem is a broken bone or an infection, because the treatments are completely different. Decide "hardware or software" first, every time.

Important misconception. "If it started after I installed something, it must be software." Usually true â€" but installing a heavy program can also reveal a *hardware* weakness (e.g., more RAM pressure causing crashes). Keep testing.

Limitation. Mixed faults exist. A failing drive can cause software symptoms; a virus can cause hardware-like crashes. When in doubt, test the hardware foundation first.

F.10 Summary of the session

- A **software fault** is a problem with instructions or data, fixed by cleaning/correcting/restoring â€" not by replacing parts.
- **Slow performance** has many causes: startup programs, full disk, malware, or failing hardware. Investigate with **Task Manager**.
- **Malware** symptoms: pop-ups, redirects, slowness, strange programs. Safe approach: disconnect, **full scan**, quarantine, restart, re-scan, update passwords, educate.
- **Boot problems**: use recovery options, **Safe Mode**, and **System Restore**. "Boot device not found" is often hardware.
- **Drivers**: use **Device Manager** to update (from the maker) or **roll back** a bad update.
- Windows tools: **Task Manager**, **Event Viewer**, `sfc /scannow`, **chkdsk**, **Disk Cleanup**, **System Restore**, **Startup Apps**.
- Always decide **software vs hardware first**, and always **back up** before destructive steps.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The pre-prepared **simulated faulty** machine (extra startup programs enabled, a full Recycle Bin, a large unnecessary file, and one device driver deliberately disabled or mis-set â€" no real malware).
- The clean machine for comparison.
- A restore point created earlier so the simulation can be undone.
- Command Prompt (Administrator) tested; Event Viewer accessible.
- The Software Diagnostic Report Form on the projector.

G.2 Demonstration 1 â€" Task Manager and startup programs (5 minutes)

Say: "Watch me find out *what* is making this computer slow â€" without guessing."

Steps:

1. Press **Ctrl+Shift+Esc** to open **Task Manager**.
2. On the **Processes** tab, click the **CPU** column header to sort by CPU use. Point at the top process.
3. **Say:** "This is the culprit *right now* â€" it is using a huge share of the CPU. On a healthy idle machine, the top numbers should be small."
4. Go to the **Startup Apps** tab. Point at the long list.
5. **Say:** "Every one of these starts automatically every time the computer boots. The customer does not need most of them. Watch: I disable the ones marked 'High' impact, but I leave security and audio alone."
6. Disable two or three clearly unnecessary startup apps.
7. **Say:** "Notice I did not delete anything. Disabling start-up does not uninstall the program; it just stops it launching by itself. That is a safe fix."

Expected results: the startup list shortens; the class can see the "Impact" column (High/Medium/Low) that Windows provides.

Reasoning to say aloud: "We are finding the *cause*, not just making the number look smaller."

Common errors to show and correct:

- Ending a critical system process and destabilising the machine. Show the warning that a program might lose data.
- Disabling a start-up item that the user actually needs.

Questions to ask: "If the CPU is high but settles down after five minutes, is that a fault?" (Not necessarily â€" start-up work is normal; it becomes a fault if it never settles or stays extreme.)

G.3 Demonstration 2 â€" Event Viewer and `sfc /scannow` (5 minutes)

Say: "Now let us find errors that already happened."

Steps:

1. Open **Event Viewer**. Expand **Windows Logs â†' System**.
2. **Say:** "Red = Error, Yellow = Warning. I look at the errors around the time of the crash."
3. Open one error, point to the **Source** and the **Event ID** (error code). **Say:** "I copy this code and search it exactly, so I learn what it means."
4. Open **Command Prompt as Administrator** (show right-click â†' Run as administrator). Type `sfc /scannow` and press Enter.
5. **Say:** "This scans every Windows system file and repairs any that are corrupted. It takes about 10 to 20 minutes. While it runs, let us do another task."
6. (Optional) Open **Disk Cleanup** and show the categories and the "Total amount of disk space you gain" figure.

Expected results: Event Viewer shows red error entries with Event IDs; `sfc /scannow` begins scanning and reports a result; Disk Cleanup shows reclaimable space.

Reasoning to say aloud: "Event Viewer is the *history*; Task Manager is the *present*. sfc repairs *system files*. Each tool answers one question."

Common errors to show and correct:

- Running `sfc` in a normal (non-Admin) Command Prompt â†' it may fail or do nothing. Show the difference.
- Panicking at *any* red error â€" some errors are harmless and old. Read the time and source.

Questions to ask: "Does `sfc /scannow` remove viruses?" (No â€" that is the antivirus's job.)

G.4 Demonstration 3 â€" Simulated malware check and Second opinion (3 minutes)

Say: "Here is how we handle a suspected infection safely."

Steps:

1. On the simulated machine, open **Windows Defender** (or the academy antivirus) â†' **Virus & threat protection**.
2. Choose **Scan options â†' Full scan** â†' **Scan now**. Explain that this checks the whole system, not just common spots.
3. While it runs, **say:** "If it finds something, we **quarantine** it â€" that moves it to a safe locker. Then we restart and scan again. If symptoms continue but the scan is clean, we use a **second-opinion** scanner, because no single tool catches everything."
4. Point out the importance of **keeping definitions updated** (there is usually an "up to date" indicator).

Expected results: Defender runs a scan; the class sees the quarantine/review screens.

Common errors to show and correct:

- Clicking fake "you have 5 viruses" web pop-ups. **Say:** "Real antivirus does not pop up in your browser demanding a click. That is a trick."
- Disabling the antivirus "just for a minute" to download something. **Say:** "Never â€" that is exactly when you get infected."

Questions to ask: "Why disconnect from the internet first?" (To stop the malware communicating or spreading.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding, learners use Windows' built-in tools on a practice machine and record findings on the Software Diagnostic Report Form.

Safety and ethics first â€" read aloud before starting:

- Only work on the **practice machines**; do not experiment on a stranger's or school's live computer.
- **Never disable the antivirus.**
- **Back up** or confirm a backup before running any tool that writes to disk (chkdsk, restore).
- **Do not click** unknown pop-ups or links.
- If in doubt about a step, ask the instructor before doing it.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Task Manager.** Open **Task Manager** (Ctrl+Shift+Esc). On the **Processes** tab, sort by **CPU**, then by **Memory**, then by **Disk**. Record the top process in each column and its percentage.
2. **Startup Apps.** Go to the **Startup Apps** tab. Record how many apps are enabled and which are marked **High** impact. Disable **one** clearly unnecessary app. Record what you disabled.
3. **Disk space.** Open **This PC** and record the free space on the C: drive and whether Windows warns of low space.
4. **Pause and predict.** "If I disable five startup programs, will the computer get more RAM or just boot faster?" Write your prediction. (Answer: it frees up some memory and CPU and speeds boot; it does not add physical RAM.)
5. **Disk Cleanup.** Open **Disk Cleanup** for C:. Record the "Total amount of disk space you gain". Do **not** click OK until the instructor confirms; then clean the **temporary files and Recycle Bin only**.
6. **Event Viewer.** Open **Event Viewer â†' Windows Logs â†' System**. Find **three** red Error entries. Record the **Event ID** and **Source** of each.
7. **System File Checker.** Open **Command Prompt as Administrator**. Run `sfc /scannow`. Record the final message (no problems / problems repaired / could not repair). This may take some minutes â€" start it and continue other checks.
8. **Driver check.** Open **Device Manager**. Look for any device with a **yellow warning triangle**. Record the device name if found.
9. **Antivirus check.** Open the antivirus. Confirm definitions are **up to date**, then start a **Quick Scan** (a Full Scan is too slow for class time; the instructor may run a Full Scan on the demo machine). Record the result.
10. Complete the **Software Diagnostic Report Form**.

Expected results:

- Task Manager shows running processes and resource percentages.
- At least one startup app disabled.
- Free disk space recorded; Disk Cleanup shows reclaimable space.
- Three Event Viewer errors recorded with IDs.
- `sfc /scannow` completes with a message.
- Device Manager checked; any yellow-triangle device noted.
- Antivirus scan completed with a result.
- A completed report form.

Points to pause on and predict:

- After step 1: "If the Disk column is pinned at 100% while nothing is happening, what should you suspect?" (A failing drive â€" check S.M.A.R.T. â€" or a heavy background task.)
- After step 4: "Will disabling start-up items change the installed programs?" (No â€" the programs remain installed; they just do not auto-launch.)
- After step 8: "A device has a yellow triangle. Is that hardware or software?" (Software â€" it needs a driver or has a conflict.)

Troubleshooting while walking the room:

- Command Prompt will not run `sfc` â†' confirm it says "Administrator" in the title; right-click and choose Run as administrator.
- Event Viewer is empty â†' expand the correct log; some machines log few errors. Use another log (Application) if System is quiet.
- Disk Cleanup shows 0 bytes â†' this machine may already be clean; note that honestly.
- Antivirus scan hangs â†' it may be waiting for updates; skip and record it as "not completed".
- Learner wants to "just format it" â†' stop them; remind them of the data-first rule.

Completion checklist (learner ticks each):

- ☐ Task Manager opened; top CPU, memory and disk processes recorded
- ☐ Startup Apps reviewed; one unnecessary app disabled and noted
- ☐ Disk free space recorded
- ☐ Disk Cleanup run (temp files and Recycle Bin only)
- ☐ Three Event Viewer errors recorded with Event IDs
- ☐ `sfc /scannow` run and its result recorded
- ☐ Device Manager checked for yellow-triangle devices
- ☐ Antivirus scan run and result recorded
- ☐ Software Diagnostic Report Form completed
- ☐ All safety and ethics steps followed (no disabling antivirus, no unknown pop-ups)

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: Diagnose a **slow/faulty software scenario** on your own, using the correct Windows tools, and write a short diagnosis. Do not copy the instructor's demonstration.

Scenario. *Mr. Bassey*, who runs a small printing business in Mile 3, Port Harcourt, brings you a Windows desktop. He reports:

- The computer takes **over five minutes** to reach the desktop in the morning.
- Once running, it is **slow to open files**.
- He sees **adverts popping up** even when his browser is closed.
- Yesterday a message said **"Low disk space on C:"**.
- Last week, **his sound stopped working** after a Windows update.

Requirements:

1. List the **three most likely software causes** for the slowness, and say which Windows tool you would use to confirm each.

2. Explain what the **pop-up adverts** most likely indicate, and describe the **safe removal approach** in numbered steps.
3. Explain how "Low disk space" makes the computer slow and name **one** Windows tool that can help.
4. The **sound stopped after an update**. Which tool do you open, and what action might fix it?
5. Decide: are these problems **all software**, or should any of them prompt a **hardware** check? Justify your answer.

Time allowance: 15 minutes.

Constraints:

- Use only the Windows tools from today's session.
- Do not disable the antivirus.
- Write in plain language a customer could follow.

Success criteria:

- The three slowness causes are plausible and each is matched to a suitable tool.
- The malware approach is safe (disconnect â†' full scan â†' quarantine â†' restart â†' re-scan â†' passwords â†' educate).
- The disk-space explanation is correct and names a real tool.
- The sound issue is correctly identified as a driver problem.
- At least one hardware check is justified (the disk-space + slow pattern should prompt a S.M.A.R.T. check).

Hints available if stuck:

- Hint 1: The slow *morning* start-up points at one particular tab in Task Manager.
- Hint 2: "Low disk space" + slow file opening is a **storage** issue â€" but is a *full* drive the same as a *failing* drive? Think about it.
- Hint 3: Sound stopping right after an update screams "the translator between Windows and the sound chip".

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

1. Likely causes of slowness + tools:

- Too many **startup programs** â†' confirm in **Task Manager â†' Startup Apps** (High-impact entries).
- **Malware** (supported by the pop-ups) â†' confirm with a **full antivirus scan**.
- **Low disk space / full drive** â†' confirm by checking **free space in This PC** and using **Disk Cleanup**. *(Also acceptable: a failing drive, since "slow + full + adverts" can hide hardware â€" confirmed by CrystalDiskInfo S.M.A.R.T.)*

2. Pop-up adverts: most likely **malware/adware**. Safe removal approach:

1. Disconnect from the internet.
2. Run a **full** antivirus scan (definitions updated).
3. Review results; **quarantine/remove** detected threats.
4. Restart if prompted.
5. **Scan again** to confirm.
6. If symptoms persist, run a **trusted second-opinion** scanner.
7. Warn the customer to change important passwords (spyware may have captured them).
8. Educate the customer: only download from trusted sources.

3. Low disk space: Windows needs free space for temporary files, updates and paging (using disk as extra memory). When the drive is nearly full, these operations slow down, so the whole computer feels slow. **Disk Cleanup** (and emptying the Recycle Bin / uninstalling unused programs) helps.

4. Sound after an update: It is a **driver** problem. Open **Device Manager**, find the sound device (look for a yellow warning triangle), then either **Update driver** (from the manufacturer's official site is best) or **Roll Back Driver** if the update caused it.

5. Software or hardware? These are **mostly software** problems. However, because the machine is slow **and** the drive was reported full, the technician should also run a **hardware** check â€" **CrystalDiskInfo S.M.A.R.T.** â€" to rule out a failing drive (the same "slow" symptom can be a dying HDD). *This is the key insight of the exercise:* never assume "software" without a quick hardware sanity check.

Marking guidance (out of 15):

- Three plausible slowness causes + matching tools (3)

- Safe malware approach, in sensible order (4)
- Correct disk-space explanation + real tool (2)
- Correct driver identification + update/rollback (3)
- Correctly notes a hardware check is warranted and why (3)

Accept any answer showing correct reasoning, even if wording differs. Penalise unsafe steps (formatting first, disabling antivirus, clicking pop-ups).

J. REAL-WORLD CASE STUDY

Situation. *Chidinma* is a final-year student at a school in Port Harcourt. She needs a "free" copy of a design program for her project. A friend gives her a downloaded installer on a USB flash drive. She installs it. Within days, her laptop slows down badly, her browser opens strange pages by itself, and pop-ups appear on the desktop. She is afraid she has "spoiled the laptop" and is about to pay a shop to "reset everything" â€” which would wipe her project files.

Relevant facts.

- The slowness and pop-ups began **right after** she installed the "free" program.
- Her project files are stored on the same drive, with **no backup**.
- The machine still boots and works, just slowly.

The decision point. Should the shop format the laptop and lose her project, or is there a safer path?

Possible approaches.

1. Format and reinstall Windows immediately (fast, but her project files would be at risk if not backed up).
2. Disconnect from the internet, back up her project files, run a full antivirus scan, quarantine the threats, restart, re-scan, and remove the "free" program politely.
3. Ignore the pop-ups and keep using the laptop.

Recommended solution (for this stage). Approach 2. First **back up her project files** (this is not optional). Then disconnect the internet and run a **full scan**, quarantine anything found, restart, and scan again to confirm. Remove the "free" program. Explain how she got infected and why "free cracked software" from a USB is one of the most common infection routes in Nigeria.

Trade-offs.

- Approach 1 is the "nuclear option": it certainly removes the malware but risks her data and wastes time. It should be the **last** resort, after backup â€” not the first reaction.
- Approach 2 is safer, cheaper, faster, and preserves her work; but it depends on removing the *cause* (the habit) or the infection will return.
- Approach 3 lets the infection and the risk grow â€” spyware could steal her passwords.

Professional connection. A technician earns a customer's trust by refusing to destroy data for convenience. The CEA graduate is trained to preserve the customer's files, remove the threat safely, and educate the customer â€” exactly the "honest assessment and recommendations" the certificate stands for.

Transfer exercise (ask the class): "Chidinma asks, 'How do I know which downloads are safe?' What three rules would you give her?"

Suggested answer: (1) Only download software from the **official** website of the maker, or from a trusted store â€” never "cracked"/"free premium" versions. (2) **Scan any USB drive** from someone else before opening its files. (3) Keep antivirus **updated** and never click pop-ups that claim to "clean your PC".

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide. When a learner has a problem, direct them to the relevant row.

K.1 The computer is slow but Task Manager looks normal

- **Symptoms:** user complains of slowness; CPU/memory/disk look modest

- **Likely causes:** failing drive (slow reads); intermittent background task; too little RAM despite low average use; overheating causing throttling.
- **How to investigate:** look at the **Disk** column over time; run **CrystalDiskInfo** (S.M.A.R.T.); check temperatures with **HWMonitor**; review Event Viewer for disk warnings.
- **Safe corrective steps:** back up; replace/upgrade the drive if failing; clean cooling if hot; add RAM if memory pressure is confirmed.
- **How to confirm resolved:** normal speed returns and the S.M.A.R.T./temperature readings are healthy.
- **Prevention:** regular health checks and cleaning (Session 8).

K.2 Pop-ups and browser redirects

- **Symptoms:** adverts appear; browser opens strange sites; new toolbars.
- **Likely cause: malware/adware**, usually from an untrusted download or an infected USB.
- **How to investigate:** check recently installed programs and browser extensions; run a full scan; check **Startup Apps** for unknown entries.
- **Safe corrective steps:** disconnect internet â†' full scan â†' quarantine â†' restart â†' re-scan â†' remove the unwanted program/extension â†' change passwords â†' educate.
- **How to confirm resolved:** pop-ups and redirects stop after a restart and a clean re-scan.
- **Prevention:** trusted downloads, scan USB drives, keep antivirus updated.

K.3 "Low disk space" and slow saving

- **Symptoms:** warning messages; slow saves; small free space on C:.
- **Likely cause:** the drive is nearly full (temp files, downloads, Recycle Bin, unused programs).
- **How to investigate:** check free space in **This PC**; run Disk Cleanup to see reclaimable space.
- **Safe corrective steps:** empty Recycle Bin; Disk Cleanup; uninstall unused programs; move personal files to external storage/cloud. **Do not delete system files or the Windows folder.**
- **How to confirm resolved:** free space rises and saving is fast again.
- **Prevention:** monthly Disk Cleanup; keep at least ~15% of the drive free.

K.4 Blue screen after a driver or program install

- **Symptoms:** blue screen ("Your PC ran into a problem"); recent install/update.
- **Likely cause:** a faulty driver or an incompatible program (sometimes faulty RAM â€" always rule it out).
- **How to investigate:** note the stop code; start **Safe Mode**; check **Device Manager**; check **Event Viewer** around the crash time.
- **Safe corrective steps:** in Safe Mode, **roll back** the recent driver or uninstall the recent program; if unstable, use **System Restore** to an earlier point; run a **memory test** to rule out RAM.
- **How to confirm resolved:** the machine runs normally for a full session without a blue screen.
- **Prevention:** get drivers from the maker; avoid untrusted software; keep System Restore points.

K.5 A device stopped working after a Windows update

- **Symptoms:** no sound / no Wi-Fi / printer not found, right after an update.
- **Likely cause:** a **driver** mismatch caused by the update.
- **How to investigate:** open **Device Manager**; look for a yellow warning triangle; read the device's error in Properties.
- **Safe corrective steps: Roll Back Driver**; or **Update driver** from the manufacturer's official site; then restart.
- **How to confirm resolved:** the device works again.
- **Prevention:** install manufacturer drivers deliberately; create restore points before big updates.

K.6 Distinguishing likely from confirmed

As in Session 5, a **likely** cause becomes **confirmed** only after evidence: an Event Viewer error with a code, a scan result, a `sfc` message, a driver roll-back that fixes the device, or a hardware reading. Teach learners to write "likely" until a test proves it. Say clearly: **"I have not confirmed this yet."**

L. SESSION ASSIGNMENT

Assignment title: "Software Diagnostic Report on a Real Machine"

Purpose: To practise using Windows' built-in diagnostic tools, to diagnose common software faults safely, and to write findings plainly for a customer.

Realistic scenario: A customer brings you a Windows computer and says: *"It is very slow to start, sometimes shows adverts, and the sound stopped working last week."* Produce a professional software diagnostic report before any destructive action.

Detailed instructions:

1. Use a real Windows computer you have permission to work on (a CEA practice machine, a family machine, or a machine at your workplace). **Do not work on a computer without permission.**
2. **Observe and record** the reported symptoms in your own words.
3. **Task Manager:** record the top CPU, memory and disk processes and the number of startup apps enabled.
4. **Startup Apps:** disable **one** unnecessary app and record what you disabled. (Re-enable it afterwards if the machine is not yours.)
5. **Disk space & Disk Cleanup:** record free space; run Disk Cleanup on temporary files and the Recycle Bin only; record space reclaimed.
6. **Event Viewer:** record three error Events (Event ID + Source).
7. `sfc /scannow` **:** run it as Administrator and record the final message.
8. **Device Manager:** list any device with a yellow warning triangle.
9. **Antivirus:** confirm definitions are up to date; run a **Quick Scan** and record the result. *(Do not run a Full Scan on a machine you do not own without permission â€" it is slow.)*
10. **Conclusion:** state the most likely software cause(s) and your recommended action, in plain language for the customer.

Required deliverables:

- One completed **Software Diagnostic Report Form** with all readings and results.
- A one-paragraph plain-language summary for the "customer".

Submission format: Written report on the CEA template (paper or class folder).

Deadline: Before Session 7.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Observation & symptom record	2	Clear, specific symptoms recorded
Task Manager & Startup findings	3	Top processes and startup app count recorded; one safely disabled
Disk space & Disk Cleanup	3	Free space and reclaimed space recorded; safe cleanup only
Event Viewer findings	3	Three errors with Event ID and Source
`sfc /scannow` result	3	Run as Administrator; final message recorded
Device Manager & antivirus	3	Yellow-triangle device noted (if any); scan result recorded
Conclusion & recommendation	2	Logical, safe, honest conclusion in plain language
Report completeness & safety	1	No unsafe actions; form fully filled

Common mistakes to avoid: running `sfc` without Administrator rights; deleting system files; disabling the antivirus; formatting before backup; writing "confirmed" without a test; ignoring a possible hardware cause.

Instructor reference solution (abbreviated example):

> ***Symptoms:*** *slow start-up; occasional adverts; sound stopped after last week's update.* ***Task Manager:*** *top process "Update Helper" at 45% CPU and 70% disk;* ***9*** *startup apps enabled, 3 marked High impact.* ***Startup Apps:*** *disabled "Photo Startup Tool" (unnecessary).* ***Disk:*** *only* ***4 GB free*** *of a 500 GB drive â€" very low. Disk Cleanup reclaimed* ***6.1 GB*** *(temp files +*

*Recycle Bin). **Event Viewer:** Error â€" Event ID **7000** (Service Control) source Service Control Manager; Error â€" Event ID **7009**; Error â€" Event ID *1001 (Windows Error Reporting). `sfc /scannow`: "Windows Resource Protection did not find any integrity violations." **Device Manager:** yellow triangle on **Realtek High Definition Audio**. **Antivirus:** definitions up to date; Quick Scan â€" no threats. **Conclusion â€" likely causes:** (1) too many startup programs + a very full disk â†' slow start-up and slow file opening; (2) the sound device has a **driver** problem after the recent update. **Recommendation:** keep three unnecessary startup apps disabled; empty the drive further and set up monthly Disk Cleanup; **roll back** the audio driver or install the correct one from the laptop maker's site; restart and re-test the sound. Also run a S.M.A.R.T. check (Session 5) because the drive is nearly full and slow reads can mask a failing drive. **Plain-language summary:** "Your computer is starting too many programs at once and the storage disk is almost full, which is why it is slow. The sound stopped because of a driver problem after an update. We can fix all three without losing your files."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer. In one sentence each, what is a **software fault** and what is a **hardware fault**? **Q2. Multiple choice.** Which Windows tool lets you see which programs are using the most CPU right now? a) Disk Cleanup b) Event Viewer c) Task Manager d) System Restore **Q3. Explain in your own words.** Why is it dangerous to "just format" a computer first when the owner complains of slowness? **Q4. Practical instruction.** What exact command checks and repairs corrupted Windows system files, and how must the window be opened to run it properly? **Q5. Matching.** Match each symptom to the most likely cause:

1. Pop-ups and browser redirects
2. Sound stopped right after a Windows update
3. "Low disk space on C:"
4. Device shows a yellow triangle in Device Manager A. Driver problem B. Malware C. Full drive D. Missing/incorrect hardware driver **Q6. Scenario.** A computer is slow and makes a soft clicking sound. Is this more likely software or hardware, and what should you check first?

INSTRUCTOR ANSWER KEY (Q1â€"Q6)

A1. A **software fault** is a problem with programs, drivers, the operating system or data; a **hardware fault** is a problem with a physical part. Software faults are fixed by cleaning/correcting/restoring the software; hardware faults are fixed by repairing or replacing a part.

A2. (c) Task Manager.

A3. Because formatting erases everything, including the customer's personal files, and it may not even fix the problem â€" the real cause could be a failing **hard drive**, in which case the fault returns *and* the data is gone. Always back up and diagnose first; format only as a last resort.

A4. `sfc /scannow`. It must be run in a **Command Prompt opened as Administrator** (right-click Command Prompt â†' Run as administrator).

A5. 1â€"B (malware), 2â€"A (driver problem), 3â€"C (full drive), 4â€"D (missing/incorrect driver). *(Note both 2 and 4 are driver issues; 2 is a driver problem triggered by an update, 4 is a general missing/wrong driver.)*

A6. More likely **hardware** â€" a soft clicking sound plus slowness is the classic sign of a **failing hard drive**. Check **CrystalDiskInfo S.M.A.R.T.** first, and **back up the data immediately** before anything else. (A memory/CPU fault is possible but the click strongly points to the drive.)

Marking: award for correct concept. For Q5, accept 2â€"D and 4â€"A if the learner explains the driver reasoning clearly.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned that a computer can have perfect parts and still be very sick â€" because the problem is in its **software**. You learned the common software faults (slow performance, malware, boot problems, driver issues, full disk) and the Windows tools that find them: **Task Manager** (what is using CPU, memory and disk right now), **Startup Apps** (what

launches automatically), **Event Viewer** (the history of errors), `sfc /scannow` (repairing corrupted system files), **chkdsk**, **Disk Cleanup**, **System Restore**, **Safe Mode**, and **Device Manager** (drivers). You also learned the safe approach to malware â€" disconnect, full scan, quarantine, restart, re-scan, change passwords, educate â€" and, most importantly, how to decide whether a fault is **software or hardware** before you act.

Five important takeaways:

1. A software fault is fixed by cleaning/correcting/restoring â€" never by replacing parts blindly.
2. **Back up first.** Never format before you have a backup and a diagnosis.
3. Task Manager shows the present; Event Viewer shows the past; `sfc` repairs system files.
4. Safe malware removal: disconnect â†' **full scan** â†' quarantine â†' restart â†' re-scan â†' passwords â†' educate.
5. Decide **software or hardware first**; a slow machine with clicking may be a failing drive.

Questions to help you recognise what you can now do:

- Can I open Task Manager and name the biggest culprit?
- Can I run `sfc /scannow` correctly as Administrator?
- Can I explain the safe malware-removal steps in order?
- Can I use Device Manager to fix a device that stopped working?
- Can I tell whether a fault is software or hardware and say why?

Safety self-check (tick honestly):

- ☐ I never disabled the antivirus.
- ☐ I backed up before any destructive step.
- ☐ I did not click unknown pop-ups or links.
- ☐ I worked only on machines I had permission to use.
- ☐ I asked the instructor when unsure.

Connection to the next session. You can now find faults â€" hardware and software. But finding a fault is only half the job. Next session, Session 7 â€" Component Replacement, you will learn to **choose compatible parts and physically replace them** â€" RAM, storage (including upgrading an old hard drive to a fast SSD), power supplies and GPUs â€" and verify the machine afterwards. Diagnosis told you *what* is wrong; replacement makes it *right*.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner opens Task Manager **first** when investigating slowness, and reads the actual percentages.
- The learner checks for a **hardware** cause (S.M.A.R.T., temperature) even when the problem "looks like software".
- The learner runs `sfc /scannow` in an **Administrator** window without being reminded.
- The learner follows the safe malware sequence in order and refuses to disable antivirus.
- The learner backs up before any destructive action and says so out loud.
- The learner explains the difference between "likely" and "confirmed".

Common misconceptions to watch for and correct:

- "Slow always means virus." â†' Could be startup programs, a full/failing disk, too little RAM, or heat.
- "Formatting is the fastest fix." â†' It destroys data and may not fix a hardware cause.
- "`sfc /scannow` removes viruses." â†' No; it repairs system files only.
- "Any free driver-updater website is fine." â†' Use Windows Update or the manufacturer's official site.
- "Safe Mode is a permanent way to use the PC." â†' It is a temporary rescue tool.

Suggested remedial exercises for learners who struggle:

1. **Tool-naming drill:** show five symptoms and have the learner name the right tool for each, repeatedly, until fluent.
2. **Order-the-steps drill:** shuffle the malware-removal steps on cards and have the learner arrange them correctly.
3. **Present-vs-past drill:** give scenarios and ask "which tool â€" Task Manager or Event Viewer â€" tells me this?"

Differentiation:

- **Level 1 (supported):** provide a step-by-step tool card with the exact clicks, and a partially completed report form; pair with a stronger learner.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** present a **mixed** fault (e.g., a full disk *and* a suspicious process *and* a yellow-triangle device) and require the learner to prioritise the fixes and justify the order, including a hardware sanity check.

Timing guidance: `sfc /scannow` and antivirus full scans are slow. Start `sfc` early in the practical and let it run while learners do other checks; do **not** let a scan consume the whole session. Protect the guided practical (Section H). If short on time, convert the case study to a two-minute discussion and keep the independent exercise, which is where the real diagnosis skill is practised.

APPENDIX 6A â€" 5-MINUTE HARDWARE-vs-SOFTWARE RECAP (only if needed)

If learners cannot distinguish hardware from software faults, do this before Section D.

1. Write on the board: **Hardware = physical part. Software = instructions and data.**
2. Say: "Hardware faults are fixed by **repairing or replacing a part**. Software faults are fixed by **cleaning, correcting, updating, restoring or reinstalling**."
3. Quick drill â€" call out and have the class answer "H or S":
    - "Hard drive making a clicking sound." (H)
    - "Pop-ups on the desktop." (S)
    - "No power at all." (H)
    - "Sound stopped after a Windows update." (S)
    - "CPU at 95 Â°C." (H)
    - "USB flash drive infected from a friend." (S) Confirm each learner can answer before continuing.

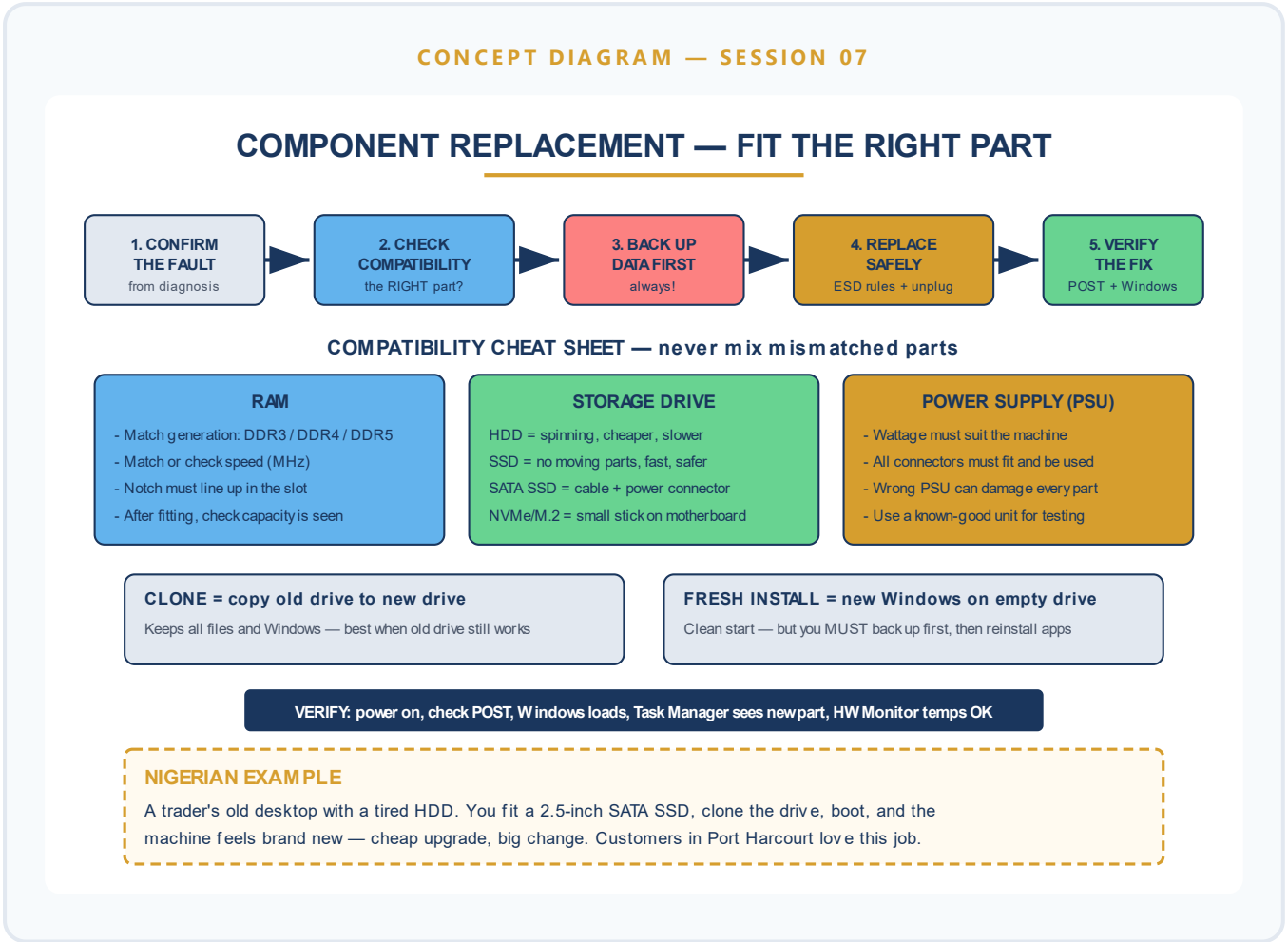
END OF SESSION 6 â€" End of Week 2 (Diagnostics)

Next: Session 7 â€" Component Replacement.

SESSION 07 OF 12

Component Replacement

CONCEPT DIAGRAM — SESSION 07



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 7 â€" COMPONENT REPLACEMENT

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	7 of 12
Session title	Component Replacement
Week / Module	Week 3 â€" Module: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (hands-on, safety-critical)

Item	Detail
Prerequisite knowledge	Sessions 1â€“6: components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware diagnostics, and software diagnostics. If a learner cannot demonstrate safe case-opening and ESD precautions, give the 10-minute safety refresher in Appendix 7A before this session.
Connection to previous work	Sessions 5 and 6 taught you to **find** the fault. This session teaches you to **fix** it by choosing compatible parts and physically replacing them, then verifying the machine works.
Preparation before class	One or more practice desktops that may be opened; matching **known-good** replacement RAM (same generation as the practice machines) and a spare known-good stick; at least one **2.5-inch SSD** and a matching **SATA cable + power cable**, or a spare HDD; a known-good PSU of suitable wattage for the practice machine; thermal paste; cleaning cloths; anti-static wrist straps (one per learner); screwdrivers with the correct heads; a **known-good** USB or SATA cable; a bootable Windows USB (for the OS-reinstall branch) and, if available, a **cloning tool** (a program that copies one drive to another) pre-installed on a bootable USB or on the running machine. Printed **Compatibility Check Sheet** and **Replacement Service Log** (one per learner).

Important wording used in this session.

- A **component** is a physical part of the computer (RAM, drive, PSU, GPU, CPU cooler).
- To **verify** a replacement means to power on the machine and *confirm* that the new part is detected and working correctly â€” never assume it just because it fitted.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** why a replacement part must be **compatible** with the machine, and check compatibility for RAM (generation, speed, size), storage (interface) and PSU (wattage and connectors).
2. **State** the difference between a **hard disk drive (HDD)** and a **solid-state drive (SSD)**, and between **SATA** and **NVMe**, and choose the correct one for a given machine.
3. **Upgrade RAM** correctly â€” match the type, seat the stick by the notch, and confirm the new capacity is recognised.
4. **Replace a storage drive** by removing the old one, fitting the new one, connecting the correct cables, and either **cloning** the old drive or **installing Windows** on the new drive.
5. **Explain "clone" and "fresh install"**, including when each is appropriate and why backing up comes first.
6. **Replace a power supply safely** using a known-good unit of suitable wattage, reconnecting every cable and double-checking before power-on.
7. **Re-apply thermal paste** correctly (clean, apply a small amount, refit the cooler) and confirm improved cooling.
8. **Verify** a machine after any replacement using the tools from Sessions 5 and 6 (POST, Windows, Task Manager, HWMonitor).
9. **Follow safety and ESD procedures** at every step, and know when a task should be handled by a qualified technician.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one practice desktop per two learners, safe to open and reassemble.
- **Replacement RAM** that matches the practice machines' generation (e.g., DDR4 if the boards use DDR4). Never mix mismatched generations.
- A **known-good** spare RAM stick kept aside as the "test" stick.
- At least one **2.5-inch SATA SSD** with a **SATA data cable** and a **power cable**; or a spare known-good HDD if SSDs are not available.
- One known-good **PSU** of suitable wattage for the practice machine, plus its cables.

- A tube of **thermal paste** (a little goes a long way).
- Cleaning cloths and **isopropyl alcohol** (for cleaning old paste).
- **Anti-static wrist straps** (one per learner), screwdrivers with correct heads, tweezers, a small parts tray, a flashlight.
- A **bootable Windows USB** (for the fresh-install branch).
- Optional: a bootable **cloning tool** (a program that copies one drive to another) on a USB or installed, and a USB-to-SATA adapter cable if available (useful for cloning from an old drive that is still readable).

Software

- Windows 10/11 installation media.
- A cloning tool (optional), used under supervision.
- CrystalDiskInfo and HWMonitor (from Sessions 5â€"6) for verification.

Printed materials

- Compatibility Check Sheet (one per learner).
- Replacement Service Log (one per learner).

Safety equipment

- Anti-static wrist straps, safety glasses, first aid kit.
- Fire safety: know the location of the fire extinguisher and the first-aid kit.

Internet

- Optional, for looking up a motherboard's supported RAM or a PSU wattage calculator. Do not depend on it; the instructor prepares compatibility notes in advance.

Alternative for learners who lack a facility

- Desktop replacement is best learnt on a desktop. Learners without access to a personal desktop share a CEA practice machine. Learners who own a laptop may observe laptop RAM/SSD replacement when the instructor demonstrates it (laptop parts are smaller and differ by model).

D. OPENING ACTIVITY (6 minutes)

Instructor setup: On the bench, place two objects side by side: an old mechanical **hard drive (HDD)** and a **solid-state drive (SSD)** of the same size. Also write on the board the price of a typical small SSD versus the cost of a new budget computer (use realistic Port Harcourt market figures the academy has prepared â€" do not invent exact prices).

What the instructor says:

> *"Good morning. This week you stop watching and start doing. Last week you learned what is wrong. Today you make it right.*
>
> *Look at these two things. One is an old spinning hard drive. The other is a solid-state drive â€" an SSD. Same size, similar price. But this one [SSD] can wake an old, tired computer up so well that the owner will swear you replaced the whole machine. Customers love this job more than almost any other, because it is cheap and the change is dramatic.*
>
> *Here is the question for today: How do you know the right part to buy? Buy the wrong RAM and it will not fit. Buy the wrong SSD and it will not connect. Replace a power supply wrongly and you can damage the whole computer â€" or hurt yourself.*
>
> *Today you learn to **choose compatible parts**, to **replace them safely**, and to **verify** that the machine is truly fixed. This is the day you become a technician, not just a diagnoser."*

Expected learner response: Learners will be interested in the SSD speed claim; some may ask "how much does an SSD cost?" or "can my laptop take one?". Accept the questions and say they are exactly what a technician must learn to answer for customers.

Why this opener works: It promises a satisfying, high-value repair (the SSD upgrade), it immediately raises the real professional skill of *choosing the right part*, and it sets the safety tone for a hands-on session where mistakes have real consequences.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1: "What are the two ESD safety steps before you touch any internal component?"

- Expected answer: **Power off and unplug** the machine; **wear an anti-static wrist strap** (or touch a bare metal part of the case first).
- Common gap: learners remember the strap but forget to unplug. Say: both, every time.

Question 2: "In what order did we remove components during teardown?"

- Expected answer: a sensible order â€" e.g., remove power cables first, then the GPU, then RAM, then storage, and the CPU cooler only when needed (or the reverse for reassembly).
- Common gap: forgetting to disconnect power first. Emphasise: **power first, always**.

Question 3: "What is the five-step diagnostic workflow?"

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).

Question 4: "A machine is slow and CrystalDiskInfo shows a healthy drive, RAM passes, and temperatures are normal. What might make it fast again without replacing it?"

- Expected answer: cleaning up software (startup programs, full disk, malware) from Session 6 â€" and, if the drive is an old HDD, replacing it with an **SSD** (today's topic) even though it is not "faulty".

Question 5: "Why must we never open a PSU, and how do we test one?"

- Expected answer: **lethal stored charge**; test by **swap** with a known-good PSU, never by opening.

Bridge to today (say this):

> *"Diagnosis is detective work. Replacement is surgery. Before a surgeon operates, they check that the new part matches the patient â€" blood type, size, and so on. Today, 'matching' means RAM generation and speed, storage interface, and PSU wattage. Get the match wrong and the operation fails."*

Gap handling: If more than a third of the class cannot state the ESD steps in Question 1, stop and run the Appendix 7A safety refresher now. This session must not proceed with learners who cannot work safely.

F. FULL LECTURE NOTES

F.1 Why compatibility matters â€" the foundation of replacement

The idea. Not every part fits every computer. Before buying or fitting any part, you must confirm it is **compatible** with the machine.

Plain-English definition. Compatible means the part is of the correct type and size to work with that specific motherboard, case or system. An incompatible part will not fit, will not connect, or will fit but not work (or even damage something).

Why it matters. Customers will lose faith if you order a part that does not fit. Worse, a wrong part (for example, the wrong voltage or a badly matched power supply) can damage the computer. Checking compatibility *first* is what separates a professional from a guesser.

Analogy. Buying a replacement part is like buying shoes: the right brand is not enough â€" you need the right **size**. A beautiful shoe that is the wrong size is useless. Compatibility is the "size" of a computer part.

What we always check, by part:

- **RAM:** the **generation** (DDR3, DDR4 or DDR5), the size (GB per stick), and ideally the **speed** (MHz) and whether it is a **desktop (DIMM)** or **laptop (SO-DIMM)** stick.
- **Storage:** the **interface** (SATA or NVMe) and the physical size/format (2.5-inch, 3.5-inch, M.2).
- **PSU:** the **wattage** (must be enough and not excessive) and the **connectors** the machine needs.
- **GPU:** the **slot type (PCIe)**, the physical length that will fit the case, and whether the PSU can supply enough power.

- **Laptop parts:** model-specific; always match the exact laptop model.

Important misconception. "Any RAM will fit any computer." False. **DDR3, DDR4 and DDR5 are different shapes and are not interchangeable.** A DDR4 stick will not go into a DDR3 slot, and forcing it will damage the slot.

Limitation. Even compatible parts can differ in quality; a cheap part of the right type may fail early. Compatibility is necessary, not sufficient â€" quality matters too.

F.2 Choosing RAM â€" generation, speed and size

Plain-English definition. RAM (Random Access Memory) is the fast working memory that holds whatever the computer is doing right now. It comes in **sticks**. More RAM lets you run more programs at once before the machine slows down.

The three things to match:

1. **Generation: DDR3**, **DDR4** or **DDR5**. These are different physical shapes and **cannot** be mixed. Check the motherboard's manual or look up the model to know which generation the machine uses.
2. **Size:** how many gigabytes (GB) per stick, and the total the machine supports. A 32-bit version of Windows can only use about 4 GB, so extra RAM is wasted there; 64-bit Windows uses more.
3. **Speed:** measured in **MHz** (e.g., 2666 MHz, 3200 MHz). Ideally use the same speed as the existing RAM. If you mix speeds, the machine usually runs all sticks at the **slowest** speed â€" it should still work, but matching is best.

Desktop vs laptop. Desktops use **DIMM** (longer) sticks; laptops use **SO-DIMM** (shorter) sticks. They are not interchangeable.

Running in pairs (dual-channel). Many motherboards run faster when two matched sticks are installed in the correct paired slots (often the same colour slots). This is optional for beginners but good practice.

Realistic Port Harcourt scenario. A shop owner's desktop with 4 GB RAM slows to a crawl when he opens his accounting program and a browser with many tabs. The technician checks the motherboard is DDR4, buys one matching 8 GB DDR4 stick of the same speed, installs it, and the machine runs smoothly. The cost is a fraction of a new computer.

Analogy. RAM is the size of a trader's work-table. A small table cluttering up with goods slows every task; a bigger table lets you spread out and work faster. But the table must fit your shop â€" you cannot buy a table bigger than the room.

Important misconception. "More RAM always makes it faster." It helps when the machine is *short* of memory (many programs open, disk light always busy). If the real problem is a slow HDD or a virus, adding RAM will not fix it.

Limitations of upgrading RAM. Some laptops have RAM **soldered** to the board and cannot be upgraded. Some machines have a maximum capacity (e.g., 16 GB) beyond which extra RAM is not recognised. Always check first.

F.3 Choosing storage â€" HDD vs SSD, SATA vs NVMe

Plain-English definitions.

- **HDD (Hard Disk Drive):** stores data on spinning metal platters with a moving head. Cheap per gigabyte, but slower and mechanical (it can click and wear out).
- **SSD (Solid-State Drive):** stores data in memory chips with **no moving parts**. Much faster, quieter and more shock-resistant. More expensive per gigabyte, but the speed difference is huge.
- **SATA:** a common connection type used by both 2.5-inch/3.5-inch HDDs and 2.5-inch SSDs. It uses a flat **SATA data cable** and a **SATA power cable**. Typical speed: around 6 Gb/s (the interface limit), though a mechanical HDD is far slower than that in real use.
- **NVMe:** a newer, much faster connection used by **M.2** SSDs that slot directly onto the motherboard, with no data or power cable. NVMe drives can be several times faster than SATA SSDs.

Why this matters for choosing a part. You must match the **interface** the machine supports:

- If the machine has SATA ports and drive bays â†' a **2.5-inch SATA SSD** is the safe, universal upgrade.
- If the motherboard has an **M.2 slot** that supports NVMe â†' an **NVMe SSD** is faster (but check the slot's key type and supported length).
- A 3.5-inch bay takes a **3.5-inch HDD**; a 2.5-inch bay takes a **2.5-inch SSD/HDD** (a 2.5-inch drive can usually be mounted in a 3.5-inch bay with an adapter bracket).

Why an SSD upgrade speeds up an old PC. An HDD must physically spin up and move a head to find data, which is slow. Windows, on start-up and when opening files, reads thousands of small pieces of data. On an HDD each read waits for the

mechanical parts; on an SSD the data is found instantly. That is why replacing an HDD with an SSD makes an old computer feel new â€" the mechanical delay disappears. **The CPU and RAM are unchanged; the speed came from removing the mechanical bottleneck.**

Analogy. An HDD is like fetching a book from a huge library by walking down long shelves. An SSD is like having the book already on your desk. The book is the same; the *time to get it* is completely different.

Important misconception. "SSD means I can never lose data." SSDs fail too, often suddenly and with less warning than HDDs. **Backups remain essential** for both.

Limitations of SSD choice.

- An old motherboard with only SATA ports cannot use NVMe â€" you must use a SATA SSD.
- Some very old machines do not support a drive large enough to hold the customer's data comfortably; check capacity needs.
- Cloning requires the new drive to have **enough capacity** to hold everything on the old drive.

F.4 Choosing a PSU â€" wattage and connectors

Plain-English definition. The **PSU (Power Supply Unit)** converts wall electricity into the low voltages the components need. **Wattage (W)** is its power capacity â€" how much total power it can supply.

The two things to match:

1. **Enough wattage:** the PSU must supply at least as much as the machine needs. A rough guide: a basic office desktop with no dedicated GPU needs far less than a gaming machine with a big graphics card. If the PSU is too weak, the machine crashes or shuts down when working hard. If it is far stronger than needed, it is not dangerous, but it is a waste of money (and efficiency is best around a moderate load).
2. **The correct connectors:** the PSU must have the connectors the machine needs â€" the big **24-pin** motherboard connector, the **CPU power** connector (often 4-pin or 8-pin), **SATA power** connectors for drives, and **PCIe power** connectors for a graphics card if fitted.

Safety â€" the golden rule. Never open a PSU. Even unplugged, its capacitors hold a **lethal charge**. A PSU is a **replace-only** item.

Realistic Port Harcourt scenario. A customer's desktop shuts down during printing (when the printer and CPU work together). Diagnosis by swap test shows the PSU is failing. The technician checks the new PSU has enough wattage and the right connectors, fits it, and the shutdowns stop.

Analogy. The PSU is like the generating set for a building. If the generator is too small for all the appliances, everything dims or the generator cuts out when many appliances run at once. It must be big enough *and* have the right sockets to plug everything in.

Important misconception. "A bigger-wattage PSU always makes the machine more reliable." Wattage is capacity, not health. A failing 700 W PSU is still failing. Match the wattage sensibly and choose a quality unit.

Limitation. Cheap "no-name" PSUs often overstate their wattage and can damage components. Buy from a reputable brand. If unsure about wattage for a big build, a technician uses a wattage calculator or the GPU maker's recommendation â€" not guesswork.

F.5 Upgrading RAM â€" the procedure

Safety first: power off, unplug, press the power button for a second to discharge leftover power, wear your anti-static wrist strap, and work on a hard flat surface.

Steps:

1. **Confirm compatibility** using the Compatibility Check Sheet: correct generation (DDR3/4/5), desktop or laptop type, size and speed.
2. **Open the case** and locate the RAM slots on the motherboard.
3. **Remove the old stick (if replacing):** push the retaining **clips** at both ends outward until the stick pops up, then lift it out by its edges. **Do not touch the gold contacts.**
4. **Align the new stick:** note the **notch** (the off-centre gap in the row of contacts). It only fits one way. Line the notch up with the slot's key.

5. **Seat the stick:** press down **firmly and evenly** at both ends until the clips snap up into place. You should hear/feel a click. If it resists, **stop** â€" the notch is probably the wrong way round; do not force it.
6. **Run dual-channel (optional):** if fitting two sticks, use the paired slots indicated in the motherboard manual (often the same colour).
7. **Close the case**, reconnect power, power on.
8. **Verify:** the machine should POST and start. Check the new total in Windows: **Task Manager â†' Performance â†' Memory**, or the **About** page, or the BIOS. Confirm the full amount is recognised (some is reserved by the system/hardware, so a small shortfall is normal).

Analogy. Seating RAM is like fitting a key into a lock â€" the key only goes in one way (the notch), and it needs a firm, straight push, not a twist or a shove.

Common mistake. Forcing the stick. If it will not seat easily, it is almost always the wrong orientation or wrong generation. Back off and re-check.

F.6 Replacing storage â€" monitoring, cloning vs fresh install

Safety first: power off, unplug, discharge, wrist strap. **If the old drive still works, back up the customer's data before doing anything** â€" cloning can fail, and a fresh install erases the old system.

Definition: cloning. Cloning means copying the entire contents of the old drive (Windows, programs, settings and files) to the new drive, so the machine starts up exactly as before but on the new hardware. It saves the customer from reinstalling everything.

Definition: fresh install. A **fresh install** means putting a brand-new copy of Windows on the new drive (Session 9), then reinstalling the customer's programs and restoring their files from backup. It is cleaner and often faster, but it is more work and the customer must have their data backed up.

When to clone vs fresh install:

Situation	Better choice
Old drive still readable, customer wants everything unchanged	**Clone**
Old drive is failing and barely reads (may fail mid-clone)	Back up files; **fresh install** is safer
Customer's Windows is full of junk/malware	**Fresh install** (cleaner)
Customer has no backup and the drive is healthy	**Clone** (but still back up first if possible)

Steps to replace the drive:

1. **Back up** the data (clone tool copies everything, but backup protects against clone failure).
2. Power off, unplug, discharge, strap on, open the case.
3. **Locate the drive bay** and the current drive.
4. **Disconnect** the **SATA data cable** and the **SATA power cable** from the drive.
5. **Remove the screws** holding the drive (note where they go; use a tray).
6. **Slide the old drive out** and set it aside carefully (handle it flat; do not drop an HDD).
7. **Fit the new drive** into the bay (use an adapter bracket if a 2.5-inch drive goes into a 3.5-inch bay). Secure the screws.
8. **Reconnect** the SATA data cable (to a motherboard SATA port) and the SATA power cable.
9. **Close the case**, reconnect power, power on.
10. **Now either clone or install Windows:**
    - **Clone path:** boot from the cloning tool (or run it from Windows, with the old drive attached via a USB adapter) and copy old â†' new. Then set the new drive as the boot drive and remove the old one. Verify it boots and everything is present.
    - **Fresh-install path:** boot from the Windows USB and install Windows on the new drive (Session 9), then reinstall programs and restore files.
11. **Verify:** the machine boots from the new drive, Windows runs, and the drive shows up as healthy (CrystalDiskInfo) and with the expected capacity.

Analogy. Cloning is like moving your whole house to a new building â€” every piece of furniture already in place. A fresh install is like moving into an empty new building and furnishing it yourself. Cloning is quicker to settle in; a fresh install gives you a clean start.

Important misconception. "Cloning always works." It can fail â€” especially with a dying source drive, a drive that is larger than the destination, or some Windows/partition setups. That is why **backing up first** is not optional.

Limitation. Cloning needs the new drive to be **at least as large** as the used space on the old drive. If the old drive is full, a same-size new drive may not fit the clone.

F.7 Replacing a PSU safely

Safety first â€” read twice. Power off, **unplug the power cable from the wall and from the PSU**, press the power button to discharge, wear your anti-static wrist strap. **Never open the PSU case. Never work with it plugged in.**

Steps:

1. **Confirm compatibility:** the new PSU has **enough wattage** and the **correct connectors** (24-pin, CPU power, SATA power, PCIe power if needed).
2. Open the case. **Photograph or note every cable** before removing it â€” a phone photo is a technician's best friend.
3. **Disconnect ALL PSU cables** from the motherboard and every component: the 24-pin, the CPU power, SATA power to each drive, and PCIe power to the GPU. Also disconnect any cables to extra fans or lights.
4. **Remove the four screws** at the back of the case holding the PSU (usually at the top-back or bottom-back).
5. **Slide the old PSU out** and set it aside. Remember: it is not repairable â€” it is going to be replaced.
6. **Slide the new PSU into place** and fit the four screws.
7. **Reconnect every cable**, following your photo/notes: 24-pin first, then CPU power, then SATA power to drives, then PCIe to the GPU. Make sure each connector is **fully seated** and not forced into the wrong socket.
8. **Double-check every connection.** Count them against the photo. A missed cable (especially the CPU power) is the number-one cause of "it will not start".
9. Keep cables tidy so they do not touch fan blades.
10. **Close the case, reconnect the power cable, and power on.**
11. **Verify:** the machine POSTs, fans spin, and it reaches Windows/shows a picture. Check that all drives are detected.

Analogy. Replacing a PSU is like replacing the main breaker of a house: you shut off the supply, label every wire, fit the new box, and reconnect each wire to exactly the right place. One wrong wire and things do not work (or worse).

Important misconception. "The PSU plugs in one way, so I cannot make a mistake." You can plug the **wrong connector into the wrong socket** with force, especially PCIe power into CPU power or vice versa. Check shapes â€” do not force.

Limitation. Modular PSUs (where cables detach from the PSU) mean you must use **only the cables supplied with that PSU model**. Mixing cables from different brands/models can damage components because the pin wiring differs. Never reuse one PSU's cables with a different PSU.

F.8 Re-applying thermal paste correctly

Plain-English definition. Thermal paste (also called thermal compound) is a grey paste applied between the CPU and its **heatsink/fan** (the cooler). It fills the tiny invisible gaps between the two metal surfaces so heat can pass from the CPU into the cooler more efficiently.

Why it exists. Metal surfaces look flat but are microscopically rough. Air is a poor heat conductor. The paste fills those gaps, giving a much better heat path. It dries out over years, so replacing it every so often restores cooling.

When to re-paste: when the CPU and cooler are separated (e.g., during a cleaning or a cooler replacement), or when temperatures have crept up over the years despite clean fans.

Steps:

1. Power off, unplug, discharge, wrist strap on.
2. Remove the CPU cooler (unscrew or unclip it â€” types differ; note the order).
3. **Clean** the old paste off both the CPU (top surface) and the cooler base using isopropyl alcohol and a lint-free cloth or cotton bud, until both are clean and dry. **Do not bend the CPU pins** (on older socket types the pins are on the CPU; on newer ones they are in the socket â€” be careful either way).

4. **Do not remove the CPU** unless you must; cleaning the top surface in place is safest for beginners.
5. Apply a **small amount** of new paste â€" a **pea-sized dot** in the centre of the CPU is the common, safe method. Some prefer a small line; the point is a *small* amount that will spread under pressure.
6. **Refit the cooler**, pressing evenly so the paste spreads. Tighten screws gently and evenly (a little on each screw in turn, not one fully tight).
7. **Reconnect the cooler's fan cable** to the CPU fan header on the motherboard.
8. **Power on and verify** with **HWMonitor**: temperatures should be normal or lower than before.

Analogy. Thermal paste is like adding a thin layer of cooking oil between a frying pan and the burner â€" it helps heat travel evenly where the surfaces do not perfectly touch. Too little and heat does not transfer; too much and it makes a mess and can squeeze out.

Important misconception. "More paste is better." No â€" **too much** can squeeze out onto the motherboard and can actually reduce contact. A small pea-sized amount is correct.

Safety limitation. Removing a cooler on some machines also removes the CPU (on a few older designs). If the CPU comes out stuck to the cooler, **do not pull it apart** â€" ask the instructor. Twisting or pulling can bend pins and destroy the CPU.

F.9 Verifying after replacement â€" the professional finish

The idea. A replacement is not finished when the part is fitted; it is finished when the machine is **verified working**. Never hand back a machine you have not tested.

A verification checklist for any replacement:

1. **POST and boot:** power on; the machine completes POST and reaches Windows (or the installer). No beeps, no errors.
2. **Detected correctly:** the OS sees the new part.
   - RAM: Task Manager â†' Performance â†' Memory shows the new total.
   - Storage: This PC / Disk Management shows the new drive with the right capacity; CrystalDiskInfo shows it as **Good**.
   - PSU: all drives and the GPU power up; no random shutdowns during a quick load test.
   - Thermal paste: HWMonitor shows normal (or improved) CPU temperature.
3. **Run for a while:** use the machine for a few minutes; open programs; watch for crashes, restarts or overheating.
4. **Document:** record the old part, the new part, the reason, and the verification results on the Replacement Service Log.
5. **Return the machine to the customer** with a plain-language explanation and any advice (e.g., back up regularly).

Important misconception. "If it powers on, it is fixed." Powering on is step one. A drive might not be detected, a RAM stick might be faulty, or the machine might overheat ten minutes later. Verify *and* test under use.

F.10 Summary of the session

- **Compatible** means the part matches the machine (RAM generation/size/speed; storage interface; PSU wattage/connectors; laptop = model-specific).
- **RAM:** match DDR generation, DIMM vs SO-DIMM, size and speed; seat by the **notch**, click the clips, verify the total.
- **Storage: HDD** is mechanical and slow; **SSD** is fast; **SATA** uses cables; **NVMe** uses an M.2 slot. An SSD upgrade speeds up an old PC by removing the mechanical bottleneck.
- **Cloning** copies everything to the new drive; a **fresh install** is cleaner but more work. **Back up first either way.**
- **PSU:** match wattage and connectors; **never open it**; reconnect **every** cable and double-check.
- **Thermal paste:** clean off the old, apply a **pea-sized** amount, refit the cooler, check temperatures.
- **Verify** every replacement before returning the machine.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- A practice desktop on the bench, opened, with wrist straps available.

- A matching replacement RAM stick and a known-good test stick.
- A 2.5-inch SSD with SATA data and power cables (drive replacement demonstration).
- A known-good PSU (for the connector explanation; the instructor may demonstrate or narrate the replacement depending on time and safety).
- Thermal paste, cleaning materials.
- A bootable Windows USB and, if available, a cloning tool.
- The Compatibility Check Sheet and Replacement Service Log on the projector.

G.2 Demonstration 1 â€” Compatibility check (3 minutes)

Say: "Before I touch a single screw, I check that the part fits."

Steps:

1. Show the **Compatibility Check Sheet**. Fill in the practice machine's details: RAM generation (e.g., DDR4), memory type (DIMM), storage interface (SATA), PSU wattage needed.
2. Hold up the replacement RAM. **Say:** "DDR4, DIMM, correct speed â€” this will fit. If it were DDR3, I would stop here, because DDR3 and DDR4 are different shapes."
3. Hold up the SSD. **Say:** "2.5-inch SATA SSD. The board has SATA ports, so this fits. If the board had an M.2 NVMe slot, I could go faster."

Expected results: the sheet completed; learners able to state one reason a part is compatible.

Reasoning to say aloud: "Checking first protects the customer's money and my reputation."

Questions: "What are the two things we match for a PSU?" (Wattage and connectors.)

G.3 Demonstration 2 â€” RAM upgrade (4 minutes)

Say: "Watch the notch â€” it is the safety feature."

Steps:

1. Power off, unplug, press power to discharge, put on the wrist strap.
2. Push the RAM clips outward; remove the old stick by its edges.
3. Hold up the new stick and point to the **notch**. **Say:** "The notch is off-centre. It only goes in one way. If I try the wrong way, it will not seat â€” and I will not force it."
4. Align and press firmly at both ends until the clips snap up. **Say:** "Both clips up = fully seated."
5. Close the case, reconnect power, power on.
6. Show the total in **Task Manager â†' Performance â†' Memory**. **Say:** "Verified. The machine sees the new amount."

Expected results: the machine boots and shows the increased memory.

Common errors to show and correct:

- Trying to force the stick the wrong way (demonstrate stopping).
- Touching the gold contacts.
- Forgetting the wrist strap.

Questions: "What do I do if the stick resists?" (Stop; check orientation/generation; do not force.)

G.4 Demonstration 3 â€” Drive replacement and clone/install (5 minutes)

Say: "This is the upgrade customers love."

Steps:

1. Show the old HDD and the new SSD side by side. **Say:** "Same size, but one has moving parts and one does not. That is the source of the speed."
2. Power off, unplug, discharge, strap on.
3. Disconnect the **SATA data** and **SATA power** cables from the old drive. Point them out clearly.
4. Remove the screws, slide out the old drive, fit the new SSD, secure it, reconnect the SATA data and power cables.
5. **Say:** "Two choices now: **clone** â€” copy everything across â€” or **fresh install**. If I clone, the customer's Windows, programs and files come along. If I install fresh, it is cleaner but I must reinstall everything."

6. If a cloning tool is available, show the first screen (or describe the process): source drive â†' destination drive, start.
7. Otherwise, begin booting the **Windows USB** and show the install screen (full install is Session 9).
8. Close up, power on, and show the new drive detected and CrystalDiskInfo reporting **Good**.

Expected results: the new drive is detected and healthy; the clone/install path is understood.

Common errors to show and correct:

- Not backing up before starting. **Say:** "If the clone fails and we wiped nothing yet, the old drive still has the data. But if we had deleted it first, the customer's files would be gone. Backup first."
- Connecting the data cable to the wrong port or leaving the power cable off.
- Handling the HDD roughly.

Questions: "When would a fresh install be better than cloning?" (When the old Windows is full of junk/malware, or the source drive is barely readable.)

G.5 Demonstration 4 â€" Thermal paste and PSU connector safety (3 minutes)

Steps (thermal paste):

1. Show the dry old paste on a spare CPU/cooler. Clean it with isopropyl alcohol and a cloth.
2. Apply a **pea-sized** dot in the centre. **Say:** "Small. Too much squeezes out and can make things worse."
3. Show how the cooler presses down and spreads it. Reconnect the fan cable.

Steps (PSU connectors â€" narration/short demonstration):

1. Show the **24-pin**, **CPU power**, **SATA power** and **PCIe** connectors. **Say:** "Each shape is different so you cannot easily connect the wrong one â€" but with force you can. Never force."
2. **Say loudly:** "We NEVER open a PSU. Even unplugged it can kill you. We replace it, we do not open it."
3. If demonstrating removal, unplug, discharge, note/photo every cable, disconnect all, remove four screws, slide out, slide in the new one, reconnect all, double-check, power on.

Expected results: learners see correct paste amount and understand PSU connector matching and the danger.

Questions: "Why must we use only the cables that come with the PSU?" (Different models wire the pins differently; mixing can damage parts.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding, each learner (or pair) replaces a part on a practice machine â€" a **RAM upgrade** and/or a **storage drive replacement** â€" and verifies the machine afterwards.

Safety first â€" read aloud before starting:

- **Power off, unplug**, press the power button to discharge, **wear your anti-static wrist strap.**
- Handle RAM and drives **by the edges**; never touch gold contacts.
- **Never open the PSU.** If a PSU task is set, it is a swap only.
- **Back up data before replacing any storage drive.**
- Work on a hard, flat surface; keep screws in a tray.
- Stop and call the instructor if a part resists or if you are unsure.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Compatibility check.** Complete the **Compatibility Check Sheet** for the practice machine: RAM generation and type, storage interface, PSU wattage needed. Confirm with the instructor before proceeding.
2. Power off, unplug, discharge, strap on. Open the case.
3. **RAM upgrade.** Locate the RAM slots. Push the clips and remove the old stick by its edges. Note the **notch**. Align the new stick and press firmly at both ends until the clips snap. Record what you did.
4. **Pause and predict.** "After this, what number will Task Manager show for Memory?" Write your prediction, then power on (after closing up) and check.
5. **Verify RAM.** Close the case, reconnect power, boot. Open **Task Manager â†' Performance â†' Memory** and record the total. Did it match your prediction?

6. **Storage drive replacement.** Power off, unplug, discharge. Disconnect the **SATA data** and **SATA power** cables from the old drive. Remove the screws and slide the old drive out. Fit the new SSD, secure it, and reconnect the **data** and **power** cables.
7. **Pause and predict.** "Will the machine boot straight into Windows, or will it ask for an operating system?" Write your prediction. (If the new drive is blank, there is no Windows on it yet.)
8. **Verify storage.** Power on. Check that the new drive is detected (BIOS and/or Windows Disk Management) and, if Windows is running, that **CrystalDiskInfo** shows the new drive as **Good** with the correct capacity.
9. **Thermal paste (if the instructor sets this task):** remove the CPU cooler, clean the old paste with isopropyl alcohol, apply a **pea-sized** dot, refit the cooler, reconnect the fan cable. Record the CPU temperature before and after using **HWMonitor**.
10. **PSU swap (instructor-supervised only, if time allows):** note every cable; disconnect all; remove four screws; slide out; fit the known-good PSU; reconnect every cable; **double-check against your notes**; power on. Verify POST. **Never open the old PSU.**
11. Complete the **Replacement Service Log**.

Expected results:

- Compatibility sheet completed and confirmed.
- RAM replaced, clips seated, new total visible in Windows.
- Drive replaced, cables connected, new drive detected and healthy (Good in CrystalDiskInfo); the machine either boots (if the drive was cloned/OS installed) or shows an OS prompt (if blank).
- Thermal paste replaced and temperatures checked (if this task was set).
- Machine POSTs and runs; service log completed.

Points to pause on and predict:

- After step 3: "What tells you the stick is fully seated?" (Both clips snap up.)
- After step 6: "Which cable is data and which is power?" (The thinner flat cable is SATA data; the wider multi-coloured connector is SATA power.)
- After step 7: "Why might the machine not boot?" (The new drive is blank until Windows is cloned or installed.)

Troubleshooting while walking the room:

- Machine will not power on â†' check the 24-pin and CPU power are fully seated; check the power cable; check the wall socket.
- No picture / beeps after RAM change â†' the stick may not be fully seated or is the wrong type; reseat it; check the notch.
- New drive not detected â†' check both SATA data and power cables; try a different SATA port; confirm cables are fully pushed in.
- Machine boots to "no operating system" â†' expected if the drive is blank; the OS must be cloned or installed (Session 9).
- Temperatures high after re-paste â†' the cooler may not be seated evenly, or too little paste; re-check the cooler fit.
- Learner forces RAM â†' stop immediately; show the notch; reseat gently.

Completion checklist (learner ticks each):

- ☐ Compatibility Check Sheet completed and confirmed
- ☐ Safety followed: unplugged, discharged, wrist strap worn, no PSU opened
- ☐ RAM removed and new stick fully seated (both clips up)
- ☐ New RAM total verified in Windows
- ☐ Old drive removed, new drive fitted and cables (data + power) connected
- ☐ New drive detected; CrystalDiskInfo shows Good (if Windows available)
- ☐ Thermal paste replaced and temperature verified (if set)
- ☐ PSU swap completed and verified (if set; instructor-supervised)
- ☐ Replacement Service Log completed
- ☐ All tools and screws accounted for; case closed properly

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: Work alone. Read a customer scenario, choose the **compatible** parts, plan the replacement, and write the plan and the verification steps. You do **not** perform a new physical task here â€" you produce the professional **plan** a technician would use. (If time allows, the instructor may let you execute part of your plan on a practice machine.)

Scenario. *Mrs. Nwosu*, who runs a small fashion business from home in Woji, brings in a desktop with the following facts:

- It is an older tower PC. The motherboard label says it supports **DDR4** memory and has **SATA** ports.
- It currently has **4 GB RAM** and a **500 GB mechanical hard drive (HDD)**.
- Windows and the customer's files are on the HDD, which is **healthy** (CrystalDiskInfo: Good).
- Complaint: "It is very slow to start and to open files, and when I open my design program and a browser it freezes."
- She wants to keep **all her files and programs exactly as they are**, and she has a **1 TB external drive she has never used** for backups.
- Her budget is limited.

Requirements:

1. **Choose the RAM:** state the generation, type (DIMM/SO-DIMM), a sensible size, and why. (Remember 32-bit vs 64-bit Windows for capacity.)
2. **Choose the storage:** decide between an **HDD and an SSD**, choose the **interface** (SATA or NVMe) based on her machine, and state the capacity you would recommend.
3. **Decide clone or fresh install**, and justify it using her wish to "keep everything exactly as they are".
4. **List the key steps** of the replacement, in order, including the two safety must-dos at the start.
5. **State your verification steps** â€" how you will prove the machine is fixed after the replacement.
6. **State one backup action** you will take before touching the drive, and why.

Time allowance: 15 minutes.

Constraints:

- You may only use parts the machine supports (DDR4, SATA).
- You must not open the PSU.
- Write in a way you could explain to Mrs. Nwosu in plain language.

Success criteria:

- RAM choice is DDR4, DIMM (desktop), sensible size, correct reasoning.
- Storage choice is a **SATA SSD** (the machine has no NVMe slot), sensible capacity.
- **Clone** is chosen and justified (she wants everything unchanged), with a backup first.
- Steps are in the correct order and include safety.
- Verification is specific (Task Manager memory, CrystalDiskInfo Good, faster boot, no freezing).

Hints available if stuck:

- Hint 1: "Keep everything exactly as they are" points strongly to one of clone/fresh install.
- Hint 2: The machine does not have an NVMe slot, so which SSD type must you choose?
- Hint 3: Before copying or installing anything, what should she do with that unused 1 TB external drive?

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

1. RAM choice. DDR4, **DIMM** (desktop form factor), a sensible **8 GB or 16 GB** stick (or a matching pair). Because the machine has only 4 GB and freezes when several programs are open, more memory will help. Note: if the Windows is **32-bit**, it can only use about 4 GB, so check the Windows type first â€" if 32-bit, explain that upgrading RAM alone will not help much and a 64-bit Windows would be needed. **(Key teaching point: check 32-bit vs 64-bit.)**

2. Storage choice. A **2.5-inch SATA SSD** (the board has SATA ports, not NVMe). Recommend a capacity at least as large as her used space (e.g., 500 GB or 1 TB). Explain that the SSD removes the mechanical bottleneck that makes the HDD slow.

3. Clone. Choose **cloning**, because she wants to keep all her files and programs exactly as they are. (A fresh install would require reinstalling everything and restoring her files.) **But back up first** â€" cloning can fail.

4. Key steps (in order):

1. **Back up** her data to the 1 TB external drive (and/or clone).

2. **Power off, unplug, discharge, wear the anti-static wrist strap.**
3. Open the case; disconnect the SATA data and power cables from the old HDD.
4. Remove the screws, slide out the HDD.
5. Fit the SSD (adapter bracket if needed), secure it, reconnect **SATA data and SATA power**.
6. Close up, power on.
7. **Clone** old HDD â†' new SSD using the cloning tool (source â†' destination).
8. Set the SSD as the boot drive; remove the old HDD (or keep it as a spare after wiping, with the customer's permission).
9. Boot from the SSD and check everything is present.

5. Verification steps. Boot successfully from the new SSD; confirm **Task Manager â†' Performance â†' Memory** shows the new RAM total; confirm **CrystalDiskInfo** shows the SSD as **Good**; measure that start-up and file-opening are clearly faster; run the design program + browser together without freezing; open a few customer files to confirm the clone worked.

6. Backup action. Back up her files to the 1 TB external drive before replacing the drive, because if the clone fails, the files are safe, and because it is simply professional practice: never touch a customer's storage without a backup.

Marking guidance (out of 15):

- Correct DDR4/DIMM RAM choice + sensible size + checks 32/64-bit (3)
- Correct SATA SSD choice + capacity reasoning (3)
- Correct clone decision + justification (3)
- Steps in correct order with safety must-dos (3)
- Specific verification steps (2)
- Backup action stated and justified (1)

Accept any answer that shows correct compatibility reasoning and safety. Penalise choosing NVMe for a SATA-only board, choosing a fresh install while ignoring "keep everything the same", or omitting the backup.

J. REAL-WORLD CASE STUDY

Situation. *Mr. Okafor* runs a small cyber cafÃ© and business centre near a school in Port Harcourt. He has five desktop PCs used for typing, printing and browsing. They are about six years old: each has 4 GB RAM and a 500 GB mechanical hard drive. Customers complain that the machines take "forever" to start and freeze when two people use them. Mr. Okafor is considering buying five new computers, which he cannot really afford.

Relevant facts.

- The hard drives are **healthy** (CrystalDiskInfo: Good) â€" they are not faulty, just slow (old mechanical technology).
- The machines are **short of RAM** for modern browsing and office work.
- The CPUs and other parts are still serviceable.
- A few machines were recently diagnosed with malware, now cleaned.

The decision point. Buy five new computers, or upgrade the five old ones?

Possible approaches.

1. Buy five brand-new desktop computers (expensive).
2. Upgrade each machine: add matching RAM and replace the HDD with a **SATA SSD** (clone the existing system so nothing is relearned).
3. Do nothing and tell customers to be patient (bad for business).

Recommended solution (for this stage). Approach **2**. For each machine: check compatibility (DDR generation, SATA ports), add a matching RAM upgrade, and clone each HDD onto a SATA SSD. The machines will start in seconds instead of minutes, and the RAM upgrade stops the freezing. The cost is a small fraction of five new computers, and each upgrade is a job Mr. Okafor can even pay a CEA graduate to do.

Trade-offs.

- Approach 1 solves everything but costs many times more; the old CPUs would still be fine for cafÃ© work.
- Approach 2 is affordable, dramatic in results, and preserves the existing setups; it requires care (compatibility, backup, cloning) and takes time per machine.

- Approach 3 loses customers.

Professional connection. This is exactly how a real repair business grows: offering a **cost-effective upgrade** that gives the customer most of the benefit of a new machine at a fraction of the price. The certificate holder who can correctly choose compatible parts and clone a drive is the person a cyber cafÃ© owner trusts with five machines.

Transfer exercise (ask the class): "Mr. Okafor asks, 'Can I just put a bigger HDD instead of an SSD?' What do you tell him, and why?"

Suggested answer: "Yes, a bigger HDD will give you more space, but it will **not** fix the slowness â€" an HDD is still mechanical. For speed, the drive type matters more than the size. An SSD of the same or smaller capacity will make the machines far faster. If you need both speed and space, use an SSD for Windows and programs and keep a large HDD for storing files."

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide.

K.1 Machine will not power on after a replacement

- **Symptoms:** no lights, no fans after closing up and pressing power.
- **Likely causes:** the 24-pin or CPU power cable is loose or not reconnected; the PSU switch is off; the wall socket/cable is faulty; a cable was left disconnected during the swap.
- **How to investigate:** re-open and check the 24-pin and CPU power connectors are fully seated; check the PSU switch and power cable; compare against your photo/notes.
- **Safe corrective steps:** reseat connectors; use a working socket/cable. **Never open the PSU.**
- **How to confirm resolved:** the machine POSTs and boots.
- **Prevention:** photograph every cable before removal; double-check all connections before power-on.

K.2 RAM not detected / wrong amount shown

- **Symptoms:** the boot fails, beeps, or Windows shows less memory than fitted.
- **Likely causes:** the stick is not fully seated; wrong orientation (notch); wrong generation; a faulty stick; or the OS is 32-bit and cannot use the full amount.
- **How to investigate:** reseat and press until both clips click; confirm generation; check the OS type (32/64-bit) in Windows.
- **Safe corrective steps:** reseat correctly; replace a faulty stick; explain 32-bit limits to the customer.
- **How to confirm resolved:** Task Manager shows the expected total (minus a small normally-reserved amount).
- **Prevention:** match generation and type on the Compatibility Check Sheet first.

K.3 New drive not detected

- **Symptoms:** BIOS/Windows does not see the new drive.
- **Likely causes:** SATA data or power cable not connected; loose cable; a bad SATA port; a dead drive (rare, if known-good).
- **How to investigate:** check both cables; try a different SATA port; try a known-good cable.
- **Safe corrective steps:** reconnect both cables fully; change port/cable.
- **How to confirm resolved:** the drive appears in BIOS and in Windows Disk Management.
- **Prevention:** label the data vs power cable; push each connector fully home.

K.4 Machine boots to "no operating system" after fitting a blank drive

- **Symptoms:** POST passes, but there is no Windows.
- **Likely cause:** the new drive is blank (expected if not cloned/no OS installed).
- **How to investigate:** confirm the drive is the new one and is empty of an OS.
- **Safe corrective steps: clone** the old drive to the new one, or install Windows (Session 9) and restore data.
- **How to confirm resolved:** Windows boots from the new drive.
- **Prevention:** plan clone vs install *before* fitting; back up first.

K.5 Random shutdowns return after fitting a new PSU

- **Symptoms:** machine still shuts down under load after a PSU swap.
- **Likely causes:** the new PSU is underpowered for the load; a connector is loose; the real cause was overheating or RAM, not the PSU.
- **How to investigate:** check the PSU wattage vs the machine's needs; reseat all connectors; re-check temperatures (HWMonitor) and RAM (memory test).
- **Safe corrective steps:** fit a correctly-rated PSU; fix the actual cause.
- **How to confirm resolved:** the machine runs a heavy load without shutting down.
- **Prevention:** size the PSU correctly; diagnose thoroughly before replacing (Session 5).

K.6 Temperatures worse after re-pasting

- **Symptoms:** CPU temperature higher than before re-pasting.
- **Likely causes:** too little paste; the cooler is not seated evenly; the fan cable is not reconnected; the cooler is loose.
- **How to investigate:** check the cooler is flat and tight; check the fan cable is on the CPU fan header; check the fan spins.
- **Safe corrective steps:** reseat the cooler evenly; add a small correct amount of paste; reconnect the fan cable.
- **How to confirm resolved:** HWMonitor shows normal/lower temperatures.
- **Prevention:** even tightening; pea-sized amount; always reconnect the fan cable.

K.7 Distinguishing likely from confirmed

As always: a replacement is only **confirmed good** after **verification** (POST, OS detection, temperature/health reading, and a short period of real use). Never hand a machine back based on "it powered on once".

L. SESSION ASSIGNMENT

Assignment title: "Compatibility and Replacement Plan for a Real Upgrade"

Purpose: To practise choosing compatible parts, planning a safe replacement, and writing a professional plan a customer could approve.

Realistic scenario: A customer brings you a working but slow desktop (or laptop) and wants it faster **without buying a new computer**. You must produce a written **upgrade plan**: which parts to buy, why they are compatible, how you will fit them, how you will keep their data, and how you will verify the result.

Detailed instructions:

1. Choose a real machine you have access to and permission to inspect (a CEA practice machine or your own). If it is a laptop, note that parts are model-specific.
2. **Compatibility Check Sheet:** record (where known) RAM generation and type, storage interface, available drive bays/M.2 slot, and the PSU wattage needed or fitted.
3. **Choose the parts:** propose a RAM upgrade and/or a storage upgrade (SSD). For each, state **why it is compatible** and what you are matching (generation, interface, connectors).
4. **Decide clone vs fresh install** (for a storage change) and justify it, including the **backup step**.
5. **Write the steps** in order, including the two safety must-dos at the start and the PSU golden rule.
6. **Write the verification plan** â€” exactly how you will prove the machine is fixed.
7. **Write a plain-language paragraph** for the customer explaining what you will do and any risks.

Required deliverables:

- One completed **Compatibility Check Sheet**.
- A written **upgrade and replacement plan** (steps + verification + plain-language summary).
- One completed **Replacement Service Log** (the plan may be a "dry run" if you do not physically perform the upgrade).

Submission format: Written plan and completed sheets (paper or class folder).

Deadline: Before Session 8.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Compatibility Check Sheet	3	Correct generation/type/interface/wattage recorded
RAM choice & reasoning	3	Correct generation, DIMM/SO-DIMM, sensible size; notes 32/64-bit
Storage choice & reasoning	3	Correct interface (SATA/NVMe) for the machine; sensible capacity
Clone vs fresh install + backup	3	Correct choice, justified; backup step included
Replacement steps in order	3	Correct order; safety must-dos; PSU never opened
Verification plan	3	Specific: POST, OS detection, CrystalDiskInfo, memory total, speed/no crashes
Plain-language summary	2	A customer could understand and approve it

Common mistakes to avoid: choosing DDR3 RAM for a DDR4 board; choosing NVMe for a SATA-only board; forgetting the backup; omitting the safety steps; forgetting to verify; forcing a part.

Instructor reference solution (abbreviated example):

> ***Machine:*** *CEA practice desktop, DDR4, SATA ports, no M.2 slot. Currently 4 GB RAM, 500 GB HDD (healthy), 64-bit Windows.* ***Compatibility sheet:*** *RAM = DDR4, DIMM; storage = SATA; PSU needed â‰ˆ (per practice-machine note) modest, no dedicated GPU.* ***RAM choice:*** *add one 8 GB DDR4 DIMM of matching speed (total 12 GB). Compatible: correct generation and desktop form factor; Windows is 64-bit so it will use the memory.* ***Storage choice: 2.5-inch SATA SSD****, 500 GBâ€"1 TB. Compatible: the board has SATA ports; the drive fits a 2.5-inch bay (adapter if needed). Reason for speed: the SSD removes the mechanical HDD bottleneck.* ***Clone vs install: Clone*** *â€" the customer wants everything unchanged; SSD capacity chosen â‰¥ used space on the HDD.* ***Backup*** *to external drive first.* ***Steps:*** *1) Back up data. 2) Power off, unplug, discharge, wear wrist strap. 3) Open case; remove RAM, fit matching stick (notch, clips click). 4) Disconnect SATA data + power from HDD; remove screws; slide out. 5) Fit SSD; reconnect SATA data + power; secure. 6) Close up; power on. 7) Clone HDD â† SSD; set SSD as boot drive. 8) Boot and check. (Never open the PSU.)* ***Verification:*** *POST passes; Task Manager shows 12 GB (minus reserved); CrystalDiskInfo shows Good; boot and file-opening clearly faster; run design program + browser together without freezing; open sample files to confirm the clone.* ***Plain-language summary:*** *"We will add more memory so the computer can run several programs at once, and replace the old spinning hard drive with a modern solid-state drive so it starts in seconds. We will copy everything across exactly as it is first, and back up your files before we begin, so nothing is lost."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer. What does it mean for a replacement part to be **compatible**? **Q2. Multiple choice.** A motherboard supports **DDR4** memory. Which RAM can you fit? a) DDR3 b) DDR4 c) DDR5 d) Any of them, RAM always fits **Q3. Explain in your own words.** Why does replacing an old mechanical hard drive with a solid-state drive make a computer faster, even though the CPU and RAM are unchanged? **Q4. Practical instruction.** What are the two safety must-dos before you open a computer to replace a part? And what is the one thing you must **never** open? **Q5. Short answer.** Give one reason to **clone** a drive and one reason to do a **fresh install** instead. **Q6. Scenario.** After replacing RAM, the machine beeps and shows nothing on screen. What is the most likely cause, and what do you do? (Remember the golden rule about forcing.)

INSTRUCTOR ANSWER KEY (Q1â€"Q6)

A1. It means the part is of the correct **type and size** to work with that specific machine â€" matching, for example, RAM generation/form factor, storage interface, or PSU wattage and connectors â€" so it will fit, connect and work (and not damage anything).

A2. (b) DDR4. DDR3 and DDR5 are different shapes and cannot be fitted.

A3. An HDD must physically spin and move a reading head to find data, so every read â€" and Windows makes thousands of small reads when starting up and opening files â€" waits for mechanical parts. An SSD has no moving parts and finds data

almost instantly. Removing that mechanical wait is what makes the machine feel new; the CPU and RAM did not change.

A4. The two must-dos: **power off and unplug** (and press power to discharge), and **wear an anti-static wrist strap** (ESD protection). The one thing you must **never** open is the **PSU** (power supply) â€" its capacitors hold a lethal charge even when unplugged.

A5. Clone: when the old drive is healthy and the customer wants Windows, programs and files kept exactly as they are (saves reinstallation). **Fresh install:** when the old Windows is full of junk/malware, or the old drive is failing and barely readable, so a clean start is safer.

A6. Most likely the RAM stick is **not fully seated**, or it is the **wrong type/orientation**. Do **not** force it. Power off and unplug, reseat the stick correctly (check the **notch** and press until both clips click), confirm it is the correct generation, then power on and re-check.

Marking: award for correct concept; deduct heavily for unsafe answers (opening the PSU, forcing parts).

N. REFLECTION AND RECAP

Plain-English summary. Today you learned to **replace components safely** â€" the skill that turns a diagnosis into a repair. You learned that every part must be **compatible**: RAM must match the motherboard's **generation** (DDR3/4/5), type (DIMM for desktop, SO-DIMM for laptop), size and speed; storage must match the **interface** (**SATA** uses cables, **NVMe** uses an M.2 slot); and a PSU must have enough **wattage** and the right **connectors**. You learned to upgrade **RAM** by seating it on the **notch** until the clips click; to replace a **drive** and choose between **cloning** (copy everything) and a **fresh install** (clean start) â€" always **backing up first**; to replace a **PSU** by reconnecting **every** cable and double-checking, while **never opening** it; and to re-apply **thermal paste** with a **pea-sized** amount. Finally, you learned that a replacement is only done when the machine is **verified** working.

Five important takeaways:

1. **Compatibility first** â€" matching saves money and prevents damage.
2. **An SSD removes the mechanical bottleneck** and makes old PCs feel new.
3. **Back up before replacing storage**; choose clone vs fresh install on purpose.
4. **Never open a PSU**; replace it and double-check every cable.
5. **Verify** every replacement before returning the machine.

Questions to help you recognise what you can now do:

- Can I check whether a RAM stick is compatible before buying it?
- Can I fit RAM and confirm the new amount in Windows?
- Can I choose between clone and fresh install and explain why?
- Can I replace a PSU safely and reconnect every cable?
- Can I re-paste a CPU and confirm the temperature improved?

Safety self-check (tick honestly):

- ☐ I powered off, unplugged and discharged before opening the machine.
- ☐ I wore my anti-static wrist strap.
- ☐ I never opened a PSU.
- ☐ I backed up before touching a storage drive.
- ☐ I did not force any part, and I verified the machine after every replacement.

Connection to the next session. You can now replace parts. Next session, Session 8 â€" Cleaning and Maintenance, you will learn to **keep** computers healthy: safe dust removal, ESD care, cleaning fans, heatsinks, keyboards, ports and screens, when to re-apply thermal paste, and how to build **preventive maintenance schedules** for homes, small offices and cyber cafÃ©s, including protection from dust, humidity and power surges (surge protectors and UPS). Prevention is the cheapest repair of all.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner checks **compatibility** (generation, interface, wattage) *before* fitting anything.
- The learner seats RAM by the **notch** and confirms both clips are up, without forcing.
- The learner correctly identifies SATA **data** vs **power** cables and reconnects both.
- The learner chooses clone or fresh install with a clear reason and **backs up first**.
- The learner reconnects **every** PSU cable and double-checks before power-on.
- The learner **verifies** (POST, OS detection, CrystalDiskInfo, temperature) before declaring success.
- The learner automatically follows ESD and power-off safety and refuses to open a PSU.

Common misconceptions to watch for and correct:

- "Any RAM fits any PC." â†' DDR3/4/5 are different shapes; DIMM â‰ SO-DIMM.
- "More RAM always makes it faster." â†' Only if the machine is short of memory.
- "An SSD cannot fail." â†' It can; back up regardless.
- "Cloning always works." â†' It can fail; back up first.
- "A bigger PSU is always more reliable." â†' Wattage is capacity, not health.
- "More thermal paste is better." â†' A pea-sized amount is correct.

Suggested remedial exercises for learners who struggle:

1. **Compatibility sorting:** give ten part cards and ten machine cards; have the learner match them and explain each match.
2. **Cable tracing:** on a real machine, have the learner point to and name SATA data, SATA power, 24-pin, CPU power and PCIe power connectors.
3. **Order-the-steps drill:** shuffle the RAM and PSU replacement steps and have the learner reorder them, including the safety steps.

Differentiation:

- **Level 1 (supported):** provide a partially completed Compatibility Check Sheet and step card; the learner completes the RAM upgrade with hands-on help.
- **Level 2 (standard):** the exercise as written (RAM + drive).
- **Level 3 (challenge):** perform the full **clone** to an SSD *and* the RAM upgrade, then verify both; or replace a PSU and GPU as well, documenting every step on the service log.

Timing guidance: This is a hands-on, safety-critical session. Protect the guided practical above all. If time is short, do the **RAM upgrade** as the core practical and demonstrate the drive replacement while learners watch; move the PSU swap to an instructor-supervised demonstration only. Never rush a safety step. Cloning can be slow â€" start it early or demonstrate the process and let learners complete it in Session 9 if needed.

APPENDIX 7A â€" 10-MINUTE SAFETY REFRESHER (only if needed)

If learners cannot demonstrate safe case-opening and ESD precautions, run this before Section D.

1. **The two must-dos** â€" the class chants them: **"Power off and unplug. Wear the anti-static wrist strap."**
2. **Discharge** â€" press the power button after unplugging to drain leftover electricity.
3. **The golden rule** â€" say it three times: **"Never open the PSU. Even unplugged, it can kill you."**
4. **Handle by the edges** â€" never touch gold contacts on RAM or a GPU.
5. **Screws and trays** â€" keep screws in a tray; know where each goes. Demonstrate once, then have each learner demonstrate the two must-dos before they touch a machine.

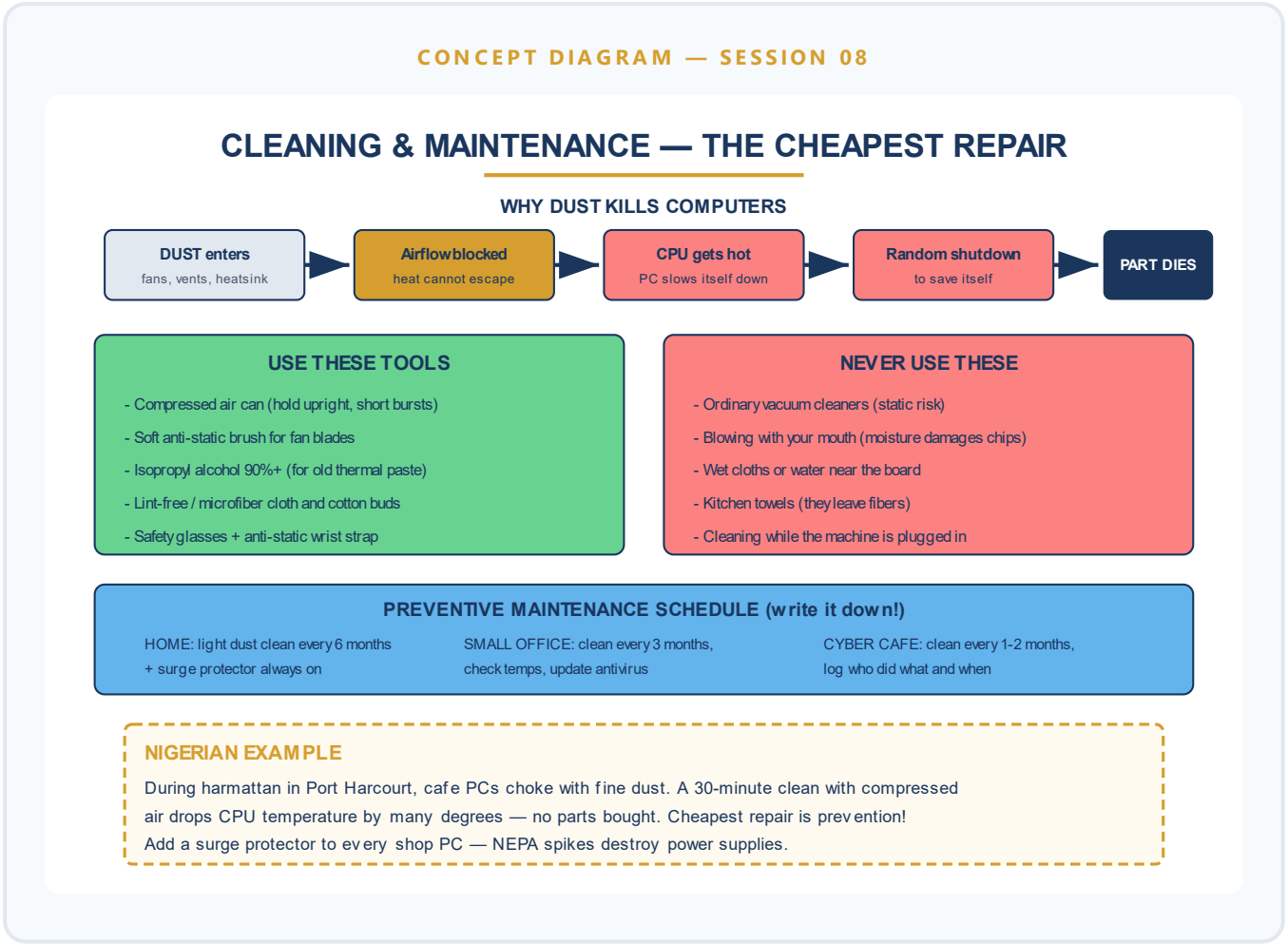
END OF SESSION 7

Next: Session 8 â€" Cleaning and Maintenance.

SESSION 08 OF 12

Cleaning Maintenance

CONCEPT DIAGRAM — SESSION 08



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 8 â€" CLEANING AND MAINTENANCE

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	8 of 12
Session title	Cleaning and Maintenance
Week / Module	Week 3 â€" Module: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (hands-on, safety-critical)

Item	Detail
Prerequisite knowledge	Sessions 1â€“7: components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware and software diagnostics, and component replacement (including thermal paste and safe handling). If a learner cannot demonstrate the two safety must-dos, run the 10-minute refresher in Appendix 8A before this session.
Connection to previous work	Session 7 taught you to **replace** parts. This session teaches you to **prevent** faults â€” keeping a machine cool, clean and protected so it lasts longer and costs the customer less. Cleaning is the cheapest repair of all.
Preparation before class	A deliberately **dusty practice desktop** (or one that can be made dusty for demonstration); canned **compressed air**; a **soft anti-static brush** (or a clean, soft paintbrush); **isopropyl alcohol** (90%+); lint-free/microfiber cloths; **cotton buds**; thermal paste; a small **screwdriver and anti-static wrist straps** (one per learner); HWMonitor installed; a **surge protector** and, if available, a **UPS** to show; an old keyboard and an old monitor/phone screen for safe cleaning practice; printed **Cleaning Checklist** and **Maintenance Plan Template** (one per learner). Optional: a copy of a PC cleaning video is not required â€” keep it live.

A note on what cleaning means here. Cleaning a computer is not just making it *look* nice. It is a **maintenance action** that removes the main causes of overheating (dust, blocked airflow, dried thermal paste) and protects the machine from environmental risks (humidity, power surges). Done wrong, cleaning can damage a computer with **static electricity** or **liquid**; done right, it extends a machine's life by years.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** how dust, heat and blocked airflow damage a computer, and how these lead to slowness, shutdowns and part failure.
2. **List** the correct cleaning tools and **state** which tools are unsafe (e.g., ordinary vacuum cleaners).
3. **Perform** a safe internal clean â€” powering off, unplugging, using ESD care, and using compressed air, brushes and isopropyl alcohol correctly.
4. **Clean** fans, heatsinks, the case interior, keyboard, ports and screens safely and appropriately for each surface.
5. **Explain** when to re-apply thermal paste and how often, and recognise the signs (rising temperatures, dust-clogged heatsink).
6. **Describe** the environmental threats to a computer â€” dust, humidity, heat and power surges â€” and how a **surge protector** and a **UPS** help.
7. **Create** a practical **preventive maintenance schedule** for a home, a small office and a cyber cafÃ©, including who does what and how it is documented.
8. **Produce** a written **maintenance recommendation** for a real machine after cleaning and a cooling check.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one **dusty practice desktop** safe to open; ideally a second, cleaner machine for comparison.
- **Compressed air** cans (one or more). Keep them upright; do not let learners shake or tilt them.
- A **soft anti-static brush** (or clean, soft paintbrush) for loosening dust from fan blades and heatsink fins.
- **Isopropyl alcohol (90% or higher)** â€” for cleaning old thermal paste and stubborn grime.
- **Microfiber / lint-free cloths** and **cotton buds**.
- **Thermal paste** (a little goes a long way).
- **Anti-static wrist straps** (one per learner), a small screwdriver set, a parts tray.
- An **old keyboard**, an **old monitor or phone screen**, and (optionally) an old keyboard with removable keys â€” for practising safe external cleaning.

- A **surge protector** and, if available, a **UPS** to demonstrate.
- **Safety glasses** and a **first-aid kit**.

Software

- **HWMonitor** (from Sessions 5â€“6) to record temperatures before and after cleaning.

Printed materials

- Cleaning Checklist (one per learner).
- Maintenance Plan Template (one per learner).

Safety equipment and environment

- Anti-static wrist straps, safety glasses, first-aid kit.
- Well-ventilated area for using compressed air (dust flies); avoid blowing dust into the room.
- Know the location of the **fire extinguisher** and the **first-aid kit**.

Internet

- Not required. Optionally, the instructor may look up a specific motherboard's clean-airflow guidance, but do not depend on it.

Alternative for learners who lack a facility

- Any working computer can be cleaned. A desktop gives the fullest practice (internal dust). Learners who own only a laptop may practise **external** cleaning (vents, keyboard, screen, ports) safely; internal laptop cleaning is more delicate and should be instructor-led beside them.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: On the bench, place the **dusty practice desktop** with its side panel removed so the class can see the inside. Beside it, place a clean, dust-free machine (or a clean fan/heatsink) for contrast. Have HWMonitor ready on the dusty machine (or a note of its earlier temperature).

What the instructor says:

> *"Good morning. Look inside this computer. What do you see? Dust. A grey blanket of dust sitting on the heatsink, in the fan blades, in every corner.*
>
> *Now here is the important part. Dust does not just look bad. That dust blanket is a **heat trap**. Heat cannot escape. The CPU gets hot, so the computer slows itself down on purpose â€" that is why a dusty machine feels slow. If it gets hotter still, it shuts itself off to save its own life. And year after year, that heat quietly shortens the life of every part inside.*
>
> *In Nigeria, in Port Harcourt, with our heat and harmattan dust, this is one of the most common reasons a computer dies young. But here is the good news: cleaning is the **cheapest repair** there is. No parts to buy. Just proper tools, proper care, and a habit of doing it regularly.*
>
> *Today you will learn to clean a computer **safely** â€" because done wrongly, cleaning can damage a computer with static or liquid â€" and you will learn to build a **maintenance schedule** so the computer never gets this bad again."*

Expected learner response: Learners often say "just blow it with the mouth" or "use a cloth" or "use a vacuum cleaner". Accept these and note them on the board. Say:

> *"By the end of today you will know exactly why a vacuum cleaner is dangerous, why blowing with your mouth is a bad idea (moisture), and what to use instead."*

Then demonstrate the hook: on the compressor or with a canned air, give one short, controlled puff across the dusty heatsink over a bin or outside airflow. The cloud of dust that rises is the moment the lesson lands.

Why this opener works: It shows the real, physical enemy (dust) instead of describing it, it names a very common Nigerian reality (heat + dust), and it challenges several common bad habits (vacuuming, blowing) with a promise to teach the correct tool.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1: "Name the six major components."

- Expected answer: **Motherboard, CPU, RAM, storage, PSU, GPU.**
- Common gap: forgetting the PSU. Probe and move on.

Question 2: "What are the two safety must-dos before opening a computer?"

- Expected answer: **Power off and unplug** (press power to discharge); **wear an anti-static wrist strap.**
- Common gap: forgetting the discharge step. Correct it â€" cleaning involves moving air and cloths, so static and loose power are real risks.

Question 3: "What is ESD, and why does it matter when cleaning?"

- Expected answer: **Electrostatic discharge** â€" a tiny static shock from your body or a cloth that can silently damage components. Cloths and brushes rub and build static, so we must ground ourselves and use anti-static tools.
- Common gap: learners think ESD is only a concern when replacing parts. Make clear it is *especially* relevant to cleaning, because rubbing a cloth on a case builds static.

Question 4: "When do we re-apply thermal paste, and what amount do we use?"

- Expected answer: when the CPU and cooler are separated, or when temperatures rise despite clean fans; use a **pea-sized** amount. (From Session 7.)

Question 5: "A computer shuts down after 30 minutes and the case is hot. What are two likely causes?"

- Expected answer: **overheating** (dust, a failed fan, dried thermal paste) and possibly a **failing PSU**. This bridges directly into today.

Bridge to today (say this):

> *"Session 5 taught you to find overheating with tools. Session 7 taught you to replace parts. Today, you learn to prevent the overheating in the first place. A surgeon can operate; a good doctor keeps you from needing the operation."*

Gap handling: If learners cannot state the two safety must-dos or explain ESD (Questions 2â€"3), stop and run the Appendix 8A refresher now. Cleaning involves rubbing and air, which build static, so this session cannot proceed safely without it.

F. FULL LECTURE NOTES

F.1 Why dust and heat destroy computers

The idea. A computer produces heat when it works and must get rid of that heat. Dust blocks the path heat takes to escape. Block the path, and the heat builds up â€" and heat is the slow killer of electronics.

Plain-English definitions.

- **Dust** is a mixture of tiny particles â€" skin, cloth fibres, and in Nigeria especially **harmattan dust** and road dust. It settles on every surface inside a computer.
- **Heat** is a by-product of electricity moving through components. It must be carried away by the cooling system (fans, heatsinks, airflow) and out of the case.
- A **heatsink** is a block of metal fins attached to a hot chip (like the CPU). The metal absorbs heat from the chip, and the fins give a large surface area for a **fan** to blow air across and carry the heat away.

Why dust causes damage â€" the chain of events:

1. Dust settles on fan blades and between heatsink fins.
2. The dust "felt" blocks airflow and insulates the metal, so heat cannot escape.
3. The CPU/GPU gets hotter than it should.

4. The computer **slows itself down on purpose** to make less heat (this is called **thermal throttling**) â€" so a dusty computer *feels slow*.
5. If it gets hotter still, the computer **shuts itself off** suddenly to protect the chip (a **thermal shutdown**) â€" which looks like random restarts or shutdowns.
6. Over months and years, the constant high heat **shortens the life** of the chip, the drive and other parts.

Analogy. Imagine doing hard labour in Port Harcourt midday heat while wrapped in a thick blanket. You cannot work well, you feel weak, and if you keep going you collapse. The dust blanket does exactly this to a computer.

Realistic Port Harcourt scenario. A cyber cafÃ© near Mile 1 has machines that keep shutting down during busy evenings. The cafÃ© owner thinks the machines are "old and spoiled". A technician opens one and finds the CPU heatsink completely packed with dust â€" the fins look like a carpet. After cleaning, the machines run cool and no longer shut down. The owner was about to buy new computers; he only needed a cleaning.

Important misconception. "Cleaning is just to make it look nice." No â€" the purpose is **cooling and reliability**. A machine can look clean on the outside and be clogged inside.

Limitation. Cleaning cannot reverse damage already done. If a fan bearing is worn out or thermal paste has dried and temperatures are already very high, cleaning helps a lot but may need to be combined with a **fan replacement** or **re-pasting** (Session 7). And cleaning will not fix a computer whose real problem is software (Session 6) or a failing part (Session 5).

F.2 Safe cleaning tools â€" and the dangerous ones

The idea. Not every cleaning method is safe. Some tools can destroy a computer. Know which is which.

SAFE tools:

- **Compressed air** (canned air or a compressor with a moisture trap) â€" for blowing loose dust out of heatsinks, fans and the case. Hold the can **upright**; short bursts; keep the nozzle a little distance away; never let it blow liquid.
- **Antistatic brush / soft paintbrush** â€" for gently loosening dust from fan blades and heatsink fins, and for brushing the motherboard surface.
- **Isopropyl alcohol (90%+)** on a **lint-free cloth or cotton bud** â€" for stubborn grime and for cleaning old thermal paste. It evaporates quickly and does not conduct electricity when dry.
- **Microfiber / lint-free cloths** â€" for wiping the case interior, screens and exterior.
- **Cotton buds** â€" for small crevices and ports.
- **Anti-static wrist strap** â€" always, to protect against ESD.

DANGEROUS / UNSAFE tools:

- **Ordinary vacuum cleaner** â€" **never** use inside a computer. The plastic nozzle and the moving air **build static electricity**, which can silently destroy components. A vacuum can also physically suck small parts and jumpers off the board. (Special anti-static vacuums exist for professional use; the ordinary household one does not.)
- **Wet cloths / water** â€" water conducts electricity and causes corrosion. Never use water inside a computer.
- **Your own mouth / blowing with breath** â€" your breath carries **moisture** (which can cause corrosion) and saliva droplets; use compressed air instead.
- **Sharp metal tools** â€" can scratch circuit traces and short connections.
- **Household cleaning sprays** (e.g., strong surface cleaners, bleach solutions) â€" can leave residues that damage components and screens. Use only approved screen cleaners or a barely-damp microfiber cloth.

Why compressed air can is safer than a vacuum. A can of compressed air is a **sealed, non-sparking** source; it does not generate static the way a vacuum's plastic nozzle does. Keep the can upright so it does not spray its liquid propellant onto the parts.

Important misconception. "A vacuum cleaner is fine if I do not touch the parts." The **static** builds on the nozzle through the moving air and can discharge onto the board even without contact. The safe rule is: **compressed air inside, vacuum only outside and never near open electronics.**

Limitation. Compressed air can push dust *deeper* into crevices if used carelessly. Blow dust **outward, toward the exit**, and work in a well-ventilated place, or ideally take the machine outside or near a window so the dust leaves the room.

F.3 The safe internal cleaning procedure

Safety first â€” read aloud and follow every time:

- **Power off, unplug** the power cable, then press the power button for a second to discharge leftover electricity.
- **Wear an anti-static wrist strap.**
- Use only **safe tools** (F.2).
- Work in a **well-ventilated area**; dust will fly.
- **Never open the PSU.**
- Handle components by the edges; do not spin fans with the air jet.
- If anything smells burnt or you see a swollen battery, **stop and call the instructor**.

Step-by-step procedure:

1. **Prepare the workspace.** Unplug everything. Move the machine to a hard, flat surface in a ventilated area (outside if possible). Put on the wrist strap.
2. **Open the case.** Remove the side panel (method depends on the case; the instructor demonstrates on the practice machine).
3. **Loosen, then blow.** Use the **brush** first to gently loosen the dust "carpet" on the heatsink fins and fan blades. Then use **compressed air** (short bursts, can upright) to blow the loosened dust **out of the case**.
4. **Clean the heatsink and fans.** Hold each fan blade **steady** with a finger or a soft object so it does **not spin** while you blow â€” spinning a fan with air can damage its bearing and can generate a tiny reverse current. Brush and blow the fins clean.
5. **Clean the case interior.** Wipe surfaces with a **microfiber cloth**; use a brush for corners. Clean the **air filters** if the case has them (many can be removed and rinsed and dried â€” only refit when fully dry).
6. **Clean the connectors and slots lightly.** Use a brush or a short gentle burst of air on RAM slots and expansion slots. Do not push dust into them.
7. **Inspect while you are in there.** Look for: a fan that does not spin when the machine runs; bulging capacitors; a swollen laptop battery; burnt smells; loose cables.
8. **Reassemble.** Replace the side panel, reconnect only the power cable and needed cables, and power on.
9. **Verify and record.** Open **HWMonitor** and record the CPU/GPU temperatures. Compare with the "before" reading. A drop of several degrees is a clear success.

Analogy. Cleaning the inside is like clearing a blocked drainage channel before the rainy season. You do not wait for the flood; you clear the channel so the water (heat) can flow away freely.

Important misconception. "I should spin the fan with the air jet to clean it faster." No â€” forced spinning damages the fan bearing. Hold the blade steady.

Limitation. Some heatsinks are so clogged that air alone cannot clear them; the cooler may need to be **removed** to clean properly (and then thermal paste re-applied on refitting â€” Session 7). Some laptops require deeper disassembly to reach the fan; that is more delicate work and should be supervised.

F.4 Cleaning keyboards, ports and screens

The idea. Not everything can be cleaned the same way. Blowing compressed air into a keyboard's electronics area, or spraying liquid on a screen, can cause damage. Match the method to the surface.

Keyboard (external cleaning, safe for learners):

- Power off the computer.
- Tip the keyboard upside down and gently shake out crumbs.
- Use **short bursts of compressed air** along the rows of keys, at a slight angle, to lift dust and crumbs out.
- Wipe the key tops and the frame with a **barely-damp microfiber cloth** (dampened with a little isopropyl alcohol or plain water â€” **not dripping**). Never spray liquid directly onto the keyboard.
- Between keys, use a **cotton bud**. Do not force it under keys and do not bend key mechanisms.
- **Removable-key keyboards:** the instructor may show removing a keycap to clean beneath; **do not** remove keys on a laptop keyboard (they break easily).

Ports (USB, HDMI, charging ports):

- Power off and unplug.

- Use **short bursts of compressed air** to blow dust out of the port. Do **not** insert sharp metal objects, which can bend the internal contacts.
- For stubborn dust, a **soft dry brush** or a **wooden toothpick** used *very* gently along the edge â€" never force anything into the port.
- **Never** put liquid into a port.

Screens (monitors, laptop screens, phone screens):

- Power off the screen (this makes dust easier to see and is safer).
- Use a **microfiber cloth**, dry for light dust, or **very slightly** dampened with **plain water or an approved screen cleaner** for marks.
- **Never spray liquid directly on the screen** â€" spray onto the cloth, not the screen, and never let liquid run into the edges.
- Use **gentle, circular** motion with light pressure. Do not press hard â€" screens are fragile.
- **No** paper towels (they scratch), **no** household glass cleaner (ammonia can damage screen coatings), **no** alcohol on screens unless the manufacturer says it is safe for that screen (alcohol can damage some anti-glare coatings).

Analogy. Different surfaces need different care, like cleaning a pair of glasses versus cleaning a car engine â€" same idea (remove dirt), completely different method.

Important misconception. "More liquid cleans better." For screens, less is more. Excess liquid is the main cause of screen and keyboard damage during cleaning.

Limitation. Some marks (scratches, screen-print wear) cannot be cleaned away. On laptops, deep cleaning behind the keyboard requires disassembly and is not a beginner task.

F.5 When to re-apply thermal paste

Plain-English definition. Thermal paste (thermal compound) is the grey paste between the CPU and its heatsink (cooler). It fills the microscopic gaps between the two metal surfaces so heat passes efficiently from the chip into the cooler.

When to re-apply:

- Whenever the **CPU and cooler have been separated** (for example, during a deep clean where the cooler is removed, or a cooler replacement).
- When temperatures have **crept up over the years** despite clean fans and good airflow.
- On a machine that has run hot for a long time and shows signs of dried, cracked paste.

How often (a guide, not a strict rule): many technicians recommend re-pasting roughly **every 1â€"2 years**, or whenever the cooler is removed. For a well-cooled home PC it may last longer; for a hot, dusty cyber-cafÃ© machine that is opened regularly, more often. Judge by **temperature readings**, not only by a calendar.

Signs the paste needs replacing:

- CPU temperatures are high even after cleaning dust and confirming fans spin.
- The old paste, when the cooler is lifted, looks dry, cracked or powdery (instead of soft and grey).
- Fan noise is high and the machine still runs hot.

The steps (recap from Session 7):

1. Power off, unplug, discharge, wrist strap on.
2. Remove the cooler carefully (do not twist if the CPU is on an older socket design where the CPU may come away stuck to the cooler â€" ask the instructor).
3. **Clean** old paste from CPU and cooler base with **isopropyl alcohol** and a lint-free cloth/cotton bud until both are clean and dry.
4. Apply a **pea-sized** dot of new paste in the centre.
5. Refit the cooler, tightening screws **evenly** (a little each, in turn).
6. Reconnect the **fan cable**.
7. Power on and verify temperatures with **HWMonitor** (should be normal or lower).

Analogy. Thermal paste is like a thin layer of oil between a pot and a hot plate. The pot and plate do not touch perfectly; the oil fills the tiny gaps so heat moves evenly. Too little and heat does not travel; too much and it makes a mess.

Important misconception. "Thermal paste lasts forever." It dries out over the years and loses effectiveness. Also, "more is better" is false â€” a **pea-sized** amount is correct.

Limitation. Not every high temperature is a paste problem. Sometimes the real issue is a worn-out cooler, a failed fan, blocked vents or an underpowered case airflow. Diagnose the whole cooling system, not just the paste.

F.6 Environmental threats â€” dust, humidity, heat and power

The idea. Cleaning protects the *inside* of the computer. But the *environment* around the computer â€” dust, humidity, heat and unstable electricity â€” can also destroy it. Managing the environment is part of maintenance.

Dust (already covered). Keep computers off the floor where dust gathers, away from windows where harmattan dust blows in, and out of the kitchen area (cooking smoke and oil stick to fans and make dust worse).

Humidity and moisture.

- **High humidity** (common in Port Harcourt during the rainy season) can allow moisture to condense on cold components, causing corrosion and short circuits over time.
- **Low humidity / dryness** (common in the harmattan) makes static electricity build up more easily â€” increasing ESD risk.
- Practical measures: keep machines in dry, ventilated rooms; avoid placing them where they get wet; allow a cold machine (e.g., brought in from the rain) to reach room temperature before powering it on; use a dehumidifier in severe cases (a professional/household solution, not essential for beginners).

Heat.

- Ambient (room) heat directly raises component temperatures. Keep computers out of direct sun, away from hot spots, and ensure the room is ventilated.
- Ensure the case has clear **intake and exhaust** vents (not against a wall or curtain) so air can flow.

Power problems â€” surges, sags and outages.

- **Surge:** a sudden spike in voltage (e.g., when power returns after an outage, or from lightning). A surge can destroy the PSU and components instantly.
- **Sag/brownout:** voltage dropping below normal (common where supply is weak). Can cause crashes, restarts and data corruption.
- **Outage:** sudden loss of power â€” can corrupt files, especially during updates.

Protection devices:

- A **surge protector** (a power strip with surge protection) absorbs sudden voltage spikes. It helps protect against surges but does **not** keep the computer running during an outage.
- A **UPS (Uninterruptible Power Supply)** is a battery backup box that keeps the computer running for a few minutes during an outage, giving time to save work and shut down safely. It also helps with sags/brownouts. It is especially valuable for cyber cafÃ©s, offices and any machine doing important work.
- **Note:** a surge protector must be replaced after a serious surge; many have an indicator light that shows when protection is lost.

Analogy. Electricity is like water coming through a pipe. A **surge** is a pressure burst that can burst the pipe; a **sag** is a drop in pressure that stalls the machines; an **outage** is the water being cut. A surge protector is a shock absorber; a UPS is a small reserve tank.

Realistic Port Harcourt scenario. A cyber cafÃ© loses power suddenly while a customer is midway through printing a document and another is saving a school project. Without a UPS, the print job is spoilt and the saved project may be corrupted. With a UPS on the key machines, there is time to save and shut down.

Important misconception. "A surge protector also keeps the computer on during a blackout." No â€” that is a **UPS**'s job. A surge protector only guards against spikes.

Limitation. No device guarantees 100% protection â€” lightning can overwhelm any protector. Good practice also includes **regular backups** (so a power problem can never destroy the only copy of the customer's work) and properly **grounded** wall sockets.

F.7 Preventive maintenance schedules

The idea. Instead of waiting for a machine to fail, we perform small tasks on a **schedule**. This keeps computers cool, healthy and reliable â€" and it is a service a technician can sell.

Plain-English definition. A **preventive maintenance schedule** is a plan listing what to check, how often, and who is responsible, so that small problems are caught before they become big ones.

Why it exists. Most computer failures are slow, not sudden: dust builds, paste dries, drives age, disks fill up. A schedule catches these while they are still cheap to fix.

A general schedule (adapt to the machine and environment):

Frequency	Task	Tool / method
Monthly	Run a full antivirus scan; check for Windows updates; run Disk Cleanup; clear the Recycle Bin	Windows tools (Session 6)
Every 3 months	Clean internal dust (compressed air, brush); check all fans spin; check temperatures	HWMonitor; compressed air
Every 6 months	Check drive health; run a memory test; check air filters; tighten/verify airflow; check surge protector	CrystalDiskInfo; Memory Diagnostic
Annually	Full diagnostic check of all components; re-apply thermal paste if temperatures warrant it; review and archive old files; check backups work	HWMonitor; paste; backup check
After events	Clean and inspect after a major dust event, a lightning/surge event, a move, or a spill	Visual inspection

Schedules for different settings:

- **Home computer:** the general schedule above, plus a note to keep it off the floor and away from windows, and to back up important photos/documents.
- **Small office (a few computers):** add a written record per machine; assign one person responsible; use a **UPS** on machines doing critical work; back up shared files regularly.
- **Cyber cafÃ© (many machines, heavy use, lots of dust):** clean **more often** (monthly internal cleaning may be needed in dusty areas); use surge protectors at minimum and UPS on key machines; keep a simple **logbook** per machine recording cleaning and faults; train staff to report symptoms early (noisy fans, shutdowns).

Documentation. Every task should be **recorded** â€" a simple logbook or table with the date, the machine, the task done, the temperature readings, and who did it. Documentation turns "I think I cleaned it" into **proof**, helps spot machines that run hotter than others, and looks professional to a customer.

Analogy. A maintenance schedule is like a car service book. You do not wait for the engine to seize; you change the oil on time. The computer is the same.

Important misconception. "If it is working, it does not need maintenance." Working machines are exactly the ones worth protecting â€" prevention keeps them working.

Limitation. Schedules must match the environment. A cyber cafÃ© in a dusty area needs far more frequent cleaning than a home office in a sealed, air-conditioned room. A schedule copied blindly from the internet will be wrong; adapt it.

F.8 Summary of the session

- **Dust + heat** cause slowness (thermal throttling), random shutdowns (thermal shutdown) and early part failure.
- **Safe tools:** compressed air, antistatic/soft brush, isopropyl alcohol, microfiber cloth, cotton buds, wrist strap. **Unsafe:** vacuum cleaner (static), water, breath (moisture), harsh sprays.
- **Internal clean:** power off, unplug, discharge, strap on; loosen with a brush, blow outward with compressed air; hold fans steady; inspect; verify with HWMonitor.
- **Keyboards/ports/screens:** air for keys/ports; barely-damp microfiber for screens, never spray on the screen; never liquid in ports.
- **Thermal paste:** re-apply when the cooler is removed or temperatures creep up (roughly every 1â€"2 years); **pea-sized** amount.
- **Environment:** manage dust, humidity, heat and power; use a **surge protector** for spikes and a **UPS** for outages/sags.
- **Preventive schedules** for home, office and cafÃ©; **document** every task.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The dusty practice desktop, opened on the bench.
- Compressed air, antistatic brush, isopropyl alcohol, microfiber cloths, cotton buds.
- An old keyboard and an old monitor/phone screen for cleaning practice.
- A surge protector and a UPS.
- HWMonitor ready; a note of the dusty machine's "before" temperature.
- The Cleaning Checklist and Maintenance Plan Template on the projector.

G.2 Demonstration 1 â€” Safe internal clean (6 minutes)

Say: "Watch the order: loosen, then blow outward, and never let the fan spin."

Steps:

1. Power off, unplug, press power to discharge, put on the wrist strap.
2. Remove the side panel. Point out the dust on the heatsink and fans.
3. Use the **brush** to gently loosen the dust on the heatsink fins and fan blades. **Say:** "I loosen it first. Dry dust sticks; brushing frees it, then the air can carry it away."
4. Take the can of **compressed air**, hold it **upright**, and give **short bursts** to blow the dust **outward** (toward the open side / out of the case). **Say:** "Upright â€” if I tilt it, it can spray cold liquid onto the parts. Short bursts, not one long blast."
5. Hold a fan blade steady while blowing it. **Say:** "If I let the air spin the fan, I can damage the bearing. Free-wheel spinning is bad for fans."
6. Wipe the case interior with a **microfiber cloth**; point out the air filter and how to clean it.

Expected results: visible dust removed; heatsink fins visibly clearer.

Reasoning to say aloud: "Loosen, blow outward, protect the fans â€” that is the whole technique."

Common errors to show and correct:

- Tilting the can (show the warning) â†’ liquid risk.
- Blowing dust *into* the machine instead of out.
- Spinning the fan with air.
- Using a vacuum cleaner â€” hold it up and **say loudly: "Never this, inside a computer â€” static!"**

Questions: "Why brush before blowing?" (Loosen it so air can carry it away.) "Why hold the fan?" (Prevent bearing damage.)

G.3 Demonstration 2 â€” Re-pasting a CPU (4 minutes)

Say: "Now the paste â€” small is powerful."

Steps:

1. Show a cooler base with **dried, cracked** old paste. **Say:** "This is what paste looks like when it has failed â€” dry and cracked. It can no longer pass heat."
2. Clean the CPU and cooler base with **isopropyl alcohol** and a lint-free cloth until both are clean. **Say:** "Clean and dry, both surfaces."
3. Apply a **pea-sized** dot in the centre. **Say:** "Small. Too much squeezes out and can make things worse."
4. Show how the cooler presses down and spreads the paste. Refit and **reconnect the fan cable**.
5. **Verify:** point to HWMonitor and describe the expected temperature drop.

Expected results: clean surfaces, a pea-sized paste dot, cooler refitted, fan cable reconnected.

Common errors to show and correct:

- Using too much paste.
- Forgetting the fan cable.
- Twisting the cooler (risk of pulling a stuck CPU â€” warn).

Questions: "How do you know the paste needs replacing?" (Dry/cracked, or high temps despite clean fans.)

G.4 Demonstration 3 â€” External cleaning and protection devices (5 minutes)

Steps (keyboard/ports/screen):

1. Power off. Show **short bursts of compressed air** along the keyboard rows; shake crumbs out first.
2. Show wiping key tops with a **barely-damp** microfiber cloth â€” **never** spraying onto the keyboard.
3. Show a **cotton bud** for between-keys cleaning.
4. On the old screen, show a **dry microfiber cloth** for dust, and a **slightly damp** cloth for marks, sprayed on the **cloth**, not the screen. **Say:** "No paper towels, no glass cleaner, no hard pressing."

Steps (surge protector and UPS): 5. Hold up the **surge protector**. **Say:** "This absorbs sudden voltage spikes â€” like when power returns after an outage. It does NOT keep the computer on during a blackout." 6. Hold up the **UPS**. **Say:** "This is a battery backup. It keeps the computer alive for a few minutes so you can save and shut down safely. This is what a cyber cafÃ© needs on its key machines." 7. Point to the surge protector's indicator light. **Say:** "When this light goes out, the protection is spent â€” replace it."

Expected results: learners see correct external cleaning technique and understand the difference between a surge protector and a UPS.

Common errors to show and correct:

- Spraying liquid directly on a screen or keyboard.
- Confusing a surge protector with a UPS.

Questions: "Which device protects against a blackout?" (UPS.) "Which protects against a spike?" (Surge protector.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding, each learner (or pair) performs a **full clean and cooling check** on a practice machine and records temperatures before and after.

Safety first â€” read aloud before starting:

- **Power off, unplug**, press power to discharge, **wear your anti-static wrist strap.**
- Use only **safe tools** (compressed air, antistatic brush, isopropyl alcohol, microfiber cloth, cotton buds). **Never** a vacuum cleaner inside a computer.
- Hold the compressed-air can **upright**; do not tilt or shake it.
- **Hold fan blades steady**; do not let the air spin them.
- **Never open the PSU.**
- Work in a **well-ventilated area** (dust will fly).
- Stop and call the instructor if you smell burning or see a swollen battery.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Before temperature.** On the running machine (before opening), open **HWMonitor** and record the **CPU and GPU temperatures** and the fan speeds. This is your "before" reading.
2. Power off, unplug, discharge, strap on. Open the case.
3. **Inspect first.** Before cleaning, note what you see: dust on heatsink? dust on fans? a fan not spinning? record it on the Cleaning Checklist.
4. **Loosen dust.** Use the **brush** to gently loosen dust from the heatsink fins and fan blades.
5. **Blow outward.** Use **compressed air** (upright, short bursts) to blow dust **out of the case**. Hold each fan blade steady.
6. **Clean the interior.** Wipe surfaces with a microfiber cloth; clean the air filter if present; gently air the RAM and expansion slots (do not push dust in).
7. **Pause and predict.** "After cleaning, will the CPU be hotter, same, or cooler? How much cooler do you expect?" Write your prediction.
8. **External cleaning.** Using the old keyboard and screen (or carefully on the practice machine's own), practise: air along the keyboard rows; wipe keys with a barely-damp cloth; clean the screen with a microfiber cloth (dry, or slightly damp with the cloth, never sprayed on the screen).

9. **Reassemble and power on.** Replace the side panel, reconnect power, power on.
10. **After temperature.** Open **HWMonitor** and record the CPU/GPU temperatures again. Compare with the "before" reading.
11. **Cooling check.** Confirm all fans spin. If any fan does not, note it â€" it may need replacement.
12. **Write the maintenance recommendation.** On the Maintenance Plan Template, write a short schedule for this machine (monthly/3-monthly/6-monthly/annual tasks), including cleaning frequency suited to the environment.
13. Complete the **Cleaning Checklist**.

Expected results:

- "Before" and "after" temperatures recorded, usually showing a **drop** after cleaning.
- Visible dust removed from heatsink and fans.
- All fans spinning confirmed (or a non-spinning fan identified).
- A completed Cleaning Checklist and a written maintenance recommendation.

Points to pause on and predict:

- After step 3: "Which part looks worst â€" the heatsink, the fans, or the case?" (Usually the heatsink/fans.)
- After step 7: "If the fins were fully clogged, what temperature change would you expect?" (A larger drop.)
- After step 11: "If one fan does not spin, what does that mean for the machine?" (Risk of overheating â€" the fan needs attention/replacement.)

Troubleshooting while walking the room:

- Dust will not come off the heatsink â†' it may need the cooler removed for a proper clean (supervised) and re-paste.
- Compressed air sprays frost/liquid â†' the can was tilted; set it upright and pause.
- A learner reaches for the vacuum â†' stop immediately; explain static risk.
- Temperature did not drop â†' the real issue may be a dead fan, dried paste, or blocked vents â€" record it and recommend the next step.
- Learner sprays liquid on a screen â†' stop; show the correct method (cloth dampened, sprayed on the cloth).
- Machine will not power on after reassembly â†' check the 24-pin and CPU power connectors are seated and the side panel did not trap a cable.

Completion checklist (learner ticks each):

- ☐ "Before" temperatures and fan speeds recorded (HWMonitor)
- ☐ Safety followed: unplugged, discharged, wrist strap worn, no PSU opened
- ☐ Dust loosened with a brush and blown outward with compressed air (can upright)
- ☐ Fan blades held steady while cleaning
- ☐ Case interior and air filter cleaned; slots lightly cleaned
- ☐ External cleaning practised (keyboard, ports, screen) safely
- ☐ "After" temperatures recorded and compared with "before"
- ☐ All fans checked for spinning
- ☐ Maintenance recommendation written on the template
- ☐ Cleaning Checklist completed; tools and screws accounted for

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I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task: Work alone. Produce a **preventive maintenance plan** and a **maintenance recommendation** for a real operating setting. Do **not** copy the instructor's general schedule â€" tailor yours to the setting given, and justify each choice.

Scenario. You have been hired as the maintenance technician for **"Bright Minds Tutorial Centre"** in Port Harcourt. The centre has:

- **12 desktop computers** used by students, in a **ground-floor room with windows that open onto a dusty road**.
- Frequent **power cuts**, and the voltage is often unstable when supply returns.
- Machines are used for typing, browsing and printing; a few are used to store student records.
- Casual staff operate the machines; there is **no maintenance habit** yet.

- The room is **not air-conditioned**.

Requirements:

1. **Cleaning schedule:** state how often the computers need internal cleaning, and justify it based on the **dusty, non-air-conditioned** environment (be specific about frequency and why it must be more frequent than a home PC).
2. **Power protection:** recommend which devices to use and on which machines, explaining the difference between a **surge protector** and a **UPS**, and which problem each solves.
3. **Software maintenance:** list the monthly software tasks (from Session 6) that keep the machines healthy.
4. **Responsibility and documentation:** say who does each task and how completed work will be **recorded**.
5. **Environmental advice:** give three practical changes to the room or setup to reduce dust, heat or humidity problems.
6. **A plain-language summary** for the tutorial centre owner explaining why this plan saves money.

Time allowance: 15 minutes.

Constraints:

- Base your frequency on the dusty, non-air-conditioned reality â€” do not just copy a magazine schedule.
- Include the **safety rules** any cleaner must follow (power off/unplug, ESD strap, no vacuum inside, never open the PSU).
- Keep the summary understandable to a non-technical owner.

Success criteria:

- Cleaning frequency is justified by the environment (more frequent than a home PC).
- Surge protector vs UPS is explained correctly, with sensible placement.
- Monthly software tasks are included.
- Responsibility and a documentation method are stated.
- Three sensible environmental changes are given.
- The summary is plain and relates to money saved.

Hints available if stuck:

- Hint 1: Dusty road + open windows + no air-conditioning = clean **more often** than a typical home.
- Hint 2: Which device keeps a computer on during a blackout â€” a surge protector or a UPS? Put that one on the machines that store records.
- Hint 3: "Failed maintenance is invisible until the machine dies." What record would let the owner *see* that the work was done?

INSTRUCTOR ANSWER KEY â€” Independent Class Exercise (do not show learners)

1. Cleaning schedule. Internal cleaning **more often than a home PC** â€” for this dusty, non-air-conditioned room, roughly **every 4â€“6 weeks** for internal blow-out, with a **quick weekly check** of vents/fans, and a "clean now" trigger whenever a machine runs loud or hot. Justification: open windows onto a dusty road and no air-conditioning mean dust accumulates fast; frequent cleaning prevents overheating, slowdowns and shutdowns. *(Accept a sensible range, e.g., monthly to every 6 weeks, provided the reasoning is environmental.)*

2. Power protection.

- **Surge protector** on **all** machines (and the printer/network equipment): blocks sudden voltage **spikes**, e.g., when power returns after an outage. Does **not** keep machines running during a blackout.
- **UPS** on the machines that store **student records** and on the printer/network if budget allows: keeps the computer alive for a few minutes during **outages** and steadies **sags/brownouts**, allowing a safe save and shutdown and preventing file corruption.
- Explain the key difference clearly: **spikes = surge protector; blackout/sag = UPS**.

3. Monthly software tasks. Full antivirus scan; check for Windows updates; run Disk Cleanup / empty Recycle Bin; review Startup Apps for unnecessary entries; check disk space and free some if low; back up the student-records machine. *(From Session 6.)*

4. Responsibility & documentation. Assign a named staff member (or the technician) per task; keep a simple **logbook/table** per machine: date, machine ID, task done (with temperature readings if cleaning), and signature. Post a printed schedule on the wall. The logbook proves the work was done and reveals machines that run hotter than others.

5. Environmental advice (any three of): keep machines off the floor and away from the windows/dust path; fit dust filters on intakes and clean them; close windows or use screens/curtains during dusty periods; place machines so vents are not against walls; ventilate the room to reduce heat; consider a fan-assisted airflow; keep drinks away from machines; allow a cold/damp machine to reach room temperature before switching on.

6. Plain-language summary (example). "Your computers are getting dusty fast because of where they are, and dust makes them slow down and shut off. If we clean them every month and protect them from power spikes and blackouts, they will last longer, break less and lose fewer student records. This small, regular cost is far less than replacing computers that died young."

Marking guidance (out of 15):

- Cleaning frequency justified by environment (3)
- Correct surge protector vs UPS use and placement (3)
- Monthly software tasks included (3)
- Responsibility and documentation stated (2)
- Three sensible environmental changes (2)
- Plain-language, money-focused summary (2)

Accept any plan that adapts sensibly to the dusty, unstable-power environment and includes safety and documentation.

J. REAL-WORLD CASE STUDY

Situation. *Ngozi* runs a small hair salon in Rumuola that also has a computer for bookings, customer records and the salon's social-media page. The computer is on the floor near the door, where dust from the street and hair clippings collects. For months it has been getting slower and louder, and twice recently it shut down in the middle of work. Ngozi assumes it is "old" and plans to buy a new one.

Relevant facts.

- The machine is only about three years old.
- It sits on the floor, close to the open door.
- It has never been opened or cleaned.
- The room is hot, and the computer shares a power strip with a hair dryer (a high-power appliance) on the same socket.
- Customer records are stored on the machine with **no backup**.

The decision point. Buy a new computer, or clean and protect the existing one?

Possible approaches.

1. Buy a new computer (expensive; the new one will also get dusty and hot in the same spot).
2. Clean the current machine (internal dust + thermal paste if needed), then fix the setup: move it off the floor, away from the door, onto a surge protector, separate from the hair dryer, and start a cleaning and backup routine.
3. Ignore it and hope it lasts.

Recommended solution (for this stage). Approach **2**. Internal cleaning clears the dust blanket (the likely cause of the slowdown and shutdowns); moving the machine off the floor and away from the door reduces future dust; a **surge protector** (and separating it from the hair dryer) protects it from power problems; setting up a **backup** for the customer records and a **monthly cleaning routine** prevents the problem returning. If temperatures stay high after cleaning, re-apply thermal paste. This costs a fraction of a new computer.

Trade-offs.

- Approach 1 is expensive and does not fix the *environment* â€" the new machine will face the same dust, heat and power problems and may also die young.
- Approach 2 is affordable, treats the actual cause, and adds lasting protection; it requires a small amount of regular effort.
- Approach 3 lets the machine die, likely losing the customer records.

Professional connection. This is a core professional insight: **most "old" computers are not old â€" they are neglected.** A technician who cleans, protects and documents earns repeat business and referrals, because the customer sees real value (a fast machine and saved records) at low cost. The certificate's promise of "honest assessments and recommendations" is exactly this.

Transfer exercise (ask the class): "Ngozi asks, 'Why did my computer shut down in the middle of work?' Give her a plain answer and one prevention step."

Suggested answer: "Dust had clogged the cooling inside, so the machine got too hot and shut itself off to protect the parts. We will clean it, and if you keep it off the floor and clean it every month, it will stay cool and stop shutting down."

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide.

K.1 Dust will not come off the heatsink / temperatures stay high after cleaning

- **Symptoms:** cleaning done, but CPU still hot.
- **Likely causes:** the heatsink is so clogged that air cannot clear it; dried thermal paste; a failed or slow fan; blocked vents; the room is too hot.
- **How to investigate:** check fan speeds (HWMonitor); check the fan spins; inspect the vents; check the paste condition.
- **Safe corrective steps:** remove the cooler for a proper clean and re-apply thermal paste (Session 7, supervised); replace a failed fan; unblock vents; improve room airflow.
- **How to confirm resolved:** temperatures drop and stay normal during use.
- **Prevention:** clean on schedule before it clogs; maintain airflow.

K.2 A fan does not spin

- **Symptoms:** one fan is stationary while the machine runs.
- **Likely causes:** the fan is clogged and stalled; the fan cable is disconnected; the fan bearing has failed; the fan is not getting power.
- **How to investigate:** check its cable is connected to the right header; clear dust; test.
- **Safe corrective steps:** clean it; reseat its cable; if it still does not spin, **replace the fan**.
- **How to confirm resolved:** the fan spins freely and quietly.
- **Prevention:** regular cleaning; replace worn fans promptly.

K.3 Machine will not start after cleaning

- **Symptoms:** no power after reassembly.
- **Likely causes:** a cable was knocked loose (24-pin, CPU power, SATA, fan); a component was dislodged; the side panel trapped a cable.
- **How to investigate:** re-open and check all connectors and component seating; look for trapped cables.
- **Safe corrective steps:** reseat connections; free trapped cables; power on.
- **How to confirm resolved:** the machine POSTs and boots.
- **Prevention:** photograph connections before cleaning; work gently; check before closing.

K.4 Static damage after cleaning

- **Symptoms:** the machine behaves strangely or dies after a clean, with no visible cause.
- **Likely cause: ESD** â€" static from a cloth, brush or body discharged into a component.
- **How to investigate:** there may be no visible sign; ESD damage can be invisible. This is why prevention is essential.
- **Safe corrective steps:** ESD damage is often permanent and part-level; identify and replace the affected part (diagnostic repeat, Session 5).
- **How to confirm resolved:** the machine runs normally after replacing the damaged part.
- **Prevention: always** wear an anti-static wrist strap; use antistatic tools; never use a vacuum cleaner inside.

K.5 Screen damage during cleaning

- **Symptoms:** streaks, smears or damage after wiping a screen.
- **Likely causes:** liquid sprayed onto the screen; harsh glass cleaner; hard pressing; paper towels.
- **How to investigate:** inspect for fluid marks or surface damage.

- **Safe corrective steps:** for smears, gently wipe with a clean, dry microfiber cloth; surface damage may not be repairable (may require replacing the panel).
- **How to confirm resolved:** the screen is clear and undamaged.
- **Prevention:** screen cloth barely damp, liquid on the cloth not the screen, gentle pressure, no harsh chemicals.

K.6 Power-related damage or corruption

- **Symptoms:** crashes, restarts, or corrupted files, especially after outages or when power returns.
- **Likely causes: surge, sag or outage**; no surge protection; no UPS.
- **How to investigate:** ask about the power supply history; check for a surge protector; check for corrupted files.
- **Safe corrective steps:** fit a **surge protector**; use a **UPS** on critical machines; **back up**; repair corrupted files (Session 6, `sfc`).
- **How to confirm resolved:** stable operation and no further corruption.
- **Prevention:** surge protector on everything, UPS on critical machines, regular backups.

K.7 Distinguishing likely from confirmed

As always, a *likely* cause becomes *confirmed* by evidence: a temperature reading that drops after cleaning, a fan observed spinning (or not), a paste visibly dry, a machine that will not POST only after a specific event. Record observations honestly and never claim "cleaned and fixed" without a **before/after temperature reading** and a working test.

L. SESSION ASSIGNMENT

Assignment title: "Cleaning and Maintenance Report with a Maintenance Plan"

Purpose: To perform a safe clean and cooling check on a real machine and to produce a written **maintenance recommendation and schedule** a customer could follow.

Realistic scenario: A customer asks you to "service" their computer â€" clean it, check the cooling, and advise them how to keep it healthy. You will clean a machine (or a dusty machine at CEA), record the results, and write the recommendation.

Detailed instructions:

1. Choose a real computer you have permission to clean (a CEA practice machine or your own). **Do not open a machine that is not yours without permission.**
2. **Before cleaning:** power on, open **HWMonitor**, and record the CPU (and GPU, if shown) temperatures and fan speeds.
3. **Clean safely:** power off, unplug, discharge, wear the wrist strap; loosen dust with a brush, blow outward with **compressed air (can upright)**, hold fans steady, wipe the interior; clean the keyboard, ports and screen appropriately for their surfaces. **Never use a vacuum inside and never open the PSU.**
4. **Inspect:** note dust levels, any fan not spinning, paste condition, and airflow.
5. **After cleaning:** power on, record temperatures again, and confirm all fans spin.
6. **Write the maintenance recommendation:** a schedule (monthly / 3-monthly / 6-monthly / annual tasks) tailored to the machine and its **environment** (home, office, or dusty cafÃ©), including power-protection advice (surge protector / UPS).
7. **Write a plain-language summary** for the customer explaining the benefit (cooler, faster, longer life) and the safety rules.

Required deliverables:

- One completed **Cleaning Checklist** with before/after temperatures.
- One written **maintenance recommendation and schedule** (using the Maintenance Plan Template).
- A one-paragraph plain-language summary for the customer.

Submission format: Completed checklist and written plan (paper or class folder).

Deadline: Before Session 9.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Before/after temperature record	3	Both recorded with HWMonitor; comparison made

Criterion	Marks	Description of a strong answer
Safe cleaning technique	4	Brush-then-air, can upright, fans held, no vacuum, no PSU
Inspection findings	3	Dust, fan, paste and airflow noted honestly
Fan/cooling check	2	All fans checked; any fault identified
Maintenance schedule	4	Tailored to environment with sensible frequencies and tasks
Power-protection advice	2	Correct use of surge protector (spikes) and UPS (outages/sags)
Plain-language summary	2	Customer can understand the benefit and safety rules

Common mistakes to avoid: using a vacuum inside; tilting the compressed-air can; spinning fans with air; spraying liquid on screens or into ports; forgetting the wrist strap; skipping the temperature reading; copying a generic schedule without adapting it to the environment.

Instructor reference solution (abbreviated example):

> ***Machine:*** *CEA practice desktop, dusty, used in a non-air-conditioned room.* ***Before (HWMonitor):*** *CPU* ***72 Â°C*** *idle; GPU* ***60 Â°C****; CPU fan 900 RPM; one case fan 0 RPM.* ***Cleaning:*** *power off, unplugged, discharged, wrist strap on. Brushed the heatsink fins and fan blades; blew outward with compressed air (can upright); held fans steady; wiped the interior; cleaned the air filter; cleaned the keyboard (air + barely-damp cloth) and the screen (dry microfiber).* ***Inspection:*** *heavy dust carpet on the CPU heatsink; one case fan not turning; paste looked dry at the edges; vents partly blocked by the machine's position against a wall.* ***After (HWMonitor):*** *CPU* ***48 Â°C*** *idle; GPU* ***45 Â°C****; CPU fan 1,200 RPM; case fan still 0 RPM (****needs replacement****).* ***Recommendation:*** *replace the stopped case fan; re-apply thermal paste if temperatures stay high after the new fan (Session 7); move the machine so vents are clear of the wall; internal clean* ***every 4â€“6 weeks*** *in this dusty room; monthly antivirus scan + Disk Cleanup + updates; check drive health every 6 months; annual full diagnostic; use a* ***surge protector*** *and back up important files.* ***Plain-language summary:*** *"Your computer was full of dust, which made it overheat and slow down. After cleaning, the temperature dropped a lot. One fan has stopped and should be replaced, and you should clean every month because the room is dusty. Keep it away from the wall so air can flow. A surge protector will protect it from power spikes."*

M. KNOWLEDGE CHECK

(Learner version â€“ answers only in the Instructor Key below.)

Q1. Short answer. In one or two sentences, explain how dust makes a computer slow. **Q2. Multiple choice.** Which of these must you **never** use to clean the inside of a computer? a) Compressed air b) An antistatic brush c) An ordinary vacuum cleaner d) A microfiber cloth **Q3. True or false, with a reason.** "You should spray screen cleaner directly onto the monitor screen." **Q4. Explain in your own words.** When does thermal paste need to be replaced, and how much do you use? **Q5. Short answer.** What is the difference between a **surge protector** and a **UPS**? **Q6. Scenario.** A cyber cafÃ©'s computers keep shutting down during busy evenings, and the room is not air-conditioned. Give two maintenance actions that would help, and explain why.

INSTRUCTOR ANSWER KEY (Q1â€“Q6)

A1. Dust settles on fan blades and between heatsink fins and blocks airflow, so heat cannot escape. The CPU gets hot and the computer slows itself down on purpose (**thermal throttling**), so it feels slow; if it gets hotter still, it shuts off to protect itself.

A2. (c) An ordinary vacuum cleaner. Its plastic nozzle and moving air build **static electricity** that can destroy components (and it can suck off small parts). Use compressed air inside instead.

A3. False. Never spray liquid directly onto a screen â€” it can run into the edges and damage the screen. Spray (or dampen) the **cloth**, then wipe gently.

A4. Replace thermal paste whenever the CPU and cooler are separated, or when temperatures creep up despite clean fans (roughly every **1â€“2 years**). Use a **pea-sized** amount in the centre of the CPU â€” not too much.

A5. A **surge protector** absorbs sudden voltage **spikes** (e.g., when power returns after an outage); it does **not** keep the computer running during a blackout. A **UPS** contains a battery and keeps the computer running for a few minutes during an **outage** and steadies **sags/brownouts**, allowing a safe save and shutdown.

A6. Any two sensible actions, e.g.: (1) **Clean the internal dust regularly** (monthly in a dusty, non-air-conditioned room) â€" dust blocks airflow and causes overheating and shutdowns. (2) **Improve cooling/airflow** â€" ensure vents are clear, all fans work (replace failed fans), and the room is ventilated; add a **surge protector + UPS** if the shutdowns relate to unstable power. *(Also acceptable: re-apply thermal paste if temperatures stay high; move machines out of direct heat.)*

Marking: award for correct concept; deduct for unsafe answers (vacuum inside, spraying liquid on screens, opening the PSU).

N. REFLECTION AND RECAP

Plain-English summary. Today you learned that cleaning is not about looks â€" it is about **cooling and reliability**. Dust blocks airflow, so heat cannot escape; the computer then **slows itself down** (thermal throttling) and can **shut itself off** (thermal shutdown), and over time the heat shortens the life of every part. You learned the **safe tools** (compressed air, antistatic brush, isopropyl alcohol, microfiber cloth, cotton buds, wrist strap) and the **dangerous ones** (vacuum cleaner, water, your breath, harsh sprays). You practised the safe internal clean (power off, unplug, discharge, brush, blow outward, hold fans steady, inspect, verify with **HWMonitor**) and external cleaning of keyboards, ports and screens. You learned when to **re-apply thermal paste** (a pea-sized amount) and how to protect a computer from its **environment** â€" dust, humidity, heat and **power surges/sags/outages** â€" using a **surge protector** and a **UPS**. Finally, you learned to build **preventive maintenance schedules** for homes, offices and cyber cafÃ©s, and to **document** every task.

Five important takeaways:

1. **Dust + heat = slowdowns, shutdowns and early death.** Cleaning is the cheapest repair.
2. **Compressed air (upright) and an antistatic brush inside; never a vacuum cleaner inside.**
3. **Hold fans steady; never spin them with air.**
4. **Re-apply thermal paste** when the cooler is removed or temperatures creep up â€" pea-sized.
5. **Protect from power:** a **surge protector** for spikes, a **UPS** for outages â€" and always **back up**.

Questions to help you recognise what you can now do:

- Can I clean a computer safely using the right tools?
- Can I record before/after temperatures and show the improvement?
- Can I clean a keyboard, ports and screen without damaging them?
- Can I explain a surge protector versus a UPS?
- Can I write a maintenance schedule tailored to an environment?

Safety self-check (tick honestly):

- ☐ I powered off, unplugged and discharged before opening the machine.
- ☐ I wore my anti-static wrist strap and used antistatic tools.
- ☐ I never used a vacuum cleaner inside the computer.
- ☐ I never opened the PSU.
- ☐ I kept liquid away from the inside of the machine and off screens.
- ☐ I recorded a before/after temperature reading and verified the machine worked.

Connection to the next session. You can now diagnose, replace and maintain hardware. The final major technical skill is the **software foundation** itself: next session, Session 9 â€" Operating System Installation, you will learn to prepare for a Windows install, install Windows on a machine (including one with a new SSD), install drivers, run updates, and activate â€" turning a blank or freshly-repaired machine into a working computer ready for the customer. Maintenance kept the machine healthy; installation gives it its brain.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner sets up correctly: **power off, unplug, discharge, wrist strap** â€" every time.
- The learner uses the **correct tools** and refuses the vacuum cleaner inside.
- The learner uses compressed air **upright** and holds **fans steady** while blowing.
- The learner records **before/after temperatures** and makes an honest comparison.
- The learner cleans screens/ports appropriately (no direct spraying, no liquid in ports).
- The learner's maintenance schedule is **tailored to the environment**, not copied.
- The learner explains the difference between a **surge protector** and a **UPS** correctly.

Common misconceptions to watch for and correct:

- "Cleaning is only cosmetic." â†' It is about cooling and reliability.
- "A vacuum cleaner is fine." â†' Static risk; never inside.
- "Spin the fan with air to clean it." â†' Damages the bearing.
- "More liquid cleans better." â†' Excess liquid damages screens/keyboards.
- "A surge protector keeps the computer on in a blackout." â†' That is a UPS.
- "If it is working, no maintenance is needed." â†' Working machines are the ones worth protecting.

Suggested remedial exercises for learners who struggle:

1. **Tool-sorting drill:** give cards for tools (compressed air, brush, vacuum, cloth, water, alcohol) and have the learner sort them into "safe" and "unsafe" with reasons.
2. **Technique rehearsal:** practise on a spare fan/heatsink: brush-then-blow, can upright, hold the fan steady â€" without powering anything on.
3. **Schedule-adaptation drill:** give three environments (home, office, cafÃ©) and have the learner adjust cleaning frequency and justify each change.

Differentiation:

- **Level 1 (supported):** provide a step-by-step cleaning card and a partially filled Maintenance Plan Template; pair with a stronger learner for the internal clean.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** perform the full clean **and** remove the cooler to re-apply thermal paste (supervised), verify with a before/after temperature graph, and design a maintenance schedule for a **cyber cafÃ©** with a per-machine logbook.

Timing guidance: This session is hands-on and can be messy. Protect the guided practical and the before/after temperature exercise â€" that is the core learning. If time is short, trim the case study to a two-minute mention and demonstrate external cleaning briefly. Ensure adequate ventilation and time to tidy up. Do **not** rush the safety steps; a rushed clean with a vacuum or a tilted can can destroy a machine.

APPENDIX 8A â€" 10-MINUTE SAFETY REFRESHER (only if needed)

If learners cannot demonstrate the two safety must-dos and ESD awareness, run this before Section D.

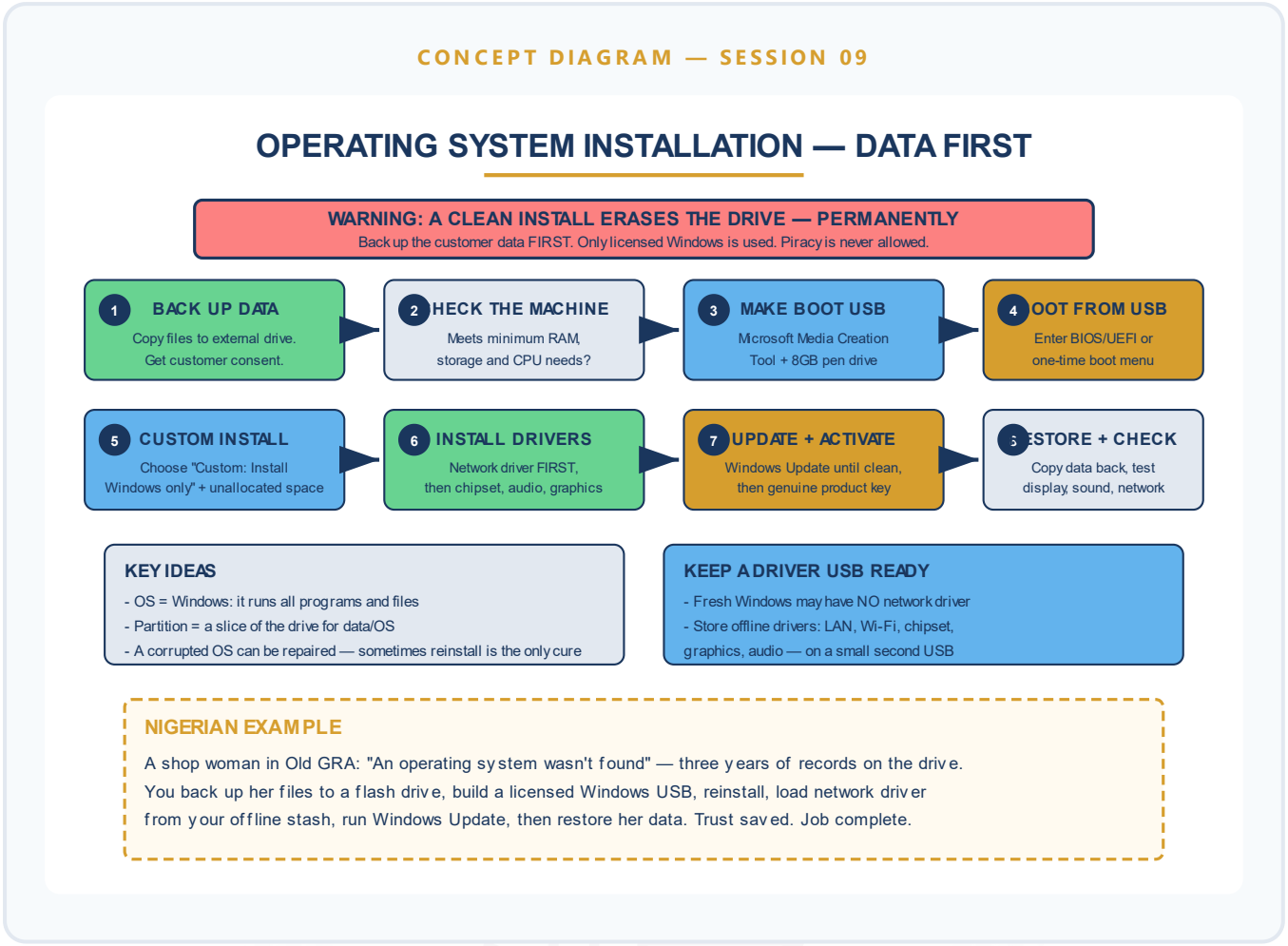
1. **The two must-dos** â€" the class chants: **"Power off and unplug. Wear the anti-static wrist strap."**
2. **Discharge** â€" press the power button after unplugging to drain leftover electricity.
3. **ESD in cleaning** â€" say: "Rubbing a cloth or brush builds **static**. Static silently kills components. That is why we ground ourselves with the strap and use antistatic tools."
4. **The tool rule** â€" say three times: **"Compressed air and brush INSIDE. Vacuum NEVER inside."**
5. **The liquid rule** â€" "Liquid goes on the **cloth**, never on the screen or into a port."
6. **The golden rule** â€" **"Never open the PSU."** Demonstrate once, then have each learner demonstrate the two must-dos and state the tool and liquid rules before they touch a machine.

END OF SESSION 8

Next: Session 9 â€" Operating System Installation.

SESSION 09 OF 12

OS Installation



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 9 â€" OPERATING SYSTEM INSTALLATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	9 of 12
Session title	Operating System Installation
Week / Module	Week 3 â€" Module 3: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (guided, hands-on, safety-critical)
Prerequisite knowledge	Sessions 1â€"8 â€" components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware and software diagnostics, component replacement, cleaning and

Item	Detail
	maintenance. A learner must already know how to open a case safely and to back up files.
Connection to previous work	In Session 6 you learned to diagnose software faults such as a corrupted operating system. In Session 7 you replaced a hard drive or SSD. Today you complete the natural next step: putting a fresh, working operating system onto a machine so the customer can use it again.
Preparation before class	Academy practice PCs ready; a **licensed** Windows ISO and a USB flash drive (8 GB or larger) per pair; the official Microsoft Media Creation Tool downloaded while the academy internet is working; a spare network/board driver folder on a second USB; a laptop connected to the projector; a printed "OS Install Checklist" and "Customer Intake Form" per learner.
Safety flag	This is a **hands-on, data-destroying** session. No drive is wiped in class until (a) data has been backed up and (b) the learner has stated out loud that they understand the wipe is permanent. Only **properly licensed** Windows is used. Piracy is never taught, demonstrated or tolerated.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain in plain English** what an operating system is and why a corrupted OS sometimes has to be reinstalled rather than repaired.
2. **Back up a customer's data correctly** before any wipe, and explain why data safety comes before every other step.
3. **Check whether a machine meets the minimum hardware requirements** for a given Windows version and say what happens if it does not.
4. **Create a bootable Windows USB drive** using the official Microsoft Media Creation Tool, in the correct order.
5. **Enter the BIOS/UEFI setup or the one-time boot menu** and set or select the correct boot device.
6. **Perform a clean install of Windows**, correctly choosing "Custom: Install Windows only (advanced)" and installing onto unallocated space.
7. **Explain in beginner terms** why we partition a drive and what a clean install removes.
8. **Install drivers in the correct order** (starting with the network driver) and run Windows Update until no updates remain.
9. **Activate Windows** using only a legitimate product key or digital licence, and state the academy's rule against piracy.
10. **Restore the user's data** and complete a **post-install verification** checklist (Device Manager, display, sound, network, essential software).

C. REQUIRED MATERIALS

Hardware

- At least one practice desktop PC that can be wiped (per pair of learners). If only one machine is available, the instructor performs the live install while pairs complete the **simulated install checklist** in parallel.
- USB flash drive, **8 GB minimum** (16 GB recommended). Note: creating the media erases the USB drive.
- A second small USB stick holding the **network/board driver** and any other offline drivers, in case the installed Windows has no working network driver.
- Tool kit and an **anti-static wrist strap** (even though we are not opening the case much, we may reseat cables).
- A power socket that learners can reach; keep liquids off the bench.

Software

- The **official Microsoft Media Creation Tool**. Download only from the genuine Microsoft website. As a general guide: search "Download Windows 11" or "Download Windows 10" from Microsoft, and use the Media Creation Tool page. (Website layout occasionally changes; the tool name stays the same.)
- A **valid, licensed** Windows product key or a machine with a **digital licence** already attached to its hardware.
- Offline driver files: network/LAN, Wi-Fi, chipset, graphics, audio.

Printed materials

- Worksheet 9.1 â€" "Pre-Installation Checklist".
- Worksheet 9.2 â€" "Post-Installation Verification".
- "Customer Intake Form" (reused from Session 10 preparation, to show the data-backup consent line).

Safety equipment

- Anti-static wrist strap per learner; safety glasses if compressed air is used during cleaning of the intake vents.

Internet

- Required for downloading the Media Creation Tool and the ISO. If the academy internet is slow, download once to a shared USB and copy it to pairs. Stress: after installation, the **machine's own** network driver may not exist â€" that is why the offline driver USB matters.

Alternative for learners who lack a facility

- A learner who owns a laptop with at least 4 GB RAM and a spare 8 GB USB may practise at home **only if they have a licensed Windows key and have backed up first**. If they have only a phone, pair them with a desktop user and have them fill the printed checklist and narrate the steps.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: On the projector, show a machine that boots to a black screen with the message *"An operating system wasn't found. Try disconnecting any drives that don't contain an operating system."* Or, if unavailable, show a photo of the same message. Do not explain it yet.

What the instructor says:

> *"Look at this screen. The customer brought this computer to the shop in tears. It worked yesterday. Today it says the operating system is gone. She has three years of her shop's records, her children's photos and her church choir videos on it.*
>
> *She wants it working again â€" today. Some of you are thinking, 'Just reinstall Windows.' That is exactly right, and it is what we do today. But look again at her face. If we reinstall carelessly, we destroy everything on that drive. So the real job is not just installing Windows. The real job is:* ***save her data first, then put the computer right.***
>
> *Today you will learn to do the whole thing properly: back up, check the machine, build the installer, boot from it, wipe and install, then bring the machine back to full working order. And we will only ever use a* ***licensed*** *copy of Windows. A repair shop that pirates software is not a repair shop you can trust."*

Expected learner response: Learners may say "just reinstall", "won't the files stay?", or "is the data already lost?". Accept all answers. Correct gently that a **clean install erases the drive** and reassure them that backing up first is precisely how we protect the customer.

Why this opener works: It reframes OS installation from a purely technical task into a *customer-trust and data-safety* task. That is the professional mindset this whole week builds. It also plants the licensing rule early.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a fast oral round-robin. Put the questions on the projector.

Question 1: "Name the six major components of a computer."

- Expected answer: motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.
- Common gap: learners forget the PSU or call the motherboard "the board". Prompt: "Which part turns wall electricity into the voltages the PC needs?"

Question 2: "What do we call the software that manages all the hardware and lets you run programs?"

- Expected answer: the operating system (Windows).

- Common gap: learners say "Windows" only. Clarify that Windows is one example of an operating system; others exist (macOS, Linux). We install Windows in this course.

Question 3: "What is a driver, and what does a yellow exclamation mark in Device Manager mean?"

- Expected answer: a driver is software that lets Windows talk to a piece of hardware; a yellow warning means the driver is missing or not working.
- Common gap: learners think drivers are only for printers. Clarify: every major component â€" chipset, network, graphics, audio â€" needs a driver.

Question 4: "According to the diagnostic workflow, what comes before you replace or wipe anything?"

- Expected answer: observe, ask questions, form a hypothesis, test. Only after confirming the fault do you act.
- Common gap: learners jump to "reinstall". Reinforce: confirm the OS itself is the problem before wiping a working system.

Bridge to today (say this):

> *"Diagnosing told us the operating system is broken and cannot be repaired cheaply. Today we treat the patient. But remember the doctor's first rule: do no harm. Before we operate, we save the patient's belongings. That is the backup."*

Gap handling: If more than a third of the class cannot explain what a driver is (Question 3), spend five extra minutes drawing a quick diagram: Windows â† driver â† hardware, and show a real Device Manager screen with one yellow icon. Do not skip this â€" the post-install half of today's session depends on it.

F. FULL LECTURE NOTES

F.1 What an operating system actually is

The idea. An **operating system (OS)** is the master software that sits between the user and the hardware. It starts the computer, manages memory, storage, files, programs and devices, and gives you the screen you click on.

Plain-English definition. The operating system is the **manager of the whole computer**. Without it, the hardware is just metal and silicon that does nothing useful when you press the power button.

Why it exists. Your music player, your browser and your documents cannot each talk directly to the disk and RAM safely â€" they would fight over the same memory and files. The OS keeps order: it decides who uses what, when, and how. It also provides the desktop, the Start menu, the taskbar and the ability to install programs.

What it does.

- **Boots** the machine (loads itself and hands control of the PC to you).
- **Manages files** in folders you can navigate.
- **Manages memory** so programs do not collide.
- **Talks to hardware** through drivers.
- **Provides security** (accounts, passwords, permissions, Windows Defender).

Everyday analogy. Think of a big **hotel**. The rooms, beds and plumbing are the hardware. The **manager** (the OS) decides who gets which room, handles keys (accounts and passwords), makes sure guests do not fight over the same room (memory), and calls the plumber (driver) when a tap leaks. A hotel with no manager is chaos.

Realistic scenario (Port Harcourt). A cyber cafÃ© in Rumuola has 8 machines. The cafÃ© owner bought them with Windows pre-installed. When a virus corrupts Windows on one machine, the cafÃ© loses a seat/hour of income. A technician who can rebuild the OS in under two hours keeps that cafÃ© earning.

Common misconception. "The operating system is stored on the hard drive, so if Windows breaks, my documents are gone too." Not automatically true â€" but a **clean install** will destroy them if you do not back up first. Files and Windows share the drive. That is the whole reason data safety is step one.

Limitation. Windows is **not free**. It requires a valid licence. Installing it without a licence is both unethical and illegal. In this course we only ever install Windows that has been properly licensed â€" usually through a key the customer already owns, or a **digital licence** tied to that specific machine's hardware.

F.2 When to reinstall the OS â€” and when not to

The idea. Reinstalling Windows is powerful but heavy-handed. It is the correct fix for some problems and the wrong fix for others.

Good reasons to do a clean install.

- The OS is so corrupted that repair tools (System Restore, Startup Repair, `sfc`, DISM) fail or the machine will not boot at all.
- The drive is being **replaced** (a new SSD has no operating system on it).
- The machine is badly infected with malware that cannot be fully removed.
- The customer explicitly wants a fresh start on a machine they own and understand the data consequences.

Reasons NOT to reinstall immediately.

- The real fault is hardware (bad RAM, failing drive, failing PSU). Reinstalling Windows on failing RAM just wastes an hour and frustrates the customer.
- The problem is only a missing or faulty driver â€” install the driver instead.
- The user simply lost a file or a program â€” try to recover or repair first.
- The data has **not** been backed up. You never wipe before you back up. Full stop.

Why this matters professionally. Reinstalling is the technician's version of "have you tried turning it off and on?" If you reinstall for every problem, customers pay for work they did not need, and their data is put at needless risk. Confirm the diagnosis first.

Common misconception. "A reinstall protects the customer's files automatically because Windows keeps an 'old Windows' folder." The so-called Windows.old folder is created only in some **upgrade** situations, not a clean custom install, and it never contains the user's personal documents. Never rely on it â€” always back up.

F.3 Data safety first â€” the backup step

The idea. Before you change, replace or wipe anything, you copy the customer's personal data somewhere safe.

What counts as "the customer's data".

- Documents: Word, Excel, PDFs.
- Photos and videos (often the most precious).
- Their Desktop, Documents, Downloads, Pictures, Music and Videos folders.
- Email data if stored locally (for example mailbox files).
- Browser bookmarks and passwords where recoverable.
- Business records, church or school files, accounting files.
- Settings the customer cares about (wallpaper, favourites).

Where to put the backup.

- An **external hard drive** or large USB drive (best for big photo/video collections).
- Cloud storage if the customer already has an account and the internet is fast enough.
- Never on the same drive you are about to wipe.

Golden rules.

1. **Back up before you touch the drive.** If the machine will not boot, take the drive out and connect it to a working PC (or a USB-to-SATA adapter) to copy the files. *This is a Session 7 skill in action.*
2. **Verify the backup** by opening a few copied files. A backup you never checked is not a backup.
3. **Tell the customer what you could and could not save**, and get them to sign the intake form saying they understand a clean install wipes the rest.
4. **Keep the customer's data confidential.** Do not share it, post it or copy it for personal use. This is a professional and legal duty (covered more in Sessions 10 and 11).

Realistic scenario. Mama Ngozi's church office PC will not boot. The secretary's only copy of the harvest records is on it. A careless technician wipes the drive and installs Windows "fresh" â€” and destroys two years of records. A professional technician first pulls the drive, copies the records to an external drive, confirms they open, *then* wipes. Same one hour of work; completely different outcome for the customer.

Analogy. The backup is like moving the family photos out of a house before demolishing and rebuilding it. You do not rebuild first and hope the photos survive.

Limitation / honesty. Sometimes a failing drive cannot be read to back up at all. That is a data-recovery situation â€" a specialised, higher-level skill **beyond this course**. Do not promise the customer data you cannot rescue. Be honest and, if needed, refer them to a specialist.

F.4 Checking hardware meets requirements

The idea. Before installing an OS, confirm the machine is powerful enough. Installing Windows on a machine below the minimum requirements leads to a slow, unstable, unhappy result.

General minimums (as a beginner guide).

Version	CPU	RAM	Free storage	Other
Windows 10 (64-bit)	1 GHz or faster	2 GB	32 GB	DirectX 9 graphics
Windows 11	64-bit, 1 GHz, 2 or more cores	4 GB	64 GB	UEFI, Secure Boot capable, TPM 2.0, DirectX 12

> ***Note:*** *Exact figures and the "PC Health Check" wording can change with Windows updates. Always confirm on Microsoft's requirements page. For this course, the figures above are a safe teaching baseline.*

How to check quickly on a working machine.

- Press `Windows key + Pause/Break` (or right-click **This PC â†' Properties**) to see processor and installed RAM.
- Open **This PC** to see drive size and free space.
- For Windows 11, Microsoft provides a **PC Health Check** tool that tells you whether a machine qualifies.

Why it exists. Windows 11 in particular refuses to install on many older machines because it requires **TPM 2.0** and **Secure Boot**, which are security features. A machine without them may be perfectly fine for Windows 10 but not eligible for Windows 11.

Realistic scenario. A student brings a 2012 laptop with 2 GB RAM and asks for "the newest Windows". The honest professional answer is: Windows 11 needs 4 GB RAM and TPM 2.0, which this machine does not have. Recommend Windows 10 (if it meets those minimums) or a RAM upgrade â€" do not force a version the hardware cannot run.

Common misconception. "More RAM, more speed, always." Often true, but a machine whose CPU is very old or whose drive is a slow HDD will still feel sluggish. Match the OS to the hardware; be honest with the customer.

Limitation. Meeting minimum requirements means Windows *will run*, not that it will run *well*. For comfortable everyday use, recommend more RAM and, ideally, an SSD.

F.5 Creating a bootable Windows USB (Media Creation Tool)

The idea. A "bootable USB" is a USB flash drive that the computer can start from, carrying the Windows installer. We build it with Microsoft's own free tool.

Why use the official Media Creation Tool.

- It is free and comes directly from Microsoft, so the installer is genuine (no tampered, pirated or virus-carrying copies).
- It downloads the correct Windows files and formats the USB so the PC can boot from it.
- It reduces user error compared with manually formatting drives.

Step-by-step (accurate as of current Windows; menu wording may vary slightly by version).

1. On a **working** internet-connected PC, go to Microsoft's official download page and get the **Media Creation Tool** for the Windows version you intend to install.
2. **Back up the USB drive** â€" everything on it will be erased.
3. Plug in a USB flash drive of **8 GB or more**.
4. Run the tool and **accept the licence terms**.
5. When asked what you want to do, choose **"Create installation media (USB flash drive, DVD, or ISO file) for another PC"**.
6. Choose **language, edition and architecture** (for most modern machines, **64-bit (x64)**). If unsure, leave the recommended defaults.

7. Choose **"USB flash drive"** as the media type.
8. **Select your USB drive** from the list â€” double-check the drive letter and capacity so you do not erase the wrong drive.
9. Click **Next** and wait. The tool downloads Windows and writes it to the USB. This can take 20â€“60 minutes depending on internet speed. **Do not unplug the USB or close the tool.**
10. When it says **"Your USB flash drive is ready"**, click **Finish** and safely remove the USB.

How the learner recognises success. The USB now contains folders such as `sources`, `boot`, and files such as `setup.exe` and `bootmgr`. If you see those, the media is likely correct.

Realistic scenario. A shop in Diobu cannot afford to keep a DVD writer. The technician keeps one good 16 GB USB with the official Windows media, plus a second USB with the machine's drivers. Those two sticks rebuild almost any customer PC.

Common misconception. "Any Windows ISO from anywhere is fine." No. Unofficial "cracked" or "activated" ISOs are pirated, often carry malware, and are against academy policy and the law. Only the official Microsoft tool/ISO and a legitimate licence are acceptable.

Limitation. The tool needs internet and time, and it **wipes the USB**. If the academy internet fails, plan ahead: download once and keep a master USB that you copy from (using a tool like Rufus is a more advanced option, not required for this course).

F.6 BIOS/UEFI and boot order

The idea. When a computer powers on, a tiny program runs **before** Windows. It checks the hardware and then decides **which device to start from** â€” usually the internal drive, but we need it to start from our USB.

Two names, one job.

- **BIOS** â€” the older system used by many older machines.
- **UEFI** â€” the modern replacement; it supports **Secure Boot**, **GPT** disks and other security features.

Both do the same basic job: they are the first code that runs and they hand control to a bootloader.

How to get in.

- Watch the very first screen for a message like *"Press DEL to enter Setup, F2 for BIOS, F12 for Boot Menu."* Keys vary by manufacturer â€” common ones are **Del, F2, F10, F12, Esc, F9**.
- The **one-time boot menu** (often `F12`, `Esc` or `F9`) lets you pick the USB for this boot only, without changing settings permanently. This is the safest and easiest method in a repair shop.
- The **BIOS/UEFI setup** screen (usually `Del` or `F2`) lets you change the permanent **boot order**.

What to change.

- Put the **USB drive first** in the boot order, or select it from the one-time boot menu.
- For modern Windows 11, ensure **UEFI mode** is used and **Secure Boot** is on (if the machine supports it). Secure Boot helps prevent malware from loading before Windows.
- Save and exit (often `F10`, then confirm **Yes**).

Analogy. The BIOS/UEFI is the **gatekeeper at the compound gate**. Every morning it decides which vehicle is allowed to drive in first. Normally it lets the family car (the internal drive) in. Today we ask it to let our delivery van (the USB) in first, because it is carrying the new furniture (Windows).

Realistic scenario. A learner builds the USB correctly but the machine still boots to the old Windows. Cause: the boot order still points to the internal drive, or the USB was plugged into a port that was not scanned at boot. Fix: use the one-time boot menu (`F12`) and select the USB explicitly.

Common misconceptions.

- "Pressing the key once at any time opens BIOS." It must be pressed **repeatedly during the first few seconds**, before Windows starts loading.
- "Legacy and UEFI are the same." They boot differently and use different partition styles (MBR vs GPT). Installing in the wrong mode can make Windows fail to boot.
- "Secure Boot is something I must switch off." Do not disable it casually; it is a security feature. Only change it if the machine genuinely cannot boot the USB and you understand the trade-off.

Limitation. Some manufacturer screens use unusual wording (for example "Boot Option Priorities", "Boot Sequence", "Legacy Support"). Teach learners that the **concept** is constant even when the words differ.

F.7 The clean install and partitioning

The idea. A **clean install** means we install Windows fresh onto a drive that has been emptied of the old system.

Step-by-step during Windows Setup (wording may vary by version).

1. The machine boots from the USB and Windows Setup opens.
2. Choose **language**, **time and currency format**, and **keyboard layout**, then click **Next â†' Install now**.
3. If asked for a **product key**, enter a valid licensed key, or choose **"I don't have a product key"** if you will activate later with the customer's key/digital licence. (Windows still needs a valid licence to be legitimate.)
4. Select the **edition** that matches the customer's licence (for example Windows 10/11 Home or Pro). Do not install an edition the machine is not licensed for.
5. **Accept the licence terms.**
6. Choose the installation type. You will see two options:
    - **Upgrade: Install Windows and keep files, settings and applications** â€" keeps the old system's personal files and apps (only offered when starting Windows from inside a working Windows).
    - **Custom: Install Windows only (advanced)** â€" this is the **clean install**. Choose this.
7. You now see a list of **drives and partitions**. To truly clean the drive:
    - Select each existing partition on the target drive and click **Delete**.
    - Repeat until only **"Drive 0 Unallocated Space"** remains. **This permanently erases the drive.** (Remember: the backup is already done.)
    - Select the **unallocated space** and click **Next**.
8. Windows copies files, installs features and updates, and **restarts several times**. **Do not remove the USB** until setup has fully completed and you are at the desktop setup screen. (In some configurations the machine may boot from the USB again; the one-time boot menu avoids this.)
9. Complete the **out-of-box setup (OOBE)**: region, keyboard, network (if available), user account and password, privacy settings. Then Windows finishes setting up.

Why partition at all? A **partition** is a defined section of a physical drive. Windows does not just need one â€" it creates several automatically when you install to unallocated space:

- An **EFI system partition** (on UEFI/GPT disks) or a small reserved partition for boot files.
- A small **Microsoft Reserved** partition.
- The main **Windows (C:)** partition where the OS and programs live.
- Usually a **Recovery** partition.

These extra partitions hold the boot files and recovery tools Windows needs. When you install to unallocated space, Windows creates them for you â€" you do not have to hand-make them at this level.

Should we split the drive into a C: and a D:? For beginners, keep it simple: let Windows create the C: partition automatically. Larger businesses sometimes create a separate data partition so that a future reinstall of C: does not touch their documents. That is a valid technique, but it is optional and must be planned **before** the wipe.

What a clean install removes.

- The old operating system.
- All installed programs.
- **All personal files on that drive** (unless backed up).
- The old user accounts and settings.

What it does NOT remove.

- Files on a **different** physical drive that you did not touch.
- The hardware itself.

Analogy. Partitioning is like drawing rooms on a large empty plot of land. Windows needs a few small utility rooms (boot, recovery) plus a big living room (C:). Installing to unallocated space is telling Windows: "Here is empty land â€" build the house your way."

Common misconception. "If I choose Custom but do not delete partitions, my files are safe." Not necessarily â€” installing over an existing partition still overwrites much of it. If you intend to keep data, you must back it up; Custom + Delete is the clean, predictable choice.

Limitation. A clean install gives you a fresh Windows but **no programs, no personal files and no drivers** â€” you must restore all of those afterwards.

F.8 Drivers, updates, activation, software, data restore and verification

The idea. Installing Windows is only half the job. A freshly installed Windows is like a new house with no furniture, no electricity connected and no paint. The rest of the work makes it usable.

1. Install drivers â€” in the right order.

- **Network driver first.** Without it, the PC cannot reach the internet, so you cannot download any other driver. This is why we keep a copy of the network/board driver on the offline USB.
- Then **chipset**, then **graphics**, then **audio**, then Wi-Fi/Bluetooth and any others.
- How to install: run the driver installer from your USB, or use Windows Update which often finds common drivers automatically.
- **Check Device Manager** (`Windows key + X â†’ Device Manager`). A **yellow exclamation mark** next to a device means a driver is missing or faulty.

2. Run Windows Update â€” repeatedly.

- Open **Settings â†’ Update & Security â†’ Windows Update** (Windows 10) or **Settings â†’ Windows Update** (Windows 11).
- Click **Check for updates**, install everything, and **restart** when asked.
- **Repeat** until it says no more updates are available. New updates often appear after the first round, so one pass is rarely enough.

3. Activate Windows â€” legally.

- Open **Settings â†’ Update & Security â†’ Activation** (Windows 10) or **Settings â†’ System â†’ Activation** (Windows 11).
- If the customer has a valid key, enter it. If the machine has a **digital licence** tied to its hardware, Windows may activate automatically after it connects to the internet.
- The academy rule: **only properly licensed Windows.** Never use cracked tools, "activators", key generators or pirated keys. They are illegal, and they carry malware and leave customers exposed.

4. Install essential software.

- A web browser if the customer wants one (Edge is built in).
- **Windows Defender** is already built in â€” it is the free, legitimate antivirus. Update it.
- Only install software the customer is **licensed** to use. Do not install pirated Microsoft Office or "cracked" versions of paid programs.

5. Restore the customer's data.

- Copy the backed-up files from your external drive back into the customer's **Desktop, Documents, Pictures** folders.
- Check a few files open correctly. Tell the customer what was restored.

6. Post-install verification (the professional finish).

- **Device Manager:** no yellow warning icons.
- **Display:** correct resolution, no weird stretching.
- **Sound:** play a short audio clip; confirm speakers/headphones work.
- **Network:** open a web page; confirm internet and, ideally, Wi-Fi.
- **Windows Update:** fully up to date.
- **Activation:** Windows reports "Windows is activated".
- **Storage:** correct drive shows its full capacity (for example a 500 GB SSD shows roughly 465 GB usable).
- **Customer sign-off:** show the customer the machine working and get them to sign the service report (Session 11).

Analogy. Drivers are like the **keys and connections** for each appliance in the new house â€” plumbing to the kitchen, electricity to every socket. Windows Update is the **maintenance schedule** that fixes known problems. Activation is the **legal title deed** proving the house is properly owned.

Common misconception. "Windows Update installs every driver, so I never need the offline USB." Sometimes true, sometimes not â€” especially for network drivers on older or unusual hardware. Always carry the offline network driver. Prepare before you need it.

Limitation. Some hardware (very new or very old) may need a driver the customer must supply, particularly for specialised devices like a fingerprint reader or an unusual printer. Be honest if you cannot find one.

F.9 Licensing and ethics â€” non-negotiable

- **Only install Windows the customer is licensed to use.** A valid product key, or a digital licence already attached to that machine, is what makes the install legitimate.
- **Never** teach, use or recommend pirated Windows, "cracked" activators, key generators or modified ISOs. They are illegal, unsafe and damage the academy's and your own reputation.
- If a customer asks you to install Windows without a licence, explain politely that you install only properly licensed software, and offer to help them buy a legitimate licence or use the free, legal alternatives where they fit their needs.
- Keep the **customer's data confidential** and handle it only for the repair.

Say this to the class:

> *"In this trade, your name is your business. One stolen licence can cost you every customer who hears about it. We do it right, or we do not do it."*

F.10 Summary of the session

- The **OS** manages the whole computer; a **clean install** is a heavy but sometimes necessary fix.
- **Back up first** â€” a clean install erases the drive.
- Check the machine **meets the OS requirements** (especially for Windows 11: RAM, TPM 2.0, Secure Boot).
- Build a bootable USB with the **official Microsoft Media Creation Tool**.
- Set the **boot order / boot menu** in **BIOS/UEFI**.
- Choose **Custom: Install Windows only (advanced)** and install to **unallocated space**; Windows creates the partitions.
- Install **drivers** (network first), run **Windows Update** until none remain, **activate legally**, add essential software, **restore data**, and **verify** with Device Manager and tests.
- Only ever use **licensed** Windows.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

What the instructor prepares:

- A practice PC that is safe to wipe (or, if it cannot be wiped live, a machine already clean, plus the prepared installer to show the first screens).
- The official **Media Creation Tool** already downloaded, plus the USB stick.
- The offline **driver USB**.
- A projector showing the target machine.

> ***Instructor note:*** *If time or hardware does not allow a full live install, demonstrate the **media creation** and the **boot/setup screens**, then walk the class through the printed checklist for the rest. Never start a wipe on a machine whose data has not been backed up.*

G.1 Demonstration 1 â€” Building the USB (5 minutes)

Say: "Watch how Microsoft's own tool builds our installer. I do this once, and this one USB can rebuild many machines."

Do this on the projector, narrating each step:

1. Plug in the USB and open the Media Creation Tool.
2. Accept the licence terms.
3. Choose **"Create installation media for another PC"**.
4. Keep the recommended **language/edition/architecture (64-bit)**.
5. Choose **"USB flash drive"**, then **select the correct drive** â€" *pause here and say:* "Look carefully at the drive letter and size. Choosing the wrong drive erases someone's backup."
6. Click **Next**; show the progress screen. **Say:** "This downloads Windows and writes it to the USB. If we unplug now, we ruin it."

Expected result: A finished screen saying the USB is ready, and (if you open it) folders such as `sources` and `boot`.

Error to watch: If the learner selected the wrong drive, stop immediately. If the tool reports not enough space, the USB is smaller than 8 GB or is not empty.

G.2 Demonstration 2 â€" Booting from the USB (4 minutes)

Say: "Now we ask the machine to start from our USB instead of its own drive."

1. Insert the USB. Power on.
2. **Repeatedly tap the one-time boot key** (on the demo machine, tell the class which key it is â€" for example `F12`).
3. In the boot menu, select the **USB drive**.

Expected result: Windows Setup appears (language screen). Point out that this is Windows Setup, not the old desktop.

Ask the class: "What if nothing happens and the old Windows loads?" â†' *The USB was not scanned, the key was pressed too late, or the USB is not bootable.*

G.3 Demonstration 3 â€" Setup and the wipe (4 minutes)

1. Choose **language**, click **Next â†' Install now**.
2. Choose **"I don't have a product key"** only to show the option exists â€" and **state clearly** that a valid licence is still required. Better still, demonstrate entering a licensed key if the academy has one.
3. Accept the licence terms.
4. Choose **"Custom: Install Windows only (advanced)"**.
5. Show the partition list. Select the old partitions, click **Delete**, and show them turn into **"Unallocated Space"**.

Say (this is the critical line): "Everything on this drive is now gone forever. This is why we backed up. If I had not backed up, this moment would have destroyed the customer's memories."

6. Select the **unallocated space â†' Next**. Show Windows begin copying files. **Say:** "It will restart several times. Do not remove the USB until we reach the desktop setup."

Questions to ask during the demo:

- "Why do we install the network driver first?" â†' Without internet, no other driver can be downloaded.
- "Why do we choose Custom, not Upgrade?" â†' Custom is the clean install; it does not keep the old broken system.
- "Which part of this demo could destroy customer data if done carelessly?" â†' Deleting partitions / wiping the drive.

G.4 Demonstration 4 â€" First boot and verification (2 minutes)

If a machine is available that is already installed (or after the demo completes), show:

- `Windows key + X â†' Device Manager` â€" point out any yellow icon and explain it.
- **Settings â†' Windows Update** â€" run it once to show the process.
- **Settings â†' Activation** â€" show the activation status.

Say: "A machine is not 'done' when Windows installs. It is done when every device works, it is updated, it is legally activated, the customer's data is back, and the customer has signed."

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding step by step, learners complete the **pre-installation preparation** on paper/real USB, then either perform or simulate a clean install and complete the post-install checklist.

Starting materials: Per pair: one practice PC, one USB (or the instructor's shared USB), the offline driver USB, Worksheet 9.1 and Worksheet 9.2. If there is only one safe-to-wipe machine, learners take turns on the live machine while others complete the simulated checklist.

> ***SAFETY AND DATA SAFETY FIRST â€" read before starting:*** *confirm with the instructor that any machine to be wiped holds* ***no data the academy needs****, and that the customer's data (in the simulated case) has already been backed up and verified.*

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **State the fault out loud.** "The operating system is corrupted; the fix is a clean install." Confirm this against the diagnostic workflow.
2. **Back up (simulate or real).** Using the Customer Intake Form, list the folders to save: Desktop, Documents, Downloads, Pictures, Music, Videos. If the machine does not boot, describe how you would remove the drive and copy the data on a working PC (Session 7 skill). **Pause and predict:** *"If we skip this step, what exactly is lost?"* â†' All personal files on the drive.
3. **Check requirements.** Record the machine's CPU, RAM and drive space on Worksheet 9.1. Decide Windows 10 or 11 appropriately.
4. **Gather drivers.** Confirm the offline network/board driver is on the second USB.
5. **(If doing a live build) create the USB.** Follow Section F.5. **Pause and predict:** *"How long will this take, and what happens if we unplug?"* â†' 20â€"60 minutes; we corrupt the USB.
6. **Set the boot device.** Power on, tap the boot key, select the USB. **Pause and predict:** *"What screen should appear if we did this correctly?"* â†' Windows Setup.
7. **Start Setup.** Language â†' Install now â†' (key or skip) â†' accept terms.
8. **Choose the clean install.** Select **Custom: Install Windows only (advanced)**.
9. **Wipe and install.** Delete existing partitions to leave **Unallocated Space**, select it, click **Next**. **Pause and predict:** *"Is the data on this drive still recoverable normally after this?"* â†' No.
10. **Wait for install.** Let it copy files; the PC restarts several times. Do not remove the USB.
11. **OOBE setup.** Region, keyboard, network, account, privacy.
12. **Install the network driver**, then chipset, graphics, audio.
13. **Run Windows Update** at least twice until none remain.
14. **Activate Windows** with a licensed key/digital licence. Confirm status.
15. **Install essential software** (browser, confirm Defender updated). No pirated software.
16. **Restore the backed-up data** and open two files to confirm they work.
17. **Verify** using Worksheet 9.2 (Device Manager clean, display, sound, network, activation, storage).

Expected result: A working, updated, activated Windows with the customer's data restored and no missing drivers.

Pause-and-predict checkpoints: after step 2 (what is lost), step 6 (which screen), step 9 (is data still recoverable), step 12 (why network driver first).

Troubleshooting while walking the room:

- USB does not boot â†' wrong boot key/order, USB not bootable, or USB in a port not scanned; try a rear USB port on a desktop.
- "No drives were found" â†' the storage controller driver (for example Intel RST) may be needed; a more advanced fix â€" tell the learner to keep the drive connected and ask the instructor.
- Setup asks for a key and refuses to continue â†' choose the "I don't have a product key" option if offered, and plan to activate later with a valid licence.
- Internet still not working after setup â†' network driver not installed; run the offline driver.

Completion checklist (learner ticks each):

- ☐ Data backed up and verified before any wipe
- ☐ Hardware requirements checked and recorded
- ☐ Bootable USB prepared correctly

- [ ] Booted from USB to Windows Setup
- [ ] Chose **Custom** and installed to unallocated space
- [ ] Network driver installed
- [ ] Windows Update run until none remain
- [ ] Windows licensed and activated
- [ ] Essential software installed (all licensed)
- [ ] Customer data restored and opened
- [ ] Device Manager shows no yellow warnings
- [ ] Post-install verification complete

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task (NEW task): You are handling **two separate customer jobs** at the same time. Decide what to do for each, in the right order, and write the exact steps. This is not a repeat of the guided install â€” it tests judgement, not just memory.

Scenario A â€” "The Slow Laptop" (Mrs. Eze, HP laptop, 4 GB RAM, 1 TB HDD, Windows 10) Mrs. Eze says the laptop "takes 10 minutes to open and is full of viruses". It still **boots and runs** Windows. She has not backed up. She asks you to "reinstall Windows".

Scenario B â€” "The Dead Desktop" (Mr. Boma, custom desktop, 500 GB SSD, will not boot) The desktop shows *"An operating system wasn't found."* The customer says his **only copy** of his business accounts is on the Desktop of that drive. He is licensed: the machine has a Windows digital licence.

Requirements:

1. For **each** scenario, write the correct order of actions, including the **data-safety decision** you make first.
2. For **Scenario A**, answer: Should you reinstall straight away, or do something else first? Justify it.
3. For **Scenario B**, describe how you get the data **if Windows will not boot**, and how you will confirm the licence before activating.
4. For each, list the driver order you will follow after Windows is installed.
5. State the academy's rule about licensing in one sentence.

Time allowance: 15 minutes.

Success criteria: Correct order, data safety addressed first in both, a sensible decision for Scenario A (not a blind wipe), a workable data-recovery-by-removing-the-drive plan for Scenario B, correct driver order, and a correct licensing statement.

Hints if stuck:

- Hint 1: Ask "is Windows actually broken, or is the problem something else?" for Scenario A.
- Hint 2: If Windows will not boot, how did you copy data from a failing drive in Session 7?
- Hint 3: The driver order always starts with **network**, then chipset, graphics, audio.

INSTRUCTOR ANSWER KEY (do not show learners):

Scenario A â€” right order:

1. **Do not wipe immediately.** First diagnose: a slow machine full of viruses may be fixable without a clean install. Confirm with the diagnostic workflow.
2. **Back up the data first** regardless of which path you take â€” copy Desktop, Documents, Pictures etc. to an external drive and verify.
3. Run a proper malware scan (Windows Defender plus, if appropriate, a reputable scanner), remove malware, run Windows Update, and consider a **RAM upgrade** or an **SSD replacement** if the 1 TB HDD is the real cause of slowness (often it is).
4. Only if the OS is genuinely unrecoverable do you proceed to a **clean install**: boot from USB â†’ Custom â†’ wipe to unallocated â†’ install.
5. Justification: the fault is likely a combination of malware and an old HDD, not necessarily a dead OS. A blind reinstall risks data and may not even fix the slowness.

- Honest note for the answer: recommend an **SSD upgrade** â€” it will change the laptop far more than a reinstall, but it is optional and must be approved by the customer first.

Scenario B â€" right order:

1. **Data safety first.** The drive holds the only copy of the accounts and Windows will not boot. Do **not** attempt a clean install until the data is saved.
2. **Remove the SSD**, connect it to a working PC (USB-to-SATA adapter or an internal SATA port), and copy the Desktop/business files to a separate external drive. Verify the files open. (Session 7 skill.)
3. If the SSD is readable and healthy, proceed with the **clean install** on that same drive (or a new drive if the SSD is failing). Use the **official Media Creation Tool** to make the USB.
4. Boot from USB â†' **Custom** â†' delete partitions to unallocated â†' install.
5. **Licence:** this machine has a **digital licence** tied to its hardware, so after installation and connection to the internet, Windows may activate **automatically**. Check **Settings â†' Activation**. If it does not auto-activate, use the customer's legitimate method (linked Microsoft account or valid key). Never use a crack.
6. **Drivers in order:** network first, then chipset, graphics, audio, then any Wi-Fi/Bluetooth. Then Windows Update until none remain.
7. Restore the business files to the Desktop; confirm with the customer.

Licensing statement (any accurate version accepted): "We install only properly licensed Windows â€" a valid key or a digital licence tied to the machine â€" and we never use pirated or 'cracked' software."

Marking guidance (out of 10):

- Data-safety decision correct in both scenarios (3)
- Scenario A judgement (diagnose/repair before wipe) correct and justified (2)
- Scenario B data-recovery-by-removing-drive method correct (2)
- Driver order correct (1)
- Licensing rule correct (1)
- Clear, ordered, professional write-up (1)

J. REAL-WORLD CASE STUDY

Situation. *Elias Tech Hub*, a repair shop on Ebony Road, Port Harcourt, receives a school laptop from **Holy Family Primary School**. The teacher says it "just stopped working"; the school's term results are on the Desktop. The laptop will not boot past a blue recovery screen. The school has a **digital licence** for Windows 10 Pro on that machine.

Relevant facts.

- The drive is a 256 GB SSD and it is **readable** when connected to another PC.
- The term results are in a single `Results.xlsx` on the Desktop.
- The school needs the laptop back **the next morning** for a PTA meeting.
- The technician has the official Media Creation Tool USB and an offline network driver USB.

Decision point. What order of operations protects the school, fixes the machine, and is honest about timing?

Possible approaches.

1. Wipe and reinstall immediately to save time â€" fastest, but destroys the only copy of the results.
2. Back up first (remove the SSD, copy the file), then clean-install, then restore and verify â€" safe and professional.
3. Tell the school the data is lost and only fix Windows.

Recommended solution. Approach 2.

- Step 1: Remove the SSD and copy `Results.xlsx` to an external drive; open it to confirm it is intact.
- Step 2: Build/boot the official USB; clean-install Windows 10 Pro to unallocated space.
- Step 3: Install the network driver, run Windows Update, confirm the **digital licence activates automatically** (licence is tied to the machine).
- Step 4: Restore `Results.xlsx` to the Desktop; confirm the teacher can open it.
- Step 5: Write the service report (Session 11) and have the teacher sign.

Trade-offs.

- Approach 1 saves 30 minutes but risks the school's results and the shop's reputation. Unacceptable.
- Approach 2 takes longer but delivers a working, licensed, data-safe result the same day â€" realistic for a one-hour-grade job.
- Approach 3 abandons the customer needlessly; the data was recoverable.

Professional connection. This is the everyday reality of a repair shop: the technical skill (installing Windows) is only half the job. The judgement to **protect data first** and the discipline to use **licensed software** are what make a technician trustworthy and keep repeat customers coming back.

Transfer question for the class: "The school's laptop had a digital licence. What would you do differently if it had no licence at all?" *(Answer: explain that you install only licensed Windows; help them obtain a legitimate licence; do not pirate.)*

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the live troubleshooting guide. When a learner hits an issue, ask them to find the row rather than handing them the answer.

K.1 Installing before backing up

Symptoms: You realise the customer's files were on the drive and you already deleted the partitions. **Likely cause:** Skipping data safety under time pressure. **How to investigate:** Stop immediately. Do not write any more data to the drive. **Safe corrective steps:** If genuinely required and the data is critical, **stop all use** and refer to a professional data-recovery service â€" this is **beyond this course** and must be handled by a specialist. Do not promise success. **How to confirm resolved:** A specialist recovers the data, or you inform the customer honestly. **Prevention:** Back up and verify **before** every wipe. Make it an unbreakable rule.

K.2 USB will not boot

Symptoms: The old system loads, or a "no bootable device" message appears. **Likely causes:** Wrong boot key, USB not first in boot order, USB not bootable, or USB in a port not scanned. **How to investigate:** Enter the boot menu again; check the USB appears in the list. Try a different (rear) USB port on a desktop. **Corrective steps:** Rebuild the USB with the official tool if it is not bootable; select the USB explicitly. **Prevention:** Use a known-good USB and the official tool; keep the one-time boot key noted.

K.3 "No drives were found" or wrong disk shown

Symptoms: Windows Setup cannot see any drive. **Likely causes:** A newer machine needs a storage/RAID driver (for example Intel RST); or the drive is genuinely not detected (loose cable or dead drive â€" a hardware fault). **How to investigate:** Check whether the drive appears in BIOS/UEFI. Reseat cables (safety: power off, anti-static). **Corrective steps:** Load the storage driver during setup (advanced), or fix the hardware fault first. **Prevention:** This is an advanced issue â€" ask the instructor; do not guess.

K.4 Internet not working after install

Symptoms: No network, cannot download drivers. **Likely cause:** Network driver not installed. **Corrective steps:** Install the offline network driver from the USB, then run Windows Update. **Prevention:** Always prepare the network driver before starting.

K.5 Yellow exclamation marks in Device Manager

Symptoms: Device Manager shows warning icons. **Likely cause:** Missing or faulty drivers. **Corrective steps:** Identify the device, install the correct driver (USB or Windows Update). **Prevention:** Install drivers in order (network, chipset, graphics, audio) and re-check Device Manager.

K.6 Windows will not activate

Symptoms: "Activate Windows" watermark; activation fails. **Likely causes:** No valid key/digital licence; not connected to the internet; edition mismatch. **Corrective steps:** Connect to the internet and retry; confirm the licence belongs to the machine; ensure the correct edition was installed. **Prevention:** Confirm licensing **before** starting, and never rely on cracks.

K.7 Only one Windows Update pass run

Symptoms: The machine still feels under-updated; some drivers missing. **Likely cause:** Windows Update was run once. **Corrective steps:** Run it repeatedly (restarting) until no updates remain. **Prevention:** Remember the first pass triggers more passes.

Distinguish likely vs confirmed: A "likely cause" becomes "confirmed" only after you check it (see the exact error, view BIOS, check Device Manager). Teach learners to confirm before fixing.

L. SESSION ASSIGNMENT

Assignment title: "My Windows Clean-Install Runbook"

Purpose: To make the learner write, in their own words, a safe, ordered, licensing-compliant procedure they could follow alone in a real shop â€” practising both technical accuracy and data-safety discipline.

Realistic scenario: You have just been hired at *Blessed Hands Computers*, a small repair shop in Port Harcourt. On your first day, your boss hands you a customer's PC with a corrupted Windows and says, "Rebuild it properly â€” and do not lose a single file." You must follow your own written runbook.

Detailed instructions:

1. Write a **pre-installation section** that begins with the backup step and states exactly which folders you save and how you verify the backup.
2. Add a **hardware requirements check** section â€” how you check CPU, RAM and storage, and how you decide between Windows 10 and Windows 11.
3. Add a **media creation** section describing the official Microsoft Media Creation Tool steps.
4. Add a **BIOS/UEFI boot** section explaining how to reach the boot menu and select the USB.
5. Add an **install** section covering Custom install, deleting partitions to unallocated space, and the warning that this erases data.
6. Add a **post-install** section covering drivers in order, Windows Update (repeated), activation, essential software, data restore and verification.
7. Add a **licensing and ethics statement** in your own words.
8. Add a **troubleshooting table** with at least five problems and their fixes.
9. Finish with a **customer sign-off** line.

Required deliverables:

- A typed document (Word or a neatly written sheet) of **at least one full page**.
- The completed Worksheet 9.1 (pre-install checklist) and Worksheet 9.2 (post-install verification).

Submission format: Printed or emailed to the instructor; filenames as `Surname_S9_Runbook.docx`.

Deadline: Before Session 10.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Pre-installation & data safety	Backup first, folders listed, verification described	Backup mentioned, some detail	Backup missing or vague
Requirements & OS choice	Requirements checked; sound Windows 10/11 decision	Checked but decision weak	Not checked
Media creation & boot	Accurate official-tool steps; correct boot method	Mostly correct	Incorrect or missing
Install & partitioning	Correct Custom steps; wipe warning clear	Mostly correct	Wrong install type or unsafe
Post-install & verification	Drivers in order, updates repeated, activation, restore, verify	Most steps present	Many steps missing

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Licensing statement & troubleshooting	Clear, honest licensing rule; 5+ good fixes	Present but thin	Missing or unsafe

Common mistakes to avoid: putting media creation before backup; omitting the wipe warning; forgetting the network driver; running Windows Update once; using or implying pirated software.

Instructor reference solution (excerpt):

> *Pre-installation. 1) Interview the customer and confirm the fault. 2) Back up Desktop, Documents, Downloads, Pictures, Music, Videos to an external drive; open three files to verify. 3) If the PC will not boot, remove the drive and copy data on a working PC. 4) Check CPU/RAM/storage against requirements and choose Windows 10 or 11 accordingly. 5) Confirm licensing (valid key or digital licence). 6) Prepare the offline network/board driver on a second USB.*
>
> *Install. 7) Build the bootable USB with the official Microsoft Media Creation Tool. 8) Boot from USB via the one-time boot menu. 9) Language â†' Install now â†' accept terms. 10) Choose **Custom: Install Windows only (advanced)**. 11) Delete all partitions to **Unallocated Space** (this permanently erases the drive). 12) Select unallocated space â†' Next.*
>
> *Post-install. 13) Install network driver, then chipset, graphics, audio. 14) Run Windows Update repeatedly until none remain. 15) Activate with the licensed key/digital licence. 16) Install only licensed essential software. 17) Restore and verify the customer's data. 18) Verify Device Manager, display, sound, network, activation, storage. 19) Customer signs the service report.*
>
> *Licensing. I install only properly licensed Windows and never use pirated or cracked software.*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key.)

Q1. Multiple choice. Which installation type should you choose for a clean install? a) Upgrade: Install Windows and keep files and apps b) Custom: Install Windows only (advanced) c) Repair Windows d) Reset this PC

Q2. Short answer. Why must you back up a customer's data before a clean install?

Q3. Multiple choice. Which driver should be installed **first** after a fresh Windows install, and why? a) Graphics â€" so the screen looks right b) Network â€" so the internet works and you can download the rest c) Audio â€" so the customer can hear d) Printer â€" so they can print

Q4. Explain in your own words. Why must a bootable USB be at least 8 GB, and what happens to whatever is already on the USB?

Q5. True or false. Once Windows is installed, running Windows Update a single time is enough. Explain your answer.

Q6. List. Name three checks you perform as part of post-install verification.

Q7. Scenario. A customer asks you to install Windows but has no licence. What do you do, and what do you **not** do?

Q8. Fill in the blank. To reach the one-time boot menu or BIOS/UEFI, you repeatedly press a key during the very first moments after pressing the power button. Common keys include _____, _____ and _____.

Q9. Identify the mistake. A learner builds the USB, boots to Setup, chooses Custom, selects the existing C: partition (without deleting it) and clicks Next, saying, "This will keep the customer's files safe." What is wrong with this reasoning?

Q10. Order the steps (1â€"6).

- ___ Install the network driver
- ___ Back up the customer's data
- ___ Run Windows Update until none remain
- ___ Create the bootable USB
- ___ Choose Custom and install to unallocated space
- ___ Verify Device Manager, sound, network and activation

INSTRUCTOR ANSWER KEY (Q1â€“Q10)

A1. (b) Custom: Install Windows only (advanced). **A2.** Because a clean install erases everything on the drive, including all personal files. Without a backup the data is permanently lost. **A3.** (b) Network â€“ without internet you cannot download the other drivers. **A4.** Windows installation media needs 8 GB or more of space to hold the installer files. Creating the USB **erases everything already on it**, so back up the USB first. **A5.** False. The first update run often reveals more updates, including driver and security updates. Run it repeatedly, restarting as prompted, until no more updates are available. **A6.** Any three of: Device Manager shows no yellow warnings; correct display resolution; sound works; network/internet works; Windows activated; Windows fully updated; storage shows correct capacity; customer's data restored and opens. (Accept any three correct checks.) **A7.** Explain politely that you install only properly licensed Windows, and offer to help them buy a legitimate licence or consider legal alternatives that fit their needs. You do **not** install a pirated copy, use a crack/activator, or use a key generator â€“ those are illegal and unsafe. **A8.** Any three correct keys, e.g. `Del`, `F2`, `F12` (also accept `F10`, `F9`, `Esc`). **A9.** Installing over an existing partition can still overwrite much of its contents and does **not** guarantee the customer's files are safe. The predictable, clean approach is to back up first, then delete partitions to unallocated space. Never assume files are safe without a verified backup. **A10.** Order: 1 Back up the data â†' 2 Create the bootable USB â†' 3 Choose Custom and install to unallocated space â†' 4 Install the network driver â†' 5 Run Windows Update until none remain â†' 6 Verify Device Manager, sound, network and activation.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned the biggest single job in software repair: putting a fresh operating system onto a machine. You learned why a corrupted OS sometimes has to be reinstalled, and â€“ most importantly â€“ that **data safety comes first**: you back up and verify before you wipe. You checked that the hardware meets the OS requirements, built a bootable Windows USB with Microsoft's own tool, booted from it through BIOS/UEFI, chose a **Custom** clean install, wiped to unallocated space, and understood why Windows creates several partitions. You then finished the job the professional way: drivers in the right order, Windows Update repeated, **legal activation**, essential software, restoring the customer's data, and a full verification. And you committed to using only **licensed** Windows.

Five important takeaways:

1. **Back up first, verify, then wipe** â€“ a clean install erases the drive.
2. Build media with the **official Microsoft Media Creation Tool**; never pirated ISOs.
3. Choose **Custom: Install Windows only (advanced)** and install to **unallocated space**.
4. Install the **network driver first**, then the rest; run **Windows Update repeatedly**.
5. A machine is finished only when it is updated, **legally activated**, verified and the customer's data is restored.

Questions to help you recognise what you can now do:

- Can I explain the whole install process from backup to verification in my own words?
- Can I build a bootable USB with the official tool?
- Can I reach the boot menu and boot from that USB?
- Can I explain why a clean install is irreversible without a backup?
- Can I state the academy's licensing rule without looking it up?

Connection to the next session: You can now fix a machine and hand back a working computer. But a repair shop is not only about machines â€“ it is about **people**. Next session you learn **customer service**: how to greet a customer, capture the fault accurately in their own words, explain it in plain language, quote honestly, and handle an unhappy customer professionally. The best technical skill in Port Harcourt will fail if you cannot talk to the person paying for it.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner states the **backup step first**, unprompted, before any mention of wiping.
- The learner creates the USB with the **official** tool and selects the correct drive carefully.
- The learner navigates the boot menu/BIOS without being told every key.

- The learner chooses **Custom** and clearly explains that deleting partitions erases data.
- The learner installs the **network driver first** and can say why.
- The learner runs Windows Update **more than once**.
- The learner uses only licensed Windows and can state the rule confidently.
- The learner completes a real verification (checking Device Manager, sound, network, activation) rather than claiming "it works".

Common misconceptions to watch for and correct:

- "A reinstall is safe without a backup." â†' A clean install erases the drive. Back up first, always.
- "Windows Update installs all drivers automatically." â†' Often, but not always â€" especially the network driver. Keep an offline driver.
- "I can use a cracked Windows to save money." â†' Correct immediately and firmly: only licensed software; piracy is illegal and unsafe.
- "Running Windows Update once is enough." â†' Run it repeatedly until none remain.
- "Choosing Custom keeps my files." â†' Only if you do not wipe; and never rely on that. Back up.
- "BIOS and UEFI are the same thing." â†' Similar job, different technologies (MBR/legacy vs GPT/UEFI + Secure Boot).

Suggested remedial exercises for learners who struggle:

1. **Order-the-steps drill:** hand out the nine stages on cards (backup, requirements, media, boot, install, drivers, update, activate, verify) and have them arrange physically, then explain each.
2. **"What could go wrong?" drill:** give five unsafe steps (wipe before backup, skip network driver, one update pass, cracked key, delete wrong partition) and ask for the correct fix.
3. **Plain-language drill:** ask the learner to explain to a "customer" why the computer needs a fresh Windows and what will happen to her files.

Differentiation:

- **Level 1 (supported):** provide a partially completed runbook and the full printed checklist; supervise the live install closely; pair the learner with a stronger peer; allow the simulated install if a live machine is not available to them.
- **Level 2 (standard):** complete the guided practical and independent exercise as written.
- **Level 3 (challenge):** plan a **dual-partition** layout (C: for Windows, D: for data) before installing; install Windows 11 on a UEFI/Secure Boot/TPM 2.0 machine; or prepare media with an alternative tool while still using a **licensed** Windows.

Timing guidance (105 min):

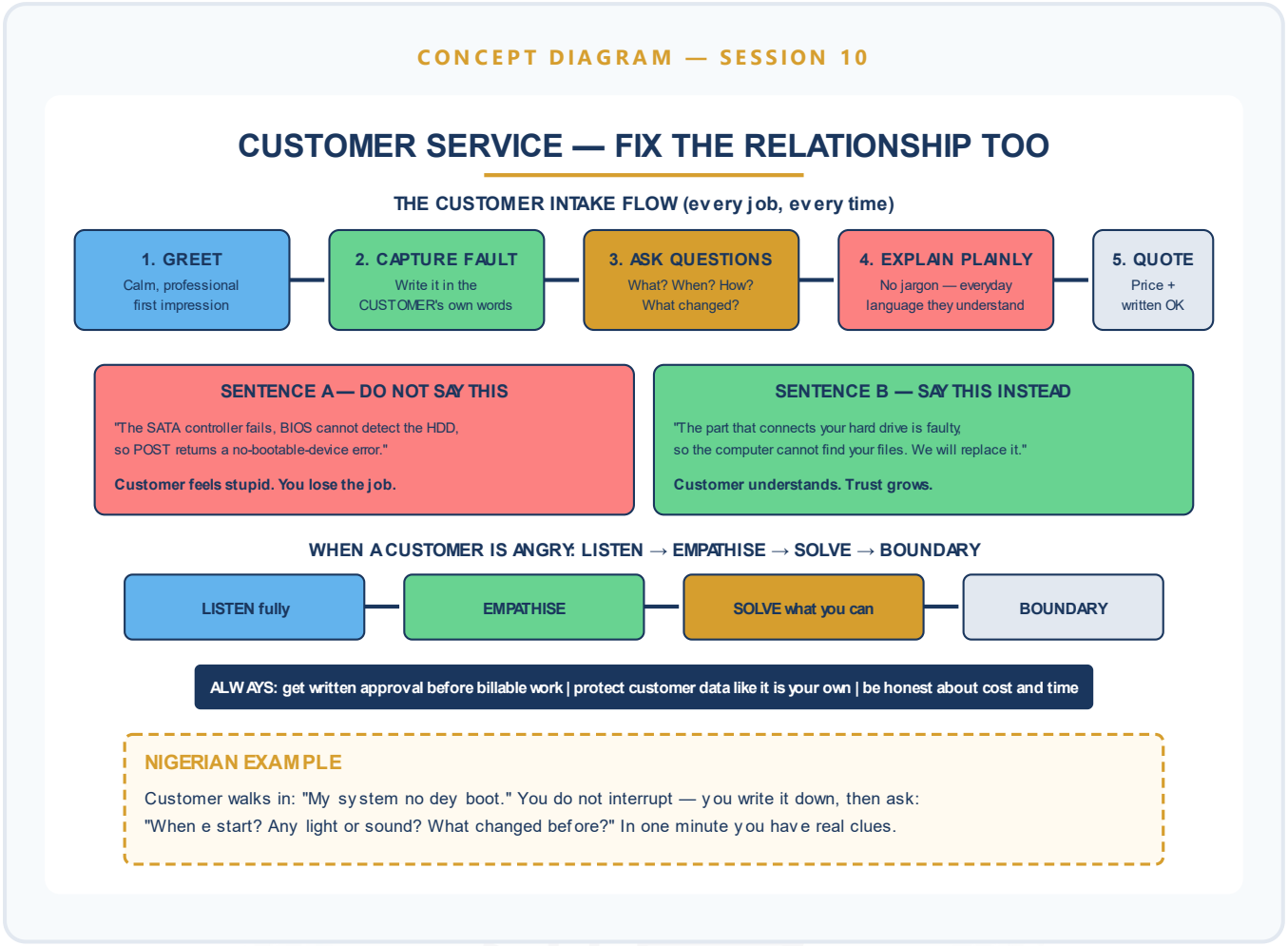
- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided practical 18, independent exercise 15, case study (can be folded into reflection) 3, reflection/recap plus knowledge check 8. If the room is slow, shorten the case study and protect the **guided practical** and the **data-safety discussion** â€" those are the priority. Never cut the backup step to save time.

END OF SESSION 9

Next: Session 10 â€" Customer Service and Communication.

SESSION 10 OF 12

Customer Service



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 10 â€" CUSTOMER SERVICE AND COMMUNICATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	10 of 12
Session title	Customer Service and Communication
Week / Module	Week 4 â€" Module 4: Professional Practice
Recommended duration	105 minutes
Difficulty level	Beginner (people skills, role-play heavy)
Prerequisite knowledge	Sessions 1â€"9 â€" the diagnostic workflow, all hardware and software repair skills, and OS installation. Learners can now fix machines; today they learn to serve the people who own

Item	Detail
	them.
Connection to previous work	In Session 4 you learned the five-step diagnostic workflow, which begins with **asking the customer good questions**. Today you turn that single step into a full professional skill: greeting, listening, capturing the fault accurately, explaining plainly, quoting honestly and handling difficult situations.
Preparation before class	Print role-play scenario cards (one set per pair); print Customer Intake Forms (one per learner); arrange the room so pairs can sit face to face; set up a "shop counter" table for realistic role-play; have two example estimates ready (one good, one bad) to display.
Professional note	This session has no dangerous hardware steps, but it carries a **confidentiality** duty: customer data and private information must never be shared, gossiped about or copied. Treat every role-play as though it were a real customer.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Greet a customer professionally** and create a calm, trustworthy first impression at a repair shop counter.
2. **Capture a fault accurately** in the customer's own words, using the Customer Intake Form, without leading or interrupting.
3. **Ask good diagnostic questions** (what, when, how, what changed) that move a vague complaint toward a real fault.
4. **Explain faults and fixes in plain language**, avoiding jargon or translating it into everyday Nigerian English.
5. **Manage expectations** about time, cost and uncertainty honestly, including saying "I don't know yet" professionally.
6. **Quote a repair and obtain written approval** before starting any billable work.
7. **Handle an unhappy or angry customer** calmly using listen â†’ empathise â†’ solve â†’ boundary.
8. **Protect customer confidentiality**, in the shop and outside it.
9. **Build trust and repeat business** through honesty, follow-up and consistent service.

C. REQUIRED MATERIALS

Hardware

- Practice computers (optional â€” one or two, so intake can reference a real device).
- A table set up as a "repair shop counter" for realism.

Printed materials

- Role-play scenario cards (one set per pair; see Section H).
- "Customer Intake Form" (one per learner) containing: customer name, phone, device details, fault in customer's words, when it started, what changed, data-backup consent, estimate approval and signature.
- "Estimate Template" (one per learner): diagnosis fee, parts, labour, total, timeline, warranty, approval line.
- Two display examples for the opening: one **good** estimate/ticket, one **poor** onethe instructor writes beforehand.

Software

- None required. If the academy has a shop-management or job-ticket app, show it briefly as an optional tool.

The instructor's own experience

- Prepare one short, true (anonymised) story of a good or bad customer interaction. Real stories teach faster than theory.

Internet

- Not required. If available, the instructor may show one short, respectful workplace-customer-service video near the end, but do not depend on it.

Alternative for learners who lack a facility

- Not applicable â€” this session needs no equipment, only people. Learners who are shy can begin as the "customer" (easier) and build up to the technician role.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: Write two sentences on the whiteboard, side by side.

- Sentence A: *"The motherboard's SATA controller is failing, causing the HDD to be undetectable by the BIOS, so the POST sequence returns a no-bootable-device error."*
- Sentence B: *"The part inside your computer that connects your hard drive is faulty, so the computer cannot find your files. We need to replace it."*

What the instructor says:

> *"Read both. They mean the same thing. Which one would you say to a market woman who brought in her shop computer? â€¦ Exactly. Sentence B.*
>
> *Here is the hard truth about this trade: you can be the best technician in Port Harcourt, but if you talk to customers in Sentence A, you will have no customers. Nobody pays a person who makes them feel stupid.*
>
> *Yesterday you learned to fix the machine. Today you learn to fix the relationship. You will learn how to greet a customer, take down their complaint in their own words, explain what is wrong in plain language, give an honest price, and stay calm when someone is shouting at you.*
>
> *This is the skill that turns a one-time repair into a customer who comes back for years â€” and tells their friends about you."*

Expected learner response: Learners will find Sentence B easier and may laugh at Sentence A. Use the moment to name the skill: **translation**.

Then demonstrate the hook: Ask a volunteer to act as a frustrated customer who says only: *"My computer is doing me somehow."* The instructor (as technician) asks three good questions and shows how a vague complaint can be turned into useful information within one minute. Say:

> *"Notice what I did: I did not guess, and I did not talk down to her. I asked good questions and let her talk. That is where every good repair begins."*

Why this opener works: It immediately shows the gap between technical language and customer language, and it proves that good questioning yields real information â€” the exact skills of today.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a fast oral round-robin. Put the questions on the projector.

Question 1: "What are the five steps of the diagnostic workflow?"

- Expected answer: observe, ask questions, form a hypothesis, test, resolve/verify.
- Common gap: learners remember "test" and "fix" but forget "ask questions". Emphasise that **questions to the customer** are step two and often solve the case before you open the case.

Question 2: "Why is it important to ask the customer what they were doing when the fault started?"

- Expected answer: it gives clues to the cause (for example a fault after a power cut, a drop, a spill, or an update).
- Common gap: learners think they can find everything from the machine alone. Clarify that customer context is free evidence.

Question 3: "What should you always do before starting a billable repair?"

- Expected answer: get the customer's approval of the cost and scope (an estimate).
- Common gap: learners skip straight to fixing. Reinforce that surprise bills destroy trust.

Question 4: "In Session 9, why did we say data safety comes first?"

- Expected answer: a clean install or wipe destroys data, so we back up and confirm with the customer first.
- Common gap: learners forget to *tell the customer* what will happen. Today's confidentiality and consent theme builds on this.

Bridge to today (say this):

> *"Look how much of today is already in what you know. The diagnostic workflow starts with asking. OS installation taught you to get the customer's consent before destroying anything. Today we simply turn these into a professional habit â€" because a repair shop lives or dies on how it treats people."*

Gap handling: If more than a third of the class cannot name the intake/approval idea in Question 3, pause and act out a 60-second "surprise bill" scene: technician fixes without quoting, then presents â‚¦60,000, customer is furious. Ask the class what went wrong. Then proceed. This makes the rest of the session land harder.

F. FULL LECTURE NOTES

F.1 Why customer service is a technical skill

The idea. Customer service is the set of behaviours that make a customer feel respected, informed and confident â€" before, during and after a repair.

Plain-English definition. Customer service is **how you treat the person while you do the job**. It is greeting them, listening, explaining, quoting honestly, keeping your word and being reachable.

Why it exists. A customer cannot see the skill in your hands. They cannot tell if your soldering is neat or your diagnosis is sharp. What they *can* judge is how you treated them. That judgement decides whether they come back, whether they recommend you, and whether they trust your price.

What it does.

- Makes a frightened customer feel safe (their data, their money).
- Prevents disputes by keeping everyone informed and getting approval.
- Builds a reputation that brings repeat and referred business.
- Reduces your own stress â€" a calm, informed customer is far easier to deal with.

Everyday analogy. Think of a **good barber**. He greets you, remembers your name, checks what style you want *before* he cuts, tells you the price, and does not start until you nod. He might cut hair no better than the next man, but you keep going back because he makes you feel attended to. A repair shop is the same.

Realistic scenario (Port Harcourt). Two technicians in Rumuola both fix a laptop for â‚¦15,000. One greets, explains and follows up. The other grunts, snatches the machine and ignores the customer. Within a year, the first has a queue and a WhatsApp list of loyal customers; the second wonders why business is slow.

Common misconception. "Customer service is only for receptionists or big companies." Wrong. In a small Nigerian repair shop, the technician **is** the business. Your people skills are part of your product.

Limitation / honesty. Good service cannot turn a bad repair into a good one. Service and skill must go together. This session assumes you are also doing the technical work well.

F.2 The first impression â€" greeting and setting the tone

The idea. The first thirty seconds set the mood of the whole transaction.

What good looks like.

- **Stop what you are doing** and turn to face the customer (or politely say "one moment, sir/ma").
- **Greet** appropriately: "Good afternoon, welcome. How can I help you today?"
- **Introduce yourself**: your name, and the shop's name if they are new.
- **Smile and use a calm voice.** Do not look distracted or annoyed.

- **Offer a seat** if they must wait, and be honest about how long.
- **Never** start with the price or with a sigh.

Analogy. Imagine you walk into a church office and the secretary does not look up. You feel unwelcome before anyone speaks. Then imagine she stands, greets you by name and offers you a chair. Same information â€” completely different feeling. Greeting is that powerful.

Realistic scenario. A school principal's laptop is failing on the day of a PTA meeting. She is already stressed. A warm "Good afternoon, ma. Sit down, let me attend to you now" lowers her panic instantly and makes her a patient, cooperative customer.

Common misconception. "Nigerians don't do small talk; just get to business." Respectful greeting is not small talk â€” in most Nigerian businesses it is the **expected** opening. Skipping it can read as rudeness.

F.3 Capturing the fault accurately â€” in the customer's own words

The idea. Before you diagnose, you record exactly what the customer experiences, in **their** words, not your technical guess.

Why their words matter. The customer's exact description often contains the clue to the fault. If you "translate" too early, you may lead yourself astray. Example: a customer says "it dey show blue screen and restart" â€” that is different from "it won't come on at all". Both could be described as "not working", but they point to completely different faults.

How to capture it well.

1. **Let them speak first** without interruption.
2. Write the complaint **as close to their wording as possible** on the intake form.
3. Use **their** phrases in quotes: *"It makes a rattling noise like something loose."*
4. Then ask your structured questions (see F.4).
5. Read it back: "So what I understand isâ€¦, is that correct?"

Useful intake questions (the "good questions" list).

- What exactly happens? What do you see on the screen?
- When did it start? Was it sudden or gradual?
- What were you doing when it started?
- Has anything changed recently â€” a drop, a spill, a power cut, an update, new software?
- Does the problem happen every time, or sometimes?
- Does anyone else use the machine?
- **Is there important data on it that must be saved?** (Always ask this.)

An analogy. You are a **detective taking a witness statement**. You write what the witness saw, not what you think happened. The conclusion comes later, after evidence.

Realistic scenario. A church secretary says, "The computer is slow." A careless technician hears "slow" and assumes a virus. Good questioning reveals: "It became slow only after the generator was changed; now it sometimes beeps and won't start." That points toward **RAM or power**, not a virus. The customer's own words saved hours of wrong work.

Common misconception. "The customer does not know anything; just look at the machine." The customer may not know the cause, but they know the **symptoms and timing** better than anyone. Use them.

Limitation. Customers can be mistaken or vague. Capture their words **accurately** but verify with tests before you commit to a diagnosis.

F.4 Explaining faults and fixes in plain language

The idea. You translate technical findings into words the customer can understand and act on.

The rules of plain language.

1. **No unnecessary jargon.** If you must use a technical word, explain it once and simply.
2. **Use analogies** to everyday things (this is the most powerful tool).
3. **Short sentences.** One idea per sentence.
4. **Check understanding:** "Does that make sense?" or "Should I explain that part again?"
5. **Be honest and specific**, not vague or scary.

The translation table.

Technical term	Plain English
RAM	The computer's short-term memory; it forgets when you switch off
Hard drive / HDD / SSD	Where your files are stored permanently
Motherboard	The main board that connects all the parts
CPU	The computer's brain
PSU	The power supply â€" it feeds electricity to the parts
GPU / graphics card	The part that draws what you see on the screen
Driver	A small program that lets Windows talk to a piece of hardware
Operating system / Windows	The main software that runs the whole computer
Malware / virus	Bad software that harms or slows the computer
Reinstall	Wipe the system clean and install it again
Boot	To start up
Partition	A section of the hard drive
Corrupted	Damaged / spoiled
Diagnostics	Tests we run to find the cause

Analogy bank (use these freely).

- RAM is a **tabletop**: small, fast, but cleared at the end of the day. The hard drive is a **filing cabinet**: bigger, slower, but keeps things forever.
- The CPU is the **brain**; RAM is the **short-term memory**; the hard drive is the **long-term memory or diary**.
- The PSU is the **heart that pumps electricity**; if it weakens, everything else misbehaves.
- A driver is an **interpreter** between Windows and a device that speaks another language.
- Overheating is like a person working in a **sealed room with no air** â€" performance drops and eventually they collapse.

Realistic scenario. A trader asks, "Wetin dey worry am?" The technician says: *"Madam, the memory that the computer uses to run your programs has gone bad. That is why it keeps restarting. We will change it â€" the same part, small cost â€" and it will work well again."* Clear, honest, calm.

Common misconception. "Using big words makes me look professional." The opposite is true. Big words make customers feel confused and suspicious, and they hide behind them when they do not understand.

Limitation. Some findings genuinely are complicated (for example data recovery from a physically failing drive). When the truth is complex, be honest about the uncertainty rather than pretending certainty.

F.5 Managing expectations â€" time, cost and uncertainty

The idea. Tell the customer the truth about what you know, what you do not know, and what it will cost and take â€" before, not after.

The three expectations to manage.

1. **Time.** "Diagnosis will take a few hours / one day." Give a realistic window, not an optimistic lie. It is far better to say "tomorrow afternoon" and deliver at midday than to promise "in one hour" and disappoint.
2. **Cost.** Explain the difference between the **diagnosis fee** and the **repair cost**, and give a range when the fault is not yet confirmed.
3. **Uncertainty.** It is professional to say: *"I cannot give the exact price until I diagnose it. I will test it, then call you with a written estimate before we start any work. Nothing will be done without your approval."*

Why honesty about uncertainty matters. A technician who says "I am not sure yet, let me test it" is **more** trustworthy than one who guesses a price on the spot. Customers respect honesty; they resent surprises.

Realistic scenario. A customer's PC does not power on. It could be a â‚¦2,000 loose cable or a â‚¦25,000 failed PSU. A good technician says: "The test will tell us. It might be small, it might be the power supply. I will know by tomorrow and call you before I spend any money."

Common misconception. "If I say I'm not sure, they will go to another shop." In reality, most customers prefer a careful technician over a confident guesser. Honesty wins the job.

Limitation. Occasionally a fault changes during repair (a second problem reveals itself). If that happens, **stop and contact the customer** before spending more of their money. Never silently increase a bill.

F.6 Quoting and getting approval before work

The idea. Before you do billable work, you give the customer a clear **estimate** and get their agreement â€" ideally in writing (or a WhatsApp message they reply to).

What an estimate should contain.

- **Diagnosis fee** (if you charge one).
- **Parts** â€" the replacement component(s), with a price each.
- **Labour** â€" your time.
- **Total** â€" the figure the customer will pay.
- **Timeline** â€" when it will be ready.
- **Warranty** â€" what is covered and for how long.
- **Approval line** â€" where the customer agrees (signature, or "Yes, proceed" on WhatsApp).

Estimates in the Nigerian shop context.

- Written or WhatsApp estimates are best because both sides keep a copy. WhatsApp is widely used and perfectly professional when worded clearly.
- Give a **range** when unsure, and state clearly what would push it to the higher end.
- State whether the diagnosis fee is **waived** if they proceed with the repair.
- **Get approval before starting.** "Nothing will be done without your approval" protects both customer and technician.

Analogy. A good **tailor** measures you and tells you the price and when to collect *before* cutting the cloth. Imagine a tailor who sews first and demands any price later â€" you would never return.

Realistic scenario. A customer approves â‚¦18,000 for a PSU replacement. Mid-repair the technician finds the PSU failure **also damaged the motherboard**. The professional stops, calls the customer, explains plainly, gives a new estimate, and only continues if approved.

Common misconception. "Verbal is fine; we trust each other." Verbal agreements are easily disputed. A written or WhatsApp estimate with a clear "Yes, go ahead" protects the technician if a customer later claims they never agreed.

Limitation. You cannot always know the full cost upfront. That is why estimates can be **revised** â€" but only with fresh customer approval.

F.7 Handling unhappy or angry customers

The idea. An upset customer is not your enemy. They are a person with a problem, and your calm professionalism can turn them into a loyal customer.

The four-step method.

1. **Listen.** Let them finish. Do not interrupt, do not defend, do not argue. Take a breath.
2. **Empathise and acknowledge.** "I understand this is frustrating." "I can see why you are upset." This costs nothing and lowers the temperature fast.
3. **Solve.** Focus on what **can** be done: "Let me see what I can do to help." Offer realistic options.
4. **Set boundaries.** Be firm but polite. If a customer becomes abusive, you may calmly end the conversation: "Sir, I want to help you, but please do not shout at me. Let us talk calmly." You are allowed to protect your dignity and that of your staff. Escalate to a manager if needed.

What NOT to do.

- Do not match their emotion or raise your voice.

- Do not blame them ("You caused this").
- Do not lie to make them happy â€" a lie today becomes a bigger fight tomorrow.
- Do not promise what you cannot deliver.

The "broken promise" moment. Sometimes the fault is genuinely your shop's (you scratched the casing, or a repair failed). The professional response is to **own it honestly**, apologise, and fix it at your own cost where reasonable. This builds more trust than any smooth talk. Hiding a mistake and hoping the customer will not notice is how shops lose their name.

Realistic scenario. A customer shouts because a repair is taking three days. The technician listens, says "I understand, sir; the part had to be ordered from Lagos and I should have told you," explains what is happening today, gives a firm pickup time, and follows up. The customer calms down and returns next time.

Common misconception. "An angry customer is always wrong." Often the customer is simply scared â€" about their data, their money or their image. Address the fear and the anger usually drops.

Limitation. Some situations cannot be resolved to full satisfaction. In those rare cases, be honest, offer options, and if needed let them take the matter to the owner or a fair third party. Never let a dispute turn into a fight.

F.8 Confidentiality of customer data

The idea. Anything on a customer's device is private and must be treated as a secret entrusted to you.

What confidentiality means in practice.

- Do not open, read or copy the customer's files except as needed for the repair.
- Do not take photos of their documents, photos or messages.
- Do not gossip about what you found on a customer's machine.
- Do not leave customer data on a shared or personal drive after the job.
- Do not use a customer's data, accounts or files for any personal purpose.
- Store intake forms and job records securely.

Why it matters.

- It is **ethical**. The customer trusted you with their private life.
- It is **professional**. Reputation spreads fast, especially in a close city like Port Harcourt.
- In serious cases, misuse of a customer's data can be a **legal** matter, not just a moral one.

Analogy. A customer's device is like a **sealed envelope handed to you to deliver**. You carry it, you do not open it, and you do not tell the neighbourhood what was inside.

Common misconception. "If they left it on the machine, it's not private." It is still private. They did not give you permission to browse it.

F.9 Building trust and repeat business

The idea. The goal of every job is not just a fixed computer â€" it is a customer who trusts you and returns.

How trust is built.

- **Keep your word** on time and price.
- **Be reachable.** Answer calls and WhatsApp messages, even to say "still working on it".
- **Follow up.** A short "How is the computer working now?" a few days later is powerful and costs nothing.
- **Give honest recommendations** â€" including "you do not need to replace this yet".
- **Be transparent** about what you did (this connects to the service report in Session 11).
- **Thank them**, and invite them back.

Realistic scenario. After replacing a laptop screen, a technician sends a WhatsApp: "Good evening, ma. Hope the screen is fine? Please let me know if anything comes up." The customer replies, then tells two friends. That one message becomes three jobs.

Common misconception. "Once they pay and leave, the job is finished." The relationship, not the transaction, is the real business.

Limitation. Trust is slow to build and fast to lose. One dishonest act can undo years of good service. Protect your name above a single sale.

F.10 Summary of the session

- Customer service is a **technical skill** in this trade â€" it wins and keeps business.
- **Greet** warmly; **listen** and capture the fault in the **customer's own words**.
- **Ask good questions** (what, when, how, what changed, important data?).
- **Explain plainly** â€" no jargon, use analogies, check understanding.
- **Manage expectations** honestly about time, cost and uncertainty.
- **Quote and get approval** before work; revise only with fresh approval.
- **Handle anger** with listen â†' empathise â†' solve â†' boundary.
- **Protect confidentiality**; **build trust** with honesty and follow-up.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

What the instructor prepares:

- The "shop counter" table and two chairs.
- Scenario card: "*My laptop will not charge and the battery was full this morning. I need it for work tomorrow.*"
- A printed Customer Intake Form on the projector or whiteboard.
- The good and bad estimate examples.

G.1 Demonstration 1 â€" The intake, done well (6 minutes)

Say: "Watch everything I do, and listen to **how** I speak as much as **what** I say."

The instructor acts as the technician. A volunteer plays the customer with the scenario above.

1. **Greet:** "Good afternoon, welcome. My name is Elias. Please sit down. How can I help you today?"
2. **Let them speak** without interrupting. Write their exact words on the form: "*My laptop will not charge and the battery was full this morning.*"
3. **Ask good questions:** When did it start? What changed? Do you use the original charger? Does the charging light come on? Is there important data on it?
4. **Read back:** "So the laptop works on battery but will not charge since this morning, and you have work due tomorrow. Have I understood correctly?"
5. **Set expectations:** "I will test the charger and the charging port first â€" that is the small possibility â€" then the battery. Diagnosis will take about one hour. I will call you with a written estimate before any repair."
6. **Mention data safety:** "Is anything on it you cannot afford to lose? Good â€" I will not wipe anything without asking you."

Then say: "Did you notice I asked before I concluded? Did you notice I quoted a **process**, not a fake price? That is the whole skill."

G.2 Demonstration 2 â€" Explaining the fault plainly (4 minutes)

Say: "Now I have a finding. Watch me translate it."

- Technical truth: "*Battery cell degradation; the battery no longer accepts charge.*"
- Plain version: "*Ma, your laptop's battery has reached the end of its life. It is like a phone battery that no longer holds charge. The charger is fine. We can replace just the battery and it will hold charge again.*"
- Check understanding: "Does that make sense? Any questions?"

Ask the class: "What did I do to make that clear?" â†' Used an everyday analogy (phone battery), avoided the word 'degradation', checked understanding, and told her what would happen next.

G.3 Demonstration 3 â€" The angry customer (3 minutes)

The instructor plays a customer who is angry that a repair is late. Narrate the four steps as you do them:

1. **Listen** â€" let them finish, no interrupting.
2. **Empathise** â€" "I understand, sir. This has delayed you and I am sorry."

3. **Solve** â€" "Here is exactly where we are and what I can do today."
4. **Boundary** â€" if shouting continues: "I want to help you, but please let us talk calmly."

Expected result: The "customer" visibly calms.

Error to watch: If the learner disagrees, defends or raises their voice, stop the scene and reset. Point out that arguing always makes it worse.

G.4 Demonstration 4 â€" Good vs bad estimate (2 minutes)

Show the two examples on screen.

- **Bad:** "Laptop repair â€" maybe â‚¦20k, come back Friday." (vague, no breakdown, no approval).
- **Good:** a clear table with diagnosis fee, parts, labour, total, timeline, warranty and an approval line.

Ask: "Which protects the customer? Which protects **you**?" â†' Both. A clear estimate is protection for everyone.

Questions to ask throughout the demo:

- "Why did I write the customer's exact words?" â†' Their words contain clues and show I listened.
- "Why did I say 'I don't know yet' about the price?" â†' Honesty about uncertainty builds trust and prevents surprises.
- "Why is a WhatsApp 'Yes, proceed' useful?" â†' It is written proof of approval.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: In pairs, learners role-play a full **customer intake** and a **plain-language fault explanation**, using a real scenario and the Customer Intake Form.

> ***Confidentiality reminder before starting:*** *everything said in role-play stays in the room. When a real customer is involved, their data and details are private.*

Setup: Pairs sit face to face. One is the **technician**, one is the **customer**. The instructor hands each pair a scenario card. The technican holds the Intake Form; the customer holds the card.

Procedure (instructor calls each numbered step; pairs follow; instructor walks the room and listens):

1. Decide who is technician and who is customer. The customer reads their scenario card silently.
2. **Greet** the customer warmly and offer a seat. *(Instructor watches for warmth and eye contact.)*
3. **Let the customer speak** first. The technician writes the fault **in the customer's own words** on the form.
4. **Ask at least four good questions** from the list (what, when, how, what changed, important data).
5. **Pause and predict:** before writing a diagnosis, the technician says out loud what the likely **area** of fault is (for example RAM, storage, power, software) and **why**.
6. **Read the intake back** to the customer and ask, "Have I understood correctly?"
7. **Explain the likely fault in plain language**, using at least one analogy. Check understanding.
8. **Set expectations**: time, a note that a written estimate will follow, and that nothing is done without approval.
9. **Offer an estimate** using the template (diagnosis fee, parts range, labour, total, timeline, warranty, approval line). Get a verbal or written approval.
10. **Switch roles** and repeat with a **different** scenario card.
11. **Give each other feedback** using the three questions below.

Expected result: A completed Intake Form that accurately captures the fault in the customer's words, a plain-language explanation the "customer" understood, and an approved estimate.

Scenario cards (use and reuse):

- **Card 1.** "My computer is very slow and takes forever to start. I have a presentation tomorrow."
- **Card 2.** "My computer is making a loud noise and I am scared it will lose my photos."
- **Card 3.** "I poured water on my laptop and now it will not come on. Please fix it quickly."
- **Card 4.** "A blue screen appears and the computer restarts by itself. I don't know what it means."

- **Card 5.** "The screen is cracked and sometimes goes black. I use it for my POS business."
- **Card 6.** "The computer works but there is no sound at all. I use it for church programmes."

Pause-and-predict checkpoints: after step 3 ("What is the likely fault area?"), after step 7 ("Did the customer understand? Ask them to repeat it back.").

Troubleshooting while walking the room:

- Technician interrupts the customer â†' gently stop the scene; remind them step 3 is "let them finish".
- Technician uses jargon ("SATA", "BIOS") â†' ask them to say the same thing to a market woman.
- No approval obtained â†' pause and show why the estimate and "yes" matter.
- Customer is too vague â†' prompt the technician to ask "What exactly do you see on the screen?"

Completion checklist (learner ticks each):

- ☐ Warm greeting given
- ☐ Fault captured in the customer's own words
- ☐ At least four good questions asked
- ☐ Fault area predicted with a reason
- ☐ Intake read back and confirmed
- ☐ Plain-language explanation with an analogy, comprehension checked
- ☐ Expectations set (time, cost, uncertainty)
- ☐ Estimate offered with an approval line
- ☐ Both roles played (two scenarios)
- ☐ Feedback given and received

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task (NEW task): This is a **writing** exercise, not a role-play. You are a technician at a Port Harcourt repair shop. Read each customer message below and produce, for **each**, (a) the exact question you would ask next, and (b) a short plain-language reply that the customer will understand â€" no jargon. Then complete the mini-estimate.

Case A â€" the vague WhatsApp message

> *Customer: "Good evening. My system dey misbehave. I need it urgent. How much?"*

Your tasks:

- A1. Write **three** good questions you would ask before naming a price.
- A2. Write a WhatsApp reply that is friendly, asks for the machine, explains that you must diagnose before quoting, and gives an honest **timeframe** for the diagnosis.

Case B â€" the frightened customer

> *Customer: "My office computer shows a scary blue screen with plenty English I don't understand. I'm worried all our staff records are gone."*

Your tasks:

- B1. Explain **plainly** what a "blue screen" is and what it might mean. Use an analogy.
- B2. Reassure them **honestly** about their data (do not promise what you cannot know).
- B3. State what you will do next and roughly how long the diagnosis will take.

Case C â€" the mini-estimate A customer's desktop will not power on. After diagnosis you find the **power supply unit (PSU) has failed**. Parts: a 500W PSU at **â‚¦18,000**. Labour: **â‚¦5,000**. Diagnosis fee: **â‚¦2,000**, waived if they approve the repair. Timeline: next day. Warranty: 30 days on the part.

- C1. Write the estimate clearly (all line items, total, timeline, warranty).

- C2. Write the sentence in which you ask for approval.

Time allowance: 15 minutes.

Success criteria: Good questions that gather useful information; replies free of jargon and respectful in tone; honest handling of uncertainty and data; a complete, clear estimate with an approval request.

Hints if stuck:

- Hint 1: Good questions start with *what*, *when*, *how* and *what changed*.
- Hint 2: Plain language means an analogy and no acronyms.
- Hint 3: An estimate always states parts, labour, total, timeline, warranty and an approval line.

INSTRUCTOR ANSWER KEY (do not show learners):

Case A

- A1. Sample questions: "What exactly do you see when you switch it on â€" a black screen, a blue screen, or does it start up slowly?" / "When did the problem start, and what changed just before it started (a power cut, an update, a new program)?" / "Is there important data on the machine that must be saved?"
- A2. Sample reply: *"Good evening, sir. Sorry for the stress. To give you a proper price I need to test the machine first â€" I don't want to guess and surprise you. Please bring it tomorrow morning; diagnosis will take about one hour, then I will send you a written estimate before any work. Nothing will be done without your approval."*

Case B

- B1. Plain explanation: *"A blue screen is the computer's way of saying it has hit a serious problem it cannot handle, so it stops and restarts. It is like a car engine cutting off to protect itself when something is wrong â€" it doesn't always mean everything is broken."*
- B2. Honest reassurance: *"I cannot promise about the files until I test, but a blue screen usually does not erase your records by itself. Please don't switch it on and off repeatedly. I will check very carefully, and if your data can be saved, I will save it before doing anything else."*
- B3. Next steps: *"I will test the memory, the hard drive and the system first. Diagnosis will take about one to two hours, and I will call you with the result and a price before I do any repair."*

Case C

- C1. Estimate:
  - Diagnosis fee: â¦2,000 (waived if repair approved)
  - Part: 500W PSU â€" â¦18,000
  - Labour: â¦5,000
  - **Total (if approved): â¦23,000**
  - Timeline: ready the next day
  - Warranty: 30 days on the PSU part
- C2. Approval sentence: *"If you are happy with this, please reply 'Yes, go ahead' and I will start the repair today. Nothing will be done until I hear from you."*

Marking guidance (out of 10):

- A1 useful questions (2)
- A2 friendly, honest, jargon-free reply (2)
- B1 plain explanation with analogy (2)
- B2 honest data handling (1)
- B3 clear next steps and timeframe (1)
- C1 complete estimate with correct total (1)
- C2 proper approval request (1)

J. REAL-WORLD CASE STUDY

Situation. *Chidi* runs a small repair corner in a Port Harcourt plaza. He is a good technician but a poor communicator. A customer brings a laptop that will not start. Chidi takes it, grunts, and says "I will check it." Two days later he calls: "It's your board; â,¦45,000." The customer is shocked â€" no warning, no explanation, no approval. She collects the machine and takes it elsewhere. The other shop diagnoses a faulty **power button cable** and fixes it for â,¦6,000. She tells everyone at her office about Chidi "who wanted to spoil her".

Relevant facts.

- Chidi's diagnosis may even have been partly right, but nobody trusts a surprise bill.
- The second shop greeted her, explained plainly, gave a written estimate and got approval.
- Same fault. Two very different outcomes â€" decided by **customer service**, not by hands-on skill.

Decision point. Where did Chidi lose the customer?

Possible approaches.

1. Blame the customer for being "unreasonable".
2. Learn to greet, explain plainly, quote honestly and get approval before work.
3. Keep doing the same but hope for the best.

Recommended solution. Approach 2. If Chidi had (a) greeted her, (b) explained what he suspected in plain language, and (c) sent a written estimate of â,¦45,000 with a clear reason *before* doing anything, the customer would either have approved it or declined politely â€" either way, no damage to his name. He also lost the chance to compete: the second shop found the real, cheaper fault.

Trade-offs.

- Approach 1 feels good but changes nothing; the shop stays empty.
- Approach 2 takes only minutes per customer but transforms reputation and repeat business.
- Approach 3 is the road to closed doors.

Professional connection. In a repair shop, **communication is part of the repair**. The intake, the plain explanation, the honest estimate and the approval are not extra chores; they are the product the customer is paying for, alongside the fix.

Transfer question for the class: "Chidi charged â,¦45,000; the second shop charged â,¦6,000 for the same-looking problem. What does that teach you about the value of a *correct diagnosis* plus *honest communication*?"

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table live. When a learner hits an issue in role-play, point them to the relevant row.

K.1 Using jargon with the customer

Symptoms: Customer looks confused; keeps asking "wetin that mean?" **Likely cause:** The technician used words like SATA, BIOS, motherboard, corrupt. **Safe corrective steps:** Translate to plain English; use an analogy; check understanding. **Prevention:** Practise your analogy bank until it is automatic.

K.2 Interrupting or not listening

Symptoms: Customer repeats themselves; technician misses key details; wrong fault area predicted. **Likely cause:** Rushing to solve before understanding. **Safe corrective steps:** Let the customer finish; write their words; read back. **Prevention:** Treat the customer as a witness â€" capture first, conclude later.

K.3 No estimate / no approval

Symptoms: Customer angry at the bill; dispute; loss of trust. **Likely cause:** Work started without written cost or consent. **Safe corrective steps:** Apologise, provide the breakdown, and agree a fair resolution; change your process going forward. **Prevention:** Always quote and get approval (signature or a clear "Yes, proceed") before billable work.

K.4 Overpromising time or outcome

Symptoms: Customer returns furious; reputation damaged. **Likely cause:** Giving an optimistic guess instead of a realistic window. **Safe corrective steps:** Own it, apologise, communicate the true status, give a firm new time. **Prevention:** Under-

promise and over-deliver. Say "I will confirm after diagnosis".

K.5 Matching the customer's anger

Symptoms: The situation escalates into an argument. **Likely cause:** Taking the anger personally and reacting. **Safe corrective steps:** Pause, lower your voice, listen, empathise, set a calm boundary. **Prevention:** Remember the customer is frustrated with the **situation**, not necessarily with you.

K.6 Breaching confidentiality

Symptoms: Customer learns their private files or situation were discussed or copied. **Likely cause:** Curiosity, gossip, or careless handling of data and records. **Safe corrective steps:** Stop immediately, secure the data, and never repeat it; where serious, this can become a legal matter. **Prevention:** Treat every device as a sealed envelope; access data only for the repair.

K.7 Hiding a shop mistake

Symptoms: Customer discovers the damage later; trust destroyed. **Likely cause:** Embarrassment and hope the customer will not notice. **Safe corrective steps:** Own it, apologise, fix it at your cost where reasonable. **Prevention:** Honesty is the cheapest long-term policy.

Distinguish likely vs confirmed: "Likely cause" is your best guess; you confirm it by listening carefully and, when relevant, checking the customer's facts. Teach learners to ask before assuming.

L. SESSION ASSIGNMENT

Assignment title: "My Shop's Customer Service Playbook"

Purpose: To turn today's skills into a practical, personal guide a learner can follow at a real counter â€” including ready-made questions, a plain-language translation table, an estimate template and scripts for difficult moments.

Realistic scenario: You have just opened a small repair shop (or are joining one). You want every customer to be greeted warmly, understood correctly, quoted honestly and treated well â€” even when they are upset.

Detailed instructions:

1. Write a **greeting script** you will actually use (2â€“3 sentences).
2. Build a **good-questions list** of at least 8 questions to gather a fault, including one mandatory question about **data**.
3. Create a **plain-language translation table** with at least **12** technical terms and their everyday meanings.
4. Write an **estimate template** with diagnosis fee, parts, labour, total, timeline, warranty and an approval line.
5. Write **scripts** for three difficult moments:
    - a customer who is angry about a delay,
    - a customer who says your price is "too much",
    - a customer who asks you to install unlicensed software.
6. Write a short paragraph on **confidentiality** â€” your personal commitment to protect customer data.
7. List three ways you will **build repeat business**.

Required deliverables:

- A typed document (Word) or a neatly handwritten booklet, **at least one full page**.
- One completed Customer Intake Form from the role-play.

Submission format: Printed or emailed; filename `Surname_S10_Playbook.docx`.

Deadline: Before Session 11.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Greeting & intake	Warm greeting; 8+ strong questions incl. data	Greeting + questions, less strong	Few questions, no data question

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Plain-language table	12+ terms, clear, everyday language	8â€“11 terms	Fewer than 8
Estimate template	Complete, professional, approval line	Mostly complete	Missing key items
Difficult-customer scripts	Realistic, calm, three scenarios handled well	Two handled adequately	One or vague
Confidentiality & repeat business	Strong commitment; 3+ concrete retention ideas	Present but thin	Missing

Common mistakes to avoid: jargon in the scripts; no data question; no approval line in the estimate; threatening or sarcastic tone in the difficult-customer scripts.

Instructor reference solution (structure):

- Greeting: "*Good afternoon, welcome to [shop]. My name is [name]. Please sit down â€“ how can I help you today?*"
- Good questions (sample): What exactly happens? / When did it start? / What were you doing? / What changed recently? / Does it happen every time? / Does anyone else use it? / **Is there important data to save?** / When do you need it back?
- Plain-language table: at least 12 rows (see F.4 migration table).
- Estimate: itemised table with total, timeline, warranty, approval line.
- Difficult-customer scripts: listen â†' empathise â†' solve â†' boundary, applied to each case.
- Unlicensed-software script: "*I'm sorry, I can only install licensed software. I can help you buy a genuine licence or suggest a free, legal option.*"
- Confidentiality paragraph: a clear, personal promise to protect customer data.
- Repeat business: WhatsApp follow-up, honest recommendations, remembering names, loyalty thank-you.

M. KNOWLEDGE CHECK

(Learner version â€“ answers only in the Instructor Key.)

Q1. List. What are the four steps for handling an unhappy customer, in order?

Q2. Multiple choice. Before starting billable repair work, you must always: a) Tell the customer to come back tomorrow b) Get the customer's approval of a written estimate c) Increase your price d) Open the machine first

Q3. Explain in your own words. Why is it better to say "I will confirm the price after diagnosis" than to guess a price on the spot?

Q4. Short answer. Why must you write the fault in the **customer's own words** rather than your technical guess?

Q5. Practical translation. Say these in plain English (no jargon): a) "The SATA controller is failing." b) "There is malware on the system."

Q6. Scenario. A customer shouts at you because a part arrived late. Give the first two things you should do and one thing you must **not** do.

Q7. Fill in the blank. Anything on a customer's device is private; we must keep it _____ and only access it for the _____.

Q8. True or false. A verbal estimate is always enough; writing it down is unnecessary. Explain your answer.

Q9. Short answer. Give two ways a technician can build repeat business after a repair is completed.

Q10. Scenario. A customer asks you to install a "cracked" copy of Windows to save money. Write your professional reply.

INSTRUCTOR ANSWER KEY (Q1â€“Q10)

A1. Listen â†' empathise/acknowledge â†' solve â†' set boundaries. **A2.** (b) Get the customer's approval of a written estimate. **A3.** Because guessing a price and being wrong causes a "surprise bill", which destroys trust. Honest uncertainty â€" "I will confirm

after diagnosis" â€" shows competence and protects both sides; you quote before doing any billable work. **A4.** The customer's words contain clues to the cause, and writing them down proves you listened. Translating too early can lead you to the wrong fault. **A5.** a) "The part that connects your hard drive to the main board is faulty, so the computer cannot find your files." b) "There is harmful software on the computer that is slowing it down or damaging it." **A6.** First: **listen** without interrupting, then **empathise/acknowledge** ("I understand; I'm sorry"). You must **not** shout back, argue or blame the customer. **A7.** â€¦ **confidential** (private) â€¦ ; â€¦ **repair** (the job it was given to you for). **A8.** False. A written estimate (or a clear WhatsApp message) gives both the customer and the technician proof of what was agreed, preventing disputes. Verbal agreements are easily denied. **A9.** Any two of: send a follow-up message asking if the machine is working well; keep your word on time and price; give honest recommendations; remember the customer's name; offer a thank-you or loyalty gesture. **A10.** Sample: *"I'm sorry, I can only install properly licensed software â€" using a cracked copy is illegal and it can carry viruses that would harm your computer. I can help you buy a genuine licence, or suggest a free, legal option that fits your needs."*

N. REFLECTION AND RECAP

Plain-English summary. Today you learned that a repair shop is a people business. You practised greeting a customer warmly; capturing their fault in their own words; asking good questions (including the all-important "is there important data?"); explaining faults and fixes in plain language with analogies; managing expectations about time, cost and uncertainty; quoting and getting approval before work; handling angry customers calmly; protecting confidentiality; and building trust for repeat business. None of this replaces good technical work â€" it **wraps around** it.

Five important takeaways:

1. **Greet and listen first**; capture the fault in the customer's own words.
2. **Explain plainly** â€" no jargon, use analogies, check understanding.
3. **Manage expectations honestly**; it is fine to say "I don't know yet".
4. **Quote and get approval** before doing billable work.
5. **Stay calm and protect confidentiality** â€" trust is your real business.

Questions to help you recognise what you can now do:

- Can I run a full customer intake and fill the form without prompting?
- Can I explain a fault to a non-technical customer and have them understand it?
- Can I write a clear estimate with an approval line?
- Can I stay calm with an angry customer and still solve the problem?
- Do I treat every customer's data as private?

Connection to the next session: You have now served the customer well and agreed the work. But a professional shop also **records** the work. Next session you learn **service reports and documentation**: what a proper report must contain, how to write it clearly and honestly, and how it protects both you and the customer. The report is also the evidence that becomes your Final Project deliverable.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner greets warmly and naturally, without a frozen script.
- The learner **lets the customer finish** and writes their exact words.
- The learner asks focused questions and follows up on vague answers.
- The learner explains a fault with an analogy and checks understanding.
- The learner is honest about uncertainty and never invents a price.
- The learner produces a clear estimate and asks for approval.
- The learner stays calm in the angry-customer scene.
- The learner treats confidentiality and data safety as non-negotiable.

Common misconceptions to watch for and correct:

- "Customers should understand technical words." â†’ It is your job to translate, not theirs to learn.
- "An angry customer is always wrong." â†’ Often they are frightened; address the fear.
- "Verbal estimates are fine." â†’ Put it in writing/WhatsApp; it protects everyone.
- "If I don't know the price, I should guess." â†’ Honest uncertainty builds more trust than a wrong guess.
- "If they left the file on the machine, it is not private." â†’ It is still private and must be protected.
- "Customer service is for receptionists." â†’ In a small shop, the technician **is** the face of the business.

Suggested remedial exercises for learners who struggle:

1. **Translation drills:** hand out ten technical sentences; ask the learner to say each one to a "market woman" in one or two simple sentences.
2. **Question-building drill:** give a one-line complaint; ask the learner to write five good questions and pick the best three.
3. **Calm-under-pressure drill:** the instructor plays a mildly difficult customer; the learner practises listen â†’ empathise â†’ solve â†’ boundary. Increase difficulty gradually.
4. **Estimate-completion drill:** give a partially filled estimate; ask the learner to complete the total, timeline, warranty and approval line.

Differentiation:

- **Level 1 (supported):** provide a **scripted** role-play the learner can follow word for word, and a heavily guided intake form; let them start as the customer and build confidence.
- **Level 2 (standard):** complete both role-plays and the written exercise as written; produce the playbook.
- **Level 3 (challenge):** handle a **two-part** difficult scenario (angry customer **plus** a revised estimate mid-repair); or, with instructor supervision, conduct a real intake with a genuine customer or staff volunteer; write a full script for a price-objection situation.

Timing guidance (105 min):

- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided role-play 18, independent written exercise 15, case study folded into reflection 3, reflection/recap + knowledge check 6. If the room is animated in role-play, protect the **role-play** and the **plain-language** practice above all â€" those are the priority. The case study can be reduced to a two-minute mention if time is short.

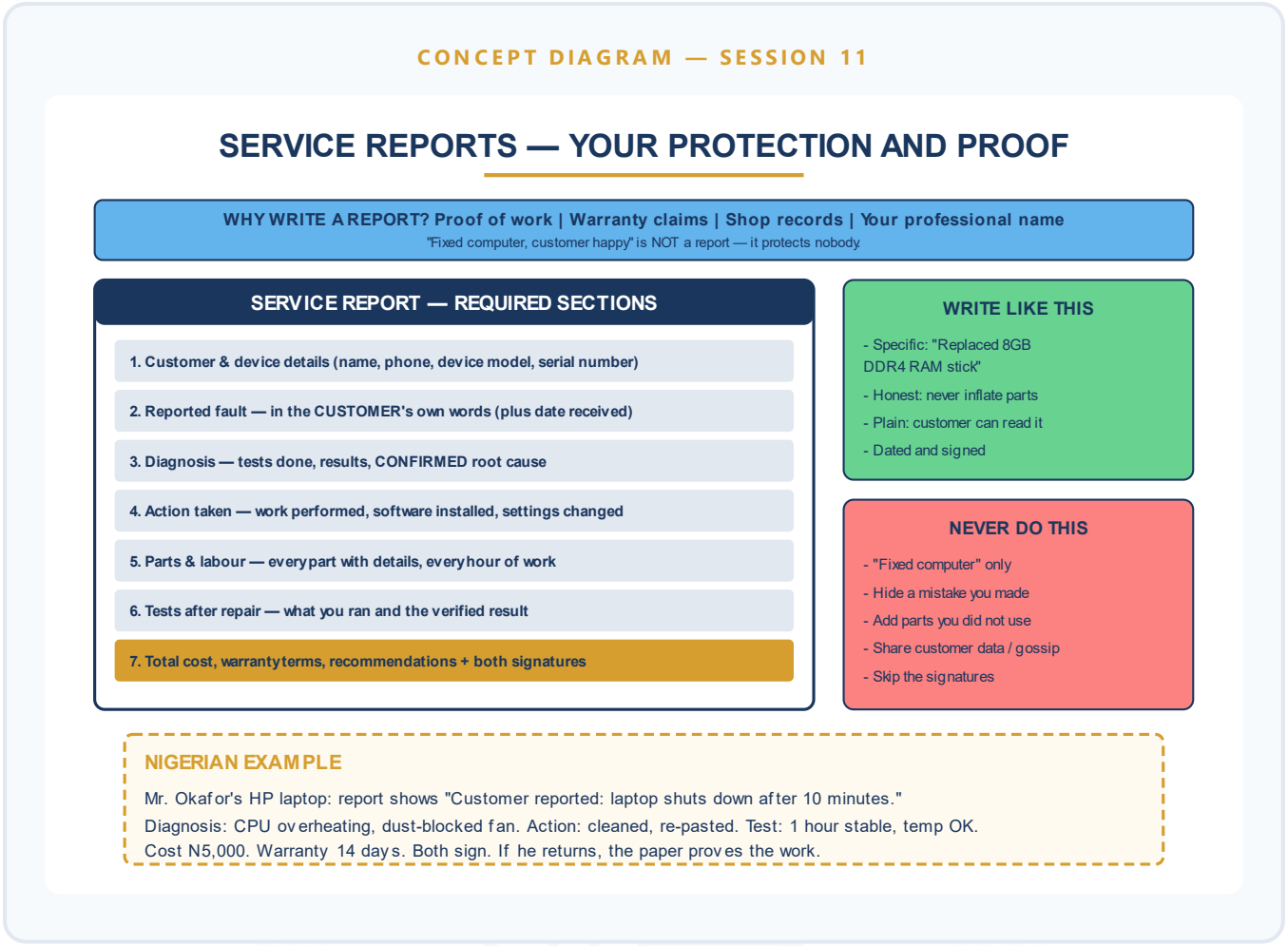
END OF SESSION 10

Next: Session 11 â€" Service Reports and Documentation.

SESSION 11 OF 12

Service Reports

CONCEPT DIAGRAM — SESSION 11



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 11 â€" SERVICE REPORTS AND DOCUMENTATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	11 of 12
Session title	Service Reports and Documentation
Week / Module	Week 4 â€" Module 4: Professional Practice
Recommended duration	105 minutes
Difficulty level	Beginner (writing and reasoning skills)
Prerequisite knowledge	Sessions 1â€"10 â€" the diagnostic workflow, hardware and software repairs, OS installation, and customer service (intake, plain language, estimates, approval).

Item	Detail
Connection to previous work	In Session 10 you captured the fault and agreed the work with the customer. Today you close the loop: you **document** exactly what was found, what was done, what it cost and what you recommend. Writing the report is the last professional step of every repair â€" and it is the deliverable that proves you completed the capstone repair in Session 12.
Preparation before class	Prepare a printed "Service Report Template" (one per learner); prepare one **strong** sample report and one **weak** sample report for analysis; have the Section G scenario written up so learners can see a report being built; ensure Microsoft Word (or a plain text editor) is available.
Professional note	A service report is a permanent record that protects the customer, the technician and the shop. It must be **accurate and honest** â€" never exaggerate work, never inflate parts, never hide a mistake.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain why service reports matter** â€" proof of work, warranty, records and professionalism.
2. **List and describe every required section** of a professional service report.
3. **Record the reported fault accurately** in the customer's own words alongside the observed facts.
4. **Document the diagnosis** â€" tests performed, results, and the confirmed root cause.
5. **Describe the action taken** â€" parts used (with details), software installed, and work performed.
6. **Record the tests performed after repair** and the verified result.
7. **Write clear, specific and honest language**, avoiding vague phrases and unnecessary jargon.
8. **Recommend preventive maintenance** and next steps for the customer.
9. **State costs, warranty terms and get the customer's signature** on the finished report.
10. **Handle customer data responsibly** while keeping and storing records.

C. REQUIRED MATERIALS

Hardware

- At least one practice computer (so learners can refer to a real device's details: serial number, components).

Software

- Microsoft Word (or a simple text editor / Google Docs if available). If no computer is available for typing, a printed template and pen are perfectly acceptable â€" the skill is the writing, not the software.

Printed materials

- "Service Report Template" (one per learner) containing all required sections.
- One **strong** sample completed report and one **weak** sample completed report for the opening activity.
- The Section G worked scenario (Mr. Okafor's HP laptop) written out for reference.
- A blank "Parts and Labour" table and a warranty-terms shortlist.

Internet

- Not required. If available, the instructor may optionally show one real-world company's published service-report format as inspiration near the end.

Alternative for learners who lack a facility

- Learners without a laptop or Word can complete the entire session on paper. The instructor types a shared example on the projector. Encourage neat, clear handwriting â€" legibility is part of professionalism.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: Display (or hold up) two service reports side by side â€" one strong, one weak. The weak one is messy: it says "Fixed computer, customer happy", with no dates, no parts, no signature and handwriting all over. The strong one is clean and complete.

What the instructor says:

> *"Look at these two papers. Both claim to be service reports for the same job. The one on the left says, 'Fixed computer, customer happy. â‚¦25,000.' The one on the right lists the customer, the device, exactly what was wrong, the tests done, the parts used, the total, the warranty and two signatures.*
>
> *Now, I want you to answer three questions with me: One â€" if the customer comes back in one month and says the fault returned, which report lets the shop prove what it did? â€¦ The right one. Two â€" if the technician is accused of damaging the machine, which report tells the truth of what happened? â€¦ The right one. Three â€" if the shop's owner wants to know how much money that week's parts cost, which report gives the answer? â€¦ The right one.*
>
> *That piece of paper is not just paperwork. It is your **protection**, your **memory** and your **professional name**. Today you will learn to write it properly, because this report is exactly what you will produce for your Final Project next session."*

Expected learner response: Learners will quickly see the messy report is untrustworthy. Some may say "the good one is more work". Acknowledge that it takes a few minutes more â€" and saves hours of argument later.

Why this opener works: It frames documentation as self-interest, not bureaucracy: the report protects the technician. That motivation makes learners take the writing seriously.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral round-robin. Put the questions on the projector.

Question 1: "What are the four steps of the customer intake process?"

- Expected answer: greet, gather information, set expectations, document.
- Common gap: learners forget the final "document" step. Point out that today **is** that step turned into a full document.

Question 2: "Why do we capture the fault in the customer's own words?"

- Expected answer: their words contain clues and prove we listened; translation too early can mislead.
- Common gap: learners think the technical finding is enough. Remind them the report must show **both** the reported fault and the observed finding.

Question 3: "What did you learn to give the customer before starting work in Session 10?"

- Expected answer: a clear written/WhatsApp estimate and approval.
- Common gap: learners forget the approval line. The estimate and the service report are two sides of the same professional habit.

Question 4: "What are the five steps of the diagnostic workflow?"

- Expected answer: observe, ask, hypothesise, test, resolve/verify.
- Common gap: learners skip "verify". The report must record the **verification** â€" the proof that the fix worked.

Bridge to today (say this):

> *"Look at everything today rides on. You already capture the fault, get approval, diagnose and verify. The service report simply **writes all of that down** â€" in a clear, honest, standard format. Nothing new to invent: just record what you already do, properly."*

Gap handling: If more than a third of the class cannot name the verification step (Question 4), spend five extra minutes on why we test after repair (open the machine, confirm the fault is gone, check no new faults). The report's "tests performed" and "result" sections depend on this. Do not proceed until it is solid.

F. FULL LECTURE NOTES

F.1 Why service reports matter

The idea. A **service report** is a professional document that records what was reported, what was found, what was done, what was tested and what the customer now owes â€” for a specific repair.

Plain-English definition. It is the **written story of one repair**, kept on record for the shop and the customer.

Why it exists. Repairs are done with hands and memory, and both are unreliable later. A report captures the facts while they are fresh, in a form that anyone can read and trust.

What it does â€” five reasons it matters.

1. **Proof of work.** If a customer disputes the repair ("you never changed that part"), the report shows exactly what was done.
2. **Warranty support.** A clear record of the part, the date and the work is what makes a warranty claim possible.
3. **A record for the future.** A later technician can see what was already done, avoiding repeated or conflicting work.
4. **Professionalism and trust.** A clean, complete report tells the customer the shop is organised and honest.
5. **Business information.** Reports track parts used, costs and recurring problems â€” useful for pricing and for spotting a customer whose machine keeps failing.

Everyday analogy. A service report is like a **hospital patient's file**. Each visit records symptoms, tests, diagnosis, treatment and result. Without that file, every doctor starts from zero â€” and mistakes multiply. Your report is the patient's file for the machine.

Realistic scenario (Port Harcourt). A school brings the same lab computer back three months after a repair, claiming the earlier fix failed. The shop checks its report: the part was replaced on a specific date, with a 90-day warranty that has expired, and the customer signed for it. The conversation is calm and factual â€” because the paper exists.

Common misconception. "Reports are just paperwork; the repair is what matters." The repair earns today's money; the report earns tomorrow's trust, resolves disputes, and supports warranty and audit.

Limitation. A report is only as good as its honesty and accuracy. A report full of vague or invented claims is worse than none â€” it destroys credibility.

F.2 What a service report must contain

The idea. Every professional report has the same standard sections, in a logical order, so nothing important is ever forgotten.

The required sections.

1. **Header / shop details.** Business name, logo (if any), address, phone, and a **unique report number** (for example `SRV-2026-0042`), the date, and the technician's name.
2. **Customer and device details.** Customer name, phone, and the device: make/model, and **serial number** or service tag where available. (Serial numbers matter for warranty and for identifying the exact machine.)
3. **Reported fault (the customer's words).** What the customer said, in their own words, quoted where helpful.
4. **Diagnosis / findings.** What you observed; the **tests performed** and their **results**; and the confirmed **root cause** â€” the real reason for the fault.
5. **Action taken (repairs performed).** Exactly what you did: parts replaced (with enough detail to identify them), software installed or updated, cleaning, reseating, etc.
6. **Parts used.** An itemised list of parts, with a description (and part number where relevant) and the quantity.
7. **Tests performed after repair / result.** What you tested to confirm the fix, and the outcome: "Fault resolved; machine boots, all devices working."
8. **Costs (parts and labour).** An itemised breakdown: parts cost, labour cost, diagnosis fee if any, and the **total**.
9. **Recommendations.** Preventive advice and next steps for the customer (for example, "use a surge protector", "back up regularly", "consider an SSD upgrade").
10. **Warranty terms.** What is covered, for how long, and any conditions.
11. **Sign-off.** Technician signature and date, and **customer signature** and date.

> *Note: different shops combine some sections (for example, "parts used" inside "action taken"), and add a diagnosis fee line. The **content** above is what matters; the exact layout can vary. Our CEA template presents them as separate, clearly labelled sections so nothing is missed.*

Analogy. The sections are like a **story with a beginning, middle and end**: the customer's complaint (beginning), the diagnosis and repair (middle), and the result, cost and recommendation (end). Skip a part and the story is incomplete.

Common misconception. "If the repair was simple, I can write a short note." Even a simple repair (for example, reseating a RAM stick) deserves the full record: reported fault, finding, action, test, cost, signature. The only thing that changes is how brief each section is â€" not whether it exists.

Limitation. The report must be readable by the customer. A report stuffed with raw technical output without explanation fails the customer, even if it is technically complete.

F.3 Recording the fault and the diagnosis

The idea. Keep the **customer's reported fault** and **your observed findings** separate but connected.

Why separate them.

- The **reported fault** is what the customer experienced ("it restarts by itself").
- The **observed finding** is what you actually measured (for example, "memory test failed with errors in slot 1").
- Sometimes they match; sometimes the customer's symptom has a different cause than expected. Recording both shows your professional reasoning.

Good practice.

- Quote the customer's words: *Customer reported: "It makes a rattling noise and then shuts down."*
- Record **each test and result**: *Test: reseated RAM; ran Windows Memory Diagnostic â€" no errors. Test: checked fan â€" fan bearing worn; CPU temperature rose to 92Â°C under load. Result: overheating caused automatic shutdown.*
- State the **root cause** plainly: *Root cause: worn CPU fan causing overheating and thermal shutdown.*

Analogy. It is like a **detective's case file**: the witness statement (reported fault) and the forensic evidence (observed findings) are both recorded, and the conclusion (root cause) follows from the evidence.

Common misconception. "The customer's description is unimportant once I find the fault." Wrong â€" the description explains *why* they came, and demonstrates you listened and understood.

Limitation. Customers can be inaccurate. Record their words faithfully, but always base the root cause on **evidence**, not on their guess.

F.4 Documenting the action, tests and result

The idea. State exactly what you did, then prove it worked.

Action taken â€" be specific.

- Instead of: *"Fixed the problem."* â†' Write: *"Replaced faulty CPU cooling fan (original part: 92 mm, 4-pin). Applied new thermal paste and reconnected the fan cable to the CPU_FAN header."*
- Include software actions: *"Installed network, chipset, graphics and audio drivers. Ran Windows Update until no updates remained. Activated Windows with the customer's licensed digital licence."*

Parts used â€" be identifiable.

- Description, quantity and (where relevant) part number or specification.
- Example: *"1 Ã— 500 GB SATA SSD"*, *"1 Ã— 8 GB DDR4 2666 MHz RAM module"*, *"1 Ã— 500 W PSU"*.

Tests performed after repair / result â€" prove the fix.

- Example: *"Booted to Windows; ran CPU temperature test under load â€" stable at 55Â°C. Ran memory diagnostic â€" no errors. Checked Device Manager â€" no warnings. Tested audio and network â€" working."*
- State the outcome plainly: *"Fault resolved. Machine returned to full working condition; customer's data restored and verified."*

Why this matters. The "action + tests + result" trio is what turns your claim of a successful repair into **evidence** of a successful repair.

Common misconception. "If it switches on, it is fixed." A machine can switch on and still have a hidden fault (for example, one of two RAM sticks failing). The report must show the tests that confirm the whole system is healthy.

Limitation. Some faults cannot be reproduced immediately (intermittent faults). Be honest: record what you tested, and state clearly if the fault was intermittent and could not be fully reproduced.

F.5 Writing clearly and honestly

The idea. The report must be understood by the customer and trusted by any reader â€” while remaining completely truthful.

Principles of clear writing.

1. **Plain language.** Write for the customer, not for other technicians. Explain any technical term you must use.
2. **Be specific.** "Replaced faulty RAM module in slot 2" beats "fixed memory".
3. **Be complete.** Include every test, part and step. No gaps.
4. **Be accurate.** Double-check serial numbers, part numbers, dates and costs. A single wrong figure undermines the whole document.
5. **Be professional.** Correct grammar and spelling; no personal opinions, no complaints about the customer, no sarcasm.
6. **Be honest.** Never claim work you did not do or parts you did not use. If you made a mistake, record the facts and the resolution.

The honesty rule. The report must describe what **actually happened**. Inflating labour hours, adding parts not fitted, or hiding damage is fraud and destroys trust. A short honest report is worth more than a long false one.

Analogy. Writing a service report is like giving **evidence in court**: you state facts you can stand behind, clearly and without exaggeration. If you are caught exaggerating once, everything you say afterwards is doubted.

Common misconception. "More technical language sounds more professional." The opposite: clear, honest language reads as professional; dense jargon reads as evasive.

Limitation. You cannot write a perfect report if the work was rushed. A good report is the natural result of careful work â€” another reason to do the repair properly.

F.6 Costs, warranty and sign-off

The idea. The report ends with the money, the promise and the signatures.

Costs. Itemise parts, labour and any diagnosis fee, then state the **total**. This must match the estimate the customer approved. If the cost changed mid-repair, note that the customer approved the revised estimate.

Warranty terms. State plainly:

- **What** is covered (for example, "the replacement PSU part only", or "parts and labour").
- **How long** (for example, "90 days from the repair date").
- **Conditions** (for example, "void if the case is opened by another person or if liquid damage is found").

Be careful not to promise a warranty you cannot honour. A warranty is a commitment.

Sign-off. The report is not finished until:

- The **technician** signs and dates it.
- The **customer** signs and dates it, confirming the work was done, the cost agreed and the device returned in the stated condition.

Why the customer's signature matters. It is the moment the customer accepts the repair and the terms. It protects the shop from later false claims and gives the customer a record of their consent.

Common misconception. "The customer's signature is a formality." It is a genuine agreement. Explain the report to the customer before they sign â€” do not just push paper at them.

Limitation. A warranty cannot cover everything (for example, a new fault unrelated to the repair, or damage caused later by liquid or drops). Say so honestly in the terms.

F.7 Keeping records and handling data

The idea. Once the report is signed, it must be **kept safely** and the customer's data protected.

Record keeping.

- Store reports in a safe place â€" a folder, a filing cabinet, or a secure digital file with a backup.
- Keep them for a sensible period (many shops keep records for at least a year, often longer for warranty).
- Give the customer a copy (paper or a photo/PDF via WhatsApp) and keep one.

Data handling notes.

- The report itself should **not** list private contents of the customer's files ("customer's bank statements found"). Record only what is relevant to the repair.
- When you back up a customer's data for a repair, delete your working copy once the job is safely returned â€" do not keep customer data longer than necessary.
- Never copy, share or gossip about a customer's data (Session 10 confidentiality).

Analogy. A service report is like a **receipt and a medical note combined**: you keep a copy for your records, you give the customer one, and you do not write their private business all over it.

Common misconception. "Once the customer takes the machine, the record is no longer needed." The record is exactly what you need **if** a problem returns. Keep it.

F.8 Summary of the session

- A **service report** is the written story of one repair; it protects the customer, the technician and the shop.
- Required sections include: **header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests/result, costs, recommendations, warranty, sign-off**.
- Keep the **customer's words** and your **observed findings** separate but connected.
- **Action + tests + result** turn a claim into evidence.
- Write **clearly, specifically and honestly** â€" no jargon, no vague phrases, no exaggeration.
- Record **costs, warranty and signatures**, and store records safely while protecting customer data.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

What the instructor prepares:

- A blank Service Report Template on the projector.
- The worked scenario below, so the class can see a report built live, section by section.

G.1 The scenario

> ***Customer:*** *Mr. Okafor.* ***Device:*** *HP laptop, serial number* `ABC123`*.* ***Reported fault:*** *"My computer is running very slowly and sometimes freezes."* ***Diagnosis:*** *The hard drive was nearly full (95% used) and running slowly; the machine also had only 4 GB of RAM, which is too little for the customer's workload. The old spinning hard drive was the main cause of slowness.* ***Repair:*** *Replaced the HDD with a 500 GB SSD; upgraded RAM from 4 GB to 8 GB; reinstalled Windows (licensed) and installed all drivers; ran Windows Update; restored the customer's data.* ***Parts:*** *500 GB SSD â€" â‚¦25,000; 8 GB DDR4 RAM â€" â‚¦15,000.* ***Labour:*** *â‚¦10,000.* ***Total:*** *â‚¦50,000.* ***Tests after repair:*** *booted in under 20 seconds; memory diagnostic clean; Device Manager clean; audio and Wi-Fi working; data restored and opened.* ***Warranty:*** *90 days on parts; labour covered for 30 days.*

G.2 Building the report live (narrate each section)

Say: "Watch how every fact from the job goes into a section. Nothing is invented; nothing is left out."

1. **Header:** instructor writes the shop name, a report number `SRV-2026-0042`, the date, and "Technician: [name]".
2. **Customer and device:** Mr. Okafor; phone; HP laptop; serial `ABC123`.

3. **Reported fault:** "*Customer reported: 'My computer is running very slowly and sometimes freezes.*'" Explain why it is quoted.
4. **Diagnosis/findings:** write the tests and the root cause.
5. **Action taken:** write the replacement, RAM upgrade, Windows reinstall and driver work.
6. **Parts used:** the itemised table.
7. **Tests/result:** the verification list. **Say:** "This is where I prove the repair worked."
8. **Costs:** the breakdown summing to â¦50,000.
9. **Recommendations:** "Back up regularly; avoid filling the drive past about 80%; consider a surge protector."
10. **Warranty:** the terms.
11. **Sign-off:** technician line; explain that the customer signs here.

Ask the class after each section: "Could a stranger read this and understand exactly what happened? Could this report defend the shop in a dispute?"

G.3 Deliberately show a vague phrase and improve it

Write: "*Fixed the slow computer.*" Then rewrite it as the specific version above. **Say:** "One of these protects you; the other is a blank cheque for an argument."

G.4 Show the weak sample and diagnose it

Display the weak sample report from the opening. Ask the class to find:

- Missing sections (for example, no serial number, no tests, no warranty).
- Vague language ("fixed the problem").
- Any dishonesty or exaggeration.

Expected result: Learners spot that a report missing key sections cannot protect anyone.

Error to watch: Learners may want to write only the parts they remember. Insist on using the **template** so no section is forgotten.

Questions to ask during the demo:

- "Why do we record the customer's exact words?" â† Clues and proof of listening.
- "Why do we write the tests we ran?" â† To prove the fix, not just claim it.
- "Why get the customer's signature?" â† It confirms agreement and protects both sides.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective: With the instructor guiding step by step, learners write a **complete professional service report** for the scenario below, using the template.

Starting materials: The Service Report Template (printed or digital), the scenario on the board, pens/Word.

> ***Data-safety/confidentiality reminder:*** *the scenario is fictional. In real work, record only what is relevant to the repair and protect the customer's private data.*

Scenario (given):

> ***Customer:*** *Adaeze Okafor.* ***Device:*** *Desktop PC, serial* `PC456`*.* ***Reported fault:*** *"The computer would not start at all â€“ no lights, nothing."* ***Diagnosis:*** *Tested power with a known-good PSU; the original power supply was dead. (Other components tested good.)* ***Repair:*** *Replaced the power supply with a 500 W unit; reconnected all cables; tested power-on.* ***Parts:*** *500 W PSU â€“ â¦18,000.* ***Labour:*** *â¦5,000.* ***Total:*** *â¦23,000.* ***Tests after repair:*** *powered on; booted to Windows; Device Manager clean; fan and drives spinning; USB ports working.* ***Recommendation:*** *Use a surge protector; clean the vents every 3 months.* ***Warranty:*** *90 days on the PSU part.*

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **Header.** Write the shop name, a unique report number, today's date and the technician name.
2. **Customer and device details.** Name, phone, device make/type, serial `PC456`.
3. **Reported fault.** Quote the customer's words exactly.
4. **Diagnosis/findings.** Write the test performed (known-good PSU test) and the root cause (dead power supply).
5. **Action taken.** Describe the replacement and reconnection precisely.
6. **Parts used.** Complete the itemised table (description, quantity, cost).
7. **Tests performed and result.** List at least three tests and state the result.
8. **Pause and predict:** "*Which section proves the repair actually worked?*" â†' Tests performed and result.
9. **Costs.** Write the breakdown and the total (â‚¦23,000). Check the arithmetic.
10. **Recommendations.** Write two or three preventive suggestions.
11. **Warranty terms.** State what is covered, how long, and conditions.
12. **Sign-off.** Write the technician sign-off line and the customer sign-off line.
13. **Review against the checklist** before saving.

Expected result: A complete, well-organised service report containing all required sections, in plain language, with correct figures (total â‚¦23,000).

Pause-and-predict checkpoints: after step 3 ("Why quote the customer?"), after step 7 ("Which section proves the fix?"), after step 9 ("Is the total correct?").

Troubleshooting while walking the room:

- Vague action ("fixed computer") â†' ask: "Could a stranger repeat exactly what you did from this sentence?"
- Missing serial number â†' point to the section; explain why it matters for warranty.
- Wrong total â†' re-add parts and labour; accuracy is non-negotiable.
- Jargon ("SATA controller", "no POST") â†' ask them to say it to a non-technical customer.
- No customer signature line â†' add it; the report is not finished without sign-off.

Completion checklist (learner ticks each):

- ☐ Header with unique report number and date
- ☐ Customer and device details incl. serial
- ☐ Reported fault in the customer's words
- ☐ Diagnosis with tests and root cause
- ☐ Action taken, specific
- ☐ Parts used, itemised
- ☐ Tests performed after repair and result
- ☐ Costs itemised, total correct (â‚¦23,000)
- ☐ Recommendations given
- ☐ Warranty terms stated
- ☐ Technician and customer sign-off lines
- ☐ Plain language throughout

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task (NEW task): You are given a repair that was done **carelessly** and a messy set of notes. Your job is to turn the chaos into a professional report **and** identify what is wrong with the original records. This tests judgement, not just copying.

Scenario â€" "The rushed repair."

A technician at another shop made these shorthand notes for a job:

> *"HP laptop. Won't charge. Changed the charging port. â‚¦15k. Customer picked up. Done."*

You later learn the true facts:

> ***Customer:*** *Musa Ibrahim, phone 0803-xxx-xxxx.* ***Device:*** *HP 250 G6 laptop, serial* `HP7890` *.* ***Reported fault:*** *"The laptop works on battery but will not charge when I plug it in."* ***Diagnosis:*** *The DC charging jack (power port) was physically loose/broken; the charger itself tested good.* ***Repair:*** *Replaced the DC charging jack; tested charging.* ***Parts:*** *DC charging jack â€" â‚¦6,000.* ***Labour:*** *â‚¦5,000.* ***Total:*** *â‚¦11,000. (Note: the original note said â‚¦15,000 â€" the technician had guessed and overcharged, then later refunded â‚¦4,000. The real agreed total is â‚¦11,000.)* ***Tests after repair:*** *confirmed charging; battery reached 100%; tested with the customer's own charger; all ports working.* ***Recommendation:*** *Do not yank the charging cable; use the cable gently.*

Your tasks:

1. Write the **complete professional service report** for Musa's job, using all required sections.
2. List **three things wrong** with the original shorthand note (think about completeness, accuracy and honesty).
3. Write **one sentence** explaining to the customer why the final cost is â‚¦11,000, not the â‚¦15,000 first mentioned.

Time allowance: 15 minutes.

Success criteria: A complete report with all sections in plain language; three valid criticisms of the shorthand note; a clear, honest cost sentence; correct total (â‚¦11,000).

Hints if stuck:

- Hint 1: Every fact in the scenario belongs in a section. Ask "which section?" for each.
- Hint 2: For task 2, think about what is **missing** (no serial, no tests, no signature), what is **vague** ("changed the charging port"), and what is **dishonest** (overcharging, guessing).
- Hint 3: For task 3, honesty and transparency about the correction build trust.

INSTRUCTOR ANSWER KEY (do not show learners):

Task 1 â€" model report (summary):

- **Header:** shop name; report number `SRV-2026-0051` ; date; technician name.
- **Customer/device:** Musa Ibrahim; phone; HP 250 G6; serial `HP7890` .
- **Reported fault:** *"The laptop works on battery but will not charge when I plug it in."*
- **Diagnosis:** Charger tested good; the DC charging jack was physically loose/broken. Root cause: broken charging jack.
- **Action taken:** Replaced the DC charging jack; tested charging with the customer's own charger.
- **Parts used:** 1 Ã— DC charging jack â€" â‚¦6,000.
- **Tests/result:** Confirmed charging; battery reached 100%; all ports working. Fault resolved.
- **Costs:** Parts â‚¦6,000 + labour â‚¦5,000 = **â‚¦11,000**. (Note the correction from the original â‚¦15,000 and the â‚¦4,000 refund, if applicable.)
- **Recommendations:** Handle the charging cable gently; avoid pulling the cord sideways.
- **Warranty:** as per shop terms (e.g. 90 days on the part).
- **Sign-off:** technician and customer signatures.

Task 2 â€" three things wrong with the shorthand note (accept any three):

- **Incomplete:** no customer name/phone, no device model or **serial number**, no date, no diagnosis details, no tests, no warranty, no signature.
- **Vague:** "changed the charging port" does not identify the part or the cost breakdown.
- **Inaccurate / dishonest:** the note said â‚¦15,000 but the correct agreed total was â‚¦11,000 â€" it was a guess that overcharged, then a refund had to be made. A professional quotes honestly and gets approval.
- (Also valid: no record of the customer's exact words; no recommendation.)

Task 3 â€" cost sentence (sample): *"Sir, the final cost is â‚¦11,000 â€" â‚¦6,000 for the charging jack and â‚¦5,000 for labour. My first figure of â‚¦15,000 was a guess before I had inspected the machine, so I have corrected it and refunded the difference. I apologise for the confusion."*

Marking guidance (out of 10):

- Completeness of the report (all sections) (4)
- Plain language and specificity (2)
- Accuracy of figures (â‚¦11,000) (1)

- Three valid criticisms of the shorthand note (2)
- Clear, honest cost sentence (1)

J. REAL-WORLD CASE STUDY

Situation. A customer, *Mr. Nwosu*, brings a desktop back to a Port Harcourt repair shop eight weeks after a repair. He claims the shop "never actually replaced the power supply" and demands a refund, threatening to post about it online. The shop owner opens the filing cabinet and finds the signed service report.

Relevant facts. The report shows:

- A unique report number and date (eight weeks earlier).
- The customer's reported fault in his own words.
- The diagnosis and the test performed (known-good PSU test).
- The action taken (500 W PSU replaced).
- The part used, with description.
- The tests performed after repair and the result.
- The cost breakdown, signed by the customer.
- A 90-day warranty on the part.

Decision point. How does the report resolve the dispute?

Possible approaches.

1. Argue from memory â€" "I remember we changed it."
2. Produce the signed report and let it speak.
3. Give the refund to avoid trouble.

Recommended solution. Approach 2.

- The technician calmly shows Mr. Nwosu the report he signed, with the date, the part and the test results.
- The shop offers to open the case and show him the new PSU, with its fitting date sticker, to confirm.
- Because the repair was within the 90-day warranty, the shop agrees to re-check the PSU at no charge if there is a genuine new fault â€" good service on top of good records.
- Mr. Nwosu withdraws the claim.

Trade-offs.

- Approach 1 leaves the shop at the mercy of a customer's memory versus the technician's; unwinnable in a public argument.
- Approach 2 is factual, calm and quick â€" and shows professionalism.
- Approach 3 rewards a false claim and may encourage others. It also admits fault the shop did not commit.

Professional connection. The report is not just for the customer â€" it is the shop's **memory and its defence**. This is why Session 11 exists, and why the Final Project requires a complete report. The documented, signed record is what separates a professional shop from a roadside guess.

Transfer question for the class: "Which single section of the report do you think was most important in defending the shop â€" and why?" *(Strong answer: the customer's signature, because it proves agreement to the work and the cost; the test results are a close second because they prove the fix.)*

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table live. When a learner's report has a problem, point them to the row.

K.1 Vague descriptions

Symptoms: The report says "fixed the problem" or "did maintenance". **Likely cause:** Rushing, or not knowing what detail matters. **Safe corrective steps:** Rewrite with the exact part, action and result. Ask: "Could a stranger repeat this from my words?" **Prevention:** Use the template prompts for each section.

K.2 Missing sections

Symptoms: No serial number, no tests, no warranty, no signature. **Likely cause:** Skipping parts of the template. **Safe corrective steps:** Compare against the full section list and fill every gap. **Prevention:** Always write from the template; never from memory.

K.3 Inaccurate numbers or serial numbers

Symptoms: Total does not add up; wrong model/serial; date errors. **Likely cause:** Careless recording. **Safe corrective steps:** Re-add the figures; re-check the serial from the device; correct the report before signing. **Prevention:** Double-check all numbers before the customer signs.

K.4 Technical jargon in the report

Symptoms: Customer cannot understand their own report. **Likely cause:** Writing for technicians, not customers. **Safe corrective steps:** Translate terms; use plain language; keep technical detail brief and explained. **Prevention:** Write as if the customer will read it aloud to a friend.

K.5 No customer signature

Symptoms: Report is unsigned; dispute later has no written agreement. **Likely cause:** Not asking the customer to sign. **Safe corrective steps:** Explain the report to the customer at collection and ask them to sign; add the date. **Prevention:** Make sign-off a required last step of every job.

K.6 Exaggeration or dishonesty

Symptoms: Labour hours or parts overstated; work claimed that was not done. **Likely cause:** Temptation to inflate a small job. **Safe corrective steps:** Correct the report to the truth immediately; refund any overcharge; apologise. **Prevention:** Remember: honesty protects your name and, in serious cases, keeps you out of legal trouble.

K.7 Customer data written into the report or carelessly handled

Symptoms: Private contents listed; backup copies kept too long. **Likely cause:** Not thinking about confidentiality. **Safe corrective steps:** Remove private details; delete working copies of customer data once the job is returned. **Prevention:** Record only repair-relevant facts; treat data as a sealed envelope.

Distinguish likely vs confirmed: Fix what you can **confirm** (missing section, wrong total). Do not guess at facts â€” if unsure of a part number or test result, check the device or your notes before writing.

L. SESSION ASSIGNMENT

Assignment title: "My Capstone-Ready Service Report"

Purpose: To produce a complete, professional service report for a repair â€” the same document you will need for the Session 12 capstone â€” and to build a reusable template you can take into a real shop.

Realistic scenario: You have completed a repair at your shop. Use either a repair you actually performed during the course (for example the PSU replacement from the guided exercise or a real practice-machine job) or the provided scenario. Write the full report as if the customer is coming to collect the machine.

Detailed instructions:

1. Use the Service Report Template. Complete **every** section:
    - header with a unique report number and date,
    - customer and device details including serial number,
    - reported fault in the customer's own words,
    - diagnosis with tests performed and the root cause,
    - action taken (specific),
    - parts used (itemised),
    - tests performed after repair and the result,
    - costs (parts, labour, total â€” correct arithmetic),
    - recommendations,

    - warranty terms,
    - technician and customer sign-off lines.
2. Write in **plain language**; if you use a technical term, explain it.
3. Provide a **second page**: a reusable blank template you designed (you may lay it out however you like, but it must include all sections plus a space for the shop logo).
4. Add a one-paragraph note on **confidentiality** and how you will store records.

Required deliverables:

- A completed service report (one to two pages).
- A blank reusable template (one page).

Submission format: Printed or emailed (Word or PDF); filename `Surname_S11_Report.docx`.

Deadline: Before Session 12 (the capstone). Bring a copy to Session 12.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
All sections present	All required sections, clearly labelled	Most sections present	Several missing
Reported fault & diagnosis	Customer's words quoted; tests and root cause clear	Present but thin	Vague or missing root cause
Action, parts & tests/result	Specific action; itemised parts; tests and result recorded	Mostly specific	Vague; no test evidence
Costs, warranty & sign-off	Correct totals; warranty stated; both signatures lines	Minor gaps	Wrong total or no sign-off
Plain language & honest writing	Clear, specific, honest; no jargon	Mostly clear	Jargony, vague or exaggerated
Reusable template (2nd page)	Professional, complete, logo space	Mostly complete	Incomplete

Common mistakes to avoid: vague action lines; missing serial number; totals that do not add up; jargon; no customer signature line; recording private customer data.

Instructor reference solution (structure/notes):

- Every section completed from the same facts used in class.
- Cost arithmetic checked: for the class PSU scenario, â‚¦18,000 + â‚¦5,000 = â‚¦23,000.
- Action line example: *"Removed the faulty 500 W power supply; fitted a new 500 W PSU; reconnected the 24-pin motherboard, 4-pin CPU and SATA power cables; confirmed power-on."*
- Tests/result example: *"Powered on; booted to Windows; Device Manager clean; fans and drives operating; USB ports tested."*
- Template includes logo space, report number, all sections and both signature lines.

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key.)

Q1. List. Name at least eight required sections of a service report.

Q2. Short answer. Give three reasons why service reports matter.

Q3. Multiple choice. Which is the better action line in a service report? a) "Fixed the slow computer." b) "Replaced faulty 8 GB RAM module in slot 2; ran memory diagnostic â€" no errors." c) "Did some work on the machine." d) "Customer's computer is okay now."

Q4. Explain in your own words. Why do we record both the customer's reported fault **and** our observed findings?

Q5. Short answer. Why should a service report be written in plain language?

Q6. True or false. The customer does not need to sign the service report. Explain your answer.

Q7. Scenario. You replaced a part and quoted â‚¦20,000, but after further testing the correct total is â‚¦17,000. What do you write on the report, and what do you tell the customer?

Q8. Fill in the blank. A good service report records not only what was done but also the _____ performed and the _____, to prove the fix worked.

Q9. Identify the mistake. A report says: "*Customer had an affair on WhatsApp; also replaced the keyboard.*" What is wrong with listing the first part?

Q10. Short answer. Give two points to include in the "recommendations" section after a repair.

INSTRUCTOR ANSWER KEY (Q1â€“Q10)

A1. Any eight of: header/shop details with unique report number; customer and device details incl. serial number; reported fault (customer's words); diagnosis/findings with tests and root cause; action taken; parts used; tests performed after repair and result; costs (parts and labour) with total; recommendations; warranty terms; technician and customer sign-off. **A2.** Any three of: proof of work; warranty support; a record for future technicians; professionalism and trust; business information (costs, recurring faults); protection in a dispute. **A3.** (b). **A4.** The reported fault shows what the customer experienced and that you listened; the observed finding is evidence-based and identifies the true root cause. They can differ, and recording both shows professional reasoning. **A5.** Because the report is written for the customer, who is not a technician; it must be understood and trusted. Clear plain language also reduces disputes. **A6.** False. The customer's signature confirms they agreed to the work, the cost and the return condition of the device. It protects both the customer and the shop. **A7.** Write the **correct** total of â‚¦17,000, with an honest note (if needed) that the final cost is lower than the estimate because less work/parts were needed than anticipated. Tell the customer plainly and refund/charge only the correct amount. Never keep an inaccurate higher figure. **A8.** â€¦ **tests** â€¦ ; â€¦ **result** â€¦ **A9.** It records private, irrelevant personal information about the customer. A service report must contain only repair-relevant facts and must protect the customer's confidentiality. Remove it. **A10.** Any two of: back up data regularly; use a surge protector; clean vents/dust every few months; avoid filling the drive past ~80%; consider an SSD upgrade; handle cables gently; bring the machine for a periodic check.

N. REFLECTION AND RECAP

Plain-English summary. Today you learned to write the professional record of a repair: the **service report**. You learned why it matters (proof of work, warranty, records, trust, business information); the **sections** every report must contain (header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off); how to keep the **customer's words** and your **observed findings** separate but connected; how to turn a claim into evidence with **action + tests + result**; how to write **clearly and honestly**; and how to record costs, warranty and signatures while protecting customer data. This report is exactly what you will produce for the Final Project.

Five important takeaways:

1. The report is your **protection, memory and professional name**.
2. Record **customer's words** and **observed findings** â€” both.
3. **Action + tests + result** proves the fix.
4. Be **specific and honest**; no jargon, no exaggeration.
5. Finish with **costs, warranty and signatures**, and store records safely.

Questions to help you recognise what you can now do:

- Can I list every required section of a service report from memory?
- Can I write a specific action line instead of "fixed it"?
- Can I prove a repair with recorded tests and results?
- Can I state costs and warranty honestly and get a signature?
- Can I protect customer data while keeping good records?

Connection to the next session: Next session is the **Final Project â€“ Capstone**. You will receive a machine with a simulated fault and, without help, diagnose it, repair it, verify the fix, write the full service report you practised today, and present your

work. Everything from Sessions 1â€“11 comes together in one job. Come prepared, rested and ready.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner uses the **template** and completes **every** section without prompting.
- The learner quotes the customer's reported fault and separates it from the observed findings.
- The learner writes specific action lines and records actual **tests and results**.
- The learner's totals are accurate and the warranty is stated honestly.
- The learner writes in plain language a customer could understand.
- The learner includes both technician and customer sign-off and can explain why.
- The learner treats customer data confidentially in the report.

Common misconceptions to watch for and correct:

- "Reports are just paperwork." â†' They protect the customer, the technician and the shop, and support warranty and disputes.
- "A short note is enough for a simple job." â†' Even a simple repair gets the full record; only the length of each section changes.
- "The customer won't read it." â†' Even if they do not, the report is your protection and your record.
- "I'll write it later from memory." â†' Write it while the details are fresh; memory fades and errors creep in.
- "My signature is enough." â†' The customer must sign too; it proves agreement.
- "Big technical words look professional." â†' Plain, honest language is what reads as professional.

Suggested remedial exercises for learners who struggle:

1. **Section-matching drill:** give a list of facts and a list of sections; ask the learner to match each fact to its section.
2. **Vague-to-specific drill:** hand out ten vague sentences ("fixed it", "cleaned it") and ask for specific rewrites.
3. **Fill-the-template drill:** provide a partially completed report with gaps; ask the learner to finish it.
4. **Honesty drill:** give a scenario with a tempting exaggeration; ask the learner to write the honest version and explain why.

Differentiation:

- **Level 1 (supported):** provide a partially completed report with prompts in each section; allow the learner to write on paper; check section by section.
- **Level 2 (standard):** complete the guided exercise and assignment as written, including the reusable template.
- **Level 3 (challenge):** write reports for **two different** repairs (one hardware, one software/OS), and add a short **customer-facing summary paragraph** at the top of the report that a non-technical customer can read in 30 seconds.

Timing guidance (105 min):

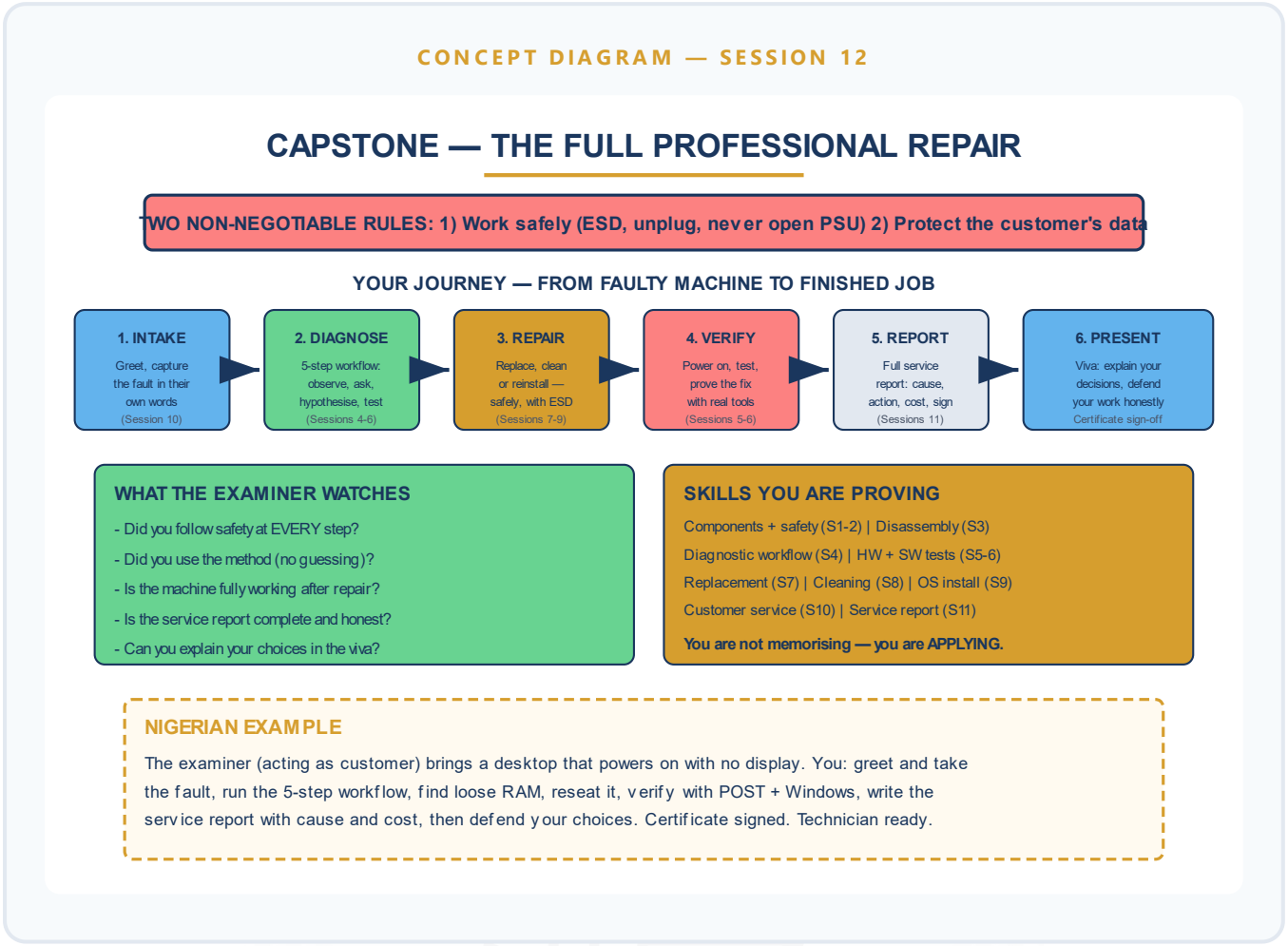
- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided report-writing 18, independent exercise 15, case study folded into reflection 3, reflection/recap + knowledge check 6. If the room is slow, shorten the case study and the knowledge check discussion, but **protect the guided report-writing** â€" the report is the capstone deliverable and must be practised. Do not cut the sign-off or confidentiality discussion.

END OF SESSION 11

Next: Session 12 â€" Final Project (Capstone).

SESSION 12 OF 12

Final Project Capstone



CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 12 â€" FINAL PROJECT (CAPSTONE): DIAGNOSE, REPAIR, DOCUMENT AND PRESENT

A. SESSION IDENTIFICATION

Item	Detail
Session number	12 of 12
Session title	Final Project (Capstone) â€" Diagnose, Repair, Document and Present
Week / Module	Week 4 â€" Module 4: Professional Practice (Capstone)
Recommended duration	105 minutes (plus a short certificate sign-off)
Difficulty level	Beginner, but it **tests all accumulated skills** (Sessions 1â€"11)

Item	Detail
Prerequisite knowledge	Sessions 1â€“11 â€” components, safety and tools, disassembly/reassembly, diagnostic workflow, hardware and software diagnostics, component replacement, cleaning and maintenance, OS installation, customer service, and service reports
Connection to previous work	This is the integration of the whole course. Every prior session taught one skill; today you combine them into a single professional repair job, exactly as a shop would.
Preparation before class	Prepare **one faulty machine per learner** (or one per pair for large classes) with a **simulated fault**; ensure a fault log exists for every machine so the examiner knows the real cause; prepare tool kits, anti-static straps, replacement parts and cleaning supplies; print the Capstone Brief, the Capstone Rubric, the Viva Question List and the Certificate Sign-off sheets; set up assessment stations; confirm each machine's **twist** is documented.
Safety and ethics flag	Safety procedures (ESD, unplug before opening, never open a PSU) are **assessed**, not optional. Data safety (back up before any wipe) is **assessed**. Only **licensed** software is used. No learner may proceed past a safety breach without correction.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Receive a faulty machine and complete a customer intake**, capturing the reported fault accurately.
2. **Diagnose the fault** using the systematic diagnostic workflow (observe, ask, hypothesise, test, resolve).
3. **Apply correct safety procedures** (ESD protection, power isolation, safe tool use) throughout.
4. **Perform the repair** â€” replacement, cleaning or OS task â€” safely and correctly.
5. **Verify the fix** with real tests and confirm the machine is fully functional.
6. **Produce a complete professional service report** documenting the entire job.
7. **Present and defend their work** in a short viva/demonstration, answering questions about their decisions.
8. **Reflect honestly** on their competence, their challenges and their next steps as a technician.

C. REQUIRED MATERIALS

Hardware (per learner or pair)

- One faulty practice machine with a **simulated fault** (see the fault bank below).
- Tool kit: correct screwdrivers, tweezers, torch, anti-static wrist strap, anti-static mat.
- Replacement components for the intended faults (RAM sticks, a known-good PSU, a known-good drive/SSD, extra fans, cleaning supplies).
- Cleaning supplies: compressed air, isopropyl alcohol, microfiber cloth, thermal paste (for cleaning/thermal faults).

Software

- Diagnostic tools as taught: Windows Memory Diagnostic / MemTest86, CrystalDiskInfo, HWMonitor, Device Manager, Windows Update, and a licensed Windows USB where an OS task is part of the fault.

Printed materials

- Capstone Brief (one per learner).
- **Capstone Rubric** (one per learner and one per assessor).
- **Viva Question List** (instructor copy, with model answers).
- Certificate Sign-off sheet (one per learner).
- Blank Service Report Template and Customer Intake Form (from Sessions 10â€“11).

Safety equipment

- Anti-static wrist straps and mats; safety glasses if compressed air is used; first aid kit available.

Room setup

- Assessment stations, each with a machine and its fault log; a clear counter for the intake role-play; a quiet corner for the viva so it is not disturbed.

D. OPENING ACTIVITY (6 minutes)

Instructor setup: Stand beside a faulty machine on the counter. Do not open it yet. Put the Capstone Brief on the projector.

What the instructor says:

> *"Four weeks ago, most of you had never opened a computer case. You did not know RAM from a hard drive. Look at you now.*
>
> *Today there is no teaching and no demonstration. Today you work. A customer â€" that is me â€" will bring you a faulty machine. You will greet the customer, take the fault, diagnose it, repair it, prove it works, write the service report, and explain your work to me.*
>
> *I will not help you. Not because I do not care, but because your future employer and your future customers will not be standing beside you. Today you prove to yourself that you can do this alone.*
>
> *Two rules stand above everything: **work safely**, and **protect the customer's data**. Break either rule and it is an automatic fail for that part, no matter how good the repair is. Everything else, you already know. Let us begin."*

Expected learner response: A mix of nerves and excitement. Reassure them: the brief, the template, the tools and the workflow posters are all in front of them. They are not being asked to memorise â€" they are being asked to **apply**.

Why this opener works: It raises the stakes (independent performance) while lowering panic (everything is provided). It also states the two non-negotiables, safety and data, before any work begins.

E. PREREQUISITE REVIEW (7 minutes)

This review is the **last-minute confidence check** before the assessment. Run it fast, as a whole-class call-and-response, so learners hear the key facts one final time.

Question 1: "What are the five steps of the diagnostic workflow?"

- Answer: observe, ask questions, form a hypothesis, test, resolve/verify.

Question 2: "What are the six major components?"

- Answer: motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.

Question 3: "What is ESD, and how do you prevent it?"

- Answer: electrostatic discharge â€" a sudden flow of static electricity that damages components. Prevent it with an anti-static wrist strap, an anti-static mat, and by touching bare metal of the case before handling components.

Question 4: "What must you do before opening any case, and what must you never open?"

- Answer: unplug the power, then hold the power button for a few seconds to discharge residual power. Never open the PSU case â€" its capacitors can hold a lethal charge.

Question 5: "What do these tools test â€" MemTest86, CrystalDiskInfo, HWMonitor?"

- Answer: MemTest86 tests RAM; CrystalDiskInfo reports hard drive/SSD health; HWMonitor shows temperatures and voltages.

Question 6: "Name the sections of a service report."

- Answer: header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off.

Question 7: "What is the academy's rule about software licences?"

- Answer: install only properly licensed software; never use pirated or cracked Windows.

Bridge to the assessment (say this):

> *"Those seven answers are your toolkit. If you can hold onto them under pressure, you will do well. Now the review is over â€" the customer is coming."*

Gap handling: If the class is visibly shaky on any single question, spend 60 seconds re-explaining it only, without restarting the whole review. Do **not** extend this into teaching: the session time belongs to the learners' independent work.

F. FULL LECTURE NOTES (Capstone Briefing)

F.1 What the capstone is

The idea. The capstone is one complete, realistic repair job done **independently**. It assesses not just whether you can fix the machine, but whether you can behave like a professional technician from first greeting to final signature.

The scenario. You are the technician at *Cyber Elias Tech*, a repair shop in Port Harcourt. A customer (the instructor) brings in a faulty computer. You must:

1. Conduct a **customer intake**.
2. **Diagnose** the fault using the systematic workflow.
3. **Perform the repair** safely.
4. **Verify** the fix.
5. Write a **service report**.
6. **Present and defend** your work in a short viva/demonstration.

What is being assessed (overview). Seven areas â€" safety adherence, diagnostic method, correct identification, repair quality, verification/testing, service report, and presentation/viva â€" worth **30 marks**, plus a **5-mark** viva/demo component. The full rubric is in Section I.

Your deliverables.

- A correctly diagnosed fault.
- A safely performed repair.
- A verified, fully functional machine.
- A complete, professional service report.
- A confident presentation/viva.

Time. Approximately 70 minutes for intake, diagnosis, repair and report; 5 minutes for the viva/demo (per learner, run in parallel or in a slot).

F.2 Milestone checkpoints â€" what you should already have done before Session 12

The capstone is not a surprise. To be ready, you should have completed these milestones in earlier sessions. Use this as your personal pre-capstone checklist.

#	Milestone	Covered in	What "done" looks like
1	Identify all major components and explain their functions	S1	You can point to and name each component on a real machine
2	Work safely: ESD, power isolation, correct tool use	S2	You always wear the strap, unplug, and use the right screwdriver
3	Disassemble and reassemble safely	S3	You can strip and rebuild without damaging or misplacing parts
4	Follow the five-step diagnostic workflow	S4	You observe, ask, hypothesise, test, and only then act

#	Milestone	Covered in	What "done" looks like
5	Test RAM, storage, temperature and power	S5	You can run MemTest86, CrystalDiskInfo and HWMonitor and read results
6	Diagnose software faults (OS, malware, drivers)	S6	You can tell an OS problem from a hardware problem
7	Replace RAM, HDD/SSD, PSU and GPU	S7	You can swap components and verify the replacement
8	Clean and maintain, replace thermal paste and fans	S8	You can remove dust, reseat coolers and apply paste correctly
9	Install an OS safely and legally	S9	You can back up, install Windows, add drivers and activate legally
10	Serve and communicate with a customer	S10	You can run an intake and explain a fault in plain language
11	Write a professional service report	S11	You can complete every section of the report, honestly

If a learner has not met these milestones, the instructor should arrange practice before the capstone (see Section O, remediation). A learner may still attempt the capstone, but the rubric will reflect the gaps.

F.3 The fault bank (instructor's simulated faults)

Each machine carries **one primary simulated fault** (and, for Level 3, possibly a second hidden fault). The fault log records the true cause so the assessor can mark accurately.

Fault	How it is simulated	Expected diagnostic path	Expected repair
Loose/unseated RAM	RAM stick not fully clipped in (or one dead stick)	No boot / beeps / memory test errors	Reseat or replace the RAM
Failing storage	A drive set to fail SMART checks or disconnected data/power cable	Slow boot, clicking, CrystalDiskInfo warnings	Reseat or replace the drive/SSD
Overheating	Dust-blocked heatsink / disconnected fan / dried thermal paste	High CPU temps on HWMonitor, thermal shutdowns	Clean, reconnect fan, replace thermal paste
Failed PSU	Known-good PSU swapped in for a dead one (or a loose 24-pin)	No power / no POST	Replace or reseat the PSU safely
Corrupted OS	Windows deliberately corrupted, or an unactivated/blank drive	Boots to recovery / no OS found	Back up, clean install Windows, drivers
Faulty GPU	GPU not seated / dead card (integrated graphics fallback)	No display / artefacts	Reseat or replace the GPU
Loose data cable	SATA cable unplugged/bad	Drive not detected	Reseat/replace the cable

Important: the learner is **not told** the fault. They must find it. The assessor follows the fault log to verify whether the learner identified the true cause.

F.4 How you will be assessed

You are assessed on **process as much as result**. A learner who fixes the machine but skips the backup step, or who diagnoses by luck with no method, will lose marks. A learner who follows the workflow, works safely, identifies the cause, repairs well, verifies and documents will score highly â€” even if a small hiccup occurs, as long as they handle it professionally.

The seven assessed criteria and the marks are in the full rubric (Section I). Read the rubric **before** you begin, so you know exactly what the examiner is looking for.

F.5 The professional mindset

- **Safety first.** Every time. The strap, the unplug, the right tool.
- **Data safety.** Back up before any wipe. Ask about data during intake.
- **Honesty.** Record what you did. Do not invent or exaggerate.
- **Method over luck.** Follow the workflow even under pressure.
- **Communication.** Explain your thinking calmly in the viva.
- **Integrity.** Only licensed software. No shortcuts that harm the customer.

Say this to the class:

> *"In ten years, the machines will change. New CPUs, new drives, new Windows. What will not change is the way a professional works: safely, methodically, honestly, and in the customer's interest. That is what we are testing today â€" the professional, not just the technician."*

G. INSTRUCTOR DEMONSTRATION

There is no instructor demonstration in this session.

The capstone is a **learner-led independent assessment**. Demonstrating the repair would defeat its purpose. Instead, the instructor:

- States the brief (Section D) and reviews prerequisites (Section E).
- Assigns each learner their machine and intake scenario.
- Observes, times and marks using the rubric (Section I).
- Runs the viva using the question list (Section J).
- Does **not** give technical hints during the independent work.

Permitted assistance: safety interventions only. If a learner is about to cause harm (for example, reaching to open a PSU, or handling a component without the strap), the instructor **must** intervene to protect them â€" and this is recorded as a safety concern against the relevant rubric criterion. No other help is given.

H. GUIDED PRACTICAL EXERCISE

There is no guided practical in this session.

The entire session is the independent capstone (Section I). However, the instructor **walks the room continuously**, observing quietly and noting evidence for the rubric, while remaining available for safety-only interventions.

I. INDEPENDENT CLASS EXERCISE â€" THE CAPSTONE (â‰ˆ70 minutes + viva)

I.1 The task

Scenario: "You are a computer repair technician at Cyber Elias Tech. A customer brings in a faulty computer. Diagnose the problem, perform the repair, verify the fix and produce a service report. Present your work to the instructor."

Procedure:

Step 1 â€" Customer intake (5 minutes).

1. Greet the customer (instructor) warmly.
2. Ask good diagnostic questions; capture the reported fault in the customer's **own words**.
3. Ask about **important data** (this is mandatory).
4. Set expectations: diagnosis first, then a written estimate before any billable work.

Step 2 â€" Diagnose (20 minutes).

1. **Observe** (look, listen, smell, feel) before opening anything.
2. **Ask** further questions if needed.
3. **Form a hypothesis** and say it out loud (for the assessor).
4. **Test** with the correct tools (memory test, SMART check, temperature, PSU test).
5. **Identify the root cause** and record every test and result.

Step 3 â€" Repair (25 minutes).

1. **Safety first:** power off, unplug, discharge, anti-static strap on, mat under the machine.
2. Perform the repair with the **correct tools** and correct technique.
3. Replace, reseat, clean or install the OS (with a backup first if data is at risk).
4. Only licensed software.

Step 4 â€" Verify (10 minutes).

1. Power on and confirm the specific fault is resolved.
2. Run diagnostic tests again (memory, SMART, temperature, Device Manager, boot).
3. Confirm all components work: display, sound, network, storage, ports.
4. State clearly: "Fault resolved" only when the evidence supports it.

Step 5 â€" Service report (10 minutes).

1. Complete **all** sections of the report.
2. Use plain language and specific details.
3. Include correct totals and warranty.
4. Get the customer's (instructor's) signature.

Step 6 â€" Present / viva (5 minutes).

1. Explain what the fault was, how you found it, what you did and how you verified it.
2. Answer the assessor's viva questions (Section J) confidently and honestly.

Time allowance: 70 minutes + 5-minute viva.

Success criteria: Fault correctly diagnosed; repair performed safely and correctly; fix verified; complete professional service report; clear, honest presentation.

I.2 CAPSTONE RUBRIC (30 marks + 5-mark viva/demo)

> ***Instructor-only marking instrument.*** *Each criterion is scored 0â€"5 (six criteria = 30 marks). The viva/demo is a separate 5 marks. Total =* ***35 marks****. Performance descriptions guide the marker; use the nearest band.*

#	Criterion	5 â€" Excellent	3 â€" Competent	1 â€" Developing	0 â€" Not demonstrated	Marks
1	**Safety adherence**	ESD strap worn correctly, power isolated and discharged, correct tools used, safe posture and workspace throughout; anticipates hazards	Safety mostly followed with one minor lapse, self-corrected	Safety procedures partial or inconsistent; needed one intervention	Ignored safety; would have caused harm (automatic 0)	/5
2	**Diagnostic method**	Full five-step workflow followed visibly; hypothesis stated and tested logically; systematic, no random part-swapping	Workflow mostly followed; minor gaps in logic	Partial method; some guessing; skipped steps	No method; random actions	/5
3	**Correct identification**	Confirmed the **exact** root cause, matching the fault log, with evidence	Identified the correct general area; some uncertainty	Wrong first guess but partially recovered, or correct area	Wrong root cause, or failed to identify	/5

#	Criterion	5 â€“ Excellent	3 â€“ Competent	1 â€“ Developing	0 â€“ Not demonstrated	Marks
				without confirmation		
4	**Repair quality**	Repair performed correctly, cleanly and durably; correct component/technique; no collateral damage	Repair correct with minor imperfections	Repair partially correct; loose fitting or messy; may need rework	Repair incorrect or caused damage	/5
5	**Verification / testing**	Specific tests re-run to prove the exact fault is gone; full-system check; honest statement of result	Basic verification done; some checks missed	Minimal verification (e.g. "it switched on")	No verification, or false claim of success	/5
6	**Service report**	All sections complete, accurate, specific, plain language, correct costs, warranty and signatures	Most sections complete; mostly clear	Several sections missing or vague; jargon; inaccurate figures	Missing or unusable report	/5
â€“	**Subtotal (30)**					**/30**
7	**Presentation / viva**	Clear, confident, honest; explains decisions and answers questions accurately; owns any limitations	Understandable; answers most questions correctly	Uncertain; struggles to explain own decisions	Cannot explain the work, or misrepresents it	**/5**
	TOTAL					**/35**

Passing standard: A learner must score at least **50% overall** across the course AND successfully complete the capstone: **correct diagnosis + safe repair + verified fix + professional service report**. A repair the learner **cannot explain** must not automatically pass. A serious safety breach (Criterion 1 = 0) fails the capstone regardless of other scores.

Assessor notes column (use on the marking sheet): record evidence, not just scores â€“ for example, "stated hypothesis: dust-blocked heatsink; ran HWMonitor showing 94Â°C; cleaned and re-pasted; temp 52Â°C after."

I.3 Milestone timing during the session

Phase	Minutes	What the assessor watches for
Intake	5	Warmth, good questions, data question asked
Diagnose	20	Workflow order, visible hypothesis, correct tools
Repair	25	Safety, technique, no collateral damage
Verify	10	Re-running the right tests, honest result
Report	10	All sections, accuracy, plain language
Viva	5	Explanation, honesty, confidence

J. VIVA / DEMONSTRATION â€“ QUESTION LIST WITH MODEL ANSWERS

> ***Instructor-only (mostly).*** *Ask 4â€“6 questions per learner, chosen to probe their actual work. Model answers are guides, not scripts; a learner who reasons correctly in their own words should be credited.*

Q1. "In your own words, what was the fault and how did you find it?"

- Model: The learner names the actual fault and walks through observe â†' ask â†' hypothesise â†' test â†' conclude. They should mention the specific test that confirmed the cause (for example, memory test errors, SMART warnings, high temperatures, or a known-good PSU test).

Q2. "What safety precautions did you take before opening the case, and why?"

- Model: Powered off and unplugged the machine; held the power button to discharge residual power; wore an anti-static wrist strap connected to bare metal; used an anti-static mat; used the correct screwdriver. These protect the technician and the components.

Q3. "Why did you test the way you did, rather than swapping parts?"

- Model: Testing finds the **root cause** and avoids wasting time and the customer's money on parts that are not faulty. Random part-swapping can also introduce new faults.

Q4. "How did you confirm the fault was truly fixed?"

- Model: Re-ran the same test that failed before (for example, memory diagnostic clean, SMART healthy, temperatures normal, machine boots to Windows), plus a full-system check (Device Manager, display, sound, network).

Q5. "Walk me through your service report â€" what would you say to the customer?"

- Model: The learner summarises the reported fault, the diagnosis, the repair, the parts, the total cost, the warranty, and the recommendation in plain language.

Q6. "If the customer's machine had important data and the fix required a reinstall, what would you have done?"

- Model: Backed up and verified the data first, informed the customer, and obtained approval before wiping; only then installed Windows. Data safety before any wipe.

Q7. "What would you do if you could not find the fault?"

- Model: Be honest â€" do not guess or pretend. Return to the workflow, re-observe, re-ask, re-test, check the simplest causes first; consult a senior technician; tell the customer the truth about the uncertainty.

Q8. "Show me where the RAM / storage / PSU is and explain what it does."

- Model: The learner points to the correct component and gives its function (RAM = short-term memory; storage = permanent files; PSU = converts wall power to the voltages the PC needs).

Q9. "Was any part of this job done legally? What about the software?"

- Model: Only properly licensed software was used; no pirated Windows. The learner can state the academy's licensing rule.

Q10. "What would you do differently next time?"

- Model: An honest reflection â€" for example, "I would ask more questions during intake", "I would check temperature earlier", or "I would keep my workspace tidier". Self-awareness is a sign of competence.

Q11. "Why did you record the customer's exact words in the report?"

- Model: They can contain clues to the cause, and they prove the technician listened and understood the customer's experience.

Q12. "A customer is angry that the repair is late. How would you handle it?"

- Model: Listen without interrupting, empathise, focus on solutions, and set calm boundaries; communicate honestly about the new time.

Marking the viva (5 marks):

- 5 â€" Clear, confident, honest; answers accurately; explains decisions; owns limitations.
- 3 â€" Understandable; answers most questions correctly; minor gaps.
- 1 â€" Uncertain; struggles to explain own work.
- 0 â€" Cannot explain the work or misrepresents it.

K. COMMON MISTAKES AND TROUBLESHOOTING (Capstone failure points)

Use this as the **remediation map** when a learner underperforms. These are the most common failure points and what to do about them.

K.1 Skipping safety steps

Symptom: Opens the case without the strap or without unplugging. **Why:** Rushing, nerves, overconfidence. **Fix:** Stop, reset the learner, restate the rule, let them continue safely (rubric reflects the lapse). **Prevent:** Make safety a habit drilled since Session 2; the strap comes out before the case.

K.2 Jumping to a repair without diagnosis

Symptom: Immediately replaces a part guessed to be at fault. **Why:** Impatience; forgetting the workflow. **Fix:** Pause; require them to state a hypothesis and one test that would confirm it before touching the machine. **Prevent:** Practise "say your hypothesis out loud" in every diagnostic session.

K.3 Fixing the symptom, not the root cause

Symptom: Machine boots after a reseat but the real fault (for example, a failing drive) remains. **Why:** Stopping at the first sign of life. **Fix:** Require verification tests; ask "how do you know the cause is gone?" **Prevent:** Always test the underlying cause, not just the symptom.

K.4 Not verifying the fix

Symptom: "It switched on, so it's fixed." **Why:** Misunderstanding what verification means. **Fix:** Re-run the specific failing test; check the whole system. **Prevent:** Use the post-repair checklist every time.

K.5 Poor or incomplete service report

Symptom: Missing sections, vague action lines, wrong totals, no signature. **Why:** Rushed at the end; forgets the template. **Fix:** Complete the template section by section; check arithmetic; add signatures. **Prevent:** Write the report from the same facts as the repair; practise in Session 11.

K.6 Data safety lapse

Symptom: Wipes or reinstalls without backing up or without customer consent. **Why:** Forgetting the customer's data in the heat of the task. **Fix:** Stop; if data may be lost, halt and refer (recovery is beyond this course); record the lapse. **Prevent:** Ask about data in intake; back up and verify before any wipe.

K.7 Cannot explain the work in the viva

Symptom: Repaired the machine but cannot say how or why. **Why:** Learnt by imitation rather than understanding; panic. **Fix:** Walk them back through their own steps; ask simpler "what" and "why" questions. **Prevent:** Narrate reasoning during practice so it becomes natural.

K.8 Licensing shortcut

Symptom: Uses a pirated or cracked Windows/software. **Why:** Convenience; not understanding the risk. **Fix:** Stop immediately; require a licensed install; explain the legal and safety consequences. **Prevent:** State the licensing rule every time software is installed.

Distinguish likely vs confirmed applies here too: do not assume a learner "does not know" â€” observe the actual behaviour and confirm with a question before judging. The rubric is about demonstrable performance.

L. SESSION ASSIGNMENT

Assignment title: "My Capstone Reflection and Next Steps"

Purpose: To close the course with honest self-assessment and a concrete plan for the learner's next stage as a technician â€” turning a certificate into a direction.

Realistic scenario: You have just completed your capstone assessment. Your instructor asks you to write a short, honest reflection and a plan, as part of your professional development file.

Detailed instructions:

1. **Describe your capstone job** in 3â€“5 sentences: the fault, how you diagnosed it, what you repaired, and how you verified it.

2. **Three things you learned** across the whole course (be specific and accurate).
3. **Two things you found challenging**, and how you handled or would handle them.
4. **One skill you want to improve** next, and one concrete step to improve it.
5. **Your next step**: choose at least one realistic path â€" apply for a junior technician role, start a small repair side-business, take a CEA advanced course, or specialise (for example, phone repair, data recovery basics, networking).
6. **An honest statement** of what your certificate does and does not mean (that it shows competence as a beginner technician, not as a certified electronics engineer).
7. Sign and date it.

Required deliverables:

- A typed or neatly handwritten reflection of **at least half a page**.
- The completed **Capstone Rubric** sheet (given to you after marking, if your instructor shares it) and your signed **Service Report**.

Submission format: Printed or emailed; filename `Surname_S12_Reflection.docx`.

Deadline: By the end of the certificate sign-off, or as instructed.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Capstone description	Clear, specific, accurate	Present but thin	Vague or missing
Three things learned	Specific, accurate, reflective	Mostly specific	Vague
Two challenges	Honest, specific, with handling	Somewhat specific	Vague
Next-step plan	Clear, realistic, concrete step	Present but general	Missing
Honest certificate statement	Accurate about what it does/does not mean	Partly accurate	Missing or overclaims

Common mistakes to avoid: overclaiming (saying the certificate makes you an "engineer"); vague lists ("I learned a lot"); no concrete next step; dishonest reflection.

Instructor reference solution (structure/notes):

- Capstone description: names the real fault and the actual test that confirmed it.
- Learned: e.g. "I can now safely replace a PSU and prove it with a power test"; "I can write a service report"; "I learned to back up data before any install."
- Challenges: e.g. "Staying calm when the machine would not boot the first time"; "Remembering to ask about data."
- Next step: e.g. "Apply to a repair shop in Port Harcourt as a junior technician, and start saving for an advanced course."
- Honest statement: e.g. "This certificate shows I can safely diagnose and repair common hardware and software faults as a beginner technician. It does not make me a certified electronics engineer and does not cover advanced repairs."

M. KNOWLEDGE CHECK

This is the **course-wide knowledge check** (20 questions across the four weeks). It is worth 5% of the overall grade. Answers are for the instructor only.

(Learner version â€" answer in the spaces provided.)

Section A â€" Components (Q1â€"Q5)

1. Name the six major components of a computer.
2. What does the CPU do?
3. What is the difference between RAM and a hard drive?
4. Which is faster, an HDD or an SSD?
5. What does the PSU do?

Section B â€" Safety (Q6â€"Q10) 6. What is ESD? 7. Name two ways to prevent ESD damage. 8. What must you do before opening a computer case? 9. Why should you never open a PSU case? 10. What tool prevents static damage to components?

Section C â€" Diagnostics (Q11â€"Q15) 11. What are the five steps of the diagnostic workflow? 12. What does a clicking noise from a hard drive usually indicate? 13. Which tool or test checks RAM? 14. What is a normal idle CPU temperature range? 15. Why should you test the simplest possible cause first?

Section D â€" Repair, OS and Professional Practice (Q16â€"Q20) 16. What should you do before a clean install of Windows? 17. Name three driver types you install after Windows. 18. What are the four steps of the customer intake process? 19. Name five sections of a service report. 20. What should you do if a repair will cost more than the estimate?

INSTRUCTOR ANSWER KEY

Section A

1. Motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.
2. It is the "brain" â€" it performs calculations and executes instructions.
3. RAM is short-term memory that empties when power is off; a hard drive stores files permanently.
4. SSD.
5. It converts electricity from the wall socket into the correct voltages for the PC components.

Section B 6. Electrostatic discharge â€" a sudden flow of static electricity that can damage components. 7. Any two of: anti-static wrist strap; anti-static mat; touching bare metal of the case before handling components; storing components in anti-static bags. 8. Unplug the power cable and press/hold the power button to discharge residual power. 9. Capacitors inside can hold a lethal charge even when the unit is unplugged. 10. Anti-static wrist strap.

Section C 11. Observe, ask questions, form a hypothesis, test, resolve/verify. 12. The drive is failing. 13. Windows Memory Diagnostic or MemTest86. 14. Roughly 30â€"50 Â°C at idle (accept a reasonable range; under load it can rise, but sustained very high temperatures indicate a cooling problem). 15. Simple causes are common and quick to test; you may solve the problem in minutes before spending time on complex checks.

Section D 16. Back up the customer's data (and verify it), then check that the hardware meets the OS requirements; prepare drivers. 17. Any three of: network, chipset, graphics, audio, Wi-Fi/Bluetooth. 18. Greet, gather information, set expectations, document. 19. Any five of: header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off. 20. Stop and contact the customer for a fresh estimate and approval before continuing work.

N. REFLECTION AND RECAP

N.1 Plain-English summary

Today you completed your **capstone**: you received a faulty machine, diagnosed it using the systematic workflow, repaired it safely, verified the fix, wrote a professional service report, and explained your work in a viva. You have finished the Computer Repairs course at Cyber Elias Academy. Whatever the mark, take a moment to appreciate how far you have come â€" from not knowing RAM from a hard drive, to diagnosing and repairing a real machine on your own.

N.2 Five important takeaways from the entire course

1. **Safety always comes first** â€" ESD protection, unplug before opening, never open a PSU.
2. **Follow the systematic diagnostic workflow** â€" observe, ask, hypothesise, test, resolve.
3. **Know your components** â€" what each part does and how to replace it is your foundation.
4. **Customer service is a technical skill** â€" communicate clearly and honestly in plain language.
5. **Professional documentation protects everyone** â€" the service report is your proof, memory and name.

N.3 Questions to help you recognise what you can now do

- Can you identify all major computer components and explain their functions?
- Can you follow ESD and electrical safety procedures correctly?
- Can you disassemble and reassemble a computer without damage?
- Can you diagnose common hardware and software faults using a workflow?
- Can you replace RAM, a hard drive/SSD and a PSU?

- Can you clean and maintain a computer, including thermal paste?
- Can you install and configure an operating system safely and legally?
- Can you run a customer intake and explain faults in plain language?
- Can you write a complete, honest service report?
- Can you complete a full repair project independently, from greeting to signature?

If you answered "yes" to most of these, you have become a **competent entry-level technician**.

N.4 Honest statement about what you have achieved

Be clear with yourself and with any employer:

- **This course makes you a competent entry-level computer repair technician.** You can safely diagnose and repair common hardware and software faults, serve customers professionally, and document your work. You can work as a junior technician or start a small repair business.
- **This course does NOT make you a certified electronics engineer or an IT professional.** It does **not** cover advanced work such as motherboard-level component repair (microsoldering), professional data recovery, or networking engineering. Those require further study.
- **This certificate does not guarantee employment.** It proves demonstrated competence; your attitude, honesty and continued practice will do the rest.

N.5 What comes next

At Cyber Elias Academy and beyond, you can continue with:

- **Cybersecurity** â€" protect computers and networks from threats (pairs naturally with repair).
- **Web Design / Web Development** â€" build and maintain websites.
- **Business & Freelancing** â€" learn to run your repair business professionally, price your work and find customers.
- **Content Creation** â€" photography, video and written content.
- **Specialisation** â€" phone repair, laptop specialist work, small-business IT support, or (with further training) data recovery and networking.

Practical next steps:

1. Keep a **job log** of every repair you do, even small ones â€" it builds your experience and your confidence.
2. Build a simple **portfolio**: before/after photos (with permission), service reports (with customer details removed), and a list of faults you have fixed.
3. Register and brand your **business** if you choose to work independently (a simple name, a phone/WhatsApp number, and an honest price list go a long way).
4. Never stop learning â€" every machine you open teaches you something new.

N.6 A closing word from CEA

> *"From Zero to Expert, Together." You started at zero and worked together to reach this point. You are not an expert yet â€" nobody becomes one in four weeks â€" but you are a **competent, safe, honest technician** with a certificate to prove it and a skill that will feed you. Use it well, treat your customers fairly, keep learning, and the rest will follow.*

O. INSTRUCTOR EVALUATION NOTES

O.1 Observable signs of genuine competence

- The learner puts on the anti-static strap and isolates power **without being reminded**.
- The learner states a clear hypothesis before testing, and tests logically.
- The learner identifies the true root cause, matching the fault log, with evidence.
- The learner repairs cleanly, with the correct component and technique.
- The learner re-runs the specific failing test and confirms the whole system works.

- The learner's service report is complete, specific, honest and signed.
- The learner explains decisions calmly in the viva and **owns any limitations honestly**.
- The learner uses only licensed software and can state the rule.

O.2 Common misconceptions to watch for and correct

- "I can just learn this on my own." â†' Structured practice prevents unsafe habits and builds method. Self-teaching without feedback often cements bad technique.
- "Repair is only about fixing broken things." â†' It is equally maintenance, customer service and documentation.
- "Passing means I am an expert." â†' Passing means you are a safe, competent **beginner**. Expertise takes years and further study.
- "If it switches on, it is fixed." â†' Verification requires re-testing the actual fault and checking the whole system.
- "Reports and safety slow me down." â†' They protect the customer, the shop and you; a careless wipe or a dishonest report can end a career.
- "A cracked Windows is fine to save money." â†' Illegal and unsafe; only licensed software.

O.3 Suggested remedial exercises / remediation

- **Did not pass the capstone:** allow supervised practice on academy machines during opening hours; reassess when ready. A second attempt is normal and expected â€" competence, not speed, is the goal.
- **Cannot diagnose:** provide additional guided diagnostic scenarios, starting with simple causes, and require the learner to say the hypothesis out loud each time.
- **Cannot follow safety:** one-on-one safety drill until the strap/unplug routine is automatic; a safety breach must be corrected before any reassessment.
- **Cannot write a report:** provide a partially completed report to finish; then progress to writing independently against the template.
- **Cannot explain the work (viva):** practise narrating reasoning during simple tasks; use "what did you do, and why?" questioning until it is natural.

O.4 Differentiation

Level 1 â€" Supported:

- Allow extra time for the capstone.
- Provide a structured step-by-step prompt sheet and gentle time reminders.
- Offer one-on-one coaching **outside** the assessment, and a retake after practice.
- Choose the simplest simulated fault (for example, loose RAM) for the first attempt.

Level 2 â€" Independent:

- Complete the capstone, knowledge check and reflection as written.

Level 3 â€" Challenge:

- Assign a machine with **two simultaneous faults** (for example, dusty overheating **plus** a failing drive), or a fault requiring an **OS installation** as part of the repair.
- Require a longer service report with a customer-facing summary paragraph.
- Ask higher-order viva questions (for example, "How would you prevent this fault recurring?").

O.5 Timing guidance (105 minutes)

Segment	Minutes
Opening + brief	6
Prerequisite review	7
Capstone: intake	5
Capstone: diagnose	20
Capstone: repair	25

Segment	Minutes
Capstone: verify	10
Capstone: report	10
Viva/demo	5 (can run in parallel slots)
Reflection, knowledge check and certificate sign-off	~12â€"17 (may shorten if needed)

Protect the **diagnose, repair and verify** time above all â€" those carry the most marks. The knowledge check can be completed during certificate sign-off or as a take-home if time runs short, but the capstone must be observed live.

CERTIFICATE SIGN-OFF (Instructor-only administrative record)

Learner Name: _____ **Date:** _____ **Capstone Result:** Pass / Not Yet Competent **Capstone Rubric Score:** _____ / 30 **Viva/Demo:** _____ / 5 **Total:** _____ / 35 **Course Overall (knowledge checks + assignments + capstone):** _____ % **Fault Diagnosed:** _____ **Repair Performed:** _____ **Service Report:** Complete / Incomplete **Presentation:** Confident / Developing **Safety Adherence:** Met / Concern noted

Instructor Signature: _____ **Certificate Awarded:** Yes / No

Comments:

Congratulations! You have completed the Computer Repairs course at Cyber Elias Academy.

You have gone **From Zero to Expert, Together** â€" from a beginner who had never opened a computer case to a competent, safe, honest entry-level repair technician. Carry the same care to every machine and every customer. Keep learning. The next level is waiting.

Need help? Contact your CEA instructor or the academy front desk on Ebony Road, Rumuola, Port Harcourt. Certificate verification is available at cea.ng/certificates/verify.

END OF SESSION 12 â€" END OF COURSE

Next steps: choose your next CEA course â€" Cybersecurity, Web Design, Business & Freelancing, or Content Creation.